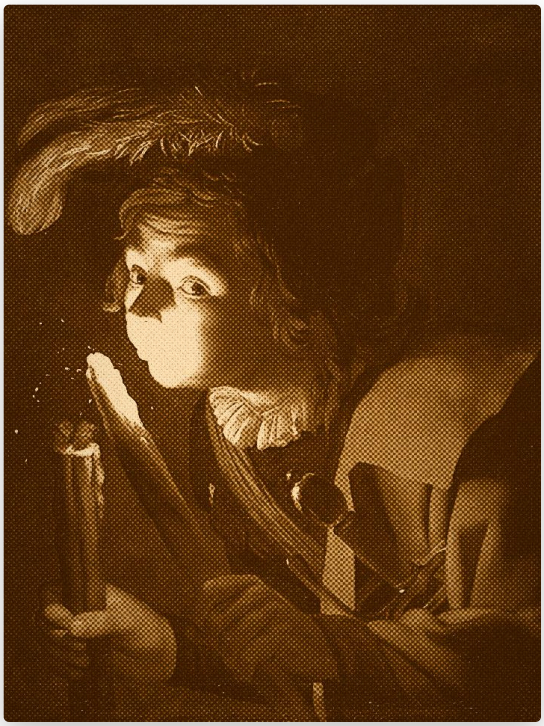
FORM S-1 · REGISTRATION STATEMENT

Ambiq Micro Inc

IPO priced 2025-07-31 · $24.00 a share · AMBQ



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As filed with the Securities and Exchange Commission on July 3, 2025.

REGISTRATION NO. 333-

UNITED STATES

SECURITIES AND EXCHANGE COMMISSION

WASHINGTON, D.C. 20549

FORM S-1

REGISTRATION STATEMENT

UNDER

THE SECURITIES ACT OF 1933

Ambiq Micro, Inc.

(EXACT NAME OF REGISTRANT AS SPECIFIED IN ITS CHARTER)

Delaware	3674	27-1911389
(State or other jurisdiction of incorporation or organization)	(Primary Standard Industrial Classification Code Number)	(I.R.S. Employer Identification Number)

6500 RIVER PLACE BLVD., BUILDING 7

SUITE 200 AUSTIN, TEXAS 78730

(512) 879-2850

(Address, including zip code, and telephone number, including area code, of Registrant's principal executive offices)

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Approximate date of commencement of proposed sale to the public: As soon as practicable after this registration statement becomes effective.

If any of the securities being registered on this Form are to be offered on a delayed or continuous basis pursuant to Rule 415 under the Securities Act, check the following box: ☐

If this Form is filed to register additional securities for an offering pursuant to Rule 462(b) under the Securities Act, please check the following box and list the Securities Act registration statement number of the earlier effective registration statement for the same offering. ☐

If this Form is a post-effective amendment filed pursuant to Rule 462(c) under the Securities Act, check the following box and list the Securities Act registration statement number of the earlier effective registration statement for the same offering. ☐

If this Form is a post-effective amendment filed pursuant to Rule 462(d) under the Securities Act, check the following box and list the Securities Act registration statement number of the earlier effective registration statement for the same offering. ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☐	Accelerated filer	☐
Non-accelerated filer	☒	Smaller reporting company	☒
		Emerging growth company	☒

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 7(a)(2)(B) of the Securities Act. ☐

The Registrant hereby amends this Registration Statement on such date or dates as may be necessary to delay its effective date until the Registrant shall file a further amendment which specifically states that this Registration Statement shall thereafter become effective in accordance with the provisions of Section 8(a) of the Securities Act of 1933 or until the Registration Statement shall become effective on such date as the Commission, acting pursuant to said Section 8(a), may determine.

Table of Contents

The information in this preliminary prospectus is not complete and may be changed. We may not sell these securities until the registration statement filed with the Securities and Exchange Commission is effective. This preliminary prospectus is not an offer to sell these securities and it is not soliciting an offer to buy these securities in any state where the offer or sale is not permitted.

PRELIMINARY PROSPECTUS

(SUBJECT TO COMPLETION, DATED , 2025)

SHARES

COMMON STOCK

This is the initial public offering of shares of common stock of Ambiq Micro, Inc. We are offering shares of our common stock. No public market currently exists for our shares. We anticipate that the initial public offering price will be between $ and $ per share.

We intend to apply to have our common stock listed on the New York Stock Exchange under the symbol "AMBQ."

We are an "emerging growth company," as defined under the federal securities laws and are subject to reduced public company reporting requirements.

Investing in our common stock involves risks. See "Risk Factors" beginning on page 15.

	Per Share	Total
Initial public offering price		$
Underwriting discounts and commissions[(1)]		$
Proceeds to Ambiq Micro, Inc. before expenses		$

(1) See "Underwriting" for a description of the compensation payable to the underwriters.

We have granted the underwriters the right to purchase up to additional shares of common stock from us, solely to cover over-allotments, if any.

Neither the Securities and Exchange Commission nor any state securities commission has approved or disapproved of these securities or determined if this prospectus is truthful or complete. Any representation to the contrary is a criminal offense.

The underwriters expect to deliver the shares of common stock to purchasers on or about , 2025.

BofA Securities	UBS Investment Bank
NEEDHAM & COMPANY	STIFEL

, 2025

Table of Contents

Table of Contents

	Page
PROSPECTUS SUMMARY	1
THE OFFERING	10
SUMMARY CONSOLIDATED FINANCIAL DATA	12
RISK FACTORS	15
SPECIAL NOTE REGARDING FORWARD-LOOKING STATEMENTS	55
INDUSTRY AND MARKET DATA	57
USE OF PROCEEDS	59
DIVIDEND POLICY	60
CAPITALIZATION	61
DILUTION	63
MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS	66
BUSINESS	81
MANAGEMENT	99

	Page
EXECUTIVE COMPENSATION	108
CERTAIN RELATIONSHIPS AND RELATED PARTY TRANSACTIONS	120
PRINCIPAL STOCKHOLDERS	124
DESCRIPTION OF CAPITAL STOCK	125
SHARES ELIGIBLE FOR FUTURE SALE	131
CERTAIN MATERIAL U.S. FEDERAL INCOME TAX CONSEQUENCES TO NON-U.S. HOLDERS OF OUR COMMON STOCK	133
UNDERWRITING	137
LEGAL MATTERS	147
EXPERTS	147
WHERE YOU CAN FIND ADDITIONAL INFORMATION	148
INDEX TO CONSOLIDATED FINANCIAL STATEMENTS	F-1

Neither we nor the underwriters have authorized anyone to provide any information or to make any representations other than those contained in this prospectus or in any free writing prospectuses we have prepared. Neither we nor the underwriters take any responsibility for, or can provide any assurance as to the reliability of, any other information that others may give you. We are not offering to sell, or seeking offers to buy, shares of our common stock in any jurisdiction where these offers and sales are not permitted. The information in this prospectus is accurate only as of the date of this prospectus, regardless of the time of delivery of this prospectus or any sale of shares of our common stock. Our business, financial condition and results of operations may have and are likely to have changed since that date.

Through and including , 2025 (the 25th day after the date of this prospectus), all dealers effecting transactions in these securities, whether or not participating in this offering, may be required to deliver a prospectus. This delivery requirement is in addition to the obligation of dealers to deliver a prospectus when acting as underwriters and with respect to their unsold allotments or subscriptions.

For investors outside the United States: Neither we nor the underwriters have done anything that would permit this offering or possession or distribution of this prospectus or any free writing prospectus in connection with this offering in any jurisdiction where action for that purpose is required, other than in the United States. Persons outside the United States who come into possession of this prospectus must inform themselves about, and observe any restrictions relating to, the offering of the shares of our common stock and the distribution of this prospectus outside of the United States. See the section titled "Underwriting" for additional information.

Table of Contents

Prospectus Summary

This summary highlights selected information that is presented in greater detail elsewhere in this prospectus. This summary does not contain all of the information you should consider before investing in our common stock. You should read this entire prospectus carefully, including the sections titled "Risk Factors" and "Management's Discussion and Analysis of Financial Condition and Results of Operations" and our consolidated financial statements and the related notes included elsewhere in this prospectus, before making an investment decision. Please see "Industry and Market Data" for information concerning certain market statistics included in this prospectus. This summary contains forward-looking statements that involve risks and uncertainties. Unless the context otherwise requires, the terms "Ambiq," "the Company," "we," "us," and "our" in this prospectus refer to Ambiq Micro, Inc. and its consolidated subsidiaries.

Our Mission

Our mission is to enable intelligence (artificial intelligence (AI) and beyond) everywhere by delivering the lowest power semiconductor solutions.

Overview

We are a pioneer and leading provider of ultra-low power semiconductor solutions designed to address the significant power consumption challenges of general purpose and AI compute – especially at the edge.

Our customers rely on Ambiq to deliver AI compute closer to end users (edge environments) where power consumption challenges are the most severe. Our leading position is built upon our hardware and software innovations that deliver two to five times lower power consumption than traditional semiconductor designs. Our products power over 270 million devices today. We shipped more than 42 million units in 2024, and we estimate that over 40% of them ran AI algorithms. We seek to drive growth in AI adoption at the edge in the personal devices, medical/healthcare, industrial edge, and smart home and building markets and continue to set new standards in edge AI performance and power efficiency. Over time, we expect to integrate our ultra-low power technology into additional chip products that benefit from greater power efficiency, including high-performance compute applications such as AI data centers and automotive.

AI is perhaps the most disruptive and revolutionary technology trend of recent history, estimated to represent $23 trillion of global annual spend by 2040, according to McKinsey. AI use cases continue to permeate our lives and improve our daily productivity by enabling us to interact with devices via voice and gestures, unlock our homes with facial recognition, track health accurately and intelligently, and hold clear calls amidst loud background noise.

To date, a majority of AI compute has been deployed in data centers due to its large physical scale and the need for wall plug energy, as AI compute requires enormous and steady energy resources. At the edge, however, power limitations have been especially acute due to small device size and limited battery life. We believe this greatly constrains the potential of AI to improve our daily, on-the-go lives. Enabling AI at the edge – where the action takes place – with vastly improved power efficiency will allow faster real-time decision-making due to data proximity, greater data privacy, higher energy efficiency from reduced network usage, and less dependence on constant costly connections to the cloud. We believe new AI use cases will only be possible if edge devices are much more power efficient.

Our proprietary Sub-threshold Power Optimized Technology (SPOT) platform is designed to fundamentally and cost-effectively reduce power consumption of battery- and wireline-powered devices alike. Depending on the application, devices incorporating SPOT demonstrate a two to five times reduction in power consumption

compared to conventional integrated circuit designs. SPOT is a ground-breaking approach at the chip design level that incorporates sub- and near-threshold hardware without using expensive manufacturing processes.

We provide a full-stack solution encompassing tightly integrated hardware and software. Our solutions include a diverse family of systems-on-chip (SoCs) and the software required to enable on-chip AI processing, general compute, sensing, security, storage, wireless connectivity, and advanced graphics. Our SoC solutions deliver compute at a very small fraction of the power consumed by our competitors' products.

Our ultra-low power SoCs serve a wide range of markets requiring on-device and real-time AI, including smartwatches and fitness trackers, augmented and virtual reality (AR/VR) glasses, smart rings, digital health monitors, security systems and access control, livestock tracking, crop monitoring, and factory automation. These devices increasingly offer on-chip AI-powered features such as speech recognition, domain-specific language models, image and video processing, and sensing, further straining power consumption, which our solutions are positioned to address.

As global demand for our SoC solutions accelerates, our sales and marketing efforts are increasingly focused on our end customers in target geographies such as the United States, Europe, and Asia (ex-Mainland China).

For the three months ended March 31, 2025 and 2024, we generated net sales of $15.7 million and $15.2 million, respectively. For the years ended December 31, 2024 and 2023, we generated net sales of $76.1 million and $65.5 million, respectively. Net sales from end customers in the United States, Europe, and Asia (outside of Mainland China) grew to $14.8 million in the three months ended March 31, 2025, a 94% increase as compared to the three months ended March 31, 2024, achieving a gross margin of 53%. Net sales from end customers in the United States, Europe, and Asia (outside of Mainland China) grew to $38.1 million in 2024, a 70% increase as compared to 2023, achieving a gross margin of 48%. We intend to continue to focus on business outside of Mainland China.

Our Industry

The State of AI Today

Since the launch of ChatGPT in 2022 by OpenAI, companies like Google and Anthropic, among others, have launched competing generative AI models underpinned by large language models (LLMs) that now serve as the foundation for an emerging class of AI-based products and services emerging as a disruption to a wide array of markets from internet search to healthcare. As with other technologies of the digital era, the speed of these advancements outpaces the rate of commercial and consumer adoption while enterprises determine if and how to leverage AI-based solutions and consumers determine whether they will adopt such solutions. Companies developing these new disruptive AI-based products and services face significant technology, financial and regulatory challenges.

The processing of AI workloads requires considerable power, and the power consumption of the servers and chips that run generative AI systems has emerged as a primary technology challenge. The capacity of a data center to fulfill generative AI queries is limited by the amount of power that can be delivered by local power plants and the cost of such power, among other things. The cost of a query and the financial viability of generative AI-based products and services are therefore limited by the availability of affordable power. As a result, improving power consumption has become a central concern for the AI industry, and is even more pronounced at the edge.

Furthermore, the rapid pace of innovation in the field of AI has led to developing and evolving regulatory frameworks globally, which are expected to become increasingly complex as AI continues to progress. Regulators and lawmakers around the world have proposed and adopted regulations and guidance addressing the use of AI. The shifting regulatory landscape for AI and the products that utilize it has created uncertainty for

companies developing AI-based solutions and could pose an impediment to near-term broad-based adoption. See "Risk Factors—Risks Related to Our Business" and "Business—Government Regulation" for a more detailed discussion of challenges associated with AI regulation.

The Growing Need for AI at the Edge

Edge AI is expanding as machine learning outperforms traditional algorithms, enabling tasks like speech recognition, and meets customer demand for cloud-independent solutions. Meanwhile, LLMs in the cloud rely on quality edge inputs, which is evident in industrial automation, healthcare, and consumer devices. To fully realize AI's potential, computation must move to the edge where data originates, enabling faster decisions, enhanced privacy, improved energy efficiency, and reduced cloud reliance.

However, realizing this vision requires overcoming a fundamental challenge: **power efficiency.**

Key Challenges of Edge-to-Cloud Data Processing

Today, advanced analytics and AI inference are often processed in the cloud due to the difficulty or impossibility of running them at the edge. However, this approach has several drawbacks:

- ***Latency.*** The round trip to the cloud adds significant latency, preventing real-time behaviors and feedback.
- ***Connectivity.*** Wireless connectivity often consumes significant power and is limited to those applications with reliable connections.
- ***Security and Privacy.*** Sending sensitive data to the cloud increases privacy breach risks.
- ***Cost.*** Cloud services represent a significant expense that usually scales with the amount of data transferred to/from the cloud.

Key Challenges to AI Inferencing at the Edge

- ***Power Consumption.*** Optimizing power consumption for all functions, from AI inferencing to wireless connectivity to sensor management, is a fundamental challenge for edge devices. This challenge is significantly amplified as processing, performance, connectivity, and AI compute needs increase.
- ***Real-Time Inferencing.*** Most of today's AI models are designed to run in the cloud, processing large batches of data. Edge AI is often called on to act on streams of data in real time, which requires a significant redesign of the data processing and inferencing algorithms.
- ***Size.*** Many edge devices are physically size-constrained, thereby limiting the capacity of the battery and battery life.
- ***Integration.*** Given device size constraints, vendors increasingly opt for highly integrated SoCs to drive their devices instead of complex designs with many discrete chips having relatively narrow functionalities.
- ***Privacy and Security.*** While bringing AI to the edge improves system security and reduces the attack surface, it creates high-security expectations for small edge devices with minimal compute resources.
- ***Resource Constrained AI Model Development.*** Most AI models today are built for the cloud, where compute performance, memory, and power are readily available. Building "right-sized" models that execute efficiently on the edge is extremely challenging.

Our ultra-low power SoC solutions are designed to fundamentally address our end customers' needs for ultra-low-power consumption for all functions, better connectivity, and increased AI compute (especially at the edge).

Table of Contents

Our Technology

SPOT® is our proprietary ultra-low power chip design platform created to solve this severe power consumption problem. SPOT consists of a set of chip design techniques that allow standard transistors to operate in an ultra-low power mode called "sub-threshold" and "near-threshold". When combined with other low-power chip architecture techniques, SPOT enables our SoCs to deliver two to five times lower power consumption than leading alternative solutions.

By significantly reducing the power required for sensing, communication, security, and AI compute, customers are given flexibility on how best to deploy the vastly improved energy budgets of their products. They can use this flexibility to run larger AI models faster, run faster inference rates, add more sensors, reduce battery size, and/or extend battery life.

Specifically, we have developed wide-ranging chip architectures, intellectual property (IP) building blocks, and integrated circuit (IC) design methodologies for digital circuits, analog circuits, and embedded memories – accumulating significant expertise and IP related to product design, validation, and production testing. Our proven expertise in, and experience with, ultra-low power design provides us with a substantial competitive advantage.

In addition, we have developed a wide range of chip architecture techniques that are designed to enable even lower power than sub-threshold and near-threshold operation alone. Many of these chip design techniques are protected by IP rights, including patents and trade secrets.

While many high-end processor chip companies can increase performance or decrease power consumption with advanced process technology nodes, these expensive manufacturing processes are typically not cost-effective for edge devices. With our SPOT platform, we are able to manufacture our products at much more cost-effective process technology nodes. Furthermore, reliance on advanced process technology nodes and Moore's Law to solve the power problem is particularly limiting as Moore's Law is viewed by many as dead or broken.

For a more detailed description of our technology, see "Business—Our Technology".

Our Products

Our SPOT platform serves as the foundation for a series of SoCs for edge AI devices that deliver ultra-low power for AI computations, general purpose computations, sensing, communications, power conversion, and more. These SoCs are paired with a full-stack software solution that is designed to enable efficient resource-constrained AI application development and a fast time-to-market.

SoC Products for Edge AI

Our portfolio includes two flagship SoC product families, both with rich sets of peripherals:

- ***Apollo:*** The Apollo family of products pair rich peripheral sets with host processors capable of software-based AI computations (e.g., Apollo3 and Apollo4 families) or vector-accelerated AI computations (e.g., Apollo5 family). For most edge AI use cases, we believe the combination of the Apollo products with our full stack software solution provides the optimal power, performance, feature, and cost trade-off.

- ***Atomiq:*** The first Atomiq family product is currently in development. This novel product targets AI applications requiring demanding edge AI compute requirements and is thus expected to provide the highest performance and lowest power ever delivered by our products. Atomiq is expected to feature a full neural processing unit (NPU) for high-performance AI acceleration along with new memory innovations – all with the goal of achieving minimum power and maximum performance on AI model execution at the edge.

Software Products for Edge AI

To truly unlock edge AI use cases, our ultra-low power SoCs must be paired with carefully optimized proprietary software. To make edge AI application development easier for our customers, Ambiq provides a series of core software products:

- ***General enablement firmware:*** Our AmbiqSuite Software Development Kit (SDK) provides low-level firmware and drivers required to operate our SoCs efficiently.

- ***Feature enablement software:*** On top of AmbiqSuite SDK, we offer a series of software modules that enable complex applications to be developed quickly and efficiently. These modules include graphiqSPOT for graphics and display management, blueSPOT for Bluetooth communications, secureSPOT for managing security, and more.

- ***AI enablement software:*** Ambiq assists our customers in developing AI features by providing novel model factories specifically designed for edge AI, optimized AI inference engines, and neuralSPOT, our cross-platform AI SDK. These tools are designed to collectively address the challenges encountered when developing AI for edge products.

For a more detailed description of our products, see “Business—Our Products”.

Our Market Opportunity

Most of our revenue today has come from ultra-low power SoCs, tightly bundled with software and various other solutions that combine 32-bit microcontrollers (MCUs) with wireless connectivity and additional circuitry, such as graphics processing units, serial interfaces, and analog-to-digital interfaces. In the near future, we believe we can expand into embedded application processors, as well as stand-alone dedicated AI processors. Even though many of our customers today run basic AI inferencing directly on our Apollo-class SoCs and on lightweight accelerators within our SoCs, we are now developing dedicated AI processor units purpose-built for accelerating AI compute. Today, we define our market opportunity as 32-bit MCUs, discrete application processors (ASICs), wireless connectivity, and AI processors built for edge applications, markets totaling $12.8 billion in 2023 and growing to $22.5 billion in 2028, according to Gartner.** This includes AI-enabled devices spanning industrial edge, smart homes and buildings, as well as body-worn AI devices such as consumer wearables and digital health devices. In the future, we believe our technology has the potential to address many semiconductor end markets that need much greater energy efficiency.

Within the edge end markets listed above, the market for 32-bit MCUs is expected to grow from $4.7 billion in 2023 to $6.0 billion in 2028, representing a compound annual growth rate (CAGR) of 5%, according to Gartner.** The market for wireless connectivity semiconductors for the same end markets is expected to grow from $2.4 billion in 2023 to $4.7 billion in 2028, representing a CAGR of 15%. The market for application processors is expected to grow from $5.7 billion in 2023 to $10.2 billion in 2028, representing a CAGR of 12%. According to Gartner, the discrete AI processors market is expected to grow from $152 million in 2023 to $1.6 billion in 2028, representing a CAGR of 60%.**

Since AI solutions are often embedded into general processor SoCs, broader AI-enabled market forecasts are also relevant to our opportunity. According to Gartner, the market for AI-enabled 32-bit MCUs and application processors in edge applications is expected to grow from $788 million in 2023 to $6.8 billion in 2028, representing a CAGR of 54%.*

Our Competitive Strengths

We believe our core competitive strengths include the following:

Leading proprietary ultra-low power chip design technology. We exclusively focus on ultra-low power, sub- and near-threshold chip design technology, which is our key differentiator and top development priority. Our SPOT platform comprises dozens of individual circuit design techniques that are difficult to replicate, complemented by overall low-power system design expertise and know-how and our proprietary AmbiqSuite SDK, making it easy for our end customers to quickly and effectively implement our products.

Extensible technology platform. We believe our SPOT platform is fully extensible to a wide variety of semiconductor applications that require substantially greater power efficiency in both battery-operated and wall-powered devices. Our technology is especially valuable for battery-operated products by significantly improving performance on a fixed power budget or by extending the time between charges in edge devices.

Robust intellectual property portfolio. Our IP is a key aspect of our business strategy, allowing us to maintain a competitive and, we believe, sustainable edge in the market. Our robust IP portfolio encompasses patents, trade secrets, and design IP that safeguard our proprietary technology and innovations related to ultra-low power and sub-threshold design. Our design IP has been proven over five generations of the Apollo family of products.

Proven demand from blue-chip end customers. We have demonstrated strong end customer growth with more than 270 million units shipped with international technology leaders such as Garmin, Google, Suunto, and others, which has validated the maturity of our technology platform and the robustness of our products.

Scalable design and manufacturing relationships. Our SoC solutions leverage existing mature semiconductor process technologies that are readily available and cost-effective. Since our founding, we have partnered with a leading semiconductor foundry, Taiwan Semiconductor Manufacturing Company Limited (TSMC), for our SoC products in 180-nanometer, 90-nanometer, 40-nanometer, and 22-nanometer process nodes. We leverage TSMC's mature fabrication processes, which are more cost-effective than the most advanced nodes, while still delivering products that consume a very small fraction of the power that our competitors require for their comparable parts.

Top industry talent, experienced management team and engineering-focused culture. We believe our team's engineering, chip design, and AI talent is critical to our success. We employ an engineering-focused workforce, as well as a highly technical senior management team with deep industry and chip design experience.

Our Strategy

Extend our presence in current markets with higher-performing and lower-power versions of our existing products. New members of the Apollo and Atomiq family of SoCs in development are designed to deliver even lower power and higher processing performance to our existing customers, potentially enabling and accelerating new innovative edge AI use cases.

Expand aggressively into new markets by broadening current product families. We intend to apply our ultra-low power value proposition to enable AI use cases in medical, digital health, industrial, security, smart home and buildings, robotics, and automotive markets. We are developing a broader portfolio of Apollo and Atomiq SoC offerings that we believe will allow us to enter these additional markets.

Bring SPOT to new classes of chip products to access both existing and new markets. SPOT is a widely applicable, ultra-low power design methodology that can be quickly applied to new chip classes such as

application processors (APs), dedicated AI processors, digital signal processors (DSPs), image processors, power management chips, communications chips, network processors, and many more. We intend to develop new ultra-lower power chip product families that we believe will provide us access to additional sockets within current customers, as well as to entirely new customers in new market segments.

Leverage our easy-to-use software and AI model suite to speed customer adoption and drive higher-margin revenue. We are working to increase our software suite to include new AI software products, such as AI model libraries, AI compilers and runtimes, AI model development utilities, AI model factories and AI training data. Complementary to our existing SoC solutions, we believe these new AI software offerings will enable us to reach a broader set of customers, will allow these customers to more quickly implement their AI functionalities, and are expected to help drive increased SoC gross margins.

Develop SPOT into a licensable technology platform. We plan to develop SPOT into an IP and chip development platform that makes it easier for other companies to license or partner with us to incorporate SPOT into their own low-power chip designs, such as building next-generation AI data center chips.

Relentlessly advance the SPOT platform to help position us for future leadership. New AI use cases enabled by novel sensors drive a growing need for ultra-low power AI chips. We plan to continue to advance the SPOT platform with innovations in circuit hardware and software.

Summary of Risk Factors

Investing in our common stock involves substantial risk. The risks described under the section titled "Risk Factors" immediately following this prospectus summary may cause us to not realize the full benefits of our objectives or may cause us to be unable to successfully execute all or part of our strategy. Some of the more significant challenges include the following:

- We have a history of net losses, and we may not achieve or maintain profitability in the future.
- We currently depend on a limited number of end customers for most of our revenue. The loss of, or a significant reduction in orders from, our key end customers would significantly reduce our revenue and adversely impact our business, financial condition, and results of operations.
- We do not have long-term commitments from our end customers and our end customers may cease purchasing our products at any time.
- The nature of the design win process requires us to incur expenses without any guarantee that research and development efforts will generate material revenue, which could adversely affect our business, financial condition and results of operations.
- We continue to invest in research and development efforts for several new markets. If we are unable to commercialize these technologies, our business, financial condition and results of operations could be negatively affected.
- We are subject to the cyclical nature of the semiconductor industry.
- Some of our end customers and other third-party agreements provide for joint or custom product development, which subjects us to a number of risks, and any failure to execute on any of these arrangements could have a material adverse effect on our business, financial condition, and results of operations.
- The AI industry is subject to complex, evolving regulatory, statutory, and other requirements that may be difficult and expensive to comply with and that could negatively impact our business.

- The market for edge AI services and products is relatively new, and may decline or experience limited growth, and our business is dependent on our clients' continuing adoption and use of our services and products.
- We have no manufacturing capabilities of our own. We are a fabless company, meaning that we do not own a semiconductor foundry, and we rely on a single third-party supplier for the fabrication of semiconductor wafers and on a limited number of suppliers of other materials, and the failure of any of our suppliers to provide us with wafers and other materials on a timely basis would harm our business, financial condition, and results of operations.
- Our dependence on TSMC as our sole supplier of wafers exposes us to certain political, social, and economic risks that may harm our business.
- Raw material and engineered material availability and price fluctuations have in the past and may in the future increase the cost of our solutions, impact our ability to meet customer commitments, and may adversely affect our business, financial condition, and results of operations.
- Our failure to comply with the Outbound Investment Security Program could subject us to penalties and other adverse consequences.
- Changes in U.S. or foreign trade policies, including the imposition of tariffs, and other factors beyond our control may adversely impact our business and operating results.
- We are a global company, which subjects us to additional business risks, including logistical and financial complexity, political instability and currency fluctuations.
- We identified material weaknesses in our internal control over financial reporting, and we may experience additional material weaknesses or otherwise fail to design and maintain effective internal control over financial reporting.
- If we are unable to obtain, maintain and enforce patent protection for our current and future proprietary technology and inventions, or if the scope of the patent protection obtained is not sufficiently broad, our ability to compete successfully and our business, financial condition, and results of operations could be adversely impacted.

Implications of Being an Emerging Growth Company and Smaller Reporting Company

We are an "emerging growth company," as defined in the Jumpstart Our Business Startups Act (the JOBS Act), enacted in April 2012. We intend to take advantage of certain exemptions under the JOBS Act from various public company reporting requirements, including not being required to have our internal control over financial reporting audited by our independent registered public accounting firm pursuant to Section 404(b) of the Sarbanes-Oxley Act of 2002, as amended (the Sarbanes-Oxley Act), reduced disclosure obligations regarding executive compensation in our periodic reports and proxy statements, and exemptions from the requirements of holding a nonbinding advisory vote on executive compensation and any golden parachute payments not previously approved.

In addition, an emerging growth company can take advantage of an extended transition period for complying with new or revised accounting standards. This provision allows an emerging growth company to delay the adoption of certain accounting standards until those standards would otherwise apply to private companies. We have elected to avail ourselves of this provision of the JOBS Act. As a result, we will not be subject to new or revised accounting standards at the same time as other public companies that are not emerging growth companies. Therefore, our consolidated financial statements may not be comparable to those of companies that comply with new or revised accounting pronouncements as of public company effective dates.

We will remain an emerging growth company and may take advantage of these exemptions until the earliest of: (i) the last day of the fiscal year following the fifth anniversary of the consummation of this offering; (ii) the last day of the fiscal year in which we have total annual gross revenue of at least $1.235 billion; (iii) the last day of the fiscal year in which we are deemed to be a "large accelerated filer" as defined in Rule 12b-2 under the Securities Exchange Act of 1934, as amended (Exchange Act), which would occur if the market value of our common stock held by non-affiliates exceeded $700.0 million as of the last business day of the second fiscal quarter of such year; or (iv) the date on which we have issued more than $1.0 billion in non-convertible debt securities during the prior three-year period.

We are also a "smaller reporting company" as defined in the Exchange Act. We may continue to be a smaller reporting company even after we are no longer an emerging growth company. We may take advantage of certain of the scaled disclosures available to smaller reporting companies and will be able to take advantage of these scaled disclosures for so long as our voting and non-voting common stock held by non-affiliates is less than $250.0 million measured on the last business day of our second fiscal quarter, or our annual revenue is less than $100.0 million during the most recently completed fiscal year and our voting and non-voting common stock held by non-affiliates is less than $700.0 million measured on the last business day of our second fiscal quarter.

Corporate Information

We were incorporated in Delaware on January 20, 2010 under the name Cubiq Microchip, Inc., and in October 2012, changed our name to Ambiq Micro, Inc. Our principal executive offices are located at 6500 River Place Blvd., Building 7, Suite 200, Austin, Texas 78730, and our telephone number is (512) 879-2850. Our corporate website address is www.ambiq.com. Information contained on or accessible through our website is not a part of this prospectus and should not be relied on in determining whether to make an investment decision. The inclusion of our website address in this prospectus is an inactive textual reference only.

Trademarks and Service Marks

We use the Ambiq logo and other marks as trademarks in the United States and other countries. The Ambiq word mark and logos, SPOT, graphiqSPOT, neuralSPOT, turboSPOT, secureSPOT, Voice-on-SPOT, and VoS, among others, are registered trademarks of Ambiq Micro, Inc. Other trademarks and trade names are those of their respective owners. This prospectus contains references to our trademarks and service marks and to those belonging to other entities. Solely for convenience, the trademarks and trade names referred to in this prospectus, including logos, artwork and other visual displays, may appear without the ®, ™ or SM symbols, but such references are not intended to indicate in any way that we will not assert, to the fullest extent under applicable law, our rights or the rights of the applicable licensor to these trademarks and trade names. We do not intend our use or display of other entities' trade names, trademarks or service marks to imply a relationship with, or endorsement or sponsorship of us by, any other entity.

The Offering

Common stock offered by us	shares
Option to purchase additional shares	shares
Common stock to be outstanding after this offering	shares (shares if the underwriters exercise their over-allotment option in full)
Use of proceeds	We estimate that the net proceeds to us from this offering will be approximately $ million (or $ million if the underwriters exercise their over-allotment option in full), based on an assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus), and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.
	We currently intend to use the net proceeds from this offering primarily for general corporate purposes, including working capital, sales and marketing activities, product development, general and administrative matters, and capital expenditures, although we do not currently have any specific or preliminary plans with respect to the use of proceeds for such purposes. We also may use a portion of the net proceeds to acquire complementary businesses, products, services, or technologies. However, we do not have agreements or commitments for any specific acquisitions at this time. See "Use of Proceeds."
Dividend policy	We do not anticipate declaring or paying, in the foreseeable future, any cash dividends on our capital stock. See "Dividend Policy."
Risk factors	You should read "Risk Factors" for a discussion of certain of the factors to consider carefully before deciding to purchase any shares of our common stock.
Listing	We intend to apply to list our common stock on the New York Stock Exchange under the symbol "AMBQ."

Unless otherwise indicated, the number of shares of our common stock to be outstanding immediately after this offering is based on shares of common stock outstanding as of March 31, 2025 on a pro forma basis, after giving effect to the adjustments set forth in footnote (2) to the consolidated statement of operations data in "—Summary Consolidated Financial Data" below, and excludes:

- up to 5,186,932 shares of our common stock issuable upon the vesting of restricted stock units (RSUs) outstanding under our 2020 Equity Incentive Plan (the 2020 Plan) as of March 31, 2025;
- 61,795,893 shares of our common stock issuable upon the exercise of outstanding stock options under our 2010 Equity Incentive Plan (the 2010 Plan) and the 2020 Plan as of March 31, 2025, with a weighted average exercise price of $0.38 per share;
- shares of our common stock issuable upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are not being exercised in connection with this offering;

Table of Contents

- shares of our common stock reserved for future issuance under our 2025 Equity Incentive Plan (the 2025 Plan), which will become effective as of immediately prior to the completion of this offering, as well as any automatic increases in the number of shares of common stock reserved for future issuance under the 2025 Plan; and
- shares of our common stock reserved for future issuance under our 2025 Employee Stock Purchase Plan (the ESPP), which will become effective as of immediately prior to the completion of this offering, as well as any automatic increases in the number of shares of common stock reserved for future issuance under the ESPP.

Unless otherwise indicated, all information contained in this prospectus assumes or gives effect to:

- the automatic conversion of all 341,496,158 shares of our outstanding redeemable convertible preferred stock into an aggregate of 356,423,629 shares of our common stock, which will occur upon the closing of this offering;
- no exercise of the outstanding options or warrants described above;
- the issuance of shares of our common stock upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are being exercised in connection with this offering;

- the filing of our amended and restated certificate of incorporation and the adoption of our amended and restated bylaws upon the closing of this offering;
- no exercise by the underwriters of their over-allotment option to purchase up to additional shares of our common stock from us to cover over-allotments, if any; and
- a -for- reverse stock split of our outstanding common stock to be effected prior to the closing of this offering.

Summary Consolidated Financial Data

The summary consolidated statements of operations data presented below for the years ended December 31, 2024 and 2023, and the consolidated balance sheet data as of December 31, 2024, are derived from our audited consolidated financial statements included elsewhere in this prospectus. The unaudited summary consolidated statements of operations data presented below for the three months ended March 31, 2025 and 2024, and the summary condensed consolidated balance sheet data as of March 31, 2025, are derived from our unaudited condensed consolidated financial statements included elsewhere in this prospectus. Our unaudited interim condensed consolidated financial statements have been prepared on a basis consistent with our audited consolidated financial statements and, in the opinion of management, reflect all adjustments, consisting only of normal recurring adjustments, necessary for a fair presentation of the financial information set forth in those statements. The following summary consolidated financial data should be read together with "Management's Discussion and Analysis of Financial Condition and Results of Operations" and our consolidated financial statements and related notes included elsewhere in this prospectus. Our historical results are not necessarily indicative of the results that may be expected for any period in the future, and our results for the three months ended March 31, 2025 may not be indicative of results to be expected for the full year or any other period.

	Year ended December 31,		Three months ended March 31,	
	2024	2023	2025	2024
	(in thousands, except share and per share data)			
			(unaudited)	
Consolidated Statements of Operations Data:				
Net sales	$76,067	$65,537	$15,732	$15,210
Cost of sales	51,776	46,096	7,343	8,976
Gross profit	24,291	19,441	8,389	6,234
Operating expenses:				
Research and development	37,168	44,020	8,687	9,436
Sales, general and administrative	27,736	26,743	8,443	5,988
Total operating expenses	64,904	70,763	17,130	15,424
Loss from operations	(40,613)	(51,322)	(8,741)	(9,190)
Other income, net	980	1,020	461	(204)
Loss before income taxes	(39,633)	(50,302)	(8,280)	(9,394)
Provision for income taxes	28	30	4	5
Net loss	$(39,661)	$(50,332)	$(8,284)	$(9,399)
Deemed dividend	(2,724)	(1,941)	—	(375)
Net loss attributable to common stockholders	$(42,385)	$(52,273)	$(8,284)	$(9,774)
Net loss per share attributable to common stockholders[(1)]:				
Basic and diluted	$(4.06)	$(5.38)	$(0.68)	$(0.97)
Weighted-average common stock outstanding[(1)]:				
Basic and diluted	10,427,915	9,722,970	12,232,916	10,102,901
Pro forma net loss per share attributable to common stockholders (unaudited)[(2)]:				
Basic and diluted				$
Pro forma weighted-average common stock outstanding (unaudited)[(2)]:				
Basic and diluted				
Comprehensive loss:				
Currency translation adjustment	319	(508)	(30)	150
Comprehensive loss:	(39,342)	(50,840)	(8,314)	(9,249)

Deemed dividends	(2,724)	(1,941)	—	(375)
Comprehensive loss attributable to common stockholders	$(42,066)	$(52,781)	$(8,314)	$(9,624)
Non-GAAP Financial Measures:				
Non-GAAP net loss[3]	$(28,607)	$(39,213)	$(5,217)	$(8,180)
Non-GAAP gross profit[3]	$23,897	$22,008	$7,411	$4,878

(1) See Notes 2 and 3 to our financial statements included elsewhere in this prospectus for a description of how we compute basic and diluted net loss per share of common stock and the number of shares used in computing these amounts.

(2) Pro forma net loss per share and the pro forma weighted-average number of shares used in computation of the per share amount for the three months ended March 31, 2025 have been computed to give effect to (i) the automatic conversion of all 341,496,158 shares of our outstanding redeemable convertible preferred stock into an aggregate of 356,423,629 shares of our common stock, which will occur upon the closing of this offering, (ii) the issuance of shares of our common stock upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are being exercised in connection with this offering, and (iii) the filing of our amended and restated certificate of incorporation upon the closing of this offering.

(3) We use non-GAAP net loss and non-GAAP gross profit, both non-GAAP financial measures, to help us make strategic decisions, establish budgets and operational goals for managing our business, analyzing our financial results, and evaluating our performance. We present the non-GAAP financial measures non-GAAP net loss and non-GAAP gross profit in this prospectus because we believe these non-GAAP financial measures provide additional tools for investors to use in comparing our core business and results of operations over multiple periods with other companies in our industry, many of which present similar non-GAAP financial measures to investors. However, our presentation of non-GAAP net loss and non-GAAP gross profit may not be comparable to similarly titled measures reported by other companies due to differences in the way that these measures are calculated. Non-GAAP net loss and non-GAAP gross profit should not be considered as the sole measures of our performance and should not be considered in isolation from, or as a substitute for, net loss and gross profit calculated in accordance with GAAP.

We define non-GAAP net loss as our net loss adjusted to exclude expenses not directly attributable to the performance of our operations, such as income taxes, depreciation and amortization, stock-based compensation expense, gain on nonmonetary transaction, severance costs, initial public offering (IPO) and other transaction costs and warrant valuation. We define non-GAAP gross profit as our gross profit adjusted to exclude expenses not directly attributable to gross profit, such as depreciation and amortization, and stock-based compensation. Refer to the additional discussion under the subsection titled "Management's Discussion and Analysis of Financial Condition and Results of Operations—Non-GAAP Financial Measures."

The following tables reconcile the most directly comparable GAAP financial measure to each of these non-GAAP financial measures.

Non-GAAP Net Loss:

	Year ended December 31,		Three months ended March 31,	
	2024	2023	2025	2024
Net loss	$(39,661)	$(50,332)	$(8,284)	$(9,399)
Add:				
Income taxes	28	30	4	5
Depreciation and amortization	6,246	7,716	1,961	1,557
Stock-based compensation	5,174	2,623	851	551
Gain on nonmonetary transaction	(1,600)	—	(1,600)	(1,600)
Severance costs	706	481	—	706
IPO and other transaction costs	551	850	1,793	—
Warrant valuation	(51)	(581)	58	—
Non-GAAP net loss	$(28,607)	$(39,213)	$(5,217)	$(8,180)

Non-GAAP Gross Profit:

	Year ended December 31,		Three months ended March 31,	
	2024	2023	2025	2024
Gross profit	$24,291	$19,441	$8,389	$6,234
Add:				
Depreciation and amortization	850	2,420	562	214
Stock-based compensation	356	147	60	30
Gain on nonmonetary transaction	(1,600)	—	(1,600)	(1,600)
Non-GAAP gross profit	$23,897	$22,008	$7,411	$4,878

	As of March 31, 2025		
	Actual	Pro Forma[1]	Pro Forma As Adjusted[2][3]
	(in thousands)		
Balance Sheet Data:			
Cash and cash equivalents	$61,418		$$
Working capital[4]	68,421		
Total assets	99,295		
Warrant liability	170		
Total liabilities	20,862		
Redeemable convertible preferred stock	378,150		
Accumulated deficit	(328,534)		
Stockholders' (deficit) equity	(299,717)		

(1) The pro forma consolidated balance sheet data gives effect to the (i) automatic conversion of all 341,496,158 shares of our outstanding redeemable convertible preferred stock into an aggregate of 356,423,629 shares of our common stock, which will occur upon the closing of this offering, and the related reclassification of the carrying value of our redeemable convertible preferred stock to permanent equity upon the closing of this offering, (ii) the issuance of shares of our common stock upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are being exercised in connection with this offering, and (iii)filing and effectiveness of our amended and restated certificate of incorporation upon the closing of this offering.

(2) The pro forma as adjusted consolidated balance sheet data gives effect to (i) the pro forma items described in footnote (1) above, and (ii) the issuance and sale by us of shares of common stock in this offering, assuming an initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus), after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

(3) The pro forma as adjusted consolidated balance sheet data is illustrative only and will change based on the actual initial public offering price and other terms of this offering determined at pricing. Each $1.00 increase (decrease) in the assumed initial public offering price of $ per share would increase (decrease) each of our pro forma as adjusted cash and cash equivalents, working capital, total assets and total stockholders' (deficit) equity by $ million, assuming the number of shares of common stock offered by us, as set forth on the cover page of this prospectus, remains the same, and after deducting underwriting discounts and commissions and estimated offering expenses payable by us. Similarly, each increase (decrease) of 1.0 million shares in the number of shares of common stock offered by us would increase (decrease) each of our pro forma as adjusted cash and cash equivalents, working capital, total assets and total stockholders' (deficit) equity by $ million, assuming the assumed initial public offering price of $ per share remains the same, and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

(4) Working capital is defined as total current assets less total current liabilities. See our consolidated financial statements and related notes included elsewhere in this prospectus for further details regarding our current assets and current liabilities.

Risk Factors

Investing in our common stock involves a high degree of risk. You should carefully consider the risks and uncertainties described below, together with all of the other information in this prospectus, including our audited consolidated financial statements and related notes and "Management's Discussion and Analysis of Financial Condition and Results of Operations," before investing in our common stock. If any of the following risks are realized, in whole or in part, our business, financial condition and results of operations could be materially and adversely affected. In that event, the price of our common stock could decline, and you could lose part or all of your investment. Additional risks and uncertainties not presently known to us or that we currently deem immaterial also may impair our business operation.

Risks Related to Our Business

We have a history of net losses, and we may not achieve or maintain profitability in the future.

We have incurred net losses and negative cash flows from operations since inception and we expect to continue to incur net losses and negative cash flows from operations for the foreseeable future, due in part to our continued investment in our business. We incurred a net loss of $8.3 million and $9.4 million for the three months ended March 31, 2025 and 2024, respectively, and $39.7 million and $50.3 million for the years ended December 31, 2024 and 2023, respectively, and had an accumulated deficit of $328.5 million and $320.3 million as of March 31, 2025 and December 31, 2024, respectively. We expect our costs to increase in future periods as we continue to expend substantial resources on research and development, expansion into new markets, marketing and general administration (including expenses related to being a public company). The net losses we incur may fluctuate significantly from quarter to quarter.

Our long-term success is dependent upon our ability to generate increased revenue, obtain additional capital when needed and, ultimately, to achieve and maintain profitable operations. We will need to generate significant additional revenue and successfully manage our research and development and other expenses to achieve and maintain profitability. It is possible that we will not achieve profitability or that, even if we do achieve profitability, we may not maintain or increase profitability in the future. Our failure to achieve or maintain profitability could negatively impact our stock price.

We depend on a limited number of end customers for most of our revenue. The loss of, or a significant reduction in orders from our key end customers that are not replaced by other orders from new or existing customers, would significantly reduce our revenue and adversely impact our business, financial condition and results of operations.

Our largest end customer historically has accounted for a large portion of our sales, representing approximately 38% of our net sales for the three months ended March 31, 2025. A different end customer accounted for approximately 28% of our net sales for the three months ended March 31, 2024. Two other single end customers directly or indirectly accounted for more than 10% of our net sales in the three months ended March 31, 2025. In addition, our top ten customers accounted for approximately 98% of our total net sales for each of the three months ended March 31, 2025 and 2024. We believe that our operating results for the foreseeable future will continue to depend to a significant extent on sales attributable to certain end customers.

In addition, historically, our sales were significantly concentrated with end customers in Mainland China. For example, in 2023, 66% of our net sales were to end customers in Mainland China, as compared to 50% in 2024. Given geopolitical concerns, subsidized competitors creating a price sensitive environment in Mainland China and our desire to service market segments outside of consumer wearable products, we have focused our management and sales efforts toward other critical geographies. As a result, we expect to continue to experience a substantial shift over the next year in our underlying customer concentration, as we seek to reduce our sales to customers located in Mainland China. However, there can be no assurance that we will be successful in our efforts to replace the revenue that we foresee from our end customers in Mainland China.

While we anticipate revenue attributable to our top end customers will fluctuate from period to period and expect to reduce our concentration of revenue from Mainland China, we expect to remain dependent on a small number of end customers for a meaningful portion of our revenue for the foreseeable future. If our end customers were to choose to reduce their orders or cease to order products from us or if our relationships with our end customers or our distributors are disrupted for any reason and we are unable to replace those orders with orders from new or existing end customers, there could be a significant negative impact on our business. Any reduction in sales attributable to our largest end customers would have a significant and disproportionate impact on our business, financial condition and results of operations.

We do not have long-term commitments from our end customers, and our end customers may cease purchasing our products at any time.

We sell our products to our end customers direct or through distributors and channel partners on a purchase order basis. While we do have cancellation and reschedule terms established with our distribution and channel partners, our end customers do not have any minimum or binding purchase obligations to us under these purchase orders and orders may be cancelled, reduced or rescheduled with little or no notice and without penalty. Cancellations of orders could result in the loss of anticipated sales without allowing us sufficient time to reduce our inventory and operating expenses. In addition, changes in forecasts or the timing of orders from our end customers expose us to the risks of inventory shortages or excess inventory. This in turn could cause our operating results to fluctuate. Certain distributor contracts do include variable consideration, such as limited price protection, return and stock rotation provisions, while direct customer contracts include general right of return provisions. These provisions require us to make judgments based on past experience and other factors. Our judgments may not be correct and our reserves in respect of these provisions may be inadequate resulting in our having to record unanticipated charges in a particular period.

Our end customers, or the distributors or channel partners through which we sell to these end customers, may choose to use solutions in addition to ours, use different solutions altogether, or develop in-house solutions that compete directly with our solutions, which could affect our end customers' future purchasing decisions. In addition, the inability of our end customers or their contract manufacturers to obtain sufficient supplies of third- party components used with our products could result in a decline in the demand for our products and a loss of sales. Any of these events could significantly harm our business, financial condition and results of operations. In addition, if our distributors' relationships with our end customers, including our larger end customers are disrupted due to our inability to deliver products and solutions in sufficient quantities or in a timely manner, or for any other reason, our end customers may cancel their purchase orders at any time, and there could be a significant negative impact on our business, financial condition and results of operations.

Our end customers regularly evaluate alternative suppliers in order to diversify their supplier bases, which increases their negotiating leverage with us and protects their ability to secure similar solutions. We believe that any expansion of our end customers' supplier bases could have an adverse effect on the prices we are able to charge and volume of product that we are able to sell to our end customers, which would negatively affect our business, financial condition and results of operations. For example, sales in Mainland China decreased $5.1 million in 2024 in part due to pricing pressures from local competitors, which resulted in our selling two of our products at substantial discounts in order to liquidate inventories that were manufactured principally for sale in Mainland China. Any reduction in sales attributable to our end customers, and in particular our largest end customers, would have a significant impact on our business, financial condition and results of operations.

The nature of the design win process requires us to incur expenses without any guarantee that research and development efforts will generate material revenue, which could adversely affect our business, financial condition and results of operations.

We focus on winning competitive bid selection processes, resulting in "design wins," to develop solutions for use in our end customers' products. These lengthy selection processes may require us to incur significant

expenditures and dedicate significant engineering resources to the development of new solutions without any assurance that we will achieve design wins. If we incur such expenditures but fail to be selected in the bid selection process, our business, financial condition and results of operations may be adversely affected. Further, because of the significant costs associated with qualifying new suppliers, end customers might use products or solutions that are functionally equivalent or that offer additional features from existing suppliers across a number of similar and successor products or solutions for a lengthy period of time. As we develop and introduce new solutions, we face the risk that end customers may not value or be willing to bear the cost of incorporating

these newer solutions into their end products, particularly if they believe their end users are satisfied with prior offerings. Regardless of the improved features or superior performance of the newer solutions, end users may be unwilling to adopt our new solutions due to such implementation hurdles, or design or pricing constraints. As a result, if we fail to secure an initial design win for any of our solutions to any particular end customer, we may lose the opportunity to make future sales of those solutions to that end customer for a significant period of time or at all, and we may also experience an associated decline in revenue relating to those components. Failure to achieve initial design wins may also weaken our position in future competitive selection processes, which would harm our business, financial condition and results of operations. Moreover, even if we achieve initial design wins with end customers, our end customers are not contractually obligated to purchase products from us in connection with such design wins.

Failure to adjust our inventory due to changing market conditions or failure to accurately estimate our end customers' demands could adversely affect our revenue and could result in charges for obsolete or excess inventories.

We make significant decisions, including determining the levels of business that we will seek, and accept production schedules, levels of reliance on outsourced contract manufacturing, personnel needs and other resource requirements, based on our estimates of end customer requirements. The lack of long-term commitments by our end customers and the possibility of rapid changes in demand for their products reduce our ability to accurately estimate future requirements of our end customers. On occasion, our end customers may require rapid increases in production, which can challenge our resources. We may not have sufficient third-party wafer fabrication capacity at any given time to meet our end customers' demands. Conversely, downturns in the end markets that our end customers serve have in the past caused and may in the future cause our end customers to significantly reduce the quantities of our products or solutions they order from us.

In addition, we base many of our operating decisions, and enter into purchase commitments, on the basis of anticipated revenue trends which are highly unpredictable. Certain of our design service agreements and purchase commitments are non-cancelable, and in some cases we may be required to recognize charges representing minimum commitments which exceed our actual requirements. These types of commitments and agreements reduce our ability to adjust our inventory to address declining market demands. If demand for our solutions is less than we expect, we may experience additional excess and obsolete inventory and be forced to incur charges for write-offs. If revenues in future periods fall substantially below our expectations, or if we fail to accurately forecast changes in demand mix, we could again be required to record substantial charges for obsolete or excess inventory or non-cancelable purchase commitments.

During a market upturn, we may not be able to purchase sufficient supplies or components to meet the increasing demand for our solutions, which could prevent us from taking advantage of business opportunities. In addition, a supplier could discontinue production of a component necessary for our products, extend lead times, limit supply or increase prices due to capacity constraints or other factors. Our failure to adjust our supply chain volume, secure sufficient supply from our third-party vendors, including our semiconductor wafer suppliers, or estimate our end customers' demand could have a material adverse effect on our business, financial condition and results of operations.

Furthermore, the risks described above have been and may be further amplified by global supply chain constraints and other macroeconomic conditions, including rising interest rates, higher rates of inflation, higher costs of fuel and

transportation, potential resulting logistics delays, market volatility, and impacts of global conflicts, such as the ongoing Russia-Ukraine and Israel-Hamas conflicts on international trade and the economy generally, all of which could have a material adverse impact on our business, financial condition and results of operations.

Decreases in average selling prices of our products and increases in input costs may reduce our gross margins.

The market for our products is generally characterized by declining average sale prices (ASP) resulting from factors such as increased competition, the introduction of new products and increased unit volumes. We anticipate that ASPs for legacy products may decrease in the future in response to the introduction of new products by us or our competitors, or due to other factors, including pricing pressures from our end customers.

We typically conduct pricing negotiations for our existing products with our largest end customers. In order to achieve and sustain profitable operations, we must continually reduce costs for our existing products and also develop and introduce new products with enhanced features on a timely basis that can be sold initially at higher ASPs. Failure to do so could cause our revenue and gross margins to decline, which would negatively affect our business, financial condition and results of operations and could significantly harm our business. In addition, in connection with the significant increase in semiconductor demand and supply shortages in recent years, the cost of certain materials used to manufacture our products, including for semiconductor wafers, has increased as demand has outpaced supply. We may be unable to reduce the cost of our products sufficiently to enable us to compete with others. Our cost reduction efforts may not allow us to keep pace with competitive pricing pressures and the increased cost of certain materials, such as semiconductor wafers and other raw materials, and could adversely affect our gross margins. We maintain a relatively small infrastructure of facilities and human capital in certain locations around the world and, as a result, have limited ability to reduce our operating costs. Accordingly, in order to remain competitive, we must continually reduce the cost of manufacturing and assembling our products through design and engineering changes. We cannot assure you that we will be successful in redesigning our products and bringing redesigned products or solutions to the market in a timely manner, or that any redesign will result in sufficient cost reductions to allow us to reduce the price of our products to remain competitive or maintain or improve our gross margins. To the extent we are unable to reduce the prices of our products and remain competitive, our revenue will likely decline, resulting in further pressure on our gross margins, which could have a material adverse effect on our business, financial condition and results of operations and our ability to grow our business.

If we fail to penetrate new markets or introduce new capabilities into our solutions, or otherwise execute our business strategy, our revenue and financial condition would be harmed.

Most of our revenue is generated from SoCs, bundled with software and various other solutions that combine 32-bit MCUs with wireless connectivity and additional circuitry, such as graphics processing units, serial interfaces, analog-to-digital interfaces, and more. In the future, we believe we can expand our solutions into embedded application processors and dedicated AI processors. We generate the majority of our revenue from the personal devices market. We believe that our future revenue growth, if any, will significantly depend on our ability to expand or further penetrate existing markets, such as medical/healthcare, industrial edge and smart homes and buildings, as well as expand into new markets such as automotive and data center and computing. In addition, we believe that our future revenue growth, if any, will significantly depend on our ability to develop dedicated processor units purpose-built for AI in the future.

Each of these new or growth markets presents distinct and substantial risks and, in many cases, requires us to develop new functionality or software to address the particular requirements of that market. We anticipate that as we continue to move into new markets, we will likely face competition from larger competitors with greater resources and more history in these markets. If any of these markets do not develop as we currently anticipate, or if the development of such markets is delayed or impacted by factors outside of our control, such as global conflicts, including the ongoing Russia-Ukraine and Israel-Hamas conflicts or other macroeconomic conditions,

or if we are unable to penetrate any of these markets successfully with our solutions, our revenue could decline, and our financial condition would be negatively impacted. Some of these markets are primarily served by only a few large, multinational OEMs and ODMs with substantial negotiating power relative to us and, in some instances, with internal solutions that are competitive to ours.

Meeting the technical requirements and securing design wins with any of the participants in these markets will require a substantial investment of our time and resources and we cannot assure you that we will secure design wins from these or other companies, that we will achieve meaningful revenue from the sales of our solutions into these markets, or that any revenue generated from these design wins will outweigh the costs of developing these designs. If we fail to penetrate these or other new markets we are targeting, our business, financial condition and results of operations would likely suffer. Moreover, even if we are successful in winning competitive bid selection processes in these new markets, it will likely take longer to generate revenue from such design wins than in our current markets.

We may also, in the future, seek to penetrate markets in new geographies. Certain of the markets we expect to target have higher barriers to entry, are more heavily regulated or favor domestic production. If we fail to penetrate these or other new markets we are targeting, or expend time and resources on entering these markets, but fail to generate sufficient revenue in such new geographies, our business, financial condition and results of operations would likely suffer.

We also plan to develop SPOT into an IP and chip development platform to make it easier for other companies to license or partner with us to incorporate SPOT into their own solutions. There is no guarantee that our efforts to develop SPOT into a licensing or partnership platform will succeed, in which case the proliferation of our SPOT platform into certain markets (e.g., data center markets) is likely to be limited.

If we are unable to manage our growth effectively, we may not be able to execute our business plan and our business, financial condition and results of operations could suffer.

In order to succeed in executing our business plan, we will need to manage our growth effectively as we make significant investments in research and development and sales and marketing, and expand our operations. If our revenue does not increase to offset these increases in our expenses, we may not achieve or maintain profitability in future periods. To manage our growth effectively, we must continue to expand our operations, engineering, accounting and finance, internal management, and other systems, procedures, and controls. This may require substantial managerial and financial resources, and our efforts may not be successful. Any failure to successfully implement systems enhancements and improvements will likely have a negative impact on our ability to manage our expected growth, as well as our ability to ensure uninterrupted operation of key business systems and compliance with the rules and regulations applicable to public companies. If we are unable to manage our growth effectively, we may not be able to take advantage of market opportunities or develop new solutions, and we may fail to satisfy end customer product or support requirements, maintain the quality of our solutions, execute our business plan or respond to competitive pressures, any of which could negatively affect our business, financial condition and results of operations.

We continue to invest in research and development efforts for several new markets. If we are unable to commercialize these technologies, our business, financial condition and results of operations could be negatively affected.

The semiconductor industry requires substantial investment in research and development in order to bring to market new and enhanced solutions. Our research and development expense was $8.7 million and $9.4 million for the months ended March 31, 2025 and 2024, respectively. We expect to increase our research and development expenditures as compared to prior periods as part of our strategy to focus on the development of new solutions, such as Apollo6, Atomiq, SoCs variants within existing Apollo families, as well as further penetration of existing markets, such as medical/healthcare, industrial edge and smart homes and buildings, and

expansion into new markets such as automotive and data center and computing. We are unable to predict whether we will have sufficient resources to achieve the level of investment in research and development required to remain competitive. For example, development of our solutions using sub-threshold and near-threshold designs at 0.4 volts, 0.5 volts, or 0.6 volts costs significantly more than development at the standard 0.8 volts or 0.9 volts. This added development cost could prevent us from being able to maintain a technological advantage over larger competitors that have significantly greater resources to invest in research and development. In addition, we cannot assure you that the technologies which are the focus of our research and development expenditures will become commercially successful or generate any revenue. If we incur increased research and development costs that do not result in revenue generation, our business, financial condition and results of operations would be adversely affected.

Our products are complex and may contain flaws which could lead to product liability, an increase in our costs and/or a reduction in our revenue.

Our products are complex and may contain software or hardware errors or defects (collectively, flaws), particularly when first introduced and when new versions are released. Our products are increasingly designed and integrated into more complex products, including higher levels of software and hardware integration in modules and system-level solutions and/or include elements provided by third parties which further increase the risk of flaws. We rely primarily on our in-house testing personnel to design test operations and procedures to detect any flaws or vulnerabilities prior to delivery of our products to our end customers.

Should challenges occur in the operation or performance of our products, we may experience delays in meeting key introduction dates or scheduled delivery dates to our end customers. These flaws could also cause significant re-engineering costs, diversion of our engineering personnel's attention from our product development efforts, and significant end customer relations and business reputation problems. Any unforeseen and undetected flaws could result in refunds, replacement, recall or other liability. Any of the foregoing could impose substantial costs and harm our business.

Product liability, data breach or cybersecurity liability claims may be asserted with respect to our products. Many of our solutions focus on security, storage, wireless connectivity, advanced graphics, and AI processing, which may make them particularly susceptible to cyberattacks. An undetected flaw, failure or vulnerability in our products could cause failure in our end customers' products, and we could face claims for damages that are disproportionately higher than the revenue we receive from the components involved, as our products are typically sold at prices that are significantly lower than the cost of the end products into which they are incorporated. Furthermore, product liability risks are particularly significant with respect to any smart home and automotive applications because of the risk of serious harm to users of these end products. There can be no assurance that any insurance or associated warranty liabilities we maintain will sufficiently protect us from such claims.

We compete against companies that have significantly greater resources and broader product lines than we do, and we may also face competition from other technological approaches to low power.

The semiconductor industry is highly competitive and many of our direct and indirect competitors have substantially greater financial, technological, manufacturing, marketing, and sales resources than we do.

Consolidation in our industry may increasingly mean that our competitors have greater resources, including the ability to attract qualified employees or pass along higher cost components into product prices, that could put us at a competitive disadvantage. We currently compete directly with numerous large MCU vendors, such as Infineon, Microchip, NXP, Renesas, Silicon Laboratories, STMicroelectronics, Texas Instruments, and others. In addition, we also compete with connectivity players such as Dialog Semiconductor (now Renesas), Nordic Semiconductor, Synaptics, and Telink or connected processor platform players, such as Qualcomm. Competition from these companies may intensify as we offer more solutions in our existing end markets or expand into new end markets. We also face competition from emerging companies.

There is no guarantee that our approach to low power will remain competitive. Our SPOT platform enables devices to consume two to five times less power than traditional technologies at the same manufacturing geometries. However, low power can be achieved using other, diversified technological approaches. There is the possibility that novel, unique approaches to low power emerge that surpass the capabilities of our technology and produce solutions that are more efficient, more cost effective, or both. Should this happen, our financial results would likely suffer.

In addition, from time to time, governments may provide subsidies or make other investments that could give competitive advantages to many semiconductor companies. For example, in August 2022, the United States enacted the U.S. CHIPS and Science Act of 2022 (the CHIPS Act), which, among other things, provides funding to increase domestic production and research and development in the semiconductor industry. Because we operate a fabless business model, we do not believe we will be eligible for such investments from the government. Many of our competitors may benefit from the investments, which will help increase their production capacity, shorten their lead time and gain market share. These competitive pressures could materially and adversely affect our business, financial condition and results of operations.

We are subject to the cyclical nature of the semiconductor industry.

The semiconductor industry is highly cyclical and is characterized by constant and rapid technological change, rapid product obsolescence, price erosion, evolving standards, short product life cycles, and wide fluctuations in product supply and demand. Historically, the industry experienced significant downturns during global recessions. These downturns have been characterized by diminished demand, production overcapacity, high inventory levels, and accelerated erosion of average selling prices. Furthermore, any significant upturn in the semiconductor industry could result in increased competition for access to third-party wafer fabrication and assembly capacity. We are dependent on the availability of this capacity to manufacture and assemble our products and we can provide no assurance that adequate capacity will be available to us in the future. Any downturns or upturns in the semiconductor industry could harm our business, financial condition and results of operations.

We depend on agreements with third party developers and licensors in order to satisfy certain end customer requirements, which subjects us to a number of risks, and any failure to execute on any of these arrangements could have a material adverse effect on our business, financial condition and results of operations.

We have entered into development, product collaboration and technology licensing arrangements with third parties, such as ARM Limited and Cadence Design Systems, Inc., and we expect to enter into new arrangements of these kinds from time to time in the future. These agreements are typically non-exclusive contracts provided under royalty-accruing or paid-up licenses. While we do not believe our business is dependent to any significant degree on any individual third-party license, we expect to continue to use and may license additional third-party technology for our solutions. These agreements may increase risks for us, such as the risks related to timely delivery of new products, risks associated with the ownership of the intellectual property developed, risks that such activities may not result in products that are commercially successful or available in a timely fashion, risks that third parties involved may abandon or fail to perform their obligations related to such agreements, and risks that certain technologies provided under such arrangements may not continue to be available on reasonable terms or at all. Any failure to timely develop commercially successful products under such arrangements as a result of any of these and other challenges could have a material adverse effect on our business, financial condition and results of operations.

If our information technology systems or data, or those of third parties, such as vendors and suppliers, with whom we work, are or were compromised, we could experience adverse consequences resulting from such compromise, including but not limited to regulatory investigations or actions; litigation; fines and penalties; disruptions of our business operations; reputational harm; loss of revenue or profits; loss of customers or sales; and other adverse consequences.

In the ordinary course of our business, we and the third parties with whom we work, routinely collect, receive, store, use, handle, transfer, disclose, make accessible, protect, secure, dispose of, transmit, share and

otherwise process proprietary, confidential, and sensitive data, including intellectual property, trade secrets, proprietary technology and information about our business customers, suppliers and partners, and proprietary technology and information owned by our customers (collectively, sensitive information). We rely upon third party service providers and technologies to operate critical business systems to process sensitive information in a variety of contexts, including, without limitation, third party providers of cloud-based infrastructure, encryption and authentication technology, employee email, content delivery to customers, and other functions. Our ability to monitor these third parties' information security practices is limited, and these third parties may not have adequate information security measures in place.

Any of the previously identified or similar threats could cause a cybersecurity incident or other interruption, which could result in unauthorized, unlawful, or accidental acquisition, modification, destruction, loss, alteration, encryption, disclosure of, or access to our sensitive information or our information systems, or those of the third parties with whom we work. Further, a cybersecurity incident or other interruption could disrupt our ability (and that of third parties with whom we work) to provide our solutions.

We, and the third parties with whom we work, are subject to a variety of evolving cybersecurity-related threats. While we have implemented various controls and defenses designed to prevent cybersecurity incidents, cybersecurity attacks and threats have continued to become more prevalent and sophisticated. These threats are constantly evolving, making it increasingly difficult to successfully defend against or implement adequate preventive measures. Geopolitical tensions or conflicts have in the past led to, and may in the future lead to, increased risk of cybersecurity attacks. Notwithstanding defensive measures, experienced programmers, hackers, state actors, or others may be able to penetrate our security controls, or those of our vendors, suppliers, or customers, through attacks such as, but not limited to, social-engineering attacks (including through deep fakes, which may be increasingly more difficult to identify as fake, and phishing attacks), malicious code (such as viruses and worms), malware (including as a result of advanced persistent threat intrusions), denial-of-service attacks, access attacks, credential stuffing, credential harvesting, personnel misconduct or error, ransomware attacks, supply-chain attacks, software bugs, server malfunctions, software or hardware failures, loss of data or other information technology assets, adware, telecommunications failures, earthquakes, fires, floods, attacks enhanced or facilitated by AI, and other similar threats.

Ransomware attacks, including those launched by organized criminal threat actors, nation-states, and nation-state-supported actors, are becoming increasingly prevalent and severe and can lead to significant interruptions in our operations, ability to provide our solutions, loss of sensitive data and income, reputational harm, and diversion of funds. Extortion payments may alleviate the negative impact of a ransomware attack, but we may be unwilling or unable to make such payments due to, for example, applicable laws or regulations prohibiting such payments. Similarly, supply-chain attacks have increased in frequency and severity, and we cannot guarantee that third parties and infrastructure in our supply chain or our third party partners' supply chains have not been compromised or that they do not contain exploitable flaws or bugs that could result in a breach of or disruption to our information technology systems (including our solutions or the third party information technology systems that support us and our solutions). We are incorporated into the supply chain of a large number of companies worldwide and, as a result, if our products are compromised, a significant number of such companies and their data could be simultaneously affected. The potential liability and associated consequences we could suffer as a result of such a large-scale event could be catastrophic and result in irreparable harm. The growth of our remote workforce, and the increase in remote working arrangement by our vendors and other third parties, poses increased risks to our information technology systems and data privacy, as more of our employees, and those of our vendors and other third parties, utilize network connections, computers and devices outside our premises or network, including working at home, while in transit and in public locations.

Future or past business transactions (such as acquisitions or integrations) could also expose us to additional cybersecurity risks and vulnerabilities, as our systems could be negatively affected by vulnerabilities present in acquired or integrated entities' systems and technologies. Furthermore, we may discover cybersecurity issues that were not found during due diligence of such acquired or integrated entities, and it may be difficult to integrate companies into our information technology environment and cybersecurity program.

While we have implemented security measures designed to protect against cybersecurity incidents, there can be no assurance that these measures will be effective. We take steps designed to detect, mitigate, and remediate vulnerabilities in our information technology systems, products and solutions (including technologies from third parties with whom we work), but we may not be able to detect, mitigate, and remediate all such vulnerabilities. We have not always been able in the past and may be unable in the future to detect vulnerabilities in our information technology systems (including our solutions) because such threats and techniques change frequently, are often sophisticated in nature, and may not be detected until after a security incident has occurred. Despite our efforts, there can be no assurances that these vulnerabilities mitigation measures will be effective. Further, we may experience delays in developing and deploying remedial measures and patches designed to address any such identified vulnerabilities. and vulnerabilities in our information technology system could be exploited and result in a cybersecurity incident.

We have in the past, and may in the future, expend significant resources or modify our business activities to try to protect against cybersecurity incidents. Certain data privacy and cybersecurity obligations may, and could in the future, require us to implement and maintain specific security measures, industry-standard or reasonable cybersecurity measures to protect our information technology systems and sensitive information. Applicable data privacy and cybersecurity obligations may require us, or we may voluntarily choose, to notify relevant stakeholders, including affected individuals, customers, regulators,

and investors, of cybersecurity incidents. Such disclosures and related actions can be costly, and the disclosure or the failure to comply with such applicable requirements could lead to adverse consequences. If we (or a third party with whom we work with) experience a cybersecurity incident or are perceived to have experienced a cybersecurity incident, we may experience adverse consequences. These consequences may include: government enforcement actions (for example, investigations, fines, penalties, audits, and inspections), additional reporting requirements and/or oversight, restrictions on processing sensitive information (including personal data), litigation (including class-action claims), indemnification obligations, negative publicity, reputational harm, monetary fund diversions, interruptions in our operations (including availability of data), financial loss, and other similar harms. Moreover, any such compromise of our information technology systems could result in the misappropriation or unauthorized publication of our confidential business or proprietary information, including trade secrets, or that of other parties with which we do business, an interruption in our operations, the unauthorized transfer of cash or other of our assets, the unauthorized release of customer or employee data or a violation of privacy or other laws. Cybersecurity incidents and attendant consequences may cause customers to stop using our solutions, deter new customers from using our solutions, and negatively impact our ability to grow and operate our business.

We must commit significant resources to ensuring that we develop secure technology, solutions and related devices. The process of ensuring the security of new technology is complex and uncertain, and if we fail to secure these devices, we could face government enforcement actions, litigation (including class claims), indemnification obligations, negative publicity, reputational harm, monetary fund diversions, interruptions in our operations, financial loss, and other similar harms.

Our contracts may not contain limitations of liability, and even where they do, there can be no assurance that limitations of liability in our contracts are sufficient to protect us from liabilities, damages, or claims related to our data privacy and cybersecurity obligations. Additionally, our insurance coverage may be inadequate, and a large data privacy or cybersecurity-related incident claim may exceed our coverage and/or prevent us from getting coverage in the future. We cannot be sure that our insurance coverage will be adequate or sufficient to protect us from or to mitigate liabilities arising out of our data privacy and cybersecurity practices, that such coverage will continue to be available on commercially reasonable terms or at all, or that such coverage will pay future claims.

In addition to experiencing a cybersecurity incident, third parties may gather, collect, or infer sensitive information about us from public sources, data brokers, or other means that reveals competitively sensitive details about our organization and could be used to undermine our competitive advantage or market position.

Our estimate of the market size for our solutions may prove to be inaccurate, and even if the market size is accurate, we cannot ensure that we will serve a significant portion of the market.

Our estimate of the market size for our solutions that we have provided in this prospectus, sometimes referred to as total addressable market (TAM), is subject to significant uncertainty and is based on assumptions and estimates, including our internal analysis and industry experience and third-party data, which may be inaccurate. These estimates are, in part, based upon the size of the markets and geographies we target and an estimated range of prices for our current solutions and future development plans. Our ability to serve a significant portion of this estimated market is subject to many factors, including our success in implementing our business strategy and ability to maintain our product pricing levels, which are subject to many risks and uncertainties.

Moreover, in order to address the entire TAM we have identified, we must continue to enhance and add to our existing markets and introduce our solutions to new markets. Accordingly, even if our estimate of the market size is accurate, we cannot ensure that we will serve a significant portion of this estimated market or that our pricing levels will not decline.

We depend on key and highly skilled personnel to operate our business, and if we are unable to retain our current personnel and/or hire additional personnel, our ability to develop and market our solutions could be harmed, which in turn could adversely affect our business, financial condition and results of operations.

Our success depends to a large extent upon the continued services of our executive officers, managers and skilled personnel, including our development engineers. In particular, we are highly dependent on the services of Scott Hanson, Ph.D. and Fumihide Esaka, our Founder and CEO, respectively, who, after over 15 and nine years of service, respectively, with our company, have been critical in the development and growth of our business and strategic direction. From time to time, there may be changes in our executive management team or other key personnel, which could disrupt our business. Our employees are not bound by obligations that require them to continue to work for us for any specified period and, therefore, they could terminate their employment with us at any time. Moreover, our employees are generally not subject to non-competition agreements. The loss of one or more of our executive officers or other key personnel or our inability to locate suitable or qualified replacements could be significantly detrimental to our product development efforts and could have a material adverse effect on our business, financial condition and results of operations. In addition, we recruit from a limited pool of engineers with specialized expertise and the competition for such personnel can be intense. Given these limitations, we may not be able to continue to attract, retain and motivate qualified personnel necessary for our business.

We also must attract and retain highly qualified personnel, including certain foreign nationals who are not U.S. citizens or permanent residents, many of whom are highly skilled and constitute an important part of our U.S. workforce, particularly in the areas of engineering and product development. Our ability to hire and retain these employees and their ability to remain and work in the United States are impacted by laws and regulations, as well as by procedures and enforcement practices of various government agencies. Changes in immigration laws, regulations or procedures may adversely affect our ability to hire or retain such workers, increase our operating expenses and negatively impact our ability to deliver our solutions, any of which would adversely affect our business, financial condition and results of operations.

Acquisitions, divestitures, strategic investments and strategic partnerships could adversely affect our business, financial condition and results of operations.

We may pursue growth opportunities by acquiring complementary businesses, products, solutions or technologies through strategic transactions, investments or partnerships. The identification of suitable acquisition, strategic investment or strategic partnership candidates can be costly and time consuming and can distract our management team from our current operations. If such strategic transactions require us to seek additional debt or equity financing, we may not be able to obtain such financing on terms favorable to us or at all, and such transaction may adversely affect our liquidity and capital structure. We may also choose to divest

certain non-core assets, which divestitures could lead to charges against earnings and may expose us to additional liabilities and risks. Any strategic transaction might not strengthen our competitive position, may increase our risks, and may be viewed negatively by our end customers, partners or investors. Even if we successfully complete a strategic transaction, we may not be able to effectively integrate the acquired business, technology, systems, control environment, products, personnel, or operations into our business or global tax structure. We may experience unexpected changes in how we are required to account for strategic transactions pursuant to accounting principles generally accepted in the United States (GAAP) and may not achieve the anticipated benefits of any strategic transaction. We may incur unexpected costs, claims or liabilities that we incur during the strategic transaction or that we assume from the acquired company, or we may discover adverse conditions post-acquisition for which we have limited or no recourse.

Downturns or volatility in general economic conditions, including as a result of international hostilities, such as the ongoing Russia-Ukraine and Israel-Hamas conflicts, could have a material adverse effect on our business, financial condition and results of operations.

Our revenue and gross margin depend significantly on general economic conditions and the demand for our solutions in the markets in which our end customers compete. Weaknesses in the global economy and financial markets, including the impact from global conflicts, such as the ongoing Russia-Ukraine and Israel-Hamas conflicts, may in the future lead to lower demand for end customers' products that incorporate our solutions, particularly in the semiconductor and battery-powered edge devices markets. A decline in end-user demand will adversely impact our end customers' demand for our solutions and can affect the ability of our end customers to obtain credit and otherwise meet their payment obligations and the likelihood of our end customers canceling or deferring existing orders. Our business, financial condition and results of operations could be negatively affected by a decline in end-user demand caused by weaknesses in the global economy and financial markets. Volatile and/or uncertain economic conditions, including increased inflation rates, can adversely impact revenue and gross margin and make it difficult for us to accurately forecast and plan our future business activities. To the extent expected favorable economic conditions do not materialize or take longer to materialize than expected, we may face an oversupply of our products and have excess inventory, which could result in lower demand, lower ASP, lower gross margin and charges for excess and obsolete inventory. Conversely, if we underestimate end customer demand, we may fail to meet end customer needs, which could impair our end customer relationships and future sales orders. In addition, any disruption in the credit markets, including as a result of the current conflicts between Ukraine and Russia or Israel and Hamas could impede our access to capital. If we have limited access to additional financing sources, we may be required to defer capital expenditures or seek other sources of liquidity, which may not be available to us on acceptable terms or at all. Similarly, if our suppliers face challenges in obtaining credit or other financial difficulties, they may be unable to provide the materials we need to manufacture our products. Additionally, global macroeconomic conditions, including those affected by ongoing Russia-Ukraine and Israel-Hamas conflicts, or any future geopolitical conflicts or pandemics, could make it difficult for our end customers and suppliers to forecast and plan future business activities and could cause global businesses to defer or reduce spending on our products or increase the costs of manufacturing our products. All of these factors relate to global economic conditions, which are beyond our control, could adversely impact our business, financial condition and results of operations.

Uncertain risks relating to the adoption, use or application of emerging technologies, including AI, by our customers, could adversely impact our financial results, cause operational challenges, and/or result in reputational harm, competitive harm and legal liability.

Our products support AI functionality implemented by our customers' products and solutions. Our latest generation of products also enable our customers to address computationally intense AI applications in edge devices. As a result, our business depends on the continued adoption of AI by our customers and their ability to commercialize their AI-based solutions. The adoption of AI solutions by our customers may not develop in the manner or in the time periods we anticipate and, as the markets for AI solutions are still developing, demand for our products that support AI may be unpredictable and vary significantly from one period to another. Even if the

demand for AI-enabled products develops in the manner or in the time periods we anticipate, if we do not have timely, competitively priced, market-accepted products available to meet our customers' needs to develop products for the AI markets, we may miss a significant opportunity and our business, operating results and financial condition could be materially and adversely affected.

Our customers are increasingly relying on our solutions to enhance their product and technology AI capabilities. However, the development, adoption, integration and use of AI remains in early stages, and ineffective or inadequate AI or generative AI governance, development, use or deployment practices by us or our customers could lead to unintended consequences. For example, AI algorithms may be flawed, or may be (or may be perceived to be) based on datasets that are insufficient. In addition, any latency, disruption or failure of our products in customers' AI solutions could result in flawed deployments, producing erroneous or harmful outputs, which could damage our reputation and lead to legal liabilities. Further, AI algorithms may be of poor quality, reflect unwanted forms of bias, or contain other errors or inadequacies, any of which may not be easily detectable. AI has been known to produce false or "hallucinatory" inferences or outputs. The use of AI applications could also increase the risk of cybersecurity incidents and the unauthorized access to confidential, proprietary or sensitive information, including through unintended or inadvertent transmission of confidential, proprietary or sensitive information by our employees or independent contractors. AI can present ethical issues and may subject us to new or heightened legal, regulatory, ethical or other challenges. Additionally, inappropriate or controversial data practices by developers and end-users, or other factors adversely affecting public opinion of AI, could impair the acceptance of AI solutions, including those incorporated in our services. Furthermore, AI technologies our customers adopt may not meet market requirements or may become obsolete earlier than planned, or third parties may deploy AI technologies in a manner that reduces customer

demand for our products and services. If we do not have sufficient rights to use the data or other material or content on which our AI solutions or other AI tools we use rely, we also may incur liability through the violation of applicable laws, third-party intellectual property, privacy or other rights, or contracts to which we are a party. Any of the foregoing may result in decreased demand for our products and services or harm to our business, financial results or reputation. If we or our customers enable or offer solutions that draw controversy due to their perceived or actual impact on society, such as AI solutions that have unintended consequences or are controversial, we may experience reputational harm, competitive harm, financial harm or legal liability.

We publish statements online and in our marketing materials describing the possible use and integration of our products in customers' AI products and solutions. Although we endeavor to be accurate with our public statements and documentation, we may at times fail to do so or be alleged to have failed to do so. If any such statements or documentation are found to be deceptive, unfair, misleading, or misrepresentative of our actual practices, even though circumstances beyond our reasonable control, we may face litigation, disputes, claims, investigations, inquiries or other proceedings which could adversely affect our business, reputation, results of operations and financial condition.

The AI industry is subject to complex, evolving regulatory, statutory, and other requirements that may be difficult and expensive to comply with and that could negatively impact broad-based adoption of AI and, as a result, our business.

The legal and regulatory framework for our products and services, including our AI computing solutions and AI model services, is rapidly evolving as many federal, state, and foreign government bodies and agencies have introduced or are currently considering additional laws, regulations and standards related to AI. Additionally, existing laws and regulations may be interpreted in ways that would affect our or our customers' operations, or our ability to offer our AI products and services in the markets in which we operate. As a result, implementation standards and enforcement practices are likely to remain uncertain for the foreseeable future, and we cannot yet determine the impact future laws, regulations, standards, or market perception of their requirements may have on our business, and we may not be able to adequately anticipate or respond to these evolving laws or regulations. In addition, because AI-related technologies are themselves highly complex and rapidly developing, it is not possible to predict all of the legal or regulatory risks that may arise relating to our

use of such technologies. New laws, guidance or decisions in this area could provide a new regulatory framework that may require us to adjust and make changes to our operations that may decrease our operational efficiency, resulting in an increase to operating costs and/or hindering our ability to improve our products and services.

In addition, compliance with evolving government regulations worldwide related to AI may increase the costs related to the development of AI products and solutions and limit global adoption, which may also adversely impact demand for our products that support AI functionality. The regulation of AI is rapidly evolving as legislators and regulators are increasingly focused on these powerful emerging technologies. The technologies underlying AI and its uses are subject to a variety of laws, including intellectual property, privacy, data protection and cybersecurity, consumer protection, competition, and equal opportunity laws, and are expected to be subject to increased regulation and new laws or new applications of existing laws. Further, a number of aspects of intellectual property protection in the field of AI are currently under development and there is uncertainty and ongoing litigation in different jurisdictions as to the degree and extent of protection warranted for AI and relevant system input and outputs. AI is the subject of ongoing review by various U.S. governmental and regulatory agencies, and various U.S. states and other foreign jurisdictions are applying, or are considering applying, their platform moderation, cybersecurity and data protection laws to AI or are considering general legal frameworks for AI. In addition, existing laws, rules and regulations may be interpreted in ways that would affect the use of AI in our business and the business of our customers.

For example, in August 2024, the EU Artificial Intelligence Act (EU AI Act), which establishes broad obligations for the development and use of AI-based technologies in the EU based on their potential risks and level of impact, came into force. The EU AI Act establishes, among other things, a risk-based governance framework for regulating AI systems operating in the European Union. This framework would categorize AI systems, based on the risks associated with such AI systems' intended purposes, as creating unacceptable or high risks, with all other AI systems being considered low risk. The EU AI Act includes requirements around transparency, conformity assessments and monitoring, risk assessments, human oversight, security, accuracy, general purpose AI, and foundation models, and provides for fines of up to the greater of €35 million or 7% of worldwide annual turnover for violations. We may not be able to anticipate how to respond to these rapidly evolving frameworks, and we may need to expend resources to adjust our offerings in certain jurisdictions if the legal frameworks are inconsistent across jurisdictions. Furthermore, because AI technology itself is highly complex and rapidly developing, it is not possible to predict all of the legal, operational or technological risks that may arise relating to the use of AI. Our customers may become subject to such upcoming AI laws, rules or regulations, which could cause a delay or impediment to the commercialization of AI technology and could lead to a decrease in demand for our customers' AI systems, which may, in turn, adversely affect our business, financial condition, and results of operations. See "Business—Government Regulation" for a more detailed discussion of challenges associated with AI regulation.

The market for edge AI services and products is relatively new, and may decline or experience limited growth, and our business is dependent on its clients' continuing adoption and use our services and products.

The edge AI market is relatively new and is subject to a number of risks and uncertainties. We believe that our future success will depend on the growth, if any, of this market and the use of our services and products.

The use of edge AI is still relatively new, and consumers may not recognize the need for or benefits of our services and products. If consumers do not recognize the need for and benefits of our services and products, then they may decide to adopt alternative services to satisfy some portion of their business needs. The market for our services and products could fail to grow significantly or there could be a reduction in demand for our services and/or products as a result of a lack of acceptance, technological challenges, competing services, a decrease in spending by current and prospective customers, weakening economic conditions and other causes. If the edge AI market does not experience significant growth, or demand for its services and/or products decreases, then our business, financial condition and results of operations could be adversely affected.

Customer demands for us to implement business practices that are more stringent than legal requirements may reduce our revenue opportunities or cause us to incur higher costs.

Some of our customers may require that we implement practices that are more stringent than those required by applicable laws with respect to labor requirements, the materials contained in our products, energy efficiency, environmental matters, or other items. To comply with such requirements, we may also require our third-party manufacturing partners to adopt such practices, as needed. Our third-party manufacturing partners may in the future refuse to implement these practices or may charge us more for complying with them. If certain of our third-party manufacturing partners refuse to implement the practices, we may be forced to source from alternate third-party manufacturing partners. The cost to implement such practices may cause us to incur higher costs and reduce our profitability, and if we do not implement such practices, such customers may disqualify us as a supplier, resulting in decreased revenue opportunities. Developing, enforcing, and auditing customer-requested practices at our own sites and in our supply chain will increase our costs and may require more personnel.

We may experience difficulties in transitioning to smaller geometry process nodes, and tapeouts in smaller nodes are more expensive.

In order to remain competitive, we have transitioned, and expect to continue to transition, our semiconductor products to be manufactured in accordance with increasingly smaller line width geometries. We periodically evaluate the benefits, on a product-by-product basis, of migrating our product designs to smaller geometry process nodes. We also evaluate the costs of migrating to smaller geometry process nodes, including both actual costs and opportunity costs related to the technologies we choose to forego. These complex transitions are imperative for us to remain competitive with the rest of the industry. We have been, and may continue to be, dependent on our relationships with our manufacturers to transition to smaller geometry processes successfully. We cannot ensure that the third parties we use will be able to effectively manage any future transitions. If we or any of our partners experience significant delays in a future transition or fail to efficiently implement a transition, we could experience reduced manufacturing yields, delays in product deliveries, and increased expenses, all of which could harm our relationships with our customers and our results of operations.

`Risks Related to Our Reliance on Third Parties`

We have no manufacturing capabilities of our own. We are a fabless company, meaning that we do not own a semiconductor foundry, and we rely on a single third-party supplier for the fabrication of semiconductor wafers and on a limited number of suppliers of other materials, and the failure of any of our suppliers to provide us with wafers and other materials on a timely basis would harm our business, financial condition and results of operations.

We rely heavily on TSMC, which is the only producer of semiconductor wafers that are used in our products. If TSMC suspends operations or limits our allocation of wafers, our ability to manufacture such wafers would be materially impaired. Furthermore, any disruption in operations at TSMC could adversely affect our ability to meet end customer demand in a timely manner, or at all, which would lead to a reduction in our revenue and may adversely affect our reputation, end customer relationships and future sales orders, potentially resulting in long-term harm to our business. In addition, the locations of TSMC and our other suppliers present additional risks. See the section "—Risks Related to the Markets We Operate in and the Regulatory Landscape of Our Business".

In addition to TSMC, we purchase a number of key material and components used in the manufacture of our products from single or a limited number of suppliers. These suppliers, including TSMC, are in great demand and the lead-time to expand capacity could be long. From time to time, we have encountered shortages and delays in obtaining wafers at desired specifications and other components and materials and price increases, and we may encounter additional shortages and delays in the future. If we cannot supply our products due to a lack of components, including semiconductor wafers, or are unable to source materials from other suppliers or to redesign products with other components in a timely manner, our business will be significantly harmed.

We do not have long-term contracts with any of our suppliers and third-party manufacturers. As a result, any such supplier or third-party manufacturer can discontinue supplying wafers, components or materials to us at any time and without penalty. Moreover, we depend on the quality of such wafers, and other components and materials supplied by such suppliers and third-party manufacturers, over which we have limited control. Any one or more of our suppliers may become financially unstable as the result of global market conditions and geopolitical developments. Moreover, our suppliers' abilities to meet our requirements could be impaired or interrupted by factors beyond their control, such as natural disasters or other disruptions. In the event that any one or more of our suppliers is unable or unwilling to deliver us products and we are unable to identify alternative sources of supply for such materials or components on a timely basis, our business, financial condition and results of operations may be adversely affected. In addition, even if we identify any such alternative sources of supply, we could experience delays in testing, evaluating and validating materials or products of potential alternative suppliers or products we obtain through outsourcing. Qualifying new contract manufacturers, and specifically semiconductor foundries, is time consuming and might result in unforeseen manufacturing and operations problems. Furthermore, financial or other difficulties faced by our suppliers, or significant changes in demand for the components or materials they use in the products they supply to us, could limit the availability of those products, components or materials to us. We are also subject to potential delays in the development by our suppliers of key components which may affect our ability to introduce new products.

Any of these challenges or delays could damage our relationships with our end customers, adversely affect our reputation and adversely affect our business, financial condition, results of operations and our ability to grow our business.

Our dependence on TSMC as our sole supplier of wafers exposes us to certain political, social and economic risks that may harm our business.

Deterioration in the political, social, business or economic conditions in the jurisdictions in which TSMC or other suppliers operate could slow or halt product shipments or disrupt our ability to manufacture, package, test or post-process products. In response, we could be forced to transfer our manufacturing, packaging, testing and post-processing activities to more stable, and potentially more costly, regions or find alternative suppliers. In

particular, because we source all of our wafers from TSMC, located in Taiwan, our supply of wafers and other critical components may be materially and adversely affected by diplomatic, geopolitical and other developments affecting the relationship between Mainland China and Taiwan. Since 1949, Taiwan and Mainland China have been separately governed. Although significant economic and cultural relations have been established between Taiwan and Mainland China in recent years, there can be no assurances that relations between Taiwan and Mainland China will not deteriorate. Any such developments could materially and adversely affect our business, financial condition and results of operations.

If we encounter sustained yield problems or other delays at our third-party wafer fabrication facilities or in the final assembly and test of our products, we may lose revenue and damage our end customer relationships.

The manufacture of our products, including the fabrication of semiconductor wafers, and the assembly and testing of our products, involve highly complex processes. For example, minute levels of contaminants in the manufacturing environment, difficulties in the wafer fabrication process or other factors can cause a substantial portion of the components on a wafer to be nonfunctional. These problems may be difficult to detect at an early stage of the manufacturing process and often are time-consuming and expensive to correct. From time to time, we have experienced problems achieving acceptable yields at our third-party wafer fabrication suppliers, resulting in delays in the availability of components. Moreover, an increase in the rejection rate of products during the quality control process before, during or after manufacture and/or shipping of such products, results in lower yields and margins. In addition, changes in manufacturing processes required as a result of changes in product specifications, changing end customer needs and the introduction of new products have historically significantly reduced our manufacturing yields, resulting in low margins on those products. Poor manufacturing

yields over a prolonged period of time could adversely affect our ability to obtain the supplies needed to deliver our components on a timely basis and harm our relationships with end customers, which could materially and adversely affect our business, financial condition and results of operations.

Raw material and engineered material availability and price fluctuations have in the past and may in the future increase the cost of our products, impact our ability to meet customer commitments, and may adversely affect our business, financial condition and results of operations.

The cost of raw and engineered materials is a key element in the cost of our products. Our inability to offset material price inflation through increased prices or improved productivity could adversely affect our business, financial condition and results of operations. Many major components, product equipment items, engineered materials, and wafers are procured or subcontracted on a single or sole-source basis. It is difficult to predict what effects shortages or price increases may have in the future. Our inability to fill our supply needs would jeopardize our ability to ship our products to our end customers on time and in the quantity required, which could, in turn, result in reduced sales and revenue and damage to our end customer relationships and future sales orders.

Furthermore, increases in the price of silicon wafers, testing costs, and commodities, which would result in increased production costs may result in a decrease in our gross margins. Moreover, our suppliers may pass the increase in engineered materials, raw materials and commodity costs to us which would further reduce the gross margin of our products. In addition, global market trends, such as a shortage of capacity to fulfill our fabrication needs, also may increase our raw material costs and thus decrease our gross margin.

We may be subject to warranty claims and other costs related to our products.

In general, we warrant our products for one year from the delivery of the product against non-conformance to our specifications and certain other flaws. Because our solutions, including hardware, software, and intellectual property cores, are highly complex and increasingly incorporate advanced technology, our quality assurance programs and those of our third party test providers may not detect all flaws, whether specific manufacturing flaws affecting individual components or systematic flaws affecting numerous shipments. Inability to detect flaws could result in a diversion of our engineering resources from component development efforts, increased engineering expenses to remediate the flaws, and increased costs due to end customer accommodation or inventory impairment charges. On occasion, we have also repaired or replaced certain components, made software fixes, reimbursed end customers for reasonable out-of-pocket production cost caused by an issue in our product or refunded the purchase price or license fee paid by our end customers due to component or software flaws. Our insurance may be unavailable or inadequate to protect against these issues. If there are significant component flaws, the costs to remediate such flaws, net of reimbursed amounts from our vendors, if any, or to resolve warranty claims, would adversely affect our reputation, which would negatively impact our business, financial condition and results of operations.

Our revenue could be materially impacted by the failure of other component suppliers to deliver required parts needed in the final assembly of our end customers' end products.

The products we supply to our end customers are typically only a portion of the many components our end customers source from multiple suppliers in order to complete the final assembly of an end product. If one or more of these other component suppliers are unable to deliver to end customers components needed to assemble end products, our end customers may delay, or ultimately cancel, their orders from us.

Risks Related to the Markets We Operate in and the Regulatory Landscape of Our Business

We are a global company, which subjects us to additional business risks, including logistical and financial complexity, political instability and currency fluctuations.

We have established international subsidiaries and have opened offices in international markets to support our activities in Asia, the Americas and Europe, and we expect to continue to conduct business with companies that are

located outside the United States, particularly in Asia. This has included the establishment of offices in Taiwan and Mainland China for non-U.S. operations. We purchase wafers from foreign entities, have our products assembled and tested by subcontractors located in Asia, and supply our products to end customers located outside of the United States. Even end customers of ours that are based in the United States often use contract manufacturers based in

Asia to manufacture their products, and these contract manufacturers typically purchase products directly from us. We derived 13% and 53% of our net sales for the three months ended March 31, 2025 and 2024, respectively, from end customers outside of the United States and 6% and 50%, respectively, from end customers located in Mainland China. Our international operations are subject to a number of risks, including:

- Complexity and costs of managing international operations and related tax obligations, including for our offices outside of the United States;
- Trade tensions, geopolitical uncertainty, or governmental actions may lead non-U.S. end customers to favor products from non-U.S. companies which could put us at a competitive disadvantage and result in decreased end customer demand for our solutions and our end customers' products;
- Restrictions imposed on sensitive technologies from certain countries and sanctions or export controls imposed on end customers or suppliers may affect our ability to sell and source our products;
- Potential political, legal and economic instability, armed conflict, and civil unrest in the countries in which we and our end customers are located, including instability caused by the conflicts between Ukraine and Russia or Israel and Hamas;
- Unanticipated restrictions on our ability to sell to foreign end customers where sales of products and the provision of services may require export licenses or are prohibited by government action, unfavorable foreign exchange controls and currency exchange rates;
- Public health crises may affect our international operations, suppliers and end customers and we may experience delays in development, a decreased ability to support our end customers and reduced design win activity if the travel restrictions or business shutdowns or slowdowns continue for an extended period of time in any of the countries in which we, our suppliers and our end customers operate and do business;
- Uncertainties related to the enforcement of intellectual property rights in various jurisdictions, including any intellectual property rights that we may license to Taiwanese or Chinese entities, including any joint ventures we may form;
- Multiple, conflicting and changing tax and other laws and regulations that may impact both our international and domestic tax and other liabilities and result in increased complexity and costs, including the impact of the Tax Cuts and Jobs Act, which we expect to increase our effective tax rate, in part due to the impact of the requirement to capitalize and amortize foreign research and development expenses that commenced in 2022;
- Longer sales cycles and greater difficulty in accounts receivable collection and longer collection periods;
- Difficulties in enforcing contracts generally;
- High levels of distributor inventory subject to price protection and rights of return to us;
- Greater difficulty in hiring and retaining qualified personnel; and
- The need to have business and operations systems that can meet the needs of our business growth and international business and operating structure.

Uncertainties with respect to the PRC legal system, including uncertainties regarding the enforcement of laws, and sudden or unexpected changes in policies, laws and regulations in the PRC could adversely affect us. Furthermore, the PRC government has significant oversight over our PRC subsidiaries' business conduct and may intervene or influence our operations in Mainland China. The PRC government has published policies that

significantly affect certain industries other than ours, and we cannot rule out the possibility that it will in the future release regulations or policies regarding our industry that could further adversely affect our business, financial condition and results of operations.

We are subject to government regulation, including import, export and economic sanctions laws and regulations that may expose us to liability and increase our costs or that may prohibit us from selling our products to specific targeted entities.

Our products are subject to export control and import laws and regulations, including the U.S. Export Administration Regulations (EAR), U.S. Customs regulations, and various economic and trade sanctions regulations administered by the U.S. Treasury Department's Office of Foreign Assets Controls. These regulations may limit the export of our products and technology, and provision of services outside of the United States, or may require export authorizations, including by license, a license exception, or other appropriate government authorization. Exports of our products and technology must be made in compliance with these laws and regulations. If we fail to comply with these laws and regulations, we and certain of our employees could be adversely affected through reputational harm, government investigations, penalties, or denial or curtailment of our ability to export our products and technology. Although we take precautions to prevent our products and technology from being provided in violation of such laws, our products and technology may have previously been, and could in the future be, provided inadvertently in violation of such laws, despite the precautions we take. In addition, changes in our products or changes in applicable export or import laws and regulations may create delays in the introduction and sale of our products in international markets, prevent our end customers from deploying our products or, in some cases, prevent the export or import of our products to certain countries, governments or persons or entities altogether. Any change in export or import laws and regulations, shift in the enforcement or scope of existing laws and regulations, or change in the countries, governments, persons or technologies targeted by such laws and regulations, could also result in decreased use of our products, or in our decreased ability to export or sell our solutions to potential end customers. Any decreased use of our products or limitation on our ability to export or sell our products would likely adversely affect our business, financial condition and results of operations.

Export control and economic sanctions laws may also include prohibitions on the sale or supply of certain of our products to specific targeted entities, or embargoed or sanctioned countries (including their governments, entities organized under their laws, and residents). Any deterioration in relations between the United States and Mainland China, Taiwan, the Middle East and other jurisdictions could lead to additional sanctions or export controls on such countries, regions, and specific individuals or entities, which could impact our ability to sell to or source components from such locales or otherwise negatively impact our business. In addition, trade regulations or other governmental actions targeted at one country or entity may impact other countries or entities. Any decreased use of our solutions or limitation on our ability to export or sell products to certain regions would adversely affect our business, financial condition and results of operations. For example, one of our largest end customers historically was Huawei, which accounted for approximately 41% of our net sales during 2024 and is designated on the Entity List administered by the U.S. Department of Commerce Bureau of Industry and Security. Consequently, our sales to Huawei are subject to a license requirement from the U.S. Government for all items subject to the licensing jurisdiction of the EAR. To date, our sales to Huawei have been authorized pursuant to licenses granted by the U.S. Government, which require among other things that our products not be used in 5G-compatible devices. If we fail to comply with the terms of these licenses, we and certain of our employees could be adversely affected through reputational harm, government investigations, penalties, or denial or curtailment of our ability to export our products and technology in the future. Moreover, such licenses could be revoked or amended by the U.S. Government at any time, and there is no guarantee that the U.S. Government will authorize future sales to Huawei of existing or new products. Further, we may lose business opportunities as Huawei (or other non-U.S. end customers affected by future U.S. export control measures or trade sanctions) may respond to U.S. export controls and trade sanctions by developing their own alternative products to replace our products or by purchasing products from non-U.S. competitors.

In addition, our association with Huawei or other end customers that are or become subject to U.S. regulatory scrutiny or export restrictions could subject us to actual or perceived reputational harm among current or prospective investors, suppliers or end customers, customers of our end customers, other parties doing business with us, or the general public. Any such reputational harm could result in the loss of investors, suppliers or end customers, which could harm our business, financial condition and results of operations.

Our failure to comply with the Foreign Corrupt Practices Act, other applicable anti-corruption and anti- bribery laws could subject us to penalties and other adverse consequences.

We have extensive international operations and a substantial portion of our business, particularly with respect to our manufacturing, assembly and testing processes, is conducted outside of the United States. Our operations are subject to the U.S. Foreign Corrupt Practices Act (FCPA), as well as the anti-corruption and anti- bribery laws in the countries where we do business. The FCPA prohibits covered parties from offering, promising authorizing or giving anything of value, directly or indirectly, to a "foreign government official" with the intent of improperly influencing the official's act or decision, inducing the official to act or refrain from acting in violation of lawful duty, or obtaining or retaining an improper business advantage. The FCPA also requires publicly traded companies to maintain records that accurately and fairly represent their transactions, and to have an adequate system of internal accounting controls. In addition, other applicable anti-corruption laws prohibit bribery of domestic government officials, and some laws that may apply to our operations prohibit commercial bribery, including giving or receiving improper payments to or from non-government parties, as well as so-called "facilitation" payments. Though we maintain policies, internal controls and other measures reasonably designed to promote compliance with applicable anti-corruption and anti-bribery laws, our employees or agents may nevertheless engage in improper conduct for which we might be held responsible. Any violations of these anti-corruption, or even allegations of such violations, can lead to an investigation and/or enforcement action, which could disrupt our operations, cause significant management distraction, and lead to significant costs and expenses, including legal fees. If we, or our employees or agents acting on our behalf, are found to have engaged in practices that violate these laws and regulations, we could suffer severe fines and penalties, profit disgorgement, injunctions on future conduct, securities litigation, bans on transacting government business, delisting from securities exchanges and other consequences that may have a material adverse effect on our business, financial condition and results of operations. In addition, our reputation and our stock price could be adversely affected if we become the subject of any negative publicity related to actual or perceived violations of anti-corruption or anti-bribery laws and regulations.

Our failure to comply with the Outbound Investment Security Program could subject us to penalties and other adverse consequences.

The Outbound Investment Security Program (OISP), issued to implement the "Executive Order on Addressing United States Investments in Certain National Security Technologies and Products in Countries of Concern", took effect on January 2, 2025. The OISP prohibits or requires notification of certain transactions involving U.S. persons and persons with a qualifying nexus to China (including Hong Kong and Macau) and specified covered activities in the semiconductors and microelectronics, quantum information technology, and artificial intelligence sectors. The OISP is a highly complex program with the potential for broad application, even with respect to entities and transactions outside of China.

At present, we are not a "covered foreign person" under the OISP such that U.S. person equity investments or other covered transactions would be implicated. However, as a U.S. company, the OISP has the potential to impact certain of our activities, including with respect to the strategic mergers and acquisitions, investments, and the development of certain JVs or new operations. Further, the OISP is likely to result in increased compliance burden and costs, which could adversely affect us.

Violations of the OISP may subject us or affiliated US persons to civil or criminal penalties, government investigations, business disruption, and reputational harm.

Changes in U.S. or foreign trade policies, including the imposition of tariffs, and other factors beyond our control may adversely impact our business and operating results.

Geopolitical tensions and trade disputes can disrupt supply chains and increase the cost of our products. This could cause our products to be more expensive for customers, which could reduce the demand for or attractiveness of such products. In addition, a geopolitical conflict in a region where we operate could disrupt our ability to conduct business operations in that region. Countries also could adopt restrictive trade measures, such as tariffs, laws and regulations concerning investments and limitations on foreign ownership of businesses, taxation, foreign exchange controls, capital controls, employment regulations and the repatriation of earnings and controls on imports or exports of goods, technology, or data, any of which could adversely affect our operations and supply chain and limit our ability to offer our products and services as intended. Changes in laws or policies governing the terms of foreign trade, and in particular increased trade restrictions, tariffs or taxes on imports from countries where we manufacture products or from where we import products or raw materials (either directly or through our suppliers) could have an impact on our competitive position, business operations and financial results. For example, the United States and other countries continue to implement restrictive trade actions, including tariffs, export controls, sanctions, legislation favoring domestic investment and other actions impacting the import and export of goods, foreign investment and foreign operations in jurisdictions in which we operate.

Recently, the United States has announced tariffs on products manufactured in several jurisdictions, including, Mexico, China and Canada, and has made announcements regarding the potential imposition of tariffs on other jurisdictions, such as the European Union. The United States has and may in the future pause, reimpose or increase tariffs, and countries subject to such tariffs have and in the future may impose reciprocal tariffs or other restrictive trade measures in response to the imposition of tariffs by the United States. We are actively monitoring and evaluating the development and potential impacts of tariffs on our supply chain and results of operations. We maintain operations worldwide, including in the jurisdictions impacted by the recently announced and contemplated tariffs. If the actual and potential tariffs and reciprocal tariffs are implemented as currently proposed, we expect that such actions could negatively impact our revenue growth and margins in future periods through increased costs, decreased demand and other adverse economic impacts. The net effect of these actions will depend on our ability to successfully mitigate and offset their impact, which may not be effective.

Trade restrictions could be adopted with little to no advanced notice, and we may not be able to effectively mitigate the adverse impacts from such measures. Political uncertainty surrounding trade or other international disputes also could have a negative impact on customer confidence and willingness to spend money, which could impair our future growth. Any of these events could increase the cost of our products, create disruptions to our supply chain and impair our ability to effectively operate and compete in the countries where we do business.

Techniques employed by short sellers may drive down the market price of our common stock.

Short selling is the practice of selling securities that the seller does not own but rather has borrowed from a third party with the intention of buying identical securities back at a later date to return to the lender. The short seller hopes to profit from a decline in the value of the securities between the sale of the borrowed securities and the purchase of the replacement shares, as the short seller expects to pay less in that purchase than it received in the sale. As it is in the short seller's interest for the price of the security to decline, many short sellers publish, or arrange for the publication of, negative opinions regarding the relevant issuer and its business prospects in order to create negative market momentum and generate profits for themselves after selling a security short. These short attacks have, in the past, led to selling of shares in the market.

Some public companies in China have been the subject of short selling in the past. Much of the scrutiny and negative publicity has centered on allegations of a lack of effective internal control over financial reporting resulting in financial and accounting irregularities and mistakes, inadequate corporate governance policies or a

lack of adherence thereto and, in many cases, allegations of fraud. If we were to become the subject of any unfavorable allegations, whether such allegations are proven to be true or untrue, we could have to expend a significant amount of resources to investigate such allegations and/or defend ourselves. While we would strongly defend against any such short seller attacks, we may be constrained in the manner in which we can proceed against the relevant short seller by principles of freedom of speech, applicable state law or issues of commercial confidentiality. Such a situation could be costly and time-consuming, and could distract our management from growing our business. Even if such allegations are ultimately proven to be groundless, allegations against us could severely impact our business operations, and any investment in our common stock could be greatly reduced or even rendered worthless.

Risks Related to Our Financial Performance

Our quarterly revenue and operating results are difficult to predict accurately and may fluctuate significantly from period to period. As a result, this could prevent us from meeting our own guidance or expectations of securities analysts or investors.

We operate in a highly dynamic industry and our future operating results could be subject to significant fluctuations, particularly on a quarterly basis. Our quarterly revenue and operating results have fluctuated significantly in the past and may continue to vary from quarter to quarter due to a number of factors, many of which are not within our control. Although some of our end customers, for example those in the personal consumer devices market, provide us with forecasts of their future requirements for our solutions, a significant percentage of our revenue in each quarter is dependent on sales that are booked and shipped during that quarter. As a result, accurately forecasting our operating results in any quarter is difficult. If our operating results do not meet the expectations of securities analysts and investors, our stock price would likely decline. Additional factors that can contribute to fluctuations in our operating results include:

- the rescheduling, increase, reduction or cancellation of significant end customer orders;
- the timing of end customer qualification of our products and commencement of volume sales by our end customers of systems that include our products;
- the timing and amount of research and development and sales and marketing expenditures;

- the rate at which our present and future end customers and end users adopt our solutions in our target end markets;
- the timing and success of the introduction of new solutions and technologies by us and our competitors, and the acceptance of our new solutions by our end customers;
- our ability to anticipate changing end customer product requirements;
- our gain or loss of one or more key end customers;
- the availability, cost and quality of wafers and other components that we purchase from third-party vendors and any problems or delays in the fabrication, assembly, testing or delivery of our products;
- the availability of production capacity at our third-party wafer fabrication facilities or other third-party subcontractors and other interruptions in the supply chain, including as a result of materials shortages, bankruptcies or other causes;
- changes in the manufacturing, assembly and testing costs of our products, including as a result of inflationary pressures;
- fluctuations in manufacturing yields;
- the changes in our product mix or end customer mix;
- competitive pressures resulting in lower than expected ASPs;
- the timing of expenses related to the acquisition of technologies or businesses;

- products rates of return or price concessions in excess of those expected or forecasted;
- the emergence of new industry standards;
- product obsolescence;
- unexpected inventory write-downs or write-offs;
- costs associated with litigation over intellectual property rights and other litigation;
- the length and unpredictability of the purchasing and budgeting cycles of our end customers;
- loss of key personnel or the inability to attract qualified management, technical, marketing and financial personnel;
- the quality of our products and any remediation costs;
- adverse changes in economic conditions in various geographic areas where we or our end customers do business;
- the general industry conditions and seasonal patterns in our target end markets, including the end user market;
- political developments related to Mainland China or Taiwan;
- other conditions affecting the timing of end customer orders or our ability to fill orders of end customers including end customers subject to export control or U.S. economic sanctions; and
- geopolitical events, such as the current conflicts between Ukraine and Russia or Israel and Hamas, threat of war or terrorist actions, or the occurrence of pandemics, epidemics or other outbreaks of disease, or natural disasters, and the impact of these events on the factors set forth above.

We may experience a delay in generating or recognizing revenue for a number of reasons. Open orders at the beginning of each quarter are typically lower than expected revenues for that quarter and are generally cancelable or reschedulable with minimal notice. Accordingly, we depend on obtaining orders during each quarter for shipment in that quarter to achieve our revenue objectives and failure to fulfill such orders by the end of a quarter may adversely affect our operating results. Furthermore, we generally rely on end customers issuing purchase orders to buy our products rather than long-term supply contracts. In addition, we maintain a small infrastructure of facilities and human capital in several locations around the world and have a limited ability to reduce the expenses. Because we base our operating expenses on anticipated revenue trends and a high percentage of our expenses are fixed in the short term, any delay in generating or recognizing forecasted revenue or changes in levels of our end customers' forecasted demand could materially and adversely

impact our business, financial condition and results of operations. Due to our limited ability to reduce expenses, in the event our revenue declines, or our revenue does not meet our expectations, it is likely that in some future quarters our operating results will decrease from the previous quarter or fall below the expectations of securities analysts and investors. As a result of these factors, our operating results may vary significantly from quarter to quarter. For the three months ended March 31, 2025 and 2024, our net sales were $15.7 million and $15.2 million, respectively.

Accordingly, we believe that period-to-period comparisons of our results of operations should not solely be relied upon as indications of future performance.

Our ability to raise capital in the future may be limited and could prevent us from executing our growth strategy.

Our ability to operate and expand our business depends on the availability of adequate capital, which in turn depends on cash flow generated by our business and the availability of borrowing through debt and equity markets or other applicable financing arrangements. We believe that our existing cash resources and anticipated cash received from sales of our products, together with the net proceeds from this offering, will be sufficient to finance our continued operations, growth strategy, planned capital expenditures and the additional expenses we

expect to incur as a public company for at least the next 12 months. However, we have based this estimate on our current operating plans and expectations, which are subject to change, and we cannot assure you that our resources will be sufficient to meet our future liquidity needs. We may require additional capital to respond to business opportunities, challenges, acquisitions or other strategic transactions and/or unforeseen circumstances. The timing and amount of our working capital and capital expenditure requirements may vary significantly depending on numerous factors, including:

- market acceptance of our products;
- the need to adapt to changing technologies and technical requirements;
- the existence of opportunities for expansion; and
- access to and availability of sufficient management, technical, marketing and financial personnel.

If our capital resources are insufficient to satisfy our liquidity requirements, we may seek to sell additional equity securities or debt securities or obtain debt financing. The sale of additional equity securities or convertible debt securities would result in additional dilution to our stockholders. Additional debt would result in increased expenses and could result in covenants that would restrict our operations and our ability to incur additional debt or engage in other capital-raising activities. We have not made arrangements to obtain additional financing and there is no assurance that financing, if required, will be available in amounts or on terms acceptable to us, if at all. If we are unable to obtain adequate financing or financing on terms satisfactory to us, when we require it, our ability to continue to grow and support our business and respond to business opportunities and challenges could be significantly limited.

We could be subject to changes in tax rates or the adoption of new tax legislation, whether in or out of the United States, or could otherwise have exposure to additional tax liabilities, which could adversely affect our business, financial condition and results of operations.

As a multinational business, we are subject to income and other taxes in both the United States and various non-U.S. jurisdictions. Changes to tax laws or regulations in the jurisdictions in which we operate, or in the interpretation of such laws or regulations, could significantly increase our effective tax rate and reduce our cash flow from operating activities, and otherwise have a material adverse effect on our financial condition. In addition, other factors or events, including business combinations and investment transactions, changes in the valuation of our deferred tax assets and liabilities, adjustments to taxes upon finalization of various tax returns or as a result of deficiencies asserted by taxing authorities, increases in expenses not deductible for tax purposes, changes in available tax credits, changes in transfer pricing methodologies, other changes in the apportionment of our income and other activities among tax jurisdictions, and changes in tax rates, could also increase our effective tax rate. Future changes in enacted tax rates could also negatively affect our results of operations. For example, on August 16, 2022, President Biden signed into law the Inflation Reduction Act of 2022, which includes a minimum tax equal to fifteen percent of the adjusted financial statement income of certain corporations as well as a one percent excise tax on share buybacks, effective for tax years beginning in 2023.

It is possible that the minimum tax could result in an additional tax liability over the regular federal corporate tax liability in a given year based on differences between book and taxable income (including as a result of temporary differences).

Our tax filings are subject to review or audit by the U.S. Internal Revenue Service (the IRS) and state, local and non-U.S. taxing authorities. We exercise significant judgment in determining our worldwide provision for taxes and, in the ordinary course of our business, there may be transactions and calculations where the proper tax treatment is uncertain. We may also be liable for taxes in connection with businesses we acquire. Our determinations are not binding on the IRS or any other taxing authorities, and accordingly the final determination in an audit or other proceeding may be materially different than the treatment reflected in our tax provisions, accruals and returns. An assessment of additional taxes because of an audit could have a material adverse effect on our business, financial condition and results of operations. Further changes in the tax laws of non-U.S. jurisdictions could arise, in particular,

as a result of the base erosion and profit shifting project that was undertaken by the Organization for Economic Co-operation and Development (the OECD). The OECD, which represents a coalition of member countries, recommended changes to numerous long-standing tax principles. These changes could increase tax uncertainty and may adversely affect our provision for income taxes and increase our tax liabilities.

Our ability to use our net operating loss carryforwards and certain other tax attributes may be limited.

We had federal and state net operating loss (NOLs) carryforwards of approximately $160.5 million as of March 31, 2025 and December 31, 2024. Under current law, our NOLs generated in tax years beginning after December 31, 2017 may be carried forward indefinitely, but the deductibility of such federal NOLs in tax years beginning after December 31, 2020, is limited to 80% of taxable income. It is uncertain if and to what extent various states will conform to the federal law. In addition, under Sections 382 and 383 of the U.S. Internal Revenue Code of 1986, as amended (the Code), if a corporation undergoes an "ownership change," generally defined as a greater than 50 percentage point change (by value) in its equity ownership by certain stockholders over a three-year period, the corporation's ability to use its pre-change NOLs and other pre-change tax attributes (such as research and development tax credits) to offset its post-change income or taxes may be limited. We may have experienced ownership changes in the past and may experience ownership changes in the future as a result of this offering and/or subsequent shifts in our stock ownership (some of which may be outside our control). As a result, our ability to use our pre-change NOLs and tax credits to offset post-change taxable income, if any, could be subject to limitations. Similar provisions of state tax law may also apply. In addition, at the state level, there may be periods during which the use of NOLs is suspended or otherwise limited, which could accelerate or permanently increase state taxes owed.

Risks Related to Our Intellectual Property

If we are unable to obtain, maintain and enforce patent protection for our current and future proprietary technology and inventions, or if the scope of the patent protection obtained is not sufficiently broad, our ability to compete successfully and our business, financial condition and results of operations could be adversely impacted.

Our future success depends, in part, upon our proprietary technology and inventions. We seek to protect our proprietary technology and inventions, particularly those relating to the design of our products, through the use of patents. As of June 6, 2025, we owned 52 issued U.S. patents, 10 issued foreign patents, 22 pending U.S. patent applications, seven pending foreign patent applications, and one pending Patent Cooperation Treaty application. The issued patents in the United States generally expire beginning in 2033 through 2042. Maintenance of patent portfolios, particularly outside of the United States, is expensive, and the process of seeking patent protection is lengthy and costly. While we intend to maintain our current portfolio of patents and to continue to prosecute our currently pending patent applications and file future patent applications when appropriate, the value of these actions may not exceed their expense. Existing patents and those that may be issued from any pending or future applications may be subject to challenges, opposition, invalidation or circumvention or designed around by our competitors or declared invalid or unenforceable in judicial or administrative proceedings, and the rights granted under our patents may not provide us with meaningful protection or any commercial advantage. It is also possible that we will fail to identify patentable aspects of our research and development output before it is too late to obtain patent protection.

Our existing issued and granted patents and any future patents we obtain may not be sufficiently broad to prevent others from using our technology or from developing competing technology. Although we enter into confidentiality agreements with parties who have access to confidential, proprietary or patentable aspects of our research and development output, such as our employees, consultants, service providers, and third parties with whom we have strategic relationships and business alliances, such agreements may be insufficient or any of these parties could breach such agreements and disclose such output before a patent application is filed, thereby jeopardizing our ability to seek patent protection and prevent competitors from using our technology. In addition, the protection afforded under the patent laws of one country may not be the same as that in other countries. This

means, for example, that our right to exclusively commercialize a product in those countries where we have patent rights for that product can vary on a country-by-country basis. We also may not have the same scope of patent protection in every country where we do business. Additionally, it is difficult and costly to monitor the use of our intellectual property. It may be the case that our intellectual property is already being infringed, misappropriated, or otherwise violated, and infringement, misappropriation or other violations of our intellectual property may occur in the future without our knowledge. The difficulty and failure to identify any violations of our intellectual property rights could materially and adversely affect our business, financial condition and result of operations and hurt our competitive advantage.

Significant litigation over intellectual property in our industry may cause us to become involved in costly and lengthy litigation which could adversely affect our business, financial condition and results of operations.

The semiconductor industry has experienced significant litigation involving patents and other intellectual property rights. From time to time, third parties, including non-practicing entities, allege intellectual property infringement by our products, our licensors' intellectual property, our end customers' products, or products using technologies or communications standards used in our industry. We periodically receive notices that claim we have infringed, misappropriated or otherwise violated other parties' intellectual property rights. As we gain greater public recognition, we may face a higher risk of being the subject of intellectual property claims. Litigation regarding intellectual property rights is inherently uncertain due to the complex issues involved, and we may not be successful in defending ourselves in such matters. Third parties may assert infringement claims against us based on existing patents or patents that may be granted in the future, regardless of their merit, and some of our competitors have, and will in the future have, extensive portfolios of issued patents. Any intellectual property claims against us, with or without merit, could be time consuming and expensive to settle or litigate, divert the attention of our management and scientific personnel, or require us to enter into costly royalty or licensing agreements, if available. Many potential litigants, including some of our competitors and patent holding companies, have the ability to dedicate substantial resources to enforcing their intellectual property rights. If such parties were to assert their intellectual property rights against us, even if we believe we would have defenses against any such assertion, there can be no assurance that any such defenses will be successful.

Patent and other types of intellectual property litigation can involve complex factual and legal questions, and their outcome is uncertain. Even if we believe such claims are without merit, a court of competent jurisdiction could hold that these third-party patents are valid, enforceable and infringed, which could adversely affect our ability to commercialize our proprietary technology. In order to successfully challenge the validity of any such U.S. patent in federal court, we would need to overcome a presumption of validity. As this burden is a high one requiring us to present clear and convincing evidence as to the

invalidity of any such U.S. patent claim, there is no assurance that a court of competent jurisdiction would invalidate the claims of any such U.S. patent or find that our proprietary technology did not infringe any such claims. As discussed above, the outcome of litigation under the laws of foreign jurisdictions is similarly uncertain. Further, even if we were successful in defending against any such claims, such claims could require us to divert substantial financial and management resources that we would otherwise be able to devote to our business. Any claims successfully brought against us could subject us to significant liability for damages, which could include treble damages and attorneys' fees if we are found to willfully infringe a third party's patent, and we may be required to stop using technology or other intellectual property alleged to be in violation of a third party's rights.

We also receive communications from end customers or suppliers requesting indemnification for allegations brought against them by third parties. Some of these allegations may result in our involvement in litigation in the future. We have certain contractual obligations to defend and indemnify our end customers from certain infringement claims.

Legal proceedings initiated by us to protect our intellectual property rights could also result in counterclaims or countersuits against us, such as claims asserting that our patents are invalid or unenforceable. In patent litigation in the United States, defendant counterclaims alleging invalidity or unenforceability are commonplace.

Grounds for a validity challenge could be an alleged failure to meet any of several statutory requirements, including lack of novelty, obviousness, non-enablement or lack of statutory subject matter. Grounds for an unenforceability assertion could be an allegation that someone connected with prosecution of the patent withheld relevant material information from the USPTO, or made a materially misleading statement, during prosecution. Third parties also may raise similar validity claims against our patents before the USPTO in post-grant proceedings such as ex parte reexaminations, inter partes review, or post-grant review, or oppositions or similar proceedings outside the United States, in parallel with litigation or even outside the context of litigation. The outcome following legal assertions of invalidity and unenforceability is unpredictable. We cannot be certain that there is no invalidating prior art, of which we and the patent examiner were unaware during prosecution. For the patents and patent applications that we have licensed, we may have limited or no right to participate in the defense of any licensed patents against challenge by a third party. If a defendant were to prevail on a legal assertion of invalidity or unenforceability, we would lose at least part, and perhaps all, of any future patent protection on our current or future proprietary technologies. Such a loss of patent protection could materially and adversely impact our business, financial condition and results of operations.

Furthermore, because of the substantial amount of discovery required in connection with intellectual property litigation, there is a risk that some of our confidential information could be compromised by disclosure during this type of litigation. There could also be public announcements of the results of hearings, motions or other interim proceedings or developments. If securities analysts or investors perceive these results to be negative, it could materially and adversely affect the trading price of our common stock. We may not have sufficient financial or other resources to adequately conduct such litigation or proceedings. Some of our competitors or other third parties may be able to sustain the costs of such litigation or proceedings more effectively than we can because of their greater financial resources and more mature and developed intellectual property portfolios. Litigation or other legal proceedings relating to intellectual property claims can be complex and could distract our scientific and management personnel from their normal responsibilities. Uncertainties resulting from the initiation and continuation of patent litigation or other proceedings could have a material adverse effect on our business, financial condition and results of operations.

Any litigation, regardless of its outcome or merit, could be time-consuming and expensive to resolve and could divert our management's time and attention. Intellectual property litigation also could force us to take specific actions, including:

- Cease using, selling or manufacturing certain products, solutions or processes;
- Attempt to obtain a license, which license may require the payment of substantial royalties or may not be available on reasonable terms or at all;
- Incur significant costs, time delays and lost business opportunities to develop alternative, non-infringing technologies or redesign solutions; or
- Pursue legal remedies with third parties to enforce our indemnification rights, which may not adequately protect our interests.

There is a risk that our trade secrets, know-how and other confidential or proprietary information will be stolen, used in an unauthorized manner, or compromised, which could materially and adversely affect our business, financial condition and results of operations.

In addition to the protection afforded by patents, we rely on trade secret protection and contractual arrangements to protect our proprietary know-how, confidential information and technology that is not covered by our patents. Our trade secrets, know-how and other proprietary information may be stolen, used in an unauthorized manner, or compromised through a direct intrusion by private parties or foreign actors, including those affiliated with or controlled by state actors, through cyber intrusions into our computer systems, physical theft through corporate espionage or other means, or through more indirect routes, including by joint venture partners, licensees that do not honor the terms of the license, potential licensees that were ultimately not licensed, or other parties reverse engineering our products, solutions, components or processes.

We generally enter into confidentiality and invention assignment agreements with our employees and consultants and confidentiality agreements with service providers and other third parties, including those with whom we have strategic relationships and business alliances. Despite the contractual provisions employed, no assurance can be given that these agreements will be effective in controlling access to our proprietary information and trade secrets, and the need to share trade secrets and other confidential information increases the risk that such trade secrets become known by our competitors or other third parties, are inadvertently incorporated into the technology of others or are disclosed or used in violation of these agreements. The

confidentiality agreements on which we rely to protect certain technologies may be breached, may not be adequate to protect our confidential information, trade secrets, and proprietary technologies, and may not provide an adequate remedy in the event of unauthorized use or disclosure of our confidential information, trade secrets, or proprietary technology. Further, these agreements do not prevent our competitors or others from independently developing products that are substantially equivalent or superior to ours. In addition, others may independently discover our trade secrets, and, in such cases, we may not be able to assert any trade secret rights against such parties.

We also rely on physical and electronic security measures to protect our confidential or proprietary information, but we cannot provide assurance that these security measures will not be breached or provide adequate protection for our property. There is a risk that third parties may obtain and improperly utilize our confidential or proprietary information to our competitive disadvantage. We may not be able to detect or prevent the unauthorized use of such information or take appropriate and timely steps to protect and enforce our intellectual property rights. Any of the foregoing factors could materially and adversely affect our business, financial condition and results of operations.

If our trademarks and trade names are not adequately protected, we may not be able to build name recognition in our markets of interest, and our competitive position may be harmed.

Our registered or unregistered trademarks or trade names in the United States and in foreign jurisdictions may be challenged, infringed, circumvented, declared generic, lapsed or determined to be infringing on or dilutive of other marks, and our current and future trademark applications may not be allowed or may subsequently be opposed. We may not be able to protect our rights in these trademarks and trade names, which we need in order to build name recognition with potential end customers. At times, competitors may adopt trade names or trademarks similar to ours, thereby impeding our ability to build brand identity and possibly leading to market confusion. In addition, there could be potential trade name or trademark infringement claims brought by owners of other trademarks or trademarks that incorporate variations of our registered or unregistered trademarks or trade names. As a means to enforce our trademark rights and prevent infringement, we may be required to file trademark claims against third parties or initiate trademark opposition proceedings. This can be expensive and time-consuming, and we may not be able to protect our rights to these trademarks and trade names, which we need to build name recognition among potential partners or end customers in our markets of interest. Over the long term, if we are unable to establish name recognition based on our trademarks and trade names, then we may not be able to compete effectively and our business may be adversely affected.

We may license our trademarks and trade names to third parties, such as distributors. Though these license agreements may provide guidelines for how our trademarks and trade names may be used, a breach of these agreements or misuse of our trademarks and trade names by our licensees may jeopardize our rights in or diminish the goodwill associated with our trademarks and trade names. Moreover, effective trademark protection may not be available or may not be sought in every country in which our products are made available, in every market in which we operate, and contractual disputes may affect the use of marks governed by private contract. Our efforts to enforce or protect our proprietary rights related to trademarks and trade names may be ineffective and could result in substantial costs and diversion of resources and could materially adversely affect our business, financial condition and results of operations.

We may be subject to claims that our employees, consultants or advisors have wrongfully used or disclosed trade secrets of their current or former employers or claims asserting ownership of what we regard as our own intellectual property.

Many of our employees were previously employed at other companies in our field, including our competitors or potential competitors, and many of our consultants are currently or were previously engaged by these companies. Although we try to ensure that our employees and consultants do not use the confidential or proprietary information or know-how of others in their work for us, we may in the future be subject to claims that we or these individuals have used or disclosed intellectual property, including trade secrets or other confidential or proprietary information, of any such individual's current or former employer. Litigation may be necessary to defend against these claims. If we fail in defending any such claims, in addition to paying monetary damages, we may lose valuable intellectual property rights or personnel or be subject to temporary or permanent injunctions against our solutions or processes. In addition, any such litigation or the threat thereof may adversely affect our ability to hire employees or contract with independent contractors. A loss of key personnel or their work product could hamper or prevent our ability to commercialize our solutions, which would have a material adverse effect on our business, financial condition and results of operations. Even if we are successful in defending against such claims, litigation could result in substantial costs and be a distraction to management and scientific personnel.

In addition, while it is our policy to require our employees and contractors who may be involved in the conception or development of intellectual property to execute agreements assigning such intellectual property to us, we may be unsuccessful in executing such an agreement with each party who, in fact, conceives or develops intellectual property that we regard as our own. The assignment of intellectual property rights may not be self- executing, or the assignment agreements may be breached, and we may be forced to bring claims against third parties or defend claims that they may bring against us to determine the ownership of what we regard as our intellectual property. Furthermore, individuals executing agreements with us may have preexisting or competing obligations to a third party, and thus an agreement with us may be ineffective in perfecting ownership of inventions developed by that individual. Disputes about the ownership of intellectual property that we may own could materially and adversely affect our business, financial condition and results of operations.

We, and the third parties with whom we work with, are subject to stringent and evolving U.S. and foreign laws, regulations, rules, standards, contractual obligations, policies, and other obligations, related to data privacy and cybersecurity. Our (or the third parties with whom we work with) actual or perceived failure to comply with such obligations could lead to regulatory investigations or actions; litigation (including class claims) and mass arbitration demands; fines and penalties; disruptions of our business operations; reputational harm; loss of revenue or profits; loss of customers or sales; and other adverse business consequences.

The legislative and regulatory framework for data privacy and protection and cybersecurity issues worldwide is rapidly evolving. In the ordinary course of business, we collect, receive, store, handle, transfer, use, make accessible, protect, secure, transmit, share, and otherwise process personal data and other sensitive information, including proprietary and confidential business data, trade secrets, intellectual property, and sensitive third-party data. Our data

processing activities subject us to numerous data privacy and cybersecurity obligations, such as various laws, regulations, rules, guidance, industry standards, external and internal privacy and cybersecurity policies, contractual requirements, and other obligations that govern the processing of personal data by us and on our behalf.

In the United States, federal, state, and local governments have enacted numerous data privacy and cybersecurity laws, including data breach notification laws, personal data privacy laws, wiretapping laws, and consumer protection laws (e.g. Section 5 of the Federal Trade Commission Act). For example, the California Consumer Privacy Act, as amended by the California Privacy Rights Act (CCPA), applies to personal data of consumers, business representatives, and employees who are California residents, and requires businesses subject to the CCPA to provide specific disclosures in privacy notices and honor requests of such individuals to exercise certain privacy rights. The CCPA, among other things, gives California residents the right to access, delete and

opt out of certain sharing of their information and imposes penalties for failure to comply, including allowing for private litigants affected by certain data breaches to recover significant statutory damages.

The enactment of the CCPA has led a wave of similar legislative developments in other states in the United States, which creates the potential for a patchwork of overlapping but different state laws and could mark the beginning of a trend toward more stringent privacy legislation in the United States, which could increase our potential liability and adversely affect our business, financial condition and results of operations. Many states have passed or are considering passing comprehensive data privacy legislation, and additional states could adopt data privacy legislation that may include more stringent data privacy requirements. Further, to the extent multiple state-level laws are introduced with inconsistent or conflicting standards, it may require costly and difficult efforts to achieve compliance with such laws. All 50 states have passed some form of legislation relating to data privacy or cybersecurity (for example, all 50 states have enacted laws requiring disclosure of certain personal data breaches). At the federal level, the United States Congress is considering various proposals for comprehensive federal data privacy legislation and, while no comprehensive federal data privacy law currently exists, we are subject to applicable existing federal laws and regulations, such as the rules and regulations promulgated under the authority of the Federal Trade Commission, which regulates unfair or deceptive acts or practices, including with respect to data privacy and security. These state statutes, and other similar state or federal laws, may require us to modify our data processing practices and policies and incur substantial compliance-related costs and expenses.

Outside the United States, an increasing number of laws, regulations, rules, and industry standards govern data privacy and cybersecurity. Many foreign countries and governmental bodies, including throughout Europe, Asia and other relevant jurisdictions where we conduct business, have laws and regulations concerning the collection, use, handling, transfer and other processing of personal data and other data obtained from their residents or by businesses operating within their jurisdictions that are more restrictive than those in the United States. For example, the European Union's General Data Protection Regulation (EU GDPR), the United Kingdom's GDPR (UK GDPR) (collectively, GDPR), and China's Personal Information Protection Law (PIPL) impose strict requirements for the collection, processing and transfer of personal data. For example, under the GDPR, government regulators may impose temporary or definitive bans on data processing, as well as fines of up to 20 million Euros (in the case of the EU GDPR), 17.5 million pounds sterling (in the case of the UK GDPR), or in each case, up to 4% of annual global revenue, whichever is greater. It may also lead to civil litigation related to processing of personal data brought by classes of data subjects or consumer protection organizations authorized at law to represent their interests, with the risks of damages or injunctive relief. In addition, an actual or asserted violation of the GDPR could result in regulatory investigations, reputational damage, orders to cease or change our processing of our data, enforcement notices and/or assessment notices (for a compulsory audit).

We also target customers in Asia and may be subject to new and emerging data privacy regimes in Asia including China's PIPL, Japan's Act on the Protection of Personal Information, and Singapore's Personal Data Protection Act. China's PIPL shares certain similarities with the GDPR, including extraterritorial application, requirements for data minimization, data localization, and purpose limitation, and obligations to provide certain notices and rights to citizens of the PRC. The PIPL allows for fines of up to 50 million renminbi or 5% of a covered company's revenue in the prior year. India's new privacy legislation, the Digital Personal Data Protection Act (DPDP), may also apply to our operations.

In the ordinary course of business, we may transfer personal data from Europe and other jurisdictions to the United States or other countries. Europe, as well as other jurisdictions, have enacted data localization laws and cross-border personal data transfer laws, which could make it more difficult to transfer information across jurisdictions. In particular, the European Economic Area (EEA) and the United Kingdom (UK) have significantly restricted the transfer of personal data to the United States and other countries whose privacy laws it generally believes are inadequate. For example, absent appropriate safeguards or other circumstances, the EU GDPR generally restricts the transfer of personal data to countries outside of the EEA, that the European Commission does not consider to provide an adequate level of data privacy and cybersecurity. Other jurisdictions may adopt similarly stringent interpretations of their data localization and cross-border data transfer laws. Additionally, existing mechanisms that facilitate cross-border personal data transfers such as the EEA standard contractual

clauses (SCC), the UK's International Data Transfer Agreement / Addendum, and the EU-U.S. Data Privacy Framework and the UK extension thereto (which allows for transfers to relevant U.S.-based organizations who self-certify compliance and participate in the Framework), are subject to legal challenges, and there is no assurance that we can satisfy or rely on these measures to lawfully transfer personal data to the United States. Further, the SCCs impose additional compliance burdens, such as conducting transfer impact assessments to determine whether additional security measures are necessary to protect the at-issue personal data.

These developments regarding cross-border data transfers have created uncertainty and increased the risk around our international operations and may require us to review and amend the legal mechanisms by which we make or receive personal data transfers to the United States and other jurisdictions. We may, among other things, be required to implement additional or different contractual and technical safeguards for any personal data transferred out of the EEA, which may increase compliance costs, lead to increased regulatory scrutiny or liability, may require additional contractual negotiations, may impose

operational burdens or limitations, and may materially and adversely impact our business, financial condition and results of operations. If we cannot implement a valid compliance mechanism for cross-border data transfers, we may face increased exposure to regulatory actions, substantial fines, and injunctions against processing or transferring personal data from Europe or other foreign jurisdictions. The inability to import personal data to the United States and other countries could significantly and negatively impact our business operations including by limiting our ability to collaborate with parties that are subject to such cross-border data transfer or localization laws, or requiring us to increase our personal data processing capabilities and infrastructure in foreign jurisdictions at significant expense.

Additionally, we have incurred, and may continue to incur, significant expenses in an effort to comply with data privacy, data protection, and cybersecurity standards, rules, and protocols imposed by law, regulation, industry standards, or contractual obligations. Publication of our data privacy statement and other policies regarding privacy, data protection, and cybersecurity may subject us to investigation or enforcement actions by regulators if those statements or policies are found to be deficient, lacking transparency, deceptive, unfair, or misrepresentative of our practices. We are also bound by contractual obligations related to data privacy, data protection, and cybersecurity and our efforts to comply with such obligations may not be successful or may have other negative consequences.

Obligations related to data privacy and cybersecurity (including and consumers' data privacy expectations) are quickly changing in an increasingly stringent fashion, creating some uncertainty as to the effective future legal framework. Additionally, these obligations may be subject to differing applications and interpretations, which may be inconsistent or conflict among jurisdictions. Preparing for and complying with these obligations requires significant resources and may necessitate changes to our solutions, information technologies, systems, and practices and to those of any third parties that process personal data on our behalf. In addition, these obligations may require us to change our business model. Despite our efforts to comply with all applicable data privacy and cybersecurity obligations, it is possible that our interpretations of the law, regulations, rules, standards, practices, or policies could be inconsistent with, or fail or be alleged to fail to meet all requirements of such obligations. Moreover, despite our efforts, our personnel or third parties with whom we work may fail to comply with such obligations, which could negatively impact our business operations and compliance posture. For example, any failure by a third-party processor to comply with applicable law, regulations, or contractual obligations could result in adverse effects, including inability to or interruption in our ability to operate our business and proceedings against us by governmental entities or others.

If we, or the third parties with whom we work, fail, or are perceived to have failed, to address or comply with data privacy and cybersecurity obligations, we could face significant consequences. These consequences may include, but are not limited to, government enforcement actions (e.g., investigations, fines, penalties, audits, inspections, and similar), litigation (including class-related claims), additional reporting requirements and/or oversight, bans on processing personal data, and orders to destroy or not use personal data. In particular, plaintiffs have become increasingly more active in bringing data privacy-related claims against companies, including class claims and mass arbitration demands. Some of these claims allow for the recovery of statutory damages on a per violation basis, and, if viable, carry the potential for monumental statutory damages, depending

on the volume of data and the number of violations. Any of these events could have a material adverse effect on our reputation, business, or financial condition, including but not limited to: loss of customers, interruptions or stoppages in our business operations, interruptions or stoppages of data collection needed to train our algorithms, inability to process personal data or to operate in certain jurisdictions, limited ability to develop or commercialize our solutions, expenditure of time and resources to defend any claim or inquiry, adverse publicity, or revision or restructuring of our operations. Even if not subject to legal challenge, the perception of concerns relating to data privacy, data protection, or cybersecurity, whether or not valid, could materially and adversely affect our business, financial condition and results of operations.

Our use of open source software could compromise the proprietary nature of our software, or the software we develop for our clients, and expose us to other legal liabilities and technological risks.

Our solutions and technology incorporate open source software, and we expect to continue to incorporate open source software in our business in the future. Few of the licenses applicable to open source software have been interpreted by courts, and there is a risk that these licenses could be construed in a manner that could impose unanticipated conditions or restrictions on our ability to commercialize our solutions. Certain open source licenses may give rise to requirements to disclose or license our proprietary source code or make available any derivative works or modifications of the open source code on unfavorable terms or at no cost, and we may be subject to such terms if such open source software is combined, linked, or otherwise integrated with our proprietary software in certain ways. If we combine our proprietary software with open source software in a certain manner, we could, under certain open source licenses, be required to release the source code of our proprietary software to the public. Although we believe that we have complied with our obligations under the applicable licenses for open source software that we use, there is little or no legal precedent governing the interpretation of many of the terms of certain of these licenses. As a result, the potential impact of these terms is uncertain and may result in unanticipated obligations or restrictions regarding those of our products, technologies or solutions affected.

If a third party were to allege that we had not complied with the conditions of one or more of these licenses, we could be required to incur significant legal expenses defending against such allegations, and could be subject to significant damages and required to comply with onerous conditions or restrictions on the use of our proprietary software. In any of these events, we could be required to seek licenses from third parties and pay royalties in order to continue using the open source software necessary to operate our business, or we could be required to discontinue use of our website and software in the event re-engineering cannot be accomplished on a timely basis. Additionally, the use of open-source software can lead to vulnerabilities that may make our solutions, software and technology susceptible to attack, and although some open-source vendors provide warranty and support agreements, it is common for such software to be available "as is" with no warranty, indemnity, support, or other contractual protection regarding infringement claims or the quality of the code. Any of the foregoing could require us to devote additional research and development resources to re-engineer our website and software, could result in customer dissatisfaction, could allow our competitors to create similar technology with lower development effort and time and may adversely affect our business, financial condition, and results of operations.

Risks Related to Ownership of Our Common Stock

Our issuance of additional capital stock in connection with financings, acquisitions, investments, our stock incentive plans or otherwise could dilute the ownership and voting power of existing stockholders.

Our amended and restated certificate of incorporation to be in effect following this offering will authorize us to issue shares of common stock or other securities convertible into or exercisable or exchangeable for shares of our common stock from time to time, for the consideration and on the terms and conditions established by our board of directors in its sole discretion, whether in connection with a financing, an acquisition, an investment, our stock incentive plans or otherwise. Such additional shares of our common stock or such other securities may be issued at a discount to the market price of our common stock at the time of issuance. In addition, our amended

and restated certificate of incorporation will authorize us to issue shares of preferred stock with such rights and preferences as may be determined by our board of directors. Our preferred stock could be issued with voting, liquidation, dividend and other rights superior to the rights of our common stock. The potential issuance of preferred stock may delay or prevent a change in control of us, discourage bids for our common stock at a premium to the market price, and materially and adversely affect the market price and the voting and other rights of the holders of our common stock. Any issuance of such securities could result in substantial dilution to our existing stockholders and cause the market price of shares of our common stock to decline.

We do not expect to declare or pay any cash dividends on our common stock for the foreseeable future.

We do not expect to pay cash dividends on our common stock for the foreseeable future. Consequently, investors must rely on sales of their shares of our common stock after price appreciation, which may never occur, as the only way to realize any future gains on their investment. Investors seeking dividends should not purchase shares of our common stock. Any future determination to pay dividends will be at the discretion of our board of directors and subject to, among other things, our compliance with applicable law, and depending on, among other things, our business prospects, financial condition, results of operations, cash requirements and availability, debt repayment obligations, capital expenditure needs, the terms of any preferred equity securities we may issue in the future, covenants in the agreements governing our current and future indebtedness, other contractual restrictions, industry trends, the provisions of the DGCL affecting the payment of dividends and distributions to stockholders and any other factors or considerations our board of directors may regard as relevant.

Our management will have broad discretion in the application of the net proceeds from this offering designated for general corporate purposes and may apply such net proceeds to uses that do not increase our market value or improve our operating results.

We intend to use the net proceeds received by us from this offering for research and development, working capital and other general corporate purposes as set forth under "Use of Proceeds." We may also use a portion of our net proceeds to acquire or invest in complementary businesses, products, solutions or technologies, though we do not have any agreements or commitments for any significant acquisitions or investments at this time. Other than as described above, we have not reserved or allocated the net proceeds for any specific purpose, and we cannot state with certainty how our management will use the net proceeds from this offering designated for general corporate purposes. Accordingly, our management will have considerable discretion in applying such net proceeds, and you will not have the opportunity, as part of your investment decision, to assess whether we are using such net proceeds appropriately. We may use our net proceeds for purposes that do not result in any improvement in our results of operations or increase the market value of our common stock. The failure by our management team to apply the net proceeds from this offering effectively could impair our growth prospects and result in financial losses that could have a material adverse effect on our business, financial condition and results of operations and cause the price of our common stock to decline. Until the net proceeds from this offering designated for general corporate purposes are used, they may be placed in investments that do not produce income or that lose value. Provisions in our amended and restated certificate of incorporation and bylaws will provide and the DGCL contains anti-takeover provisions that could prevent or discourage a takeover.

Provisions in our amended and restated certificate of incorporation and bylaws to be in effect upon the closing of this offering may discourage, delay or prevent an unsolicited merger, acquisition or other change in control of our company that stockholders may consider to be in their best interest, including transactions in which you might otherwise receive a premium for your shares. These provisions could also limit the price that investors might be willing to pay in the future for shares of our common stock, thereby depressing the market price of our common stock. In addition, because our board of directors is responsible for appointing the members of our management team, these provisions may frustrate or prevent any attempts by our stockholders to replace or remove our current management by making it more difficult for stockholders to replace members of our board of directors. Among other things, these provisions include those establishing:

- a classified board of directors with three-year staggered terms, which may have the effect of deferring, delaying or discouraging hostile takeovers, or changes in control of us or our management;

- no cumulative voting in the election of directors, which limits the ability of minority stockholders to elect director candidates;
- the exclusive right of our board of directors to elect a director to fill a vacancy created by, among other things, the expansion of the board of directors or the resignation, death or removal of a director, which prevents stockholders from filling vacancies on our board of directors;
- the ability of our board of directors to authorize the issuance of shares of preferred stock and to determine the terms of those shares, including preferences and voting rights, without stockholder approval, which could be used to significantly dilute the ownership of a hostile acquirer;

- the required approval of the holders of at least two-thirds of the shares of common stock entitled to vote at an election of directors to amend or repeal our bylaws or amend the provisions of our amended and restated certificate of incorporation regarding the election and removal of directors;
- a prohibition on stockholder action by written consent, which forces stockholder action to be taken at an annual or special meeting of our stockholders;
- a prohibition on stockholders calling a special meeting and the requirement that a meeting of stockholders may be called only by a majority of our board of directors, which may delay the ability of our stockholders to force consideration of a proposal or to take action, including the removal of directors; and
- advance notice procedures that stockholders must comply with in order to nominate candidates to our board of directors or to propose matters to be acted upon at an annual meeting or special meeting of stockholders, which may discourage or delay a potential acquirer from conducting a solicitation of proxies to elect the acquirer's own slate of directors or otherwise attempting to obtain control of us until the next stockholder meeting or at all.

Our amended and restated certificate of incorporation includes forum selection clauses, which could discourage claims or limit stockholders' ability to make a claim against us, our directors, officers, other employees or stockholders.

Our amended and restated certificate of incorporation will provide that, unless we consent in writing to the selection of an alternative forum, the Court of Chancery in the State of Delaware shall be the sole and exclusive forum for any stockholder (including a beneficial owner) to bring: (a) any derivative action or proceeding brought on behalf of the Company; (b) any claim or cause of action for breach of a fiduciary duty owed by any current or former director, officer or other employee of the Company, to the Company or the Company's stockholders; (c) any claim or cause of action against the Company or any current or former director, officer or other employee of the Company, arising out of or pursuant to any provision of the DGCL or our certificate of incorporation or bylaws; (d) any claim or cause of action seeking to interpret, apply, enforce or determine the validity of our certificate of incorporation or bylaws (as each may be amended from time to time, including any right, obligation, or remedy thereunder), (e) any claim or cause of action as to which the DGCL confers jurisdiction on the Court of Chancery of the State of Delaware; and (f) any claim or cause of action against the Company or any current or former director, officer or other employee of the Company, governed by the internal- affairs doctrine, in all cases to the fullest extent permitted by law and subject to the court having personal jurisdiction over the indispensable parties named as defendants. In addition, our amended and restated certificate of incorporation will provide that, unless we consent in writing to the selection of an alternative forum, the federal district courts of the United States will be the exclusive forum for resolving any complaint asserting a cause of action arising under the Securities Act.

Notwithstanding the foregoing, the Securities Act forum selection clause will not apply to suits brought to enforce any liability or duty created by the Exchange Act or any other claim for which the federal district courts of the United States of America shall be the sole and exclusive forum. These forum selection clauses may discourage claims or limit stockholders' ability to submit claims in a judicial forum that they find favorable and

may result in additional costs for a stockholder seeking to bring a claim. While we believe the risk of a court declining to enforce these forum selection clauses is low, if a court were to determine a forum selection clause to be inapplicable or unenforceable in an action, we may incur additional costs in conjunction with our efforts to resolve the dispute in an alternative jurisdiction, which could have a negative impact on our business, financial condition and results of operations.

Future sales of our common stock in the public market could cause the price of our common stock to decline.

Sales of a substantial number of shares of our common stock in the public market, particularly sales by our directors, executive officers, and principal stockholders, or the perception that these sales might occur, could cause the price of our common stock to decline and could impair our ability to raise capital through the sale of additional equity securities.

We, all of our directors and executive officers and the holders of substantially all of our outstanding shares of common stock and other equity securities, including options to acquire shares of our common stock, have entered or will enter into lock-up agreements with the underwriters and/or agreements with market standoff provisions that restrict our and their ability to sell or transfer shares of our common stock and/or securities convertible into or exercisable or exchangeable for our common stock, for a period of 180 days after the date of this prospectus (the Lock-Up Period), subject to certain limited exceptions, without first obtaining the written consent of BofA Securities, Inc. and UBS Securities LLC. For a further description of these lock-up agreements and market standoff agreements, please see "Underwriting." Unless earlier released, all of the securities subject to lock-up or market standoff agreements will become eligible for sale upon expiration of the lock-up period. Shares held by directors, executive officers, and other affiliates will be subject to volume limitations under Rule 144 under the Securities Act.

In addition, as of March 31, 2025, we had options outstanding that, if fully exercised, would result in the issuance of 61,795,893 shares of our common stock and 5,186,932 shares of our common stock issuable upon vesting of outstanding RSUs. We intend to file one or more registration statements on Form S-8 under the Securities Act to register the shares of our common stock subject to outstanding stock options and RSUs, as of the date of this prospectus and shares that will be issuable pursuant to future awards granted under our equity incentive plans. Once we register these shares, they can be freely sold in the public market upon issuance, subject to applicable vesting requirements, compliance by affiliates with Rule 144, and other restrictions provided under the terms of the applicable plan and/or the award agreements entered into with participants.

Following this offering, the holders of up to shares of our common stock will have rights, subject to certain conditions, to require us to file registration statements for the public resale of shares of our common stock or to include such shares in registration statements that we may file for us or other stockholders. Any registration statement we file to register additional shares, whether as a result of registration rights or otherwise, could cause the price of our common stock to decline or be volatile.

New investors in our common stock will experience immediate and substantial dilution in book value after this offering.

The initial public offering price is expected to be substantially higher than the pro forma as adjusted net tangible book value per share of our common stock. If you purchase common stock in this offering, you will incur immediate dilution of $ per share, representing the difference between the assumed initial public offering price of $ per share, the midpoint of the price range set forth on the cover page of this prospectus, and the pro forma as adjusted net tangible book value per share of our common stock as of March 31, 2025. For additional information on the dilution you may experience as a result of investing in this offering, see "Dilution."

Our stock price may be volatile, and investors in our common stock may not be able to resell shares of our common stock at or above the price paid, or at all.

There has been no public market for our common stock prior to this offering. The initial public offering price for our common stock will be determined through negotiations between the underwriters and us and may vary from the market price of our common stock following this offering. If you purchase our common stock in this offering, you may not be able to resell those shares at or above the initial public offering price, or at all. An active market in our common stock may not develop upon the closing of this offering or, if it does develop, it may not be sustainable or liquid enough for you to sell your shares. We have applied to list our common stock on , but we cannot assure you that an active trading market will develop. The trading price and volume of our common stock is likely to be volatile and could fluctuate significantly in response to numerous factors, many of which are beyond our control, including:

- variations in our actual or anticipated annual or quarterly operating results or those of others in our industry;
- results of operations that otherwise fail to meet the expectations of securities analysts and investors;
- changes in earnings estimates or recommendations by securities analysts, or other changes in investor perceptions of the investment opportunity associated with our common stock relative to other investment alternatives;
- market conditions in the semiconductor industry;
- publications, reports or other media exposure of our solutions or those of others in our industry, or of our industry generally;
- announcements by us or others in our industry including our competitors, or by our or their respective suppliers, distributors or other business partners, regarding, among other things, announcements of new solutions, significant contracts, price reductions, capital commitments or other business developments, the entry into or termination of strategic transactions or relationships, securities offerings or other financing initiatives, and public reaction thereto;
- general economic and political conditions, such as the effects of the conflicts between Russia and Ukraine or Israel and Hamas, global health crises, recessions, interest rates, local and national elections, fuel prices, international currency fluctuations, corruption, political instability and acts of war or terrorism;
- imposition of trade or similar sanctions by the United States or other countries in which we operate or do business;
- commencement of, or involvement in, litigation, in particular, litigation concerning our intellectual property;
- additions or departures of key management personnel;
- regulatory actions involving us or others in our industry, or actual or anticipated changes in applicable government regulations or enforcement thereof;
- the development and sustainability of an active trading market for our common stock;
- sales, or anticipated sales, of large blocks of our common stock; and
- technical factors in the public trading market for our common stock that may produce price movements that may or may not comport with macro, industry or company-specific fundamentals, including, without limitation, the sentiment of retail investors (including as may be expressed on financial trading and other social media sites), the amount and status of short interest in our securities, access to margin debt, trading in options and other derivatives on our common stock and any related hedging or other technical trading factors.

Furthermore, the stock market in general has experienced extreme price and volume fluctuations that have often been unrelated or disproportionate to the operating performance of particular companies. Broad market and industry factors may significantly affect the market price of our common stock, regardless of our actual operating performance. These and other factors may cause the market price and demand for our common stock to fluctuate significantly, which may limit or prevent investors from readily selling their shares of common stock and may otherwise negatively affect the liquidity of our common stock. In addition, in the past, when the market price of a stock has been volatile, holders of that stock have sometimes instituted securities class action litigation against the company that issued the stock. If any of our stockholders were to bring a lawsuit against us, we could incur substantial costs defending the lawsuit. Such a lawsuit could also divert the time and attention of our management from our core business operations.

We will incur significantly increased costs as a result of operating as a company whose common stock is publicly traded in the United States, and our management will be required to devote substantial time to new compliance initiatives.

As a public company in the United States, we will incur significant legal, accounting, and other expenses that we did not incur previously. These expenses will likely be even more significant after we no longer qualify as an emerging growth company. The Sarbanes-Oxley Act, the Dodd-Frank Wall Street Reform and Consumer Protection Act, the listing requirements of the New York Stock Exchange, and other applicable securities rules and regulations impose various requirements on public companies in the United States, including the establishment and maintenance of effective disclosure and financial controls and corporate governance practices. Our senior management and other personnel will need to devote a substantial amount of time to these compliance initiatives. Moreover, these rules and regulations will increase our legal and financial compliance costs and will make some activities more time- consuming and costly. For example, we expect that these rules and regulations may make it more difficult and more expensive for us to obtain director and officer liability insurance, which in turn could make it more difficult for us to attract and retain qualified senior management personnel or members for our board of directors. We cannot predict or estimate the amount of additional costs we may incur or the timing of such costs.

Compliance with these rules and regulations can be burdensome. Our management and other personnel need to devote a substantial amount of time to these compliance initiatives. Moreover, these rules and regulations have increased, and will continue to increase, our historical legal and financial compliance costs and will make some activities more time-consuming and costly. For example, we expect that these rules and regulations may make it more difficult and more expensive for us to attract and retain qualified members of our board of directors as compared to a private company. In particular, we incur significant expenses and devote substantial management effort toward ensuring compliance with the requirements of Section 404 of the Sarbanes-Oxley Act. We have hired additional accounting and financial staff, and engaged outside consultants, all with appropriate public company experience and technical accounting knowledge, which increased our operating expenses.

Our executive officers have limited experience in the management of a publicly traded company. Their limited experience in dealing with the increasingly complex laws pertaining to public companies could be a significant disadvantage in that it is likely that an increasing amount of their time may be devoted to these activities, which will result in less time being devoted to the management and growth of the post-combination company. We may not have adequate personnel with the appropriate level of knowledge, experience and training in the accounting policies, practices or internal control over financial reporting required of public companies. Our management will need to continually assess our staffing and training procedures to improve our internal control over financial reporting. Further, the development, implementation, documentation and assessment of appropriate processes, in addition to the need to remediate any potential deficiencies, will require substantial time and attention from management. The development and implementation of the standards and controls necessary for us to achieve the level of accounting standards required of a public company may require costs greater than

expected. It is possible that we will be required to expand our employee base and hire additional employees to support our operations as a public company which will increase our operating costs in future periods.

If we fail to maintain an effective system of disclosure controls and internal control over financial reporting, our ability to produce timely and accurate financial statements or comply with applicable regulations could be impaired.

As a public company, we will be subject to the reporting requirements of the Exchange Act, the Sarbanes- Oxley Act and the rules and regulations of the applicable listing standards of the New York Stock Exchange. We expect that the requirements of these rules and regulations will continue to increase our legal, accounting and financial compliance costs, make some activities more difficult, time-consuming and costly and place significant strain on our personnel, systems and resources. The Sarbanes-Oxley Act requires, among other things, that we maintain effective disclosure controls and procedures and internal control over financial reporting. We are continuing to develop and refine our disclosure controls and other procedures that are designed to ensure that information required to be disclosed by us in the reports that we will file with the SEC is recorded, processed, summarized and reported within the time periods specified in SEC rules and forms and that information required to be disclosed in reports under the Exchange Act is accumulated and communicated to our principal executive and financial officers. We are also continuing to improve our internal control over financial reporting.

In order to maintain and improve the effectiveness of our disclosure controls and procedures and internal control over financial reporting, we have expended, and anticipate that we will continue to expend, significant resources, including accounting-related costs and significant management oversight. Our current controls and any new controls that we develop may become inadequate because of changes in conditions in our business. In addition, changes in accounting principles or interpretations could also challenge our internal controls and require that we establish new business processes, systems and controls to accommodate such changes. We have limited experience with implementing the systems and controls that will be necessary to operate as a public company, as well as adopting changes in accounting principles or interpretations mandated by the relevant regulatory bodies. Additionally, if these new systems, controls or standards and the associated process changes do not give rise to the benefits that we expect or do not operate as intended, it could adversely affect our financial reporting systems and processes, our ability to produce timely and accurate financial reports or the effectiveness of internal control over financial reporting. Moreover, our business may be harmed if we experience problems with any new systems and controls that result in delays in their implementation or increased costs to correct any post-implementation issues that may arise.

Further, weaknesses in our disclosure controls and internal control over financial reporting may be discovered in the future. Any failure to develop or maintain effective controls or any difficulties encountered in their implementation or improvement could harm our business or cause us to fail to meet our reporting obligations and may result in a restatement of our financial statements for prior periods. Any failure to implement and maintain effective internal control over financial reporting also could adversely affect the results of periodic management evaluations and annual independent registered public accounting firm attestation reports regarding the effectiveness of our internal control over financial reporting that we will eventually be required to include

in our periodic reports that will be filed with the SEC. Ineffective disclosure controls and procedures and internal control over financial reporting could also cause investors to lose confidence in our reported financial and other information, which would likely have a negative effect on the trading price of our common stock. In addition, if we are unable to continue to meet these requirements, we may not be able to remain listed on .

We are not currently required to comply with the SEC rules that implement Section 404 of the Sarbanes- Oxley Act and are therefore not required to make a formal assessment of the effectiveness of our internal control over financial reporting for that purpose. As a public company, we will be required to provide an annual management report on the effectiveness of our internal control over financial reporting commencing with our second annual report on Form 10-K. Our independent registered public accounting firm is not required to formally attest to the effectiveness of our internal control over financial reporting until our first annual report filed with the SEC where we are an accelerated filer or a large accelerated filer, which will not occur until at least our second annual report on Form 10-K. At such time, our independent registered public accounting firm may issue a report that is adverse in the event it is not satisfied

with the level at which our internal control over financial reporting is documented, designed or operating. Any failure to maintain effective disclosure controls and internal control over financial reporting could harm our business and could cause a decline in the trading price of our common stock.

We identified a material weakness in our internal control over financial reporting, and we may experience additional material weaknesses or otherwise fail to design and maintain effective internal control over financial reporting.

In the course of preparing the financial statements that are included in this prospectus, management has determined that a material weakness exists within the internal controls over financial reporting. The material weakness identified relate to controls to address segregation of certain accounting duties and information technology controls. We concluded that the material weakness in our internal control over financial reporting occurred because we did not have the necessary business processes, systems, and related internal controls necessary to satisfy the accounting and financial reporting requirements of a public company.

As part of our plan to remediate this material weakness, we have made progress in several actions and will continue to implement our remediation plans. To remediate the material weakness, we have implemented a new financial system, are implementing a new operational system and have enhanced control procedures to mitigate segregation of duties. We have also implemented additional review controls and processes that require additional levels of review by those charged with financial governance of our company.

We will not be able to fully remediate the identified material weakness until the ongoing steps described above have been completed and our internal controls have been operating effectively for a sufficient period of time. We cannot assure you that we will be able to fully remediate the material weakness in a timely manner. We also may incur significant costs to execute various aspects of our remediation plan but cannot provide a reasonable estimate of such costs at this time.

We cannot assure you that we have identified all material weaknesses. Material weaknesses may still exist when we report on the effectiveness of our internal control over financial reporting as required under Section 404 of the Sarbanes-Oxley Act after the completion of this offering.

In the future, it is possible that additional material weaknesses or significant deficiencies may be identified. Our ability to comply with the annual internal control reporting requirements will depend on the effectiveness of our financial reporting and data systems and controls across our company. Any weaknesses or deficiencies or any failure to implement new or improved controls, or difficulties encountered in the implementation or operation of these controls, could harm our operating results and cause us to fail to meet our financial reporting obligations, or result in material misstatements in our consolidated financial statements, which could adversely affect our business, financial condition and results of operations and reduce our stock price.

If we are unable to conclude on an ongoing basis that we have effective internal control over financial reporting in accordance with Section 404, our independent registered public accounting firm may not issue an unqualified opinion. If we are unable to conclude that we have effective internal control over financial reporting, investors could lose confidence in our reported financial information, which could have a material adverse effect on the trading price of our common stock. Failure to remedy any material weakness in our internal control over financial reporting, or to implement or maintain other effective control systems required of public companies, could also restrict our future access to the capital markets.

Our management team has limited experience managing a public company.

Our management team has limited experience managing a publicly traded company, interacting with public company investors and securities analysts, and complying with the increasingly complex laws pertaining to public companies. These new obligations and constituents require significant attention from our management team and could divert their attention away from the day-to-day management of our business, which could harm our business, financial condition and results of operations.

We are an "emerging growth company," and a "smaller reporting company" and we cannot be certain if the reduced disclosure requirements applicable to emerging growth companies and smaller reporting companies will make our common stock less attractive to investors.

We are an "emerging growth company," as defined in the JOBS Act, and we could remain an emerging growth company until the last day of our fiscal year following the fifth anniversary of the closing of the IPO. For as long as we continue to be an emerging growth company, we may choose to take advantage of certain exemptions from various reporting requirements that are applicable to other public companies that are not emerging growth companies, including, but not limited to:

- not being required to engage an auditor to report on our internal controls over financial reporting pursuant to Section 404(b) of the Sarbanes-Oxley Act;

- not being required to comply with any requirement that may be adopted by the Public Company Accounting Oversight Board regarding mandatory audit firm rotation or a supplement to the auditor's report providing additional information about the audit and the financial statements (i.e., an auditor discussion and analysis);
- not being required to submit certain executive compensation matters to stockholder advisory votes, such as "say-on-pay," "say-on-frequency," and "say-on-golden-parachutes"; and
- not being required to disclose certain executive compensation related items such as the correlation between executive compensation and performance and comparisons of the chief executive officer's compensation to median employee compensation.

In addition, as an emerging growth company, we are only required to provide two years of audited financial statements and two years of selected financial data (in addition to any required interim financial statements and selected financial data) in this prospectus, and to present correspondingly reduced disclosure in "Management's Discussion and Analysis of Financial Condition and Results of Operations." We have elected to take advantage of this reduced disclosure obligation and certain of the other exemptions described above and we may elect to take advantage of these and other reduced reporting requirements in the future. As a result, the information that we provide to our stockholders may be different than the information you might receive from other public reporting companies in which you hold equity interests. In addition, the JOBS Act permits emerging growth companies to delay adopting new or revised accounting standards until such time as those standards apply to private companies. We have elected to use this extended transition period for complying with new or revised accounting standards until the earlier of the date we are no longer an emerging growth company or affirmatively and irrevocably opt out of the extended transition period provided in the JOBS Act. As a result, our consolidated financial statements and the reported results of operations contained herein may not be directly comparable to those of other public companies. We cannot predict whether investors will find our common stock less attractive because of our reliance on these exemptions. If some investors do find our common stock less attractive, there may be a less active trading market for our common stock and our stock price may be reduced or more volatile. We will remain an emerging growth company, and will be able to take advantage of the foregoing exemptions, until the last day of our fiscal year following the fifth anniversary of the closing of this offering or such earlier time that we otherwise cease to be an emerging growth company, which will occur upon the earliest of (i) the last day of the first fiscal year in which our annual gross revenues are $1.235 billion or more; (ii) the date on which we have, during the previous three-year period, issued more than $1.0 billion in non-convertible debt securities; and (iii) the date that we become a "large accelerated filer" as defined in Rule 12b-2 under the Exchange Act, which will occur as of the end of any fiscal year in which (x) the market value of our common equity held by non-affiliates is $700 million or more as of the last business day of our most recently completed second fiscal quarter, (y) we have been required to file annual and quarterly reports under the Exchange Act for a period of at least 12 months and (z) have filed at least one annual report pursuant to the Exchange Act.

We are also a "smaller reporting company" as defined in the Exchange Act. We may continue to be a smaller reporting company even after we are no longer an emerging growth company. We may take advantage of

certain of the scaled disclosures available to smaller reporting companies and will be able to take advantage of these scaled disclosures for so long as our common stock held by non-affiliates is less than $250.0 million measured on the last business day of our second fiscal quarter, or our annual revenue is less than $100.0 million during the most recently completed fiscal year and our common stock held by non-affiliates is less than $700.0 million measured on the last business day of our second fiscal quarter.

If equity research analysts or industry analysts do not publish research or reports about our business, or if they adversely change their recommendations regarding our stock, our stock price and trading volume could decline.

The trading market for our common stock is influenced by the research and reports that industry or equity research analysts publish about us or our business. If any of the analysts who cover us issue an adverse or misleading opinion regarding us, our business model, financial performance, stock price or otherwise, our stock price would likely decline. In addition, if we fail to meet the expectations of our results published by these analysts, our stock price would likely decline. If one or more of these analysts ceases coverage of our company or fails to regularly publish reports on us, we could lose visibility in the financial markets, which in turn could cause our stock price or trading volume to decline and result in the loss of all or a part of your investment in us.

Special Note Regarding Forward-looking Statements

This prospectus, including "Prospectus Summary," "Risk Factors," "Management's Discussion and Analysis of Financial Condition and Results of Operations," and "Business," contains forward-looking statements. We may, in some cases, use words such as "anticipate," "believe," "could," "estimate," "expect," "intend," "may," "plan," "potential," "predict," "project," "should," "will," "would," or the negative of those terms, and similar expressions that convey uncertainty of future events or outcomes to identify these forward- looking statements. Any statements contained herein that are not statements of historical facts may be deemed to be forward-looking statements. Forward-looking statements in this prospectus include, but are not limited to, statements about:

- the timing and success of new features, integrations, capabilities, and enhancements by us, or by our competitors to their products, including the successful integration of AI;
- our end customer relationships and our ability to retain and expand our end customer relationships and to achieve design wins;
- the success, cost, and timing of new products;

- our ability to compensate for decreases in average selling prices of our products;
- our ability to address market and end customer demands and to timely develop new or enhanced products to meet those demands;
- anticipated trends, challenges and growth in our business and the markets in which we operate, including pricing expectations;
- our expectations regarding our revenue, gross margin and expenses;
- the size and growth potential of the markets for our products, and our ability to serve those markets;
- our plans to expand sales and marketing efforts through increased collaboration with our distributors, resellers, and contracted sales representatives;
- the loss of one or more significant end customers;
- our expectations regarding competition in our existing and new markets, including our expectations concerning our mix of revenue by geography;
- regulatory developments in the United States and foreign countries, the deterioration in economic factors arising from trade disputes and the imposition of trade sanctions or increased tariffs;
- our dependence on international end customers and operations;
- the performance of our third-party suppliers and manufacturers;
- our and our end customers' ability to respond successfully to technological or industry developments;
- the cyclical nature of the semiconductor industry;
- our ability to attract and retain key management personnel;
- intellectual property and related litigation;
- the accuracy of our estimates regarding capital requirements and needs for additional financing;
- our expectations regarding the period during which we qualify as an emerging growth company under the JOBS Act;
- our expectations regarding our ability to obtain, maintain, protect, and enforce intellectual property protection for our technology;
- our use of the net proceeds from this offering; and

- the potential impact of macroeconomic conditions and geopolitical conflicts, including the conflicts between Russia and Ukraine or Israel and Hamas, recession fears, fluctuations in global interest rates, foreign exchange volatility and inflationary pressures, on our business and the businesses of our suppliers and end customers.

These forward-looking statements reflect our management's beliefs and views with respect to future events and are based on estimates and assumptions as of the date of this prospectus and are subject to risks and uncertainties. We discuss many of these risks in greater detail under "Risk Factors." Moreover, we operate in a very competitive and rapidly changing environment. New risks emerge from time to time. It is not possible for our management to predict all risks, nor can we assess the impact of all factors on our business or the extent to which any factor, or combination of factors, may cause actual results to differ materially from those contained in any forward-looking statements we may make. Given these uncertainties, you should not place undue reliance on these forward-looking statements.

You should read this prospectus and the documents that we reference in this prospectus and have filed as exhibits to the registration statement, of which this prospectus is a part, completely and with the understanding that our actual future results may be materially different from what we expect. We qualify all of the forward- looking statements in this prospectus by these cautionary statements.

You should not rely upon forward-looking statements as predictions of future events. Although we believe that the expectations reflected in the forward-looking statements are reasonable, we cannot guarantee that the future results, levels of activity, performance, or events and circumstances reflected in the forward-looking statements will be achieved or occur. Moreover, neither we nor any other person assumes responsibility for the accuracy and completeness of the forward-looking statements. We undertake no obligation to update publicly any forward-looking statements for any reason after the date of this prospectus to conform these statements to actual results or to changes in our expectations, except as required by law.

Industry and Market Data

This prospectus includes statistical and other industry and market data that we obtained from industry publications and research, surveys, and studies conducted by third parties, as well as estimates by our management based on such data. Certain information in this prospectus is contained in independent industry publications. These independent industry publications are provided below:

1. McKinsey Report titled " The Next Big Arenas of Competition," C. Bradley, M. Chui et al., October 2024.

2. Gartner®, Forecast: AI Semiconductors, Worldwide, 2022-2028, 3Q24 Update, Alan Priestley et al., 1 October 2024, Revenue basis.

- *The Market for AI-enabled 32-bit MCUs and Application Processors in edge applications = Sorted by Agriculture, Appliances, Automation, Head Mounted Displays, Hearable, Industrial Smart Robot, Medical/Healthcare, Personal Asset Tracker, Personal Robots, Smart Speakers, Smart Watches, Surveillance, Test/Measurement, Transportation and Wristbands.

3. Gartner®, Forecast: Semiconductors and Electronics, Worldwide, 2022-2028, 4Q24 Update, Rajeev Rajput et al., 16 December 2024 Revenue basis.

- **The Edge End Market/End markets=Sorted by Head Mounted Displays, Hearable, Smart Watches, Wristbands, Appliances, Smart Speakers, Personal Asset Tracker, Surveillance, Other Security, Automation, Medial/Healthcare, Energy Management, Transportation, Test/Measurement, Agriculture, Personal Robots and Industrial Smart Robots. Wireless Connectivity=Wireless Connectivity (NFC; WiFi; BT; GPS; Combo), Application Processors=Discrete Application/Multimedia Processor, 32-bit MCU=Microcontroller 32-bit and Above.

- ***Personal Devices=Wearables, Medical and Healthcare = Medical / Healthcare, Smart Home/Buildings=Sorted by Appliances, Smart Speakers, Personal Asset Tracker and Personal Robots, Industrial Edge=Sorted by Surveillance, Other Security, Automation, Energy Management, Transportation, Test/Measurement, Agriculture and Industrial Smart Robots.

- ****Personal Device Market = Sorted by Wearables. Medical and Healthcare = Medical / Healthcare. Industrial Edge Device Market = Sorted by Surveillance, Other Security, Automation, Energy Management, Transportation, Test/Measurement, Agriculture and Industrial Smart Robots. Smart Home and Smart Buildings Market = Sorted by Appliances, Smart Speakers, Personal Asset Tracker and Personal Robots.

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In presenting this information, we have made certain assumptions that we believe to be reasonable based on such data and other similar sources and on our knowledge of, and our experience to date in, the markets in which we operate. While we believe the estimated market and industry data included in this prospectus are generally reliable, such information, which is derived in part from management's estimates and beliefs, is inherently uncertain and imprecise, and you are cautioned not to give undue weight to such estimates. Market and industry data are subject to change and may be limited by the availability of raw data, the voluntary nature of the data gathering process and other limitations inherent in any statistical survey of such data. In addition, projections,

assumptions and estimates of the future performance of the markets in which we operate are necessarily subject to uncertainty and risk due to a variety of factors, including those described in "Risk Factors" and "Cautionary Note Regarding Forward-Looking Statements." These and other factors could cause results to differ materially from those expressed in the estimates made by third parties and by us. Accordingly, you are cautioned not to place undue reliance on such market and industry data or any other such estimates. The content of the sources identified herein, except to the extent specifically set forth in this prospectus, does not constitute a part of this prospectus and is not incorporated herein.

Use of Proceeds

We estimate that we will receive net proceeds of approximately $ million (or approximately $ million if the underwriters' over-allotment option is exercised in full) from the sale of the shares of common stock offered by us in this offering, based on an assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus), and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

A $1.00 increase (decrease) in the assumed initial public offering price of $ per share would increase (decrease) the net proceeds to us from this offering by approximately $ million, assuming that the number of shares offered by us, as set forth on the cover page of this prospectus, remains the same and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

Similarly, a 1.0 million share increase (decrease) in the number of shares offered by us, as set forth on the cover of this prospectus, would increase (decrease) the net proceeds to us by $ million, assuming no change in the assumed initial public offering price per share, and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

The principal purposes of this offering are to increase our capitalization and financial flexibility, establish a public market for our common stock, and facilitate future access to the public equity markets by us, our employees and our stockholders, obtain additional capital to support our operations, and increase our visibility in the marketplace.

We intend to use the net proceeds from this offering primarily for general corporate purposes, including working capital, sales and marketing activities, product development, general and administrative matters, and capital expenditures, although we do not currently have any specific plans with respect to use of proceeds for such purposes. We also may use a portion of the net proceeds to acquire complementary businesses, products, services, or technologies. However, we do not have agreements or commitments for any specific acquisitions at this time. Our expected use of the net proceeds from this offering represents our current intentions based upon our present plans and business condition. As of the date of this prospectus, we cannot predict with certainty all of the particular uses for the net proceeds to be received upon completion of this offering, or the amounts that we will actually spend on the uses set forth above.

Pending the uses described above, we intend to invest the net proceeds from this offering in short term, interest-bearing securities such as money market accounts, certificates of deposit, commercial paper and guaranteed obligations of the U.S. government.

The amounts and timing of our actual use of the net proceeds will vary depending on numerous factors, including our ability to gain access to additional financing and the relative success and cost of our research and development programs. As a result, our management will have broad discretion in the application of the net proceeds, and investors will be relying on our judgment regarding the application of the net proceeds of this offering. In addition, we might decide to postpone or not pursue certain development activities if the net proceeds from this offering and any other sources of cash are less than expected.

Table of Contents

Dividend Policy

We have never declared or paid cash dividends on our capital stock. We do not anticipate declaring or paying, in the foreseeable future, any cash dividends on our capital stock. We currently intend to retain all available funds and any future earnings to support our operations and finance the growth and development of our business. Any future determination related to our dividend policy will be made at the discretion of our board of directors and will depend upon, among other factors, our results of operations, financial condition, capital requirements, contractual restrictions, business prospects, and other factors our board of directors may deem relevant.

Table of Contents

Capitalization

The following table sets forth our cash and cash equivalents, and our capitalization as of March 31, 2025:

- on an actual basis;
- on a pro forma basis, giving effect to (i) the automatic conversion of all 341,496,158 shares of our outstanding redeemable convertible preferred stock into an aggregate of 356,423,629 shares of our common stock, which will occur upon the closing of this offering, (ii) the issuance of shares of our common stock upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are being exercised in connection with this offering, and (iii) the filing of our amended and restated certificate of incorporation upon the closing of this offering; and
- on a pro forma as adjusted basis, giving effect to the pro forma adjustments discussed above, and giving further effect to the sale of shares of our common stock by us in this offering at an assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover of this prospectus), after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

The pro forma and pro forma as adjusted information below is illustrative only, and our capitalization following the completion of this offering will be adjusted based on the actual initial public offering price and other terms of this offering determined at pricing. You should read this table together with "Management's Discussion and Analysis of Financial Condition and Results of Operations" and our consolidated financial statements and the related notes appearing elsewhere in this prospectus.

	As of March 31, 2025		
	Actual	Pro Forma	Pro Forma as Adjusted[1]
	(in thousands, except per share data)		
Cash and cash equivalents	$61,418		$$
Redeemable convertible preferred stock (Series Seed, A, B, C, D, E, F, F-1 and G), $0.000001 par value; 347,983,712 shares authorized; 341,496,158 shares issued and outstanding, actual; no shares authorized, issued or outstanding, pro forma and pro forma as adjusted	378,150		
Stockholders' (deficit) equity:			
Preferred stock, $0.000001 par value; no shares authorized, issued or outstanding, actual; shares authorized and no shares issued or outstanding, pro forma and pro forma as adjusted	—		
Common stock, $0.000001 par value; 500,000,000 shares authorized, 12,560,868 shares issued and outstanding, actual; shares authorized, shares issued and outstanding, pro forma; shares authorized, shares issued and outstanding, pro forma as adjusted	—		
Additional paid-in capital	29,367		
Accumulated other comprehensive income (loss)	(550)		

Accumulated deficit	(328,534)	
Total stockholders' (deficit) equity:	(299,717)	
Total capitalization	$78,433	$$

(1) Each $1.00 increase (decrease) in the assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus) would increase (decrease) each of the amount of cash and cash equivalents, additional paid-in capital, total stockholders' (deficit) equity and total capitalization by approximately $ million, assuming the number of shares offered by us, as set forth on the cover page of this prospectus, remains the same and after deducting underwriting discounts and commissions and estimated offering costs payable by us. Each 1.0 million increase (decrease) in the number of shares offered by us as set forth on the cover page of this prospectus, would increase (decrease) each of our cash and cash equivalents, additional paid-in capital, and total stockholders' (deficit) equity by approximately $ million, assuming no change in the assumed initial public offering price per share, and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

Unless otherwise indicated, the number of shares of our common stock to be outstanding immediately after this offering is based on shares of common stock outstanding as of March 31, 2025 on a pro forma basis, after giving effect to the adjustments set forth in footnote (2) to the consolidated statement of operations data in "Summary Consolidated Financial Data", and excludes:

- up to 5,186,932 shares of our common stock issuable upon the vesting of RSUs under our 2020 Plan outstanding as of March 31, 2025;
- 61,795,893 shares of our common stock issuable upon the exercise of outstanding stock options under our 2010 Plan and the 2020 Plan as of March 31, 2025, with a weighted-average exercise price of $0.38 per share;
- shares of our common stock issuable upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are not being exercised in connection with this offering;
- shares of our common stock reserved for future issuance under the 2025 Plan, as well as any automatic increases in the number of shares of common stock reserved for future issuance under the 2025 Plan; and
- shares of our common stock reserved for future issuance under the ESPP, as well as any automatic increases in the number of shares of common stock reserved for future issuance under the ESPP.

Dilution

If you invest in our common stock in this offering, your ownership interest will be immediately diluted to the extent of the difference between the initial public offering price per share of our common stock and the pro forma as adjusted net tangible book value per share of our common stock after this offering.

Our historical net tangible book value (deficit) as of March 31, 2025 was approximately $67.6 million, or $5.40 per share of our common stock. Our historical net tangible book value (deficit) is the amount of our total tangible assets less our total liabilities. Historical net tangible book value (deficit) per share is our historical net tangible book value (deficit) divided by the number of shares of common stock outstanding as of March 31, 2025.

Our pro forma net tangible book value as of March 31, 2025 was $ million, or $ per share of common stock. Our pro forma net tangible book value per share represents the amount of our total tangible assets reduced by the amount of our total liabilities and divided by the total number of shares of our common stock outstanding as of March 31, 2025, assuming (i) the automatic conversion of all 341,496,158 shares of our outstanding redeemable convertible preferred stock into an aggregate of 356,423,629 shares of our common stock, which will occur upon the closing of this offering, (ii) the issuance of shares of our common stock upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are being exercised in connection with this offering, and (iii) the filing of our amended and restated certificate of incorporation upon the closing of this offering.

After giving effect to the sale by us of shares of common stock in this offering at the assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus) and after deducting underwriting discounts and commissions and estimated offering expenses payable by us, our pro forma as adjusted net tangible book value as of March 31, 2025 would have been $ million, or $ per share. This amount represents an immediate increase in pro forma as adjusted net tangible book value of $ per share to our existing stockholders and an immediate dilution in pro forma as adjusted net tangible book value of $ per share to investors purchasing common stock in this offering. We determine dilution by subtracting the pro forma as adjusted net tangible book value per share after this offering from the amount of cash paid by an investor for a share of common stock in this offering.

The following table illustrates this dilution on a per share basis:

Assumed initial public offering price per share	$
Historical net tangible book value (deficit) per share as of March 31, 2025	$5.40
Pro forma increase in net tangible book value (deficit) per share as of March 31, 2025 before giving effect to this offering	
Pro forma net tangible book value per share as of March 31, 2025	

Increase in pro forma net tangible book value per share attributable to investors participating in this offering	
Pro forma as adjusted net tangible book value per share as of March 31, 2025 after giving effect to this offering	
Dilution per share to investors participating in this offering	$

A $1.00 increase (decrease) in the assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus) would increase (decrease) the pro forma as adjusted net tangible book value per share after this offering by approximately $ per share and the dilution in pro forma per share to investors participating in this offering by approximately $ per share, assuming that the number of shares offered by us, as set forth on the cover page of this prospectus, remains the same and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

Similarly, a 1.0 million share increase (decrease) in the number of shares offered by us, as set forth on the cover of this prospectus, would increase (decrease) the pro forma as adjusted net tangible book value per share after this offering by approximately $ and decrease (increase) the dilution in pro forma per share to investors participating in this offering by approximately $, assuming the assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover of this prospectus) remains the same, and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

If the underwriters exercise their over-allotment option in full, the pro forma as adjusted net tangible book value will increase to $ per share, representing an immediate increase in pro forma as adjusted net tangible book value to our existing stockholders of $ per share and an immediate decrease of dilution of $ per share to new investors participating in this offering.

The following table summarizes, on a pro forma as adjusted basis as of March 31, 2025, the number of shares purchased or to be purchased from us, the total consideration paid or to be paid to us, and the average price per share paid or to be paid to us by our existing stockholders, and investors participating in this offering at an assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover of this prospectus), before deducting underwriting discounts and commissions and estimated offering expenses payable by us. As the table below shows, investors participating in this offering will pay an average price per share substantially higher than our existing stockholders paid.

	Shares Purchased		Total Consideration		Average Price Per Share
	Number	Percent	Amount	Percent	
Existing stockholders		%		$%	$
Investors participating in this offering					
Total		100%		100%	

The table above assumes no exercise of the underwriters' over-allotment option to purchase up to an additional shares in this offering. If the underwriters' over-allotment option is exercised in full, the number of shares of our common stock held by the existing stockholder would be reduced to % of the total number of shares of our common stock outstanding after this offering, and the number of shares of common stock held by new investors participating in the offering would be increased to % of the total number of shares outstanding after this offering.

A $1.00 increase (decrease) in the assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus) would increase (decrease) the total consideration paid by investors participating in this offering, total consideration paid by all stockholders and the average price per share paid by all stockholders by approximately $ million, $ million and $, respectively, assuming that the number of shares offered by us, as set forth on the cover page of this prospectus, remains the same and before deducting underwriting discounts and commissions and estimated offering expenses payable by us.

Similarly, a 1.0 million increase (decrease) in the number of shares offered by us, as set forth on the cover of this prospectus, would increase (decrease) the total consideration paid by investors participating in this offering, total consideration paid by all stockholders and the average price per share paid by all stockholders by approximately $ million, $ million and $, respectively, assuming the assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover of this prospectus) remains the same, and after deducting underwriting discounts and commissions and estimated offering expenses payable by us.

Unless otherwise indicated, the foregoing discussion and table are based on shares of common stock outstanding as of March 31, 2025 on a pro forma basis, after giving effect to the adjustments set forth in

footnote (2) to the consolidated statement of operations data in "—Summary Consolidated Financial Data", and excludes:

- up to 5,186,932 shares of our common stock issuable upon the vesting of RSUs under our 2020 Plan, outstanding as of March 31, 2025;
- 61,795,893 shares of our common stock issuable upon the exercise of outstanding stock options under our 2010 Plan and the 2020 Plan as of March 31, 2025 with a weighted-average exercise price of $0.38 per share;
- shares of our common stock issuable upon the exercise of warrants to purchase our common stock outstanding as of March 31, 2025 at a weighted average exercise price of $ per share that are not being exercised in connection with this offering;

- shares of our common stock reserved for future issuance under the 2025 Plan, as well as any automatic increases in the number of shares of common stock reserved for future issuance under the 2025 Plan; and

- shares of our common stock reserved for future issuance under the ESPP, as well as any automatic increases in the number of shares of common stock reserved for future issuance under the ESPP.

We may choose to raise additional capital through the sale of equity or convertible debt securities due to market conditions or strategic considerations even if we believe we have sufficient funds for our current or future operating plans. To the extent we issue additional shares of common stock or other equity or convertible debt securities in the future, there will be further dilution to investors participating in this offering.

Management's Discussion and Analysis of Financial Condition and Results of Operations

The following discussion and analysis is intended to highlight and supplement data and information presented elsewhere in this prospectus, including our historical consolidated financial statements and related notes, and should be read in conjunction with the information presented in our historical financial statements and related notes thereto included elsewhere in this prospectus. This discussion and analysis contains forward- looking statements based upon current beliefs, plans and expectations that involve risks, uncertainties and assumptions, such as statements regarding our plans, objectives, expectations, intentions and projections. Our actual results and the timing of selected events could differ materially from those anticipated in these forward- looking statements as a result of several factors, including those set forth under "Risk Factors" and elsewhere in this prospectus. You should carefully read the "Risk Factors" section of this prospectus to gain an understanding of the important factors that could cause actual results to differ materially from forward-looking statements. Please also see the section titled "Cautionary Note Regarding Forward-Looking Statements."

Overview

We are a pioneer and leading provider of ultra-low power semiconductor solutions designed to address the significant power consumption challenges of general purpose and AI compute – especially at the edge.

Our customers rely on Ambiq to deliver AI compute closer to end users (edge environments) where power consumption challenges are the most severe. We seek to drive growth in AI adoption at the edge in the personal devices, medical/healthcare, industrial edge, and smart home and building markets and continue to set new standards in edge AI performance and power efficiency. Over time, we expect to integrate our ultra-low power technology into additional chip products that benefit from greater power efficiency, including high-performance compute applications such as AI data centers and automotive.

To date, a majority of AI compute has been deployed in data centers due to its large physical scale and the need for wall plug energy, as AI compute requires enormous and steady energy resources. At the edge, however, power limitations have been especially acute due to small device size and limited battery life. We believe this greatly constrains the potential of AI to improve our daily on-the-go lives. Enabling AI at the edge, where the action takes place, with vastly improved power efficiency will allow faster real-time decision-making due to data proximity, greater data privacy, higher energy efficiency from reduced network usage, and less dependence on constant costly connections to the cloud. We believe new AI use cases will only be possible if edge devices are much more power efficient.

Our proprietary Sub-threshold Power Optimized Technology (SPOT) platform is designed to fundamentally and cost-effectively reduce power consumption of battery- and wireline-powered devices alike. Depending on the application, devices incorporating SPOT demonstrate a two to five times reduction in power consumption compared to conventional integrated circuit designs. SPOT is a ground-breaking approach at the chip design level that incorporates sub- and near-threshold hardware, without using expensive manufacturing processes.

We provide a full stack solution encompassing tightly integrated hardware and software. Our solutions include a diverse family of systems-on-chip (SoCs) and the software required to enable on-chip AI processing, general compute, sensing, security, storage, wireless connectivity, and advanced graphics. Our SoC solutions deliver compute at a very small fraction of the power consumed by our competitors' products.

Our ultra-low power SoCs serve a wide range of markets requiring on-device and real-time AI, including smartwatches and fitness trackers, augmented and virtual reality (AR/VR) glasses, smart rings, digital health monitors, security systems and access control, livestock tracking, crop monitoring, and factory automation. These devices increasingly offer on-chip AI-powered features such as speech recognition, domain-specific language models, image and video processing, and sensing, further straining power consumption, which our solutions are positioned to address.

As global demand for our SoC solutions accelerates, our sales and marketing efforts are increasingly focused on our end customers in target geographies such as the United States, Europe and Asia (ex-Mainland China).

For the three months ended March 31, 2025 and 2024, we generated net sales of $15.7 million and $15.2 million, respectively, and a net loss of $8.3 million and $9.4 million, respectively. As of March 31, 2025, we had an accumulated deficit of $328.5 million.

Key Factors Affecting Our Business

We believe that the growth of our business and our future success are dependent upon many factors including those described above under "Risk Factors" and elsewhere in this prospectus and the following key factors. While these factors present significant opportunities for us, they also pose challenges that we must successfully address in order to sustain the growth of our business and enhance our results of operations.

Geographical Concentration

As we focus on creating meaningful benefits to our end customers for their edge AI capabilities, we are shifting our geographic concentration. Historically, our sales were significantly concentrated with end customers in Mainland China. Given geopolitical concerns, subsidized competitors creating a price sensitive environment in Mainland China and our desire to service new markets in medical/healthcare, industrial edge, and smart home and buildings, we continue to prioritize our management and sales efforts toward other meaningful geographies. As a result, we expect to continue to experience a substantial shift over the next year in our underlying geographical concentration. For example, in 2023, 66% of our net sales were to end customers in Mainland China, as compared to 50% in 2024. During the first quarter of 2025, 6% of our net sales were to end customers in Mainland China, as compared to 50% during the first quarter of 2024. We currently anticipate this trend will continue in 2025 and beyond. We believe this shift will result in a significant margin improvement opportunity given the benefits of other markets outside of Mainland China.

End Customer Concentration

We believe that our operating results for the foreseeable future will continue to depend to a significant extent on sales attributable to certain end customers. Our largest end customer historically has accounted for a large portion of our sales, representing approximately 38% of our net sales for the three months ended March 31, 2025. A different end customer accounted for approximately 28% of our net sales for the three months ended March 31, 2024. In addition, our top ten end customers for the three months ended March 31, 2025 and 2024, accounted for 98% and 98% of our net sales, respectively.

For the years ended December 31, 2024 and 2023, our largest end customer accounted for approximately 41% and 30% of our sales, respectively. In addition, our top ten end customers for the years ended December 31, 2024 and 2023, accounted for 97% and 95% of our total sales, respectively.

We have demonstrated strong end customer growth with technology leaders validating our technology platform and our robust product offerings. We work with our end customers at the front end of their design cycles, helping them develop next-generation products. The collaborative nature of these relationships provides us with enhanced visibility into our end customers' future requirements, allowing us to expand our business and increase our content in future products.

Product Development and Adoption

We develop and sell leading-edge ultra-low power SoCs, tightly bundled with software and various other solutions that combine 32-bit microcontrollers (MCUs) with wireless connectivity and additional circuitry, such

as graphics processing units, serial interfaces, and analog-to-digital interfaces. Our success is dependent on end customers adopting our new technology and preferring our products over competing offerings or technologies.

Our current end customer products are characterized by rapidly changing technologies, industry standards and technological obsolescence. We work closely with our end customers to understand their product roadmaps and strategies to forecast their future needs, which significantly influence our technology roadmap and development priorities. Our revenue performance is dependent on our ability to continually develop and introduce new products to meet the changing technology and performance requirements of the market and our end customers. Maintaining our competitive advantage is critical to our financial performance. We continue to expect to make significant investments in research and development, and our research and development expenses in a particular period may be significantly impacted by a specific product launch or engineering initiatives that we have undertaken to maintain our competitiveness or expand our product portfolio.

Unit Price and Volume and Gross Margins

Our revenue is driven by the number of units and average selling price (ASP) of our products, which can fluctuate from period to period based on the timing of our product lifecycle. The ASPs of our products vary significantly. While the ASP of any individual product generally decreases over time, our average ASPs have historically increased as we continue to introduce new higher-end products with higher ASPs.

Our product gross margins may fluctuate from period to period due to changes in our average selling price per unit due to new product launches and existing product mix with our end customer base. Our gross margins are also impacted by any changes to our manufacturing yield and wafer assembly and testing costs. We routinely experience increased prices for silicon wafers, packaging, printed circuit boards, and testing costs, which are used in our manufacturing process. As a result, our gross margins are impacted by our ability to offset any increases in our cost structure through increased prices, productivity improvements, or other means.

Cyclical Nature of the Semiconductor Industry

The semiconductor industry is highly cyclical and is characterized by constant and rapid technological change, rapid product obsolescence, price erosion, evolving standards, short product life cycles and wide fluctuations in product supply and demand. Historically, the industry has experienced significant downturns during global recessions. These downturns have been characterized by diminished product demand, production overcapacity, high inventory levels and accelerated erosion of average selling prices. Furthermore, any significant upturn in the semiconductor industry could result in increased competition for access to third-party wafer fabrication and assembly capacity. We are dependent on the availability of this capacity to manufacture and assemble our products and we can provide no assurance that adequate capacity will be available to us in the future. Any downturns or upturns in the semiconductor industry could harm our business, financial condition and results of operations. Our revenue has historically been subject to some seasonal variation. However, with rapid changes in technology development and our markets, the seasonal factors that affect our business may change from time to time.

Economic Volatility

Our sales and gross margin depend significantly on general economic conditions and the demand for products in the markets where our end customers compete. Weaknesses in the global economy and financial markets, including the impact of ongoing global conflicts, such as the Russia-Ukraine and Israel-Hamas conflicts, may in the future lead to lower demand for our end customers' products that incorporate our products. Volatile and/or uncertain economic conditions, including increased inflation rates, can adversely impact sales and gross margin and make it difficult for us to accurately forecast and plan our future business activities. In addition, any disruption in the credit markets could impede our access to capital, which could be further adversely affected if we are unable to obtain or maintain favorable credit ratings. If we have limited access to additional financing sources, we may be required to defer capital expenditures or seek other sources of liquidity, which may not be available to us on acceptable terms or at all.

Components of our Operating Results

Net Sales

We are a products-focused business. Our net sales are recognized when control of our products is transferred to our customers for consideration that we expect to receive for our products, net of returns and allowances. Our net sales are driven by the average selling price of our products, product volumes, and mix of products sold. Our end customers represent the actual user of our product, whether sold directly to or through a distributor.

Cost of Sales

Our cost of sales includes the cost of purchasing finished wafers manufactured by independent foundries and costs associated with the assembly, testing, shipping and handling of products along with allocated costs for salary, stock compensation and related benefits for personnel involved in the manufacturing of our products. Cost of sales also includes depreciation for equipment and photomasks supporting the manufacturing process, write downs of inventory, sell-through of products previously reserved for, IP royalties, licensing fees, logistics, quality assurance, warranty, and other costs incurred by us.

Gross Profit

Gross profit is calculated as net sales less cost of sales.

Operating Expenses

Our operating expenses are categorized as research and development costs or selling, general, and administrative expenses and classified based on the descriptions below:

Research and Development Costs

Research and development costs are expensed as incurred. Research and development costs consist primarily of compensation-related expenses, including salaries, benefits, and stock-based compensation expense for employees that support our research and development organization, external consulting and services costs, licensing fees, equipment tooling and allocations of other costs we incur. Assets purchased to support our ongoing research and development activities are capitalized when related to products that have achieved technological feasibility or have an alternative future use and are amortized over their estimated useful lives. We expect research and development costs to increase as a public company as we intend to reinvest our proceeds into our future product development and the expansion of our current product offerings.

Selling, General, and Administrative Expenses

Our selling, general, and administrative expenses consist of compensation-related expenses, including salaries, benefits, and stock-based compensation expense for employees that support our sales, finance, human resources, marketing, and other corporate functional support. Selling, general, and administrative also includes insurance costs, rent and lease expenses, travel and entertainment, and general corporate expenses, such as accounting, audit, legal, regulatory, and tax compliance. We expect selling, general, and administrative expenses to increase in absolute dollars as we incur increased accounting, legal and professional fees and other costs associated with being a public company.

Other Income (Expense), net

Other income (expense), net reflects interest income generated from our cash on hand being invested in short-term money market account activity. Our other expenses are principally the mark-to-market valuation of our warrant liabilities and the impact of foreign exchange gains and losses on our results.

Provision for Income Taxes

Our provision for income taxes includes federal, foreign and state taxes. Income taxes are accounted for using the asset and liability method.

Results of Operations

The results of operations data in the following tables for the periods presented have been derived from the audited financial statements and the unaudited condensed financial statements included elsewhere in this prospectus.

Comparison of Three Months Ended March 31, 2025 and 2024

The following table summarizes our results of operations for each of the three months ended March 31, 2025 and 2024:

	Three months ended March 31,	
	2025	2024
	(in thousands)	
Net sales	$15,732	$15,210
Cost of sales	7,343	8,976
Gross profit	8,389	6,234
Operating expenses:		
Research and development	8,687	9,436
Selling, general and administrative	8,443	5,988
Loss from operations	(8,741)	(9,190)
Other income (expense), net	461	(204)
Loss before income taxes	(8,280)	(9,394)
Provision for income taxes	4	5
Net loss	$(8,284)	$(9,399)

The following table summarizes the results of our operations for each of the three months ended March 31, 2025 and 2024 as a percentage of net sales. All percentage amounts were calculated using the underlying data:

	Three months ended March 31,	
	2025	2024
	(in thousands)	
Net sales	100.0%	100.0%
Cost of sales	46.7%	59.0%
Gross profit	53.3%	41.0%
Operating expenses:		
Research and development	55.2%	62.0%
Selling, general and administrative	53.7%	39.4%
Loss from operations	(55.6)%	(60.4)%
Other income (expense), net	2.9%	(1.3)%
Loss before income taxes	(52.6)%	(61.8)%
Provision for income taxes	0.0%	0.0%
Net loss	(52.7)%	(61.8)%

Net Sales

	Three months ended March 31,		
	2025	2024	% Change
	(in thousands, except percentages)		
Net sales	$15,732	$15,210	3.4%

Net sales for the three months ended March 31, 2025 increased $0.5 million, or 3.4%, to $15.7 million in the first quarter of 2025 from $15.2 million in the first quarter of 2024. Net sales to our end customers outside of Mainland China were $14.8 million and $7.6 million for the three months ended March 31, 2025 and 2024, respectively. The increase in net sales in the first quarter of 2025 was driven by the return of our customers' purchasing patterns subsequent to the recent semiconductor downturn. Net sales to our end customers in Mainland China for the three months ended March 31, 2025 and

2024, were $1.0 million and $7.6 million, respectively. The decrease was primarily driven by the strategic prioritization of non-Mainland China opportunities. In addition, net sales of our top ten end customers in the first quarter of 2025 increased by 3.4% to $15.4 million, compared to $14.9 million in the first quarter of 2024.

Gross Profit and Gross Margin

	Three months ended March 31,		
	2025	2024	% Change
	(in thousands, except percentages)		
Gross profit	$8,389	$6,234	34.6%
Gross margin	53.3%	41.0%	

Gross profit for the three months ended March 31, 2025 increased $2.2 million, or 34.6%, to $8.4 million in the first quarter of 2025 from $6.2 million in the first quarter of 2024, representing an overall gross margin increase of 1,230 basis points, to 53.3% in the first quarter of 2025 from 41.0% in the first quarter of 2024. Gross profit from our end customers outside of Mainland China for the three months ended March 31, 2025 and 2024, were $7.9 million and $3.7 million, respectively, driven by a more favorable product and customer mix compared to the first quarter of 2024. Gross profit from our end customers in Mainland China for the three months ended March 31, 2025 and 2024, were $0.5 million and $2.5 million, respectively. The decrease was driven by lower net sales in the first quarter of 2025 as we continue to execute on our strategic prioritization of non-Mainland China.

Research and Development Expenses

	Three months ended March 31,		
	2025	2024	% Change
	(in thousands, except percentages)		
Research and development	$8,687	$9,436	(7.9)%

Research and development expenses for the three months ended March 31, 2025 decreased $0.7 million, or 7.9%, to $8.7 million in the first quarter of 2025 from $9.4 million in the first quarter of 2024. The overall decrease in research and development was primarily attributable to a reduction in compensation-related expenses driven by a net headcount decrease of seven, driven by the geographical realignment of certain engineering functions.

Selling, General and Administrative Expenses

	Three months ended March 31,		
	2025	2024	% Change
	(in thousands, except percentages)		
Selling, general and administrative	$8,443	$5,988	41.0%

Selling, general and administrative expenses for the three months ended March 31, 2025 increased $2.5 million, or 41.0%, to $8.4 million in the first quarter of 2025 from $6.0 million in the first quarter of 2024. The increase was primarily attributable to increased transaction costs during the first quarter of 2025.

Other Income (Expense), Net

	Three months ended March 31,		
	2025	2024	% Change
	(in thousands, except percentages)		
Other income (expense), net	$461	$(204)	(326.0)%

Other income (expense), net for the three months ended March 31, 2025 increased $0.7 million, or 326.0%, to income on a net basis of $0.5 million in the first quarter of 2025 from a net expense of $0.2 million in the first quarter of 2024. The increase was primarily related to higher interest income related to our cash being held in interest bearing accounts beginning in the second quarter 2024.

Comparison of Years Ended December 31, 2024 and 2023

The following table summarizes our results of operations for the years ended December 31, 2024 and 2023:

	Year ended December 31,	
	2024	2023
	(in thousands)	
Net sales	$76,067	$65,537

Cost of sales	51,776	46,096
Gross profit	24,291	19,441
Operating expenses:		
Research and development	37,168	44,020
Selling, general and administrative	27,736	26,743
Loss from operations	(40,613)	(51,322)
Other income, net	980	1,020
Loss before income taxes	(39,633)	(50,302)
Provision for income taxes	28	30
Net loss	$(39,661)	$(50,332)

The following table summarizes the results of our operations for the years ended December 31, 2024 and 2023, as a percentage of net sales. All percentage amounts were calculated using the underlying data:

	Year ended December 31,	
	2024	**2023**
	(in thousands)	
Net sales	100.0%	100.0%
Cost of sales	68.1%	70.3%
Gross profit	31.9%	29.7%
Operating expenses:		
Research and development	48.9%	67.2%
Selling, general and administrative	36.5%	40.8%
Loss from operations	(53.4)%	(78.3)%
Other income, net	1.3%	1.6%
Loss before income taxes	(52.1)%	(76.8)%
Provision for income taxes	0.0%	0.0%
Net loss	(52.1)%	(76.8)%

Net Sales

	Year ended December 31,		
	2024	**2023**	**% Change**
	(in thousands, except percentages)		
Net sales	$76,067	$65,537	16.1%

Net sales increased $10.5 million, or 16.1%, to $76.1 million in 2024 from $65.5 million in 2023. Net sales to our end customers outside of Mainland China for 2024 and 2023 were $38.1 million and $22.4 million, respectively. The increase in 2024 was driven by the return of our customers purchasing patterns subsequent to the recent semiconductor downturn. Net sales to our end customers in Mainland China for 2024 and 2023 were $38.0 million and $43.1 million, respectively. The decrease was primarily driven by the strategic prioritization of non-Mainland China opportunities and increased pricing pressures that resulted in fewer customer product launches. In addition, net sales for our 2024 top ten end customer increased by 18.9% to $73.6 million, from $61.9 million in 2023.

Gross Profit and Gross Margin

	Year ended December 31,		
	2024	**2023**	**% Change**

	(in thousands, except percentages)		
Gross profit	$24,291	$19,441	24.9%
Gross margin	31.9%	29.7%	

Gross profit increased $4.9 million, or 24.9%, to $24.3 million in 2024 from $19.4 million in 2023, representing an overall gross margin increase of 220 basis points, to 31.9% in 2024 from 29.7% in 2023. Gross profit from our end customers outside of Mainland China for 2024 and 2023 were $18.4 million and $10.0 million, respectively, driven by the return of our customers' purchasing patterns subsequent to the recent inventory correction within the semiconductor industry. Gross profit from our end customers in Mainland China for 2024 and 2023 were $5.9 million and $9.4 million, respectively. The decrease was driven by pricing pressures from local competitors which led to two products being sold at a discount in 2024.

Research and Development Expenses

	Year ended December 31,		
	2024	2023	% Change
	(in thousands, except percentages)		
Research and development	$37,168	$44,020	(15.6)%

Research and development expenses decreased $6.9 million, or 15.6%, to $37.2 million in 2024 from $44.0 million in 2023. The overall decrease in research and development was primarily attributable to a reduction in compensation-related expenses driven by the geographical realignment of certain engineering functions, which led to a decrease in net headcount of 16 since the third quarter of 2023.

Selling, General and Administrative Expenses

	Year ended December 31,		
	2024	2023	% Change
	(in thousands, except percentages)		
Selling, general and administrative	$27,736	$26,743	3.7%

Selling, general and administrative expenses increased $1.0 million, or 3.7%, to $27.7 million in 2024 from $26.7 million in 2023. The increase was primarily attributable to a warrant issuance to an investor of $1.9 million and increased stock-based compensation expenses, partially offset by a reduction in other compensation-related expenses driven by a net headcount reduction of nine during 2024 as a result of cost control initiatives and attrition.

Other Income (Expense), Net

	Year ended December 31,		
	2024	2023	% Change
	(in thousands, except percentages)		
Other income, net	$980	$1,020	(3.9)%

Other income, net, decreased $40,000, or 3.9%, to $1.0 million in 2024. The decrease was primarily related to the mark to market adjustment of our warrant liability, partially offset by higher interest income related to our cash proceeds from the Series G redeemable convertible preferred stock financing, which were held in interest-bearing accounts in 2024.

Provision for Income Taxes

We recorded minimal income tax expense in the first quarter of 2025 on a pre-tax loss of $8.3 million, yielding an effective tax rate of (0.05)%. Our effective tax rate was lower than the U.S. statutory rate of 21%, primarily due to the change in valuation allowance. We recorded income tax expense in 2024 on a pre-tax loss of $39.7 million, yielding an effective tax rate of (0.07)%, primarily due to the change in valuation allowance.

Liquidity and Capital Resources

Since inception, we had negative cash flows and losses from operations. We have funded operations primarily through equity financing and cash from operations. We anticipate incurring losses and negative cash flows from operations until such time, if ever, that we can generate significant sales of our products currently in development and are highly dependent on our ability to find additional sources of funding in the form of debt and equity financing. As of March 31, 2025, we had $61.4 million in cash. Our principal use of cash is to fund our operations, invest in research and development to support our growth and other general corporate needs.

We believe that our cash on hand and anticipated cash from operations, together with the expected net proceeds from this offering, will be sufficient to finance our operations for at least the next twelve months from the date of this prospectus.

Our future capital requirements will depend on many factors including our growth rate, the timing and extent of our selling, general and administrative and research and development expenditures, and the continuing market acceptance of our products. Additionally, we anticipate a significant increase in accounting, legal, and professional fees and other costs associated with being a public company. If our current financial resources are not sufficient to satisfy our liquidity requirements, we may be required to seek additional financing. If we raise additional funds by issuing equity securities, our stockholders will experience dilution. Debt financing, if available, may contain covenants that significantly restrict our operations or our ability to obtain additional debt financing in the future. In the event that we need to borrow funds or issue additional equity, we cannot assure you that any such additional financing will be available on terms acceptable to us, if at all. If we are unable to raise additional capital when we need it, our business, results of operations, and financial condition would be adversely affected.

Cash Flows from Operating, Investing and Financing Activities

Changes in the net cash provided by and (used in) our operating, investing, and financing activities for the three months ended March 31, 2025 and 2024 and the years ended December 31, 2024 and 2023, are set forth in the following table:

	Three months ended March 31,		Year ended December 31,	
	2025	2024	2024	2023
	(in thousands)			
Net cash provided by (used in) operating activities	$1,406	(2,857)	$(21,428)	$(26,958)
Net cash used in investing activities	(1,116)	(685)	(3,731)	(3,021)
Net cash provided by financing activities	148	8,151	58,843	31,774

Operating Activities

For the three months ended March 31, 2025, cash flow generated from operations was $1.4 million. Operating cash flow during the quarter was related to the cash components of our net loss and $8.3 million of favorable changes in working capital, driven primarily by a reduction of $0.3 million in inventory, offset by $8.6 million resulting from the timing of our sales to customers and payments to our vendors.

For the three months ended March 31, 2024, cash flow used in operations was $2.9 million. Operating cash flow use during the quarter was related to the cash components of our net loss and by $5.8 million of favorable changes in working capital, driven primarily by $1.2 million of lower inventory purchases, and $4.6 million resulting from the timing of our sales to customers and payments to our vendors.

In 2024, cash flow used in operations was $21.4 million. Operating cash flow during 2024 was related to the cash components of our net loss and $5.1 million of favorable changes in working capital, driven primarily by a reduction of $6.0 million in inventory, offset by $0.9 million resulting from the timing of our sales to customers and payments to our vendors.

In 2023, cash flow used in operations was $27.0 million. Operating cash flow use during 2023 was related to the cash components of our net loss and by $12.9 million of favorable changes in working capital, driven primarily by $15.2 million of lower inventory purchases.

Investing Activities

For the three months ended March 31, 2025, we used $1.1 million in cash for investing activities, which related to $1.0 million of technology investments in intangible assets and $0.1 million in capital expenditures.

For the three months ended March 31, 2024, we used $0.7 million in cash for investing activities, which related primarily to $0.6 million of technology investments in intangible assets.

In 2024, we used $3.7 million in cash for investing activities, which related to $3.1 million of technology investments in intangible assets and $0.7 million in capital expenditures.

In 2023, we used approximately $3.0 million in cash for investing activities, which related to $2.6 million of technology investments in intangible assets and $0.5 million in capital expenditures.

Financing Activities

For the three months ended March 31, 2025, we generated $0.1 million related to financing activities, driven by proceeds from the exercise of stock options of $0.1 million.

For the three months ended March 31, 2024, we generated approximately $8.2 million in financing activities, driven primarily by $7.9 million from our Series G preferred stock issuance and proceeds from the exercise of stock options of $0.2 million.

In 2024, we generated $58.8 million related to financing activities, driven primarily by $58.0 million from our Series G preferred stock issuance and proceeds from the exercise of stock options of $0.9 million.

In 2023, we generated $31.8 million in financing activities, driven primarily by $31.3 million from our Series G preferred stock issuance and proceeds from the exercise of stock options and warrants of $0.3 million and $0.2 million, respectively.

Non-GAAP Financial Measures

We use non-GAAP net loss and non-GAAP gross profit, both non-GAAP financial measures, to help us make strategic decisions, establish budgets and operational goals for managing our business, analyze our financial results, and evaluate our performance. We define non-GAAP net loss as our net loss adjusted to exclude expenses not directly attributable to the performance of our operations, such as income taxes, depreciation and amortization, stock-based compensation, gain on nonmonetary transaction, severance costs, transaction costs and

warrant valuation. We define non-GAAP gross profit as our gross profit adjusted to exclude expenses not directly attributable to gross profit, such as depreciation and amortization and stock-based compensation.

We present the non-GAAP financial measures non-GAAP net loss and non-GAAP gross profit in this prospectus because we believe these non-GAAP financial measures provide additional tools for investors to use in comparing our core business and results of operations over multiple periods with other companies in our industry, many of which present similar non-GAAP financial measures to investors. However, our presentation of non-GAAP net loss and non-GAAP gross profit may not be comparable to similarly titled measures reported by other companies due to differences in the way that these measures are calculated. Non-GAAP net loss and non-GAAP gross profit have limitations, and should not be considered as the sole measures of our performance and should not be considered in isolation from, or as a substitute for, net loss and gross profit calculated in accordance with GAAP.

Some of these limitations are that non-GAAP net loss and non-GAAP gross profit:

- do not reflect incomes taxes, which are necessary costs incurred in connection with our operations and reduce cash available to us;
- exclude depreciation and amortization, and although these are non-cash expenses, the assets being depreciated may have to be replaced in the future, increasing our cash requirements;
- do not reflect stock-based compensation expenses, which represent a significant cost of attracting and retaining qualified employees, and excluding them may underestimate the true economic cost of our workforce;
- do not reflect gain on nonmonetary transactions;
- do not reflect severance costs which represent costs associated with reductions in force;
- exclude IPO and related transaction costs which represent non-recurring professional fees for advisory, legal, accounting, valuation and other professional or consulting services incurred related to this offering; and
- exclude warrant valuation costs, which represent the mark-to-market valuation of liability-classified warrants.

Because of these limitations, we consider, and you should consider, non-GAAP net loss and non-GAAP gross profit alongside other financial performance measures, including net loss and gross profit and our other GAAP results. A reconciliation of our non-GAAP net loss to net loss and non-GAAP gross profit to gross profit, the most directly comparable financial measures stated in accordance with GAAP, are provided below. Investors are encouraged to review the related GAAP financial measures and the reconciliation of the non-GAAP financial measure to their most directly comparable GAAP financial measure.

The following tables reconcile the most directly comparable GAAP financial measure to each of these non-GAAP financial measures.

Non-GAAP Net Loss:

	Three months ended March 31,		Year ended December 31,	
	2025	**2024**	**2024**	**2023**
	(in thousands)			
Net loss	$(8,284)	$(9,399)	$(39,661)	$(50,332)
Add:				
Income taxes	4	5	28	30
Depreciation and amortization	1,961	1,557	6,246	7,716
Stock-based compensation	851	551	5,174	2,623
Gain on nonmonetary transaction	(1,600)	(1,600)	(1,600)	—
Severance costs	—	706	706	481
IPO and other transaction costs	1,793	—	551	850
Warrant valuation	58	—	(51)	(581)
Non-GAAP net loss	$(5,217)	$(8,180)	$(28,607)	$(39,213)

During the three months ended March 31, 2025 and 2024, we received nonreciprocal transfer of assets with a vendor. The total fair value of nonmonetary transactions recorded during the period was approximately $1.6 million, which was recognized as a gain in cost of sales.

Non-GAAP Gross Profit:

	Three months ended March 31,		Year ended December 31,	
	2025	2024	2024	2023
	(in thousands)			
Gross profit	$8,389	$6,234	$24,291	$19,441
Add:				
Depreciation and amortization	562	214	850	2,420
Stock-based compensation	60	30	356	147
Gain on nonmonetary transaction	(1,600)	(1,600)	(1,600)	—
Non-GAAP gross profit	$7,411	$4,878	$23,897	$22,008

Critical Accounting Policies and Estimates

Our discussion and analysis of financial condition are based upon the consolidated financial statements of this business, which have been prepared in accordance with GAAP. The preparation of these financial statements requires us to make estimates and judgments that affect the reported amounts of assets, liabilities, sales and expenses, and related disclosure of contingent assets and liabilities. On an on-going basis, we evaluate our estimates, including those related to bad debts, revenue reserves, inventory valuation, stock-based compensation, taxes on income, warranty obligations and contingencies and litigation. We based our estimates on historical experience and on various other assumptions that are believed to be reasonable under the circumstances, the results of which form the basis for making judgments about the carrying values of assets and liabilities that are not readily apparent from other sources. Actual results may differ from these estimates and such differences may be material to the financial statements. We believe that the accounting policies and estimates described below are the most meaningful to our operations or require management's most difficult, subjective or complex judgments. Judgments or uncertainties affecting the application of these policies may result in materially different amounts being reported under different conditions or using different assumptions. Accordingly, we believe these are the most critical to aid in fully understanding and evaluating our financial condition and results of operations. Our significant accounting policies are fully described in Note 2 to our consolidated financial statements included in the footnotes of this prospectus.

Revenue Recognition

Our sales consist of the sale of our products to our customers. Our customers are distributors, large global OEMs and other direct customers who design and manufacture devices for the consumer and industrial markets.

Revenues are recognized when control of the product is transferred to a customer in an amount that reflects the consideration we are expected to be entitled to in exchange for those products, in accordance with ASC 606, *Revenue from Contracts with Customers*. For direct customers and large global OEMS, revenue is recognized either when the product is shipped to the customer or upon delivery, depending on the prevailing contract terms. For distributor customers, revenues are recognized when the product is shipped to the distributor, as in these cases, the distributor is the contracted customer. Revenues are recognized net of sales credits and allowances for returns and discounts when applicable. Revenue is recognized net of any taxes collected from customers, which are subsequently remitted to governmental authorities.

The transaction price is the amount of consideration to which we expect to be entitled in exchange for transferring the promised goods to the customer. The transaction price is specified in the standard price list or contract and is confirmed at the time the order for products is placed by the customer and acknowledged by us.

Certain distributor contracts do include variable consideration, such as limited price protection, return and stock rotation provisions, while direct customer contracts include general right of return provisions. We estimate variable consideration using all available information, including historical experience and current expectations of return and pricing credits and records an allowance to reduce revenue recognized. These estimates take into consideration a range of possible outcomes that are probability-weighted in accordance with the expected value method in ASC 606 for relevant factors such as current contractual and statutory requirements, specific known market events and trends, industry data, and forecasted customer buying and payment patterns. Overall, these reserves reflect our best estimates of the amount of consideration to which it is entitled based on the terms of the respective underlying contracts.

Inventories, net

Inventories are stated at the lower of cost or net realizable value. Cost is calculated on a weighted moving average basis. In estimating net realizable value, we consider (among other things) the age of its inventories and its product market acceptance based on known business factors and conditions by comparing forecasted customer unit demand for products over a specific future period, or demand horizon, to quantities on hand at the end of each accounting period. These estimates may include uncertain elements such as our demand forecast which are developed based on current backlog, inputs from our customer and internal analysis of customer purchasing trends and level of inventory in the distribution channel, actual and anticipated design wins, market and economic conditions, technological changes, new product introductions, and other factors. If our demand forecast for specific products is greater than actual demand, we could be required to write down additional inventory, which would have a negative impact on our gross margin.

Stock-Based Compensation and Fair Value of Equity Components

We record stock-based compensation expense based upon the grant date fair value for all stock options issued to all persons to the extent such options vest. That expense is determined under the fair value method using the Black-Scholes option pricing model. We recognize the compensation cost for awards on a straight-line basis over the vesting period.

The Black-Scholes option pricing model used to compute stock-based compensation expense requires extensive use of accounting judgment and financial estimates. Items requiring estimation include the fair value of our common stock, expected term option-holders will retain their vested stock options before exercising them, and the estimated volatility of our common stock price over the expected term of a stock option.

The absence of an active market for our equity components, including our preferred stock, common stock and related securities, requires our board of directors, the members of which we believe have extensive business, finance and venture capital experience, to determine the fair value of our common stock for purposes of granting options, calculating stock-based compensation expense, valuing warrants and other equity transactions for the periods presented. We obtained contemporaneous third-party valuations to assist the board of directors in determining fair value. These contemporaneous third-party valuations used the methodologies, approaches and assumptions consistent with the American Institute of Certified Public Accountants Practice Guide, *Valuation of Privately-Held-Company Equity Securities Issued as Compensation*.

In valuing our common stock, our board of directors determined the equity value of our business using various valuation methods, including combinations of income and market approaches with input from management. The income approach estimates value based on the expectation of future cash flows that a company will generate. These future cash flows are discounted to their present values using a discount rate derived from an analysis of the cost of capital of comparable companies in our industry or similar business operations as of each valuation date and is adjusted to reflect the risks inherent in our cash flows.

In addition, we considered any secondary transactions involving our preferred and common stock. In our evaluation of those transactions, we considered the facts and circumstances of each transaction to determine the

extent to which they represented a fair value exchange. Factors considered include transaction volume and timing, whether the transactions occurred among willing and unrelated parties, and whether the transactions involved investors with access to our financial information.

Application of these approaches involves the use of estimates, judgment and assumptions that are highly complex and subjective, such as those regarding our expected future net sales, gross margin, operating expenses, future cash flows, discount rates, market multiples, the selection of comparable companies, and the probability of possible future events. Changes in any or all of these estimates and assumptions or the relationships between those assumptions impact our valuations as of each valuation date and may have a material impact on the valuation of our common stock, which could impact various estimates in the financial statements including stock-based compensation, valuation of the deemed dividend for preferred stock, and impairment assessments of our long-lived tangible and intangible assets.

For valuations after the completion of this offering, our board of directors will determine the fair value of the common stock underlying our stock-based awards based on the closing price of our common stock on as reported on the date of grant. Future expense amounts for any particular period could be affected by changes in our assumptions or market conditions.

Recently Issued and Adopted Accounting Pronouncements

For more information regarding recently issued accounting pronouncements, see Note 2 to our consolidated financial statements included elsewhere in this prospectus.

Controls and Procedures

We are not currently required to fully comply with the Sarbanes-Oxley Act. We will be required to fully comply with the internal control requirements of the Sarbanes-Oxley Act during the second annual filing following the initial public offering. Only in the event that we are deemed to be a large accelerated filer or an accelerated filer and no longer qualify as an emerging growth company will we be required to fully comply with the independent registered public accounting firm attestation requirement on internal control over financial reporting. Further, for as long as we remain an emerging growth company as defined in the JOBS Act, we intend to take advantage of certain exemptions from various reporting requirements that are applicable to other public companies that are not “emerging growth companies,” including, but not limited to, not being required to comply with the independent registered public accounting firm attestation requirement. However, we do have internal controls in place in key areas of risk.

Quantitative and Qualitative Disclosures about Market Risk

We are exposed to market risk in the ordinary course of business, which consists primarily of interest rates risk associated with our cash and cash equivalents, foreign currency risk and impact of inflation. We do not engage in speculative trading activities. The following analysis provides additional information regarding these risks.

Interest Rate Risk

Our assets have limited exposure to market risk. As of the three months ended March 31, 2025 and 2024, and the years ended December 31, 2024 and 2023, we maintained a portfolio of cash and cash equivalents, including money market funds. None of these investments have a maturity date in excess of three months. Certain interest rates are variable and fluctuate with current market conditions. Because of the short-term nature of these instruments, we would not expect a sudden change in market interest rates to have a material impact on our financial condition or results of operations.

Foreign Currency Exchange Risk

Operating in international markets involves exposure to possible volatile movements in currency exchange rates. A majority of our sales and expenses are transacted in U.S. dollars and our assets and liabilities together with our cash holdings are predominately denominated in U.S. dollars reducing the exposure to currency fluctuations. Due to our international operations, a portion of our cost of sales and operating expenses is denominated in currencies other that the U.S. dollar, principally the Chinese RMB, the Taiwanese dollar, and the Singapore dollar. Our international operations give rise to transactional market risk associated with exchange rate movements of the U.S. dollar, Chinese RMB, Taiwanese dollar and Singapore dollar.

In addition, we are exposed to foreign currency translation risk for those subsidiaries whose functional currency is not the U.S. dollar as changes in the value of their functional currency relative to the U.S. dollar can adversely affect the translated amounts of our sales, expenses, net income, assets and liabilities. This can, in turn, affect the reported value and relative growth of sales and net income from one period to the next. In addition, changes in the translated value of assets and liabilities due to changes in functional currency exchange rates relative to the U.S. dollar result in foreign currency translation adjustments that are a component of other comprehensive income or loss. Foreign currency derivative instruments can be used to hedge exposures and reduce the risks of certain foreign currency transactions; however, these instruments provide only limited protection and can carry significant cost. We had no foreign currency derivative instrument hedges as of the three months ended March 31, 2025 and 2024, and the years ended December 31, 2024 and 2023. We will continue to analyze our exposure to currency exchange rate fluctuations and may engage in financial hedging techniques in the future to attempt to minimize the effect of these potential fluctuations. Exchange rate fluctuations may adversely affect our financial results in the future.

If the volume of our international operations increases and foreign currency exchange rates change, the impact to our consolidated statements of operations could be significant and may affect the comparability of our operating results. The impact from foreign currency transactions during the three months ended March 31, 2025 and 2024, and the years ended December 31, 2024 and 2023 were not material. We do not believe a 10% increase or decrease in foreign exchange rates would have resulted in a material impact to our operating results.

Emerging Growth Company Status

We are an "emerging growth company," as defined in the JOBS Act, enacted in April 2012. We intend to take advantage of certain exemptions under the JOBS Act from various public company reporting requirements, including not being required to have our internal control over financial reporting audited by our independent registered public accounting firm pursuant to Section 404(b) of the Sarbanes-Oxley Act, reduced disclosure obligations regarding executive compensation in our periodic reports and proxy statements, and exemptions from the requirements of holding a nonbinding advisory vote on executive compensation and any golden parachute payments not previously approved. In addition, an emerging growth company can take advantage of an extended transition period for complying with new or revised accounting standards. This provision allows an emerging growth company to delay the adoption of certain accounting standards until those standards would otherwise apply to private companies. We have elected to avail ourselves of this provision of the JOBS Act. As a result, we will not be subject to new or revised accounting standards at the same time as other public companies that are not emerging growth companies. Therefore, our consolidated financial statements may not be comparable to those of companies that comply with new or revised accounting pronouncements as of public company effective dates.

We will remain an emerging growth company and may take advantage of these exemptions until the earliest of: (i) the last day of the fiscal year following the fifth anniversary of the consummation of this offering; (ii) the last day of the fiscal year in which we have total annual gross revenue of at least $1.235 billion; (iii) the last day of the fiscal year in which we are deemed to be a "large accelerated filer" as defined in Rule 12b-2 under the Exchange Act, which would occur if the market value of our common stock held by non-affiliates exceeded $700.0 million as of the last business day of the second fiscal quarter of such year; or (iv) the date on which we have issued more than $1.0 billion in non-convertible debt securities during the prior three-year period.

Business

Our Mission

Our mission is to enable intelligence (AI and beyond) everywhere by delivering the lowest power semiconductor solutions.

Overview

We are a pioneer and leading provider of ultra-low power semiconductor solutions designed to address the significant power consumption challenges of general purpose and AI compute – especially at the edge.

Our customers rely on Ambiq to deliver AI compute closer to end users (edge environments) where power consumption challenges are the most severe. Our leading position is built upon our hardware and software innovations that deliver two to five times lower power consumption than traditional semiconductor designs. Our products power over 270 million devices today. We shipped more than 42 million units in 2024, and we estimate that over 40% of them ran AI algorithms. We seek to drive growth in AI adoption at the edge in the personal devices, medical/healthcare, industrial edge, and smart home and building markets and continue to set new standards in edge AI performance and power efficiency. Over time, we expect to integrate our ultra-low power technology into additional chip products that benefit from greater power efficiency, including high-performance compute applications such as AI data centers and automotive.

AI is perhaps the most disruptive and revolutionary technology trend of recent history, estimated to represent $23 trillion of global annual spend by 2040, according to McKinsey. AI use cases continue to permeate our lives and improve our daily productivity by enabling us to interact with devices via voice and gestures, unlock our homes with facial recognition, track health accurately and intelligently, and hold clear calls amidst loud background noise.

To date, a majority of AI compute has been deployed in data centers due to its large physical scale and the need for wall plug energy, as AI compute requires enormous and steady energy resources. At the edge, however, power limitations have been especially acute due to small device size and limited battery life. We believe this greatly constrains the potential of AI to improve our daily on-the-go lives. Enabling AI at the edge – where the action takes place – with vastly improved power efficiency will allow faster real-time decision-making due to data proximity, greater data privacy, higher energy efficiency from reduced network usage, and less dependence on constant costly connections to the cloud. We believe new AI use cases will only be possible if edge devices are much more power efficient.

Our proprietary Sub-threshold Power Optimized Technology (SPOT®) platform is designed to fundamentally and cost-effectively reduce power consumption of battery- and wireline-powered devices alike. Depending on the application, devices incorporating SPOT demonstrate a two to five times reduction in power consumption compared to conventional integrated circuit designs. SPOT is a ground-breaking approach at the chip design level that incorporates sub- and near-threshold hardware without using expensive manufacturing processes.

We provide a full-stack solution encompassing tightly integrated hardware and software. Our solutions include a diverse family of systems-on-chip (SoCs) and the software required to enable on-chip AI processing, general compute, sensing, security, storage, wireless connectivity, and advanced graphics. Our SoC solutions deliver compute at a very small fraction of the power consumed by our competitors' products.

Our ultra-low power SoCs serve a wide range of markets requiring on-device and real-time AI, including smartwatches and fitness trackers, augmented and virtual reality (AR/VR) glasses, smart rings, digital health monitors, security systems and access control, livestock tracking, crop monitoring, and factory automation. Body-worn AI devices drive a significant portion of our revenue today and often require weeks of battery life while running advanced AI-driven features. These devices increasingly offer on-chip AI-powered features such

as speech recognition, domain-specific language models, image and video processing, and sensing, further straining power consumption, which our solutions are positioned to address.

We leverage our internal direct sales force and our global network of distributors and sales representatives to pursue and support our core end markets. For our largest end customers, dedicated sales personnel work with the manufacturers to help ensure our solutions address their go-to-market needs. As global demand for our SoC solutions accelerates, our sales and marketing efforts are increasingly focused on our end customers in target geographies such as the United States, Europe, and Asia (ex-Mainland China).

For the three months ended March 31, 2025 and 2024, we generated net sales of $15.7 million and $15.2 million, respectively. For the years ended December 31, 2024 and, 2023, we generated net sales of $76.1 million and $65.5 million, respectively. Net sales from end customers in the United States, Europe, and Asia (outside of Mainland China) grew to $14.8 million in the three months ended March 31, 2025, a 94% increase as compared to the three months ended March 31, 2024, achieving a gross margin of 53%, and grew to $38.1 million in 2024, a 70% increase as compared to 2023, achieving a gross margin of 48%. We intend to continue to focus on business outside of Mainland China.

Our Industry

The State of AI Today

Since the launch of ChatGPT in 2022 by OpenAI, companies like Google and Anthropic, among others, have launched competing generative AI models underpinned by large language models (LLMs) that now serve as the foundation for an emerging class of AI-based products and services emerging as a disruption to a wide array of markets from internet search to healthcare. As with other technologies of the digital era, the speed of these advancements outpaces the rate of commercial and consumer adoption while enterprises determine if and how to leverage AI-based solutions and consumers determine whether they will adopt such solutions. Companies developing these new disruptive AI-based products and services face significant technology, financial and regulatory challenges.

The processing of AI workloads requires considerable power, and the power consumption of the servers and chips that run generative AI systems has emerged as a primary technology challenge. The capacity of a data center to fulfill generative AI queries is limited by the amount of power that can be delivered by local power plants and the cost of such power, among other things. The cost of a query and the financial viability of generative AI-based products and services are therefore limited by the availability of affordable power. As a result, improving power consumption has become a central concern for the AI industry, and is even more pronounced at the edge.

Furthermore, the rapid pace of innovation in the field of AI has led to developing and evolving regulatory frameworks globally, which are expected to become increasingly complex as AI continues to progress. Regulators and lawmakers around the world have proposed and adopted regulations and guidance addressing the use of AI. The shifting regulatory landscape for AI and the products that utilize it has created uncertainty for companies developing AI-based solutions and could pose an impediment to near-term broad-based adoption. See "Risk Factors—Risks Related to Our Business" and "Business—Government Regulation" for a more detailed discussion of challenges associated with AI regulation.

The Growing Need for AI at the Edge

Edge AI is expanding as machine learning outperforms traditional algorithms, enabling tasks like speech recognition, and meets customer demand for cloud-independent solutions. Meanwhile, large language models (LLMs) in the cloud rely on quality edge inputs, which is evident in industrial automation, healthcare, and consumer devices. To fully realize AI's potential, computation must move to the edge, where data originates, ensuring faster decisions, enhanced privacy, improved energy efficiency, and reduced cloud reliance.

However, realizing this vision requires overcoming a fundamental challenge: **power efficiency.**

Despite the substantial advantages of deploying AI directly on edge devices, power limitations have predominantly relegated AI inferencing to the cloud. While power efficiency is crucial for all markets, it is existential for battery-powered devices, which simultaneously provide compute, connectivity, graphics, battery management, and sensing within a stringent power envelope. Neural networks, the backbone of modern AI, demand significantly more computation than traditional algorithms.

Key Challenges of Edge-to-Cloud Data Processing

Today, advanced analytics and AI inference are often processed in the cloud due to the difficulty or impossibility of running them at the edge. However, this approach has several drawbacks:

- ***Latency.*** The round trip to the cloud adds significant latency, preventing real-time behaviors and feedback.
- ***Connectivity.*** Wireless connectivity often consumes significant power and is limited to those applications with reliable connections.
- ***Security and Privacy.*** Sending sensitive data to the cloud increases privacy breach risks.
- ***Cost.*** Cloud services represent a significant expense that usually scales with the amount of data transferred to/from the cloud.

Key Challenges to AI Inferencing at the Edge

- ***Power Consumption.*** Optimizing power consumption for all functions, from AI inferencing to wireless connectivity to sensor management, is a fundamental challenge for edge devices. This challenge is significantly amplified as processing, performance, connectivity, and AI compute needs increase.
- ***Real-Time Inferencing.*** Most of today's AI models are designed to run in the cloud, processing large batches of data. Edge AI is often called on to act on streams of data in real time, which requires a significant redesign of the data processing and inferencing algorithms.
- ***Size.*** Many edge devices are physically size-constrained, thereby limiting the capacity of the battery and battery life.
- ***Integration.*** Given device size constraints, vendors increasingly opt for highly integrated SoCs to drive their devices instead of complex designs with many discrete chips having relatively narrow functionalities.
- ***Privacy and Security.*** While bringing AI to the edge improves system security and reduces the attack surface, it creates high-security expectations for small edge devices with minimal compute resources.
- ***Resource Constrained AI Model Development.*** Most AI models today are built for the cloud, where compute performance, memory, and power are readily available. Building "right-sized" models that execute efficiently on the edge is extremely challenging.

Our ultra-low power SoC solutions are designed to fundamentally address our end customers' needs for ultra-low-power consumption for all functions, better connectivity, and increased AI compute (especially at the edge).

Our Technology

Power consumption is the most critical challenge for edge AI devices. SPOT® is our proprietary ultra-low power chip design platform created to solve this severe power consumption problem. As described below, SPOT consists of a set of chip design techniques that allow standard transistors to operate in an ultra-low power mode called "sub-threshold" and "near-threshold". When combined with other low-power chip architecture techniques (also described below), SPOT enables our SoCs to deliver two to five times lower power consumption than leading alternative solutions.

Table of Contents

Traditional Approach to Design Comes at an Energy Cost

Every electronic circuit is made up of a number of transistors that work together to perform a set of functions. In conventional digital design, these transistors operate like a switch and change between "on" and "off" states to signal digital "0" and "1" values. This "super-threshold" operating mode is achieved by switching the transistors between a voltage of 0 volts and a much higher supply voltage (e.g., 0.8 volts or 1.2 volts) that is well above the turn-on voltage of the transistor (known as the threshold voltage). This "super-threshold" operating mode consumes significant power, given the extremely high dependence of switching power on voltage.

Sub-threshold and Near-threshold Design Offers Substantially Improved Power Efficiency

Power consumption can be significantly reduced by dropping supply voltage from the conventional high value (e.g., 0.8 volts or 1.2 volts) to a lower value (e.g., 0.4 volts, 0.5 volts, or 0.6 volts). Though the conventional high supply voltages have long been considered fundamental limits, it is possible to operate at supply voltages near or below the threshold voltage (i.e., the "turn on" voltage) of the transistor. When the supply voltage is close to the threshold voltage of the transistor, the circuit is operating in "near-threshold" mode. When the supply voltage drops below the threshold voltage of the transistor, the circuit is operating in "sub-threshold" mode. The benefits and challenges are similar for both near-threshold and sub-threshold operations, so we use the term "sub-threshold" to encompass both modes of operation. Challenges for operating in the sub-threshold region include:

- Difficulty modeling and simulating transistors accurately in the non-standard sub-threshold region;

- High sensitivity to small manufacturing variations, which can lead to significant performance deviations and functional failure;
- High sensitivity to environmental variations (e.g., temperature, voltage, and light variations), which can lead to performance issues and functional failure in the field; and
- Lack of circuit designers familiar with sub-threshold concepts and lack of design IP and computer-aided design tool flows capable of supporting sub-threshold operation.

Ambiq SPOT Platform is Built to Solve the Challenges of Sub-threshold and Near-threshold

Our SPOT platform is a collection of dozens of proprietary techniques designed to solve the typical challenges associated with sub-threshold technologies. Given the disruptive approach to basic circuit building block design, we developed new analog, digital, and memory architectures while utilizing standard communication interfaces, packages, design flows, and manufacturing processes.

We have spent 15 years building our sub-threshold expertise, reusable design IP, and patent portfolio. Specifically, we have developed wide-ranging chip architectures, IP building blocks, and IC design methodologies for digital circuits, analog circuits, and embedded memories—accumulating significant expertise and IP related to product design, validation, and production testing. Our proven expertise in, and experience with, ultra-low power design provides us with a substantial competitive advantage.

In addition, we have developed a wide range of chip architecture techniques that are designed to enable even lower power than sub-threshold and near-threshold operation alone. Many of these chip design techniques are protected by IP rights, including patents and trade secrets.

While many high-end processor chip companies can increase performance or decrease power consumption with advanced process technology nodes, such as 7- and 5-nanometers, these expensive manufacturing processes are typically not cost-effective for edge devices. With our SPOT platform, we are able to manufacture our products at much more cost-effective process technology nodes, such as 12 nanometers, 22 nanometers, and 40 nanometers. Furthermore, reliance on advanced process technology nodes and Moore's Law to solve the power problem is particularly limiting as Moore's Law is viewed by many as dead or broken.

Our Products

Our SPOT platform serves as the foundation for a series of SoCs for edge AI devices that deliver ultra-low power for AI computations, general purpose computations, sensing, communications, power conversion, and

more. These SoCs are paired with a full-stack software solution that is designed to enable efficient resource-constrained AI application development and a fast time-to-market. Details about both hardware and software products are described below:

SoC Products for Edge AI

Our SoCs are built on the SPOT platform to enable every function to achieve ultra-low power. Based on our experience, the ideal edge AI SoC (shown below) includes a host processor, power conversion, wireline and wireless communications, sensor interfaces and user interfaces, memories, and a variety of accelerated compute blocks for graphics, security, and more. Importantly, for the most demanding edge AI use cases, neural processing units (NPUs, or AI accelerators) are also included. These elements are all critical for edge AI use cases, and SPOT is designed to reduce energy consumption for each of them.

Our portfolio includes two flagship SoC product families, both with rich sets of peripherals as shown in the diagram above:

- ***Apollo:*** The first Apollo family of products was launched in 2015. New families have been launched every few years with the newest Apollo5 family launching in 2024. The Apollo family of products pair rich peripheral sets with host processors capable of software-based AI computations (e.g., Apollo3 and Apollo4 families) or vector-accelerated AI computations (e.g., Apollo5 family). For most edge AI use cases, we believe the combination of the Apollo products with our full-stack software solution provides the optimal power, performance, feature, and cost trade-off.
- ***Atomiq:*** The first Atomiq family product is currently in development. This novel product targets AI applications requiring demanding edge AI compute requirements and is thus expected to provide the highest performance and lowest power ever delivered by our products. Atomiq is expected to feature a full NPU for high-performance AI acceleration along with new memory innovations — all with the goal of achieving minimum power and maximum performance on AI model execution at the edge.

Software Products for Edge AI

To truly unlock edge AI use cases, our ultra-low power SoCs must be paired with carefully optimized proprietary software. To make edge AI application development easier for our customers, Ambiq provides a series of core software products:

- ***General enablement firmware:*** Our AmbiqSuite Software Development Kit (SDK) provides low-level firmware and drivers required to operate our SoCs efficiently.
- ***Feature enablement software:*** On top of AmbiqSuite SDK, we offer a series of software modules that enable complex applications to be developed quickly and efficiently. These modules include graphiqSPOT for graphics and display management, blueSPOT for Bluetooth communications, secureSPOT for managing security, and more.

- ***AI enablement software:*** Ambiq assists our customers in developing AI features by providing novel model factories specifically designed for edge AI, optimized AI inference engines, and neuralSPOT, our cross-platform AI SDK. These tools are designed to collectively address the challenges encountered when developing AI for edge products.

Value Proposition for Edge AI Customers

The multi-fold energy improvement delivered by SPOT-enabled products is indispensable for edge AI use cases. By reducing the energy required for sensing, communication, security, and AI, customers are given flexibility in their energy budgets. They can use this flexibility in a number of key ways without compromising battery life, including:

- Moving AI from the cloud to the edge to the benefit of security, cloud costs, response time, and radio overhead;
- Running larger AI models and running them faster;
- Running faster inference rates;
- Adding more sensors and sampling them faster;
- Extending battery life (e.g., from hours to days or from days to weeks); or
- Reducing the size of batteries and industrial designs for compact use cases.

Our Market Opportunity

Most of our revenue has come from ultra-low power SoCs, tightly bundled with software and various other solutions that combine 32-bit microcontrollers (MCUs) with wireless connectivity and additional circuitry, such

as graphics processing units, serial interfaces, and analog-to-digital interfaces. In the near future, we believe we can expand into embedded application processors, as well as stand-alone dedicated AI processors. Even though many of our customers today run basic AI inferencing directly on our Apollo-class SoCs and on lightweight accelerators within our SoCs, we are now developing dedicated AI processor units purpose-built for accelerating AI compute. Today, we define our market opportunity as 32-bit MCUs, discrete application processors (ASICs), wireless connectivity, and AI processors built for edge applications, markets totaling $12.8 billion in 2023 and growing to $22.5 billion in 2028.** This includes AI-enabled devices spanning industrial edge, smart homes and buildings, as well as body-worn AI devices such as consumer wearables and digital health devices. In the future, we believe our technology has the potential to address many semiconductor end markets that need much greater energy efficiency.

Within the edge end markets listed above, the market for 32-bit MCUs is expected to grow from $4.7 billion in 2023 to $6.0 billion in 2028, representing a compound annual growth rate (CAGR) of 5%, according to Gartner.** The market for wireless connectivity semiconductors for the same end markets is expected to grow from $2.4 billion in 2023 to $4.7 billion in 2028, representing a CAGR of 15%. The market for application processors is expected to grow from $5.7 billion in 2023 to $10.2 billion in 2028, representing a CAGR of 12%. According to Gartner, the discrete AI processors market is expected to grow from $152 million in 2023 to $1.6 billion in 2028, representing a CAGR of 60%.**

Since AI solutions are often embedded into general processor SoCs, broader AI-enabled market forecasts are also relevant to our opportunity. According to Gartner, the market for AI-enabled 32-bit MCUs and application processors in edge applications is expected to grow from $788 million in 2023 to $6.8 billion in 2028, representing a CAGR of 54%.*

Projected Addressable Market Size in Our End Markets ($ billion)**

Projected Addressable Market Size for Our Products ($ billion)***

Opportunities in Core Markets

Our core end markets are divided into two broad categories: Body-worn AI and Ambient AI. Body-worn AI comprises personal devices and medical / healthcare applications, while Ambient AI comprises industrial edge devices and smart homes/buildings. Each of our end markets is described in detail below:

- ***Personal Devices.*** Personal devices, including smartwatches and fitness bands, smart rings, AR/VR glasses, and hearable products, such as wireless earphones and hearing aids, must be comfortable, fashionable, and portable while maintaining long battery life and small size. Our low-power MCU-based SoC products act as central control units of many such devices or are implemented as co-processing products. They offer power-efficient communication, display and audio capabilities, always-on, real-time information, and seamless connectivity to other smart devices like mobile phones and home appliances. In addition, our solutions incorporate AI processing, such as voice activation, activity recognition, and tracking. For instance, in most smartwatches, heart health and sleep quality analyses are now done with neural networks rather than traditional filters. For the three months ended March 31, 2025, we generated approximately 98% of our revenue from the personal devices market. According to Gartner, the personal device market was $3.7 billion in 2023 and is expected to grow to $5.1 billion in 2028, representing a CAGR of 7%.****

- ***Medical / Healthcare Applications.*** The healthcare industry is experiencing a revolution as it enters the digital age through enhanced diagnostics and precise treatments designed to improve patient outcomes. Patient data collection in applications like heart health monitors, pulse oximeters, glucose monitors, and blood pressure monitors are using AI to better predict the likelihood of developing certain conditions and diseases. Our low-power semiconductor products can enable the tracking of myriad patient data, transforming disease diagnosis and allowing earlier, more accurate, and personalized treatment choices. Innovative products based on novel data sensing such as in-ear blood flow are creating new opportunities to improve health, both on a personal level and by involving healthcare professionals. According to Gartner, the market for medical and healthcare semiconductors is expected to grow from $887 million in 2023 to $1.3 billion in 2028, representing a CAGR of 8%.****

- ***Industrial Edge Devices.*** Industrial edge devices cater to commercial applications, such as manufacturing, supply chain logistics, smart metering, and other end-use applications. These products typically help perform critical functions such as predictive maintenance, productivity monitoring, asset tracking, and inventory management. Our solutions enable AI-driven predictive maintenance for

hard-to-access legacy industrial assets, connected devices for workers at large industrial sites, livestock management, and low power wide area (LPWA) solutions to deploy remotely without relying on existing telecom infrastructure. For instance, one of our customers utilizes our solution in wireless sensor devices for anomaly detection. This device utilizes a neural network to identify when that signature has changed and thus, when a failure may be looming. According to Gartner, the industrial edge device market is expected to grow from $5.8 billion in 2023 to $11.0 billion in 2028, representing a CAGR of 14%.****

- ***Smart Homes and Buildings.*** Smart homes, or fully connected household environments, enable voice-activated remote control of heating and thermostats, smart meters, lighting, security, and a multitude of other appliances from their smart devices. Our solutions enable various applications, such as AI-enabled voice-controlled smart remotes, fast identify online (FIDO) fingerprints biometric security cards, biometric credentials for physical and logical access, and smart locks. Today's battery-powered smart doorbells and security cameras use a passive infrared (PIR) sensor to detect motion and then turn on the camera to record interesting events. Our solution can eliminate the "dumb" PIR sensor and constantly monitor video using AI models to identify not just motion but specific events like "a delivery person knocked on the door." According to Gartner, the smart home and smart buildings market is expected to grow from $2.5 billion in 2023 to $5.2 billion in 2028, representing a CAGR of 16%.****

Future Market Opportunities

- ***Automotive.*** We believe we can further design and optimize our technology for use in automobiles, which inherently require fundamentally lower power products throughout than have historically been created using conventional power techniques. While we do not currently address this market, we intend to develop products for the automotive market using our expertise in low-power processing and complex integrated design.

- ***Data Center and Compute.*** We believe our ultra-low power SPOT platform can be extended for use in wall-powered applications including data centers and compute, enabling power efficiency gains. While we do not currently address this market, we intend to develop products using our expertise in low-power processing.

Our Competitive Strengths

We believe our core competitive strengths include the following:

Leading proprietary ultra-low power chip design technology. We exclusively focus on ultra-low power, sub-, and near-threshold chip design technology, which is our key differentiator and top development priority. We have spent nearly 15 years developing our SPOT platform, enabling our products to consume a demonstrated two to five times less power than traditional super-threshold technologies at the same manufacturing geometries. Our SPOT platform comprises dozens of individual circuit design techniques that are difficult to replicate, complemented by overall low-power system design expertise and know-how and our proprietary AmbiqSuite SDK, making it easy for our end customers to quickly and effectively implement our products. Our exclusive focus on low-power technologies and our culture of continuous innovation gives us a significant competitive advantage, as many of our competitors have focused on other priorities.

Extensible technology platform. We believe our SPOT platform is fully extensible to a wide variety of semiconductor applications that require substantially greater power efficiency in both battery-operated and wall-powered devices. We can apply our technology to products such as MCUs, application processors, MPUs, GPUs, AI accelerators, connectivity, and complex chipsets that combine a number of products into complex SoCs. Our technology is especially valuable for battery-operated products by significantly improving performance on a fixed power budget or by extending the time between charges in body-worn applications such as wristbands, smart clothing, hearables, smart glasses, and digital health devices. In addition, our ultra-low power solutions can be applied to wall-powered applications.

Robust intellectual property portfolio. Our IP is a key aspect of our business strategy, allowing us to maintain a competitive and, we believe, sustainable edge in the market. Our robust IP portfolio encompasses patents, trade secrets, and design IP that safeguard our proprietary technology and innovations related to ultra-low power and sub-threshold design. Our design IP is proven over five generations of the Apollo family of products. Our team is focused on developing new inventions, for which we seek patent protection to the extent practicable. As of June 6, 2025, we owned a portfolio of approximately 62 issued patents and had approximately 30 patent applications pending worldwide.

Proven demand from blue-chip end customers. We have demonstrated strong end customer growth across more than 270 million units shipped with international technology leaders such as Garmin, Google, Suunto, and others, which has validated the maturity of our technology platform and the robustness of our solutions offering. With our SPOT and SDK solutions, we have successfully enabled our customers to achieve much higher energy efficiency levels with increased battery life, allowing them to add new functions and deliver better performance. Our customers typically spend multiple years working with us on their product design, optimization, and productization. Moving to a competitor would typically require similar resources and time commitment along with hardware and software redesign, which makes transferring vendors more difficult. Furthermore, we work with our customers throughout their design cycles to help develop next-generation products. The long-lasting and collaborative nature of these relationships provides us with enhanced visibility into our customers' future requirements, allowing us to expand our business and increase our content in future products.

Scalable design and manufacturing relationships. Our SoC solutions leverage existing mature semiconductor process technologies that are readily available and cost-effective. Since our founding, we have partnered with a leading semiconductor foundry, TSMC, for our SoC products in 180-nanometer, 90-nanometer, 40-nanometer, and 22-nanometer process nodes. We leverage TSMC's mature fabrication processes, which are more cost-effective than the most advanced nodes, while still delivering products that consume a very small fraction of the power that our competitors require for their comparable parts. We also work closely with top-tier back-end partners such as ASE Group to test and assemble our products. Through these long-standing strategic partnerships, we are able to quickly scale production using mainstream semiconductor manufacturing and wafer scale integration and reduce our capital expenditures without compromising the quality of our products.

Top industry talent, experienced management team and engineering-focused culture. We believe our team's engineering, chip design, and AI talent is critical to our success. We employ an engineering-focused workforce, as well as a highly technical senior management team with deep industry and chip design experience. As of March 31, 2025, we had 190 employees, with engineers comprising approximately 66% of our total

workforce. In recognition of our engineering focus and talent base, we have won several prestigious awards, including "IoT Semiconductor Company of the Year" by Tech Breakthrough Organization in 2021, 2023 and 2024, "Artificial Intelligence Excellence" award by The Business Intelligence Group in both 2021 and 2023, "Big Innovation" Award by the Business Intelligence Group in both 2021 and 2022 for our SPOT platform, "2024 Distinguished Golden Bull" award by Business Media International, and "Embedded Award for the Apollo510 MCU" for 2024 under the Hardware category.

Our Strategy

Extend our presence in current markets with higher-performing and lower-power versions of our existing products. We have a strong foothold in the battery-powered personal devices market, which we believe will be further boosted by the growing adoption of edge AI. Our customers choose our ultra-low power SoC products because of their differentiated power efficiency. New members of the Apollo and Atomiq family of SoCs in development are designed to deliver even lower power and higher processing performance to our existing customers, potentially enabling and accelerating new innovative edge AI use cases. This will allow customers to innovate and accelerate edge AI adoption without compromising device usability or battery life.

Expand aggressively into new markets by broadening current product families. The success of the Apollo family in battery-powered personal devices has proven the value proposition of SPOT. We intend to apply our ultra-low power value proposition to enable AI use cases in medical, digital health, industrial, security, smart home and buildings, robotics, and automotive markets. We are developing a broader portfolio of Apollo and Atomiq SoC offerings that we believe will allow us to enter these additional markets. These SPOT-enabled variants are expected to be offered at different price points, different radio standards, and different peripheral sets alongside a full AI software solution stack. We anticipate that these products require only incremental investment and development time, so a broader portfolio can be built out quickly. For most new products, we will attach high-volume "alpha" customers to the product launch, a practice that has been successful for us to date.

Bring SPOT to new classes of chip products to access both existing and new markets. SPOT is a widely applicable, ultra-low power design methodology that can be quickly applied to new chip classes such as application processors (APs), dedicated AI processors, digital signal processors (DSPs), image processors, power management chips, communications chips, network processors, and many more. By building on 15 years of pre-existing SPOT IP, we intend to develop new ultra-lower power chip product families that we believe will provide us access to additional sockets within current customers, as well as to entirely new customers in new market segments.

Leverage our easy-to-use software and AI model suite to speed customer adoption and drive higher-margin revenue. Our customers face challenges in edge AI development, hindering the rollout of new features and products. We are working to increase our software suite to include new AI software products, such as AI model libraries, AI compilers and runtimes, AI model development utilities, AI model factories, and AI training data. Complementary to our existing SoC solutions, we believe these new AI software offerings will enable us to reach a broader set of customers, will allow these customers to more quickly implement their AI functionalities, and are expected to help drive increased SoC gross margin.

Develop SPOT into a licensable technology platform. We are presently using the SPOT platform to reach a high-growth set of edge AI markets. However, there are specialized compute and AI markets that greatly depend on power efficiency (e.g., AI processing in data centers and automotive) but which we are not resourced to address today. To reach these markets, we plan to develop SPOT into an IP and chip development platform that makes it easier for other companies to license or partner with us to incorporate SPOT into their own low-power chip designs, such as building next-generation AI data center chips.

Relentlessly advance the SPOT platform to help position us for future leadership. The pace of edge AI development is breathtaking. New sensors, new AI model architectures, and new AI use cases from our

customers drive a growing need for low power. We must therefore continue to advance the SPOT platform itself in parallel with Apollo and Atomiq product development. Innovation will focus on delivering even lower power and higher performance than is possible today. Key hardware innovation areas will include scaling SPOT to the 12 nanometer process node and beyond, developing ultra-low power sensing interfaces, developing new memories, and more. We expect our software innovations to include compiler development, runtime development, among others.

Key Investors

Our team is supported by a strong group of investors that share our commitments. Our investors include Kleiner Perkins, EDB Investments Pte Ltd, Arm Holdings plc, VentureTech Alliance, Conductive Ventures and RAMXEED Corporation (formerly known as Fujitsu Semiconductor Memory Solutions). Prospective investors should not rely on past investment decisions of our investors, as our investors may have different risk tolerances and have received their shares in prior offerings at prices lower than the price offered to the public in this offering.

Our End Customers

We work closely with our OEM and original design manufacturers (ODM) end customers throughout their design cycles, allowing us to develop long-term relationships with them as our technology becomes embedded in their products. As a result, we believe we are well-positioned to be designed into their current products and to continue developing next-generation AI platforms and low-power products for their future products. During the three months ended March 31, 2025, we sold our products to more than 100 end customers via third-party distributors and, to a lesser extent, through resellers and direct sales. During the year ended December 31, 2024, we sold our products to more than 200 end customers via third-party distributors and, to a lesser extent, through resellers and direct sales. Our major end customers accounting for 10% or more of our net sales during the first quarter of 2025 included Garmin, Google and another confidential customer, and in 2024 included Huawei, Garmin, Google and Xiaomi.

We typically fulfill the needs of our OEM and ODM end customers through direct relationships and select distributors. Net sales from Future Electronics Inc. (Distribution) Pte Limited, WT Microelectronics (Hong Kong) Limited and Kaga FEI Electronics Pacific Asia Ltd. comprised 23%, 16% and 10% of our net sales for the three months ended March 31, 2025, respectively. For the three months ended March 31, 2024, Max Electron (Hong Kong) Limited and Excelpoint Systems (Hong Kong) Limited accounted for approximately 20% and 18% of our net sales, respectively. For the year ended December 31, 2024, Kaga FEI Electronics Pacific Asia Ltd. accounted for approximately 11% of our net sales. No other distributor accounted for more than 10% of our net sales for the year ended December 31, 2024. For the year ended December 31, 2023, Max Electron (Hong Kong) Limited and Kaga FEI Electronics Pacific Asia Ltd. accounted for approximately 32% and 17% of our net sales, respectively. No other distributor accounted for more than 10% of our net sales for the year ended December 31, 2023. In addition to our direct OEM and ODM relationships, we also address a significant portion of our business through demand creation by our distribution and reseller network, which we intend to continue to broaden in the future.

We currently rely, and expect to continue to rely, on a limited number of end customers for a significant portion of our net sales. For the three months ended March 31, 2025 and 2024, our top five end customers accounted for 92% and 94% of our net sales, respectively, based on the sell-through information provided to us by our distributors. For the three months ended March 31, 2025, net sales from Garmin, Google and another confidential customer comprised approximately 38%, 25% and 23% of our net sales, respectively. No other single end customer directly or indirectly accounted for more than 10% of our net sales in the three months ended March 31, 2025. For the three months ended March 31 2024, net sales from Huawei, Google and Xiaomi comprised approximately 28%, 26% and 19% of our net sales, respectively. One other single end customer directly or indirectly accounted for more than 10% of our net sales in the three months ended March 31, 2024.

End customers in Mainland China accounted for 6% and 50% of our net sales in the three months ended March 31, 2025 and 2024, respectively, as we expanded our sales and marketing efforts outside of Mainland China. We expect to continue to increase our sales and marketing efforts outside of Mainland China.

For the years ended December 31, 2024 and 2023, our top five end customers accounted for 92% and 86% of our net sales, respectively, based on the sell-through information provided to us by our distributors. For 2024, revenues from Huawei, Garmin and Google comprised approximately 41%, 24% and 21% of our net sales, respectively. No other single end customers directly or indirectly accounted for more than 10% of our net sales in the year ended December 31, 2024. For 2023, net sales from Xiaomi, Huawei and Google comprised approximately 31%, 30% and 13% of our net sales, respectively. No other single end customers directly or indirectly accounted for more than 10% of our net sales in the year ended December 31, 2023. Sales to end customers in Mainland China accounted for 50% and 66% of our net sales in the years ended December 31, 2024 and 2023, as we expanded our sales and marketing efforts outside of Mainland China.

Sales and Marketing

We primarily engage with our end customers directly using a combination of a direct sales force and field application engineers, fulfilling sales directly or through third-party distributors. Our direct sales force and field applications engineers are located near our existing OEM and ODM end customers in Germany, Japan, Mainland China, Poland, Singapore, Taiwan, and the United States. Our sales force has varied end-market expertise.

We have contracts in place with distributors and third-party sales representatives throughout Asia, particularly in Taiwan, Korea, Japan, Singapore and Mainland China, as well as in the United States and Europe. We selected these independent distributors and representatives based on their ability to provide effective field sales, marketing communications, and technical support for our products. Following initial sales efforts, these third-party distributors assist predominantly in the fulfillment of orders from our end customers.

Our sales have historically been made based on purchase orders rather than end customer-specific long-term agreements. Our material terms and conditions are generally consistent with general industry practice but vary from distributor and end customer to end customer. We typically receive purchase orders 16 weeks ahead of the end customer's desired delivery date; however, this can extend up to 36 weeks, particularly in times of global supply constraints.

Manufacturing

We operate a fabless business model and use third-party foundries, packaging, and testing contractors to manufacture, assemble, and test our semiconductor products. This outsourced manufacturing approach allows us to focus our resources on the design, sale, and marketing of our products. In addition, we believe that outsourcing our manufacturing activities provides us with the flexibility and scalability needed to respond to new market opportunities, simplifies our operations, and significantly reduces our capital commitments.

We subject our third-party manufacturing contractors to qualification requirements to meet our solutions' high quality and reliability standards. We qualify each of our partners and their processes before applying the technology to our solutions. Our engineers work closely with all our manufacturing partners to keep material flowing quickly and consistently, with high quality. All sides look for continuous improvements to lower product costs by, for example, optimizing flow steps and improving yields.

- ***Wafer Fabrication.*** We use various generations of semiconductor processes to develop and manufacture our products. We contract with TSMC to produce semiconductor wafers.

- ***Packaging and Testing.*** Upon the completion of processing at the foundry, we use third-party contractors for packaging and testing, including the ASE group, King Yuan Electronics Group (KYEC) and Sigurd in Taiwan and Mainland China.

- ***Suppliers.*** We also rely upon a limited number of other manufacturing partners for our modules, boards, and other components.

Given the disruptive nature of our technology and design, we have historically worked very closely with our manufacturing partners to ensure they provide high-quality fabrication and an outstanding support team for continuous improvement. For instance, we worked with TSMC to identify the source of excessive circuit leakage and quickly optimized the process to resolve it. In addition, to ensure the highest reliability, sub-threshold circuits built on our SPOT platform have been subjected to the same testing standards of super-threshold technologies, including those that measure performance in extreme conditions over extended periods. We believe our ability to work closely with our third-party partners to optimize the production processes used by our technologies provides us with a meaningful competitive advantage.

Research and Development

We believe that our success depends in large part on our ability to enhance our existing proprietary SPOT platform, expand our low-power SoC product portfolio, develop new innovative solutions based on our SPOT platform, and integrate additional capabilities to serve our existing and future target markets. We engage in research and development efforts in the following core areas:

- Core SPOT platform;

- Digital, analog, and radio-frequency chipset design (including MCUs, GPUs, AI accelerators, wireless connectivity, power management, security, imaging, audio, and sensing);

- General and feature enablement software development (including connectivity, graphics, security, and audio); and

- AI enablement software development (including AI model factories, AI runtimes, and AI libraries); and

- System-level solutions development (including algorithms, reference hardware, reference software, and end customer design references).

Our research and development team comprises highly skilled engineers and technologists with extensive experience in digital, mixed-signal, RF chipset design, system-level architecture, and software development. We have assembled our engineering team in the United States (our headquarters in Austin, Texas, and our California location), Mainland China (Shanghai and Shenzhen), Taiwan, and Singapore.

For the three months ended March 31, 2025 and 2024, our research and development expenses were $8.7 million and $9.4 million, representing 55.2% and 62.0% of first quarter sales for 2025 and 2024, respectively. We intend to continue to invest in research and development to support and enhance our existing low-power sub-threshold solutions and design and develop future product offerings.

Intellectual Property

We rely on a combination of intellectual property rights, including patents, trade secrets, copyrights and trademarks, and contractual protections, to protect our core technology and intellectual property. As of June 6, 2025, we owned 52 issued U.S. patents, 10 issued foreign patents, 22 pending U.S. patent applications, seven pending foreign patent applications, and one pending Patent Cooperation Treaty application. The issued patents in the United States generally expire beginning in 2033 through 2042. Our issued patents and pending patent applications relate to key sub-threshold analog and digital techniques and algorithms, memory techniques, circuits, and system-level optimization.

In addition to our own intellectual property, we also use third-party licenses for certain technologies embedded in our solutions. These are typically granted to us in non-exclusive contracts provided under royalty-accruing or paid-up licenses. While we do not believe our business is dependent to any significant degree on any

individual third-party license, we expect to continue to use and may license additional third-party technology for our solutions. We also invest in the latest commercially available software design and simulation tools, which enable us to leverage our intellectual property portfolio, improve time to commercialization, and deliver high-performance products.

We generally control access to and use of our proprietary and confidential information by employing internal and external controls, including contractual protections with employees, consultants, contractors, customers, partners, and suppliers. Our employees and consultants are required to execute confidentiality and invention assignment agreements in connection with their employment or consulting relationships with us. We also require them to agree to disclose and assign to us inventions conceived or made in connection with the employment or consulting relationship. Despite our efforts to protect our IP, such agreements may be insufficient or breached, and unauthorized parties may copy or otherwise obtain and use our software, technology, or other information that we regard as proprietary intellectual property.

The semiconductor industry is characterized by vigorous protection and pursuit of intellectual property rights and positions, resulting in protracted and expensive litigation for many companies. We have in the past received, and we expect that in the future we may receive, allegations of infringement, misappropriation or other violations or liability for damages that may invalidate our proprietary rights and harm our business and our ability to compete. Any litigation, regardless of success or merit, could cause us to incur substantial expenses, reduce our sales, and divert the efforts of our technical and management personnel. In the event we receive an adverse result in any litigation, we could be required to pay substantial damages, seek licenses from third parties, which may not be available on reasonable terms or at all, cease the sale of our solutions, expend significant resources to develop alternative technology or discontinue the use of processes requiring the relevant IP or technology.

Government Regulation

We are subject to the laws and regulations of various jurisdictions and governmental agencies affecting our operations and the sale of our solutions in areas including, but not limited to: AI; intellectual property; tax; import and export requirements, including the U.S. Export Administration Regulations, U.S. Customs regulations, and various economic and trade sanctions regulations administered by the U.S. Treasury Department's Office of Foreign Assets Controls; anti-corruption; economic and trade sanctions; national security and foreign investment; foreign exchange controls and cash repatriation restrictions; data privacy and security requirements (as described below); competition; advertising; employment; product regulations; environment, health and safety requirements; and consumer laws. These regulations may limit the export of our products and technology, and provision of services outside of the United States, or may require export authorizations, including by license, a license exception, or other appropriate government authorization. Legislation related to AI has also been introduced at the federal level and is advancing at the state level. For example, Colorado recently passed the Colorado Artificial Intelligence Act, which is set to go into effect on February 1, 2026, and which regulates the development, deployment, and use of AI systems and the California Privacy Protection Agency is currently in the process of finalizing regulations under the California Consumer Privacy Act regarding the use of AI and automated decision-making. Additionally, the EU AI Act and other AI regulations could have a material impact on planned business activities involving the development and/or use of AI technologies and/or increase compliance costs. The EU AI Act establishes, among other things, a risk-based governance framework for regulating AI systems operating in the European Union. This framework would categorize AI systems, based on the risks associated with such AI systems' intended purposes, as creating unacceptable or high risks, with all other AI systems being considered low risk. The EU AI Act includes requirements around transparency, conformity assessments and monitoring, risk assessments, human oversight, security, accuracy, general purpose AI, and foundation models, and provides for fines of up to the greater of €35 million or 7% of worldwide annual turnover for violations. We may not be able to anticipate how to respond to these rapidly evolving frameworks, and we may need to expend resources to adjust our offerings in certain jurisdictions if the legal frameworks are inconsistent across jurisdictions.

As a multinational business, the import and export of our technology are subject to laws and regulations including international treaties, U.S. export controls and sanctions laws, customs regulations, and local trade rules around the world. The scope, nature, and severity of such controls varies widely across different countries and may change frequently over time. Such laws, rules, and regulations may delay the introduction of our infrastructure and services or impact our competitiveness through restricting our ability to do business in certain places or with certain entities and individuals. For example, the U.S. Department of Commerce continues to tighten export controls and add firms to the Entity List. These export restrictions, which would require that we obtain licenses from the U.S. Department of Commerce to allow us to export our solutions to such listed firms, could limit or prevent us from doing business with certain potential customers or potential suppliers. These restrictive governmental actions and any similar measures that may be imposed on U.S. companies by other governments could limit our ability to conduct business globally.

See "Risk Factors—Risks Related to Our Business" for additional information regarding risks we face related to government regulation.

Data Privacy and Cybersecurity

Numerous data privacy and security obligations, including state, federal, and foreign laws, regulations, rules and standards, govern the collection, use, access to, handling, transmission, processing, confidentiality, and security of personal data and could apply now or in the future to our operations or the operations of our partners. Data privacy and cybersecurity laws, regulations, rules, standards, and other obligations are constantly evolving, may conflict with each other to complicate compliance efforts, and can result in investigations, proceedings, or actions that lead to significant civil penalties and restrictions on data collection, transfers, and processing. Such federal U.S. and foreign obligations may include, without limitation, the Federal Trade Commission Act, the EU General Data Protection Regulation (EU GDPR), the UK General Data Protection Regulation (UK GDPR), and Mainland China's Personal Information Protection Law (PIPL). Additionally, several states within the United States have enacted or proposed data privacy laws. For example, California passed the California Consumer Privacy Act, as amended by the California Privacy Rights Act (collectively, CCPA), Virginia passed the Consumer Data Protection Act, Colorado passed the Colorado Privacy Act, and Utah passed the Consumer Privacy Act.

Additionally, we are, or may become, subject to various U.S. federal and state consumer protection laws, which require us to publish statements that accurately and fairly describe how we handle personal data and the choices individuals may have about how we handle their personal data.

The CCPA and EU GDPR are increasingly stringent and evolving regulatory frameworks related to personal data processing that may increase our compliance obligations and exposure to non-compliance. For example, the CCPA imposes obligations on covered businesses to provide specific disclosures related to a business's collecting, using, and disclosing personal data, including certain sensitive personal data, and to respond to certain requests from California residents related to their personal data (for example, requests to know of the business's personal data processing activities, to delete the individual's personal information, and to opt out of certain personal data disclosures). Also, the CCPA provides civil penalties and a private right of action for data breaches, which may include the award of statutory damages.

Foreign data privacy and cybersecurity laws (including, but not limited to, the EU GDPR, UK GDPR, and PIPL) impose significant and complex compliance obligations on entities subject to those laws. For example, the EU GDPR applies to any company established in the European Economic Area (EEA) and to companies based outside the EEA that process personal data in connection with offering goods or services to data subjects or monitoring the behavior of

data subjects in the EEA. These obligations may include limiting personal data processing to only what is necessary for specified, explicit, and legitimate purposes, requiring a legal basis for personal data processing, requiring the appointment of a data protection officer in certain circumstances, imposing transparency obligations to data subjects, requiring data protection impact assessments in certain

circumstances, limiting the collection and retention of personal data, providing certain rights for data subjects related to their personal data, imposing specific standards for data subject consents, requiring the implementation and maintenance of technical and organizational safeguards for personal data, mandating notice of certain personal data breaches to the relevant supervisory authority(ies) and affected individuals, and mandating the appointment of representatives in the UK and/or the EU in certain circumstances. Some of these laws and regulations authorize the governing agencies to investigate companies under their jurisdiction to ensure compliance and to impose fines and other measures against companies who are not in compliance. See the section titled "Risk Factors – Risks Related to Our Intellectual Property" for additional information about the laws and regulations to which we may become subject and about the risks to our business associated with such laws and regulations.

Cyber-attacks, malicious internet-based activity, online and offline fraud, and other similar activities threaten the confidentiality, integrity, and availability of our sensitive information and information technology systems, as well as those of the third parties we work with. Such threats are prevalent and continue to rise, are increasingly challenging to detect, and come from a variety of sources, including traditional computer "hackers," threat actors, "hacktivists," organized criminal threat actors, personnel (such as through theft or misuse), sophisticated nation states, and nation-state-supported actors. Some actors now engage and are expected to continue to engage in cyber-attacks, including, without limitation nation-state actors for geopolitical reasons and in conjunction with military conflicts and defense activities. See the section titled "Risk Factors – Risks Related to Our Business" for additional information about the cybersecurity risks to our business.

Competition

The semiconductor market is intensely competitive. We anticipate that the market for our products will continually evolve and be subject to rapid technological change. We believe the principal competitive factors in our industry are:

- Power consumption and battery life;
- Product capabilities, functionality and performance;
- Product size;
- Level of integration;
- End customer support;
- Reliability;
- Ability to rapidly introduce new products to market;
- Price;
- Intellectual property protection; and
- Software capabilities.

We believe that we are competitive with respect to these factors, particularly because our SPOT platform is designed to enable our products to have significantly lower power consumption than alternative products, allowing for greater adoption of AI at the edge. Our products are manufactured in standard CMOS, which generally allows us to supply them relatively rapidly so that end customers can meet their product introduction schedules. However, the disadvantages we face include our relatively short operating history in certain of our markets and our smaller size and customer base relative to many of our competitors.

Today, we generally compete with large MCU vendors, such as Infineon, Microchip, NXP, Renesas, Silicon Laboratories, STMicroelectronics, Texas Instruments, and others. In addition, we also compete with connectivity players such as Dialog Semiconductor (now Renesas), Nordic Semiconductor, Synaptics, and Telink. In addition, we sometimes compete (and other times work alongside certain designs) with connected processor platform players such as Qualcomm.

Employees

As of March 31, 2025, we had 190 full-time equivalent employees, including 100 located in the United States, 43 located in Mainland China, 30 located in Taiwan, 15 located in Singapore, one located in the United Kingdom and one located in Germany. Our full-time equivalent employees include 126 employees in research and development (including 41 in software and system-level design and 12 in operations), 43 in sales and marketing and 21 in

general and administrative. We consider relations with our employees to be good and have never experienced a work stoppage. None of our employees are either represented by a labor union or subject to a collective bargaining agreement.

Facilities

Our principal executive offices are located in a leased facility in Austin, Texas, consisting of approximately 12,500 square feet of office space under a lease expiring in June 2026. This facility accommodates our principal engineering, sales, marketing, operations, finance, and administrative activities. We are working closely with our current landlord in Austin to renew the lease or move offices to a larger space within the same office complex. We also lease offices in Taiwan, Singapore, and Mainland China. We do not own any real property. We believe that our leased facilities are adequate to meet our current needs and that additional facilities will be available on commercially reasonable terms for lease to meet future needs.

Legal Proceedings

We are currently not a party to any material legal proceedings. We may, from time to time, become involved in litigation relating to claims arising from our ordinary course of business. These claims, even if not meritorious, could result in the expenditure of significant financial and managerial resources.

Management

Executive Officers and Directors

The following table sets forth information regarding our executive officers and directors as of June 2, 2025:

Name	Age	Position
Executive Officers		
Fumihide Esaka	61	Chief Executive Officer and Director
Sean Chen	56	President and Chief Operating Officer
Scott Hanson, Ph.D.	41	Chief Technology Officer and Director
Jeff Winzeler	65	Chief Financial Officer
Non-Employee Directors		
Wen Hsieh, Ph.D.	52	Chairman
Ker Zhang, Ph.D.	62	Director
Joseph Tautges	49	Director
Timothy Chen	50	Director

Executive Officers

Fumihide Esaka has served as our Chief Executive Officer and as a member of our board of directors since December 2015. Prior to joining Ambiq, Mr. Esaka served as chief executive officer of Transphorm Inc., a global semiconductor development company, from June 2013 to November 2015 and president and chief executive officer of Nihon Inter Electronics Corporation (NIEC) from June 2010 to May 2013. Mr. Esaka holds a B.A. in electrical engineering and computer science from the University of California San Diego. We believe that Mr. Esaka's current role as our Chief Executive Officer and his extensive executive leadership and management experience at semiconductor companies qualify him to serve on our board of directors.

Scott Hanson, Ph.D. founded Ambiq and has served as our Chief Technology Officer since February 2013. Previously, Dr. Hanson served as our CEO from January 2010 to January 2013. Dr. Hanson has served on our board of directors since January 2010. Dr. Hanson has won numerous awards, including the ECE Alumni Rising Star Award and Arbor Networks Ph.D. Research Impact Award from University of Michigan. Dr. Hanson holds a BSE in electrical engineering, a MSE in electrical engineering and a Ph.D. in electrical engineering from the University of Michigan. We believe that Dr. Hanson's role as our founder and Chief Technology Officer and his extensive knowledge of our technology qualify him to serve on our board of directors.

Sean Chen has served as our President since November 2016 and is currently also serving as Chief Operation Officer. Previously, Mr. Chen served as our Group V.P., Greater China and Global Operations from November 2015 to October 2016. Mr. Chen has over 20 years of semiconductor experience and has previously served as Vice President of Marketing and Business Development at mCube, and held several management positions at various startups and at Nagevtech, Intel, IDT, and TSMC. Mr. Chen holds a M.S. in Chemical Engineering from the National Cheng-Kung University.

Jeff Winzeler has served as our Chief Financial Officer since June 2025. Prior to joining Ambiq, Mr. Winzeler served as the Chief Financial Officer of Kandou S.A., a semiconductor company, from June 2020 to May 2025. Mr. Winzeler has also held Chief Financial Officer and Chief Operating Officer roles at several technology firms, including Everspin Technologies, Avnera, Solar Power Inc., and International DisplayWorks. Earlier in his career, Mr. Winzeler held several senior finance positions at Intel Corporation. Mr. Winzeler holds a B.S. in Finance from the University of Idaho and completed the Strategic Finance Leadership Program at Stanford University.

Non-Employee Directors

Dr. Wen Hsieh has been a member of our board of directors since November 2014. He co-founded Matter Venture Partners (MVP) in January 2023. MVP is a $311 million Silicon Valley-based venture fund focused on early-stage investments in semiconductors, robotics/Robotics-as-a-Service, AI infrastructure, manufacturing tech, energy building blocks and automation tools for life sciences. Prior to MVP, Dr. Hsieh was a General Partner of Kleiner Perkins (KP), from February 2006 to January 2023, where he led the HardTech practice. Before joining KP, Dr. Hsieh was an Associate Principal at McKinsey & Co. San Francisco, and a leader of McKinsey's semiconductor practice. Prior to McKinsey, Dr. Hsieh founded OnChip Technologies, a startup developing MEMS microfluidic biochips. Dr. Hsieh currently sits on the public company boards of Desktop Metal and Amprius Technologies. He earned a B.S. with honors, an M.S. and Ph.D. in electrical engineering, plus a Ph.D. minor in biology, all from the California Institute of Technology. We believe that Dr. Hsieh's extensive experience as an investor in the semiconductor and other HardTech industries, as well as his experience serving on boards of directors of various technology companies qualifies him to serve on our board of directors.

Ker Zhang, Ph.D. has been a member of our board of directors since August 2017. Dr. Zhang is a technology expert with over 30 years of hard tech experience as an engineer, executive, entrepreneur, CEO, Chairman, and investor. Dr. Zhang is the Executive Chairman of Crossbar Inc., a Kleiner Perkins portfolio company that develops innovative memory technology since April 2019. Dr. Zhang also has been a member of the board of Valens Semiconductor Co. since September 2021. Dr. Zhang also advises Intel Capital and served on the board of Keyssa on behalf of Intel. Previously, from February 2018 to September 2023, Dr. Zhang was an Entrepreneur In Residence at Kleiner Perkins. In addition, he was Vice President and General Manager of Intel's CDMA product and development group from October 2015 to February 2018. Dr. Zhang was the CEO of VIA Telecom from 2002 until it was acquired by Intel in 2015. Dr. Zhang holds an M.S. in physics from the University of Massachusetts and a Ph.D. in Electrical Engineering from Worcester Polytechnic Institute. He holds 3 U.S. patents. We believe that Dr. Zhang's experience and expertise in hardware engineering and years of senior management experience qualify him to serve on our board of directors.

Joseph Tautges has been a member of our board of directors since November 2022. Mr. Tautges is currently the EVP and Chief Operating Officer at Worldpay Inc. and has served in this role since March 2024 to present. Mr. Tautges previously served as the Executive Vice President, Chief Operating Officer of CDK Global from January 2021 to July 2022, and as its Executive Vice President, Chief Financial Officer from August 2017 to January 2021. Prior to joining CDK, Mr. Tautges served as Chief Financial Officer of the $18 billion Enterprise Services segment of Hewlett Packard Enterprise (HPE) from May 2014 to April 2017. Prior to HPE, Mr. Tautges held various levels of increasing responsibility in both operations and financial management with Sears Holdings from 2011 to 2014 and Aon Hewitt from 2002 to 2011. Mr. Tautges is a Certified Public Accountant. Mr. Tautges holds a degree in accountancy from Northern Illinois University. We believe that Mr. Tautges' decades of experience in various leadership roles at CDK Global and HPE and his experience as a COO and CFO qualify him well to serve on our board of directors.

Timothy Chen has been a member of our board of directors since November 2023. Mr. Chen is a founding partner of ACHI and KQ Capital, and has served as the CEO of VIA China since September 2015. Prior to serving as CEO, Mr. Chen was the Head of Sales and Business Development at VIA China, where he began in 2007. From October 2019 to February 2022, Mr. Chen served as Chairman for Via Telecom, a CDMA alternative chip supplier to Qualcomm, Inc. In addition, Mr. Chen served as Chief Financial Officer of PTK Acquisition Corp., who, in 2021, announced a SPAC business combination with Valens Semiconductor, Ltd., resulting in their listing on the New York Stock Exchange. In December 2014, Mr. Chen founded EMQ, an Asian cross-border startup developing global payment network solutions, where he now serves as Chairman and board member. In addition, since June 2006, Mr. Chen has served as a board member of CatchPlay, a leading Taiwanese multimedia and digital content provider. Mr. Chen holds a degree in Engineering from the University of California Berkely. We believe that Mr. Chen's experience in various leadership roles and his expertise in hardware engineering qualify him to serve on our board of directors.

Family Relationships

There are no family relationships among any of our directors or executive officers.

Board Composition

Our business and affairs are organized under the direction of our board of directors, which currently consists of six members. Our board of directors may establish the authorized number of directors from time to time by resolution and the authorized number of directors comprising our board of directors is currently set at ten with four vacancies. The primary responsibilities of our board of directors are to provide oversight, strategic guidance, counseling, and direction to our management. Our board of directors meets on a regular basis and additionally as required.

Certain members of our board of directors were elected under the provisions of our Amended and Restated Voting Agreement entered into in August 2023 (the Voting Agreement), which will terminate upon the closing of this offering. Under the terms of our Voting Agreement, the stockholders who are party to the Voting Agreement have agreed to vote their respective shares to elect: (i) one director designated by a majority of holders of our Series A redeemable convertible preferred stock, which seat is currently vacant; (ii) two directors designated by a majority of holders of our Series C redeemable convertible preferred stock, which individuals are initially Wen Hsieh, Ph.D. and Timothy Chen; (iii) one director designated by a majority of holders of our Series E redeemable convertible preferred stock, which seat is currently vacant; (iv) three directors designated by a majority of holders of our common stock, voting as a separate class, which individuals are initially Fumihide Esaka as CEO Director and Scott Hanson, Ph.D., with one vacancy; and (v) three directors designated by a majority of holders of our common stock and preferred stock, voting together as a single class on an as-if-converted to common stock basis, which individuals are currently Ker Zhang, Ph.D. and Joseph Tautges, with one vacancy. The Voting Agreement will terminate upon the closing of this offering, and upon the closing of the offering no stockholder will have any special rights regarding the election or designation of the members of our board of directors. Our current directors elected to our board of directors pursuant to the Voting Agreement will continue to serve as directors until their successors are duly elected and qualified by holders of our common stock.

Our board of directors has determined that four of our six directors are independent directors, as defined under the New York Stock Exchange listing rules.

In accordance with the terms of our amended and restated bylaws, which will be effective immediately upon the closing of this offering, our board of directors will be divided into three classes, Class I, Class II, and Class III, with members of each class serving staggered three-year terms.

Effective upon the closing of this offering, our board of directors will consist of six directors divided into the following classes:

- Class I, which will consist of Ker Zhang and Timothy Chen, whose terms will expire at our first annual meeting of stockholders to be held after the completion of this offering;
- Class II, which will consist of Wen Hsieh and Joseph Tautges, whose terms will expire at our second annual meeting of stockholders to be held after the completion of this offering; and
- Class III, which will consist of Fumihide Esaka and Scott Hanson, whose terms will expire at our third annual meeting of stockholders to be held after the completion of this offering.

At each annual meeting of stockholders to be held after the initial classification, the successors to directors whose terms then expire will be elected to serve from the time of election and qualification until the third annual meeting following their election and until their successors are duly elected and qualified. The authorized size of our board of directors is currently six members and may be changed only by resolution by a majority of our board of directors. We currently anticipate that our board of directors will consist of six members prior to the

completion of this offering. This classification of our board of directors may have the effect of delaying or preventing changes in our control or management. Our directors may be removed for cause by the affirmative vote of the holders of at least two-thirds (2/3) of our voting stock.

Director Independence

Under the New York Stock Exchange listing rules, independent directors must comprise a majority of our board of directors as a listed company within one year of the listing date.

Our board of directors has undertaken a review of the independence of each director. Based on information provided by each director concerning her or his background, employment and affiliations, including family relationships, our board of directors has determined that none of our directors, other than Fumihide Esaka and Scott Hanson, Ph.D., has any relationship that would interfere with the exercise of independent judgment in carrying out the responsibilities of a director and that each of these directors is "independent" as that term is defined under the New York Stock Exchange listing rules. Our board of directors has determined that Fumihide Esaka and Scott Hanson, Ph.D., by virtue of their positions as our Chief Executive Officer and Chief Technology Officer, respectively, are not independent under applicable rules and regulations of the U.S. Securities and Exchange Commission (the SEC) and the New York Stock Exchange listing rules. In making these determinations, our board of directors considered the current and prior relationships that each non-employee director has with our company and all other facts and circumstances our board of directors deemed relevant in determining their independence, including the beneficial ownership of our shares by each non-employee director and the transactions described in the section titled "Certain Relationships and Related Person Transactions."

Role of our Board of Directors in Risk Oversight/Risk Committee

One of the key functions of our board of directors is informed oversight of our risk management process. Our board of directors does not have a standing risk management committee but rather administers this oversight function directly through our board of directors as a whole, as well as through various standing committees of our board of directors that address risks inherent in their respective areas of oversight. In particular, our board of directors is responsible for monitoring and assessing strategic risk exposure and our audit committee has the responsibility to consider and discuss our major financial risk exposures and the steps our management has taken to monitor and control these exposures, including guidelines and policies to govern the process by which risk assessment and management is undertaken. The audit committee also monitors compliance with legal and regulatory requirements. Our compensation committee also assesses and monitors whether our compensation plans, policies, and programs comply with applicable legal and regulatory requirements.

Board Committees

Our board of directors has established a compensation committee, and will establish an audit committee and a nominating and corporate governance committee. Our board of directors has adopted a charter for each of these committees, which complies with the applicable requirements of current New York Stock Exchange listing rules. We intend to comply with future requirements to the extent they are applicable to us. Following the completion of this offering, copies of the charters for each committee will be available on the investor relations portion of our website.

Audit Committee

Upon the effectiveness of the registration statement of which this prospectus forms a part, our audit committee will consist of Joseph Tautges, Ker Zhang and Timothy Chen. Our board of directors has determined that each of the members of our audit committee satisfies the independence requirements of the New York Stock Exchange and Rule 10A-3 under the Exchange Act. Each member of our audit committee can read and understand fundamental financial statements in accordance with New York Stock Exchange audit committee requirements. In arriving at this determination, our board of directors has examined each audit committee member's scope of experience and the nature of their prior and/ or current employment.

Joseph Tautges will serve as the chair of our audit committee. Our board of directors has determined that qualifies as an audit committee financial expert within the meaning of SEC regulations and meets the financial sophistication requirements of the New York Stock Exchange listing rules. In making this determination, our board has considered Joseph Tautges's formal education and previous experience in financial roles. Both our independent registered public accounting firm and management periodically meet privately with our audit committee.

The functions of this committee include, among other things:

- evaluating the performance, independence, and qualifications of our independent auditors and determining whether to retain our existing independent auditors or engage new independent auditors;
- reviewing our financial reporting processes and disclosure controls;
- reviewing and approving the engagement of our independent auditors to perform audit services and any permissible non-audit services;
- reviewing the adequacy and effectiveness of our internal control policies and procedures, including the responsibilities, budget, staffing, and effectiveness of our internal audit function;
- reviewing with the independent auditors the annual audit plan, including the scope of audit activities and all critical accounting policies and practices to be used by us;
- obtaining and reviewing at least annually a report by our independent auditors describing the independent auditors' internal quality control procedures and any material issues raised by the most recent internal quality-control review;
- monitoring the rotation of partners of our independent auditors on our engagement team as required by law;
- prior to engagement of any independent auditor, and at least annually thereafter, reviewing relationships that may reasonably be thought to bear on their independence, and assessing and otherwise taking the appropriate action to oversee the independence of our independent auditor;
- reviewing our annual and quarterly financial statements and reports, including the disclosures contained in "Management's Discussion and Analysis of Financial Condition and Results of Operations," and discussing the statements and reports with our independent auditors and management;
- reviewing with our independent auditors and management significant issues that arise regarding accounting principles and financial statement presentation and matters concerning the scope, adequacy, and effectiveness of our financial controls and critical accounting policies;
- reviewing with management and our auditors any earnings announcements and other public announcements regarding material developments;
- establishing procedures for the receipt, retention, and treatment of complaints received by us regarding financial controls, accounting, auditing, or other matters;
- preparing the report that the SEC requires in our annual proxy statement;
- reviewing and providing oversight of any related person transactions in accordance with our related person transaction policy and reviewing and monitoring compliance with legal and regulatory responsibilities, including our code of ethics;
- reviewing our major financial risk exposures, including the guidelines and policies to govern the process by which risk assessment and risk management is implemented; and
- reviewing and evaluating on an annual basis the performance of the audit committee and the audit committee charter.

We believe that the composition and functioning of our audit committee complies with all applicable requirements of the Sarbanes-Oxley Act and all applicable SEC rules and regulations. We intend to comply with future requirements to the extent they become applicable to us.

Compensation Committee

Upon the effectiveness of the registration statement of which this prospectus forms a part, our compensation committee will consist of Timothy Chen and Wen Hsieh. Wen Hsieh will serve as the chair of our compensation committee. Our board of directors has determined that each of the members of our compensation committee is a non-employee director, as defined in Rule 16b-3 promulgated under the Exchange Act, and satisfies the independence requirements of the New York Stock Exchange. The functions of this committee include, among other things:

- reviewing and approving the corporate objectives that pertain to the determination of executive compensation;
- reviewing and approving the compensation and other terms of employment of our executive officers;
- reviewing and approving performance goals and objectives relevant to the compensation of our executive officers and assessing their performance against these goals and objectives;

- making recommendations to our board of directors regarding the adoption or amendment of equity and cash incentive plans and approving amendments to such plans to the extent authorized by our board of directors;
- reviewing and making recommendations to our board of directors regarding the type and amount of compensation to be paid or awarded to our non-employee board members;
- reviewing and assessing the independence of compensation consultants, legal counsel, and other advisors as required by Section 10C of the Exchange Act;
- administering our equity incentive plans, to the extent such authority is delegated by our board of directors;
- reviewing and approving the terms of any employment agreements, severance arrangements, change in control protections, indemnification agreements, and any other material arrangements for our executive officers;
- reviewing with management our disclosures under the caption "Compensation Discussion and Analysis" in our periodic reports or proxy statements to be filed with the SEC, to the extent such caption is included in any such report or proxy statement;
- preparing an annual report on executive compensation that the SEC requires in our annual proxy statement; and
- reviewing and evaluating on an annual basis the performance of the compensation committee and recommending such changes as deemed necessary with our board of directors.

We believe that the composition and functioning of our compensation committee complies with all applicable requirements of the Sarbanes-Oxley Act and all applicable SEC and New York Stock Exchange rules and regulations. We intend to comply with future requirements to the extent they become applicable to us.

Nominating and Corporate Governance Committee

Upon the effectiveness of the registration statement of which this prospectus forms a part, our nominating and corporate governance committee will consist of Joseph Tautges and Wen Hsieh. Our board of directors has determined that each of the members of our nominating and corporate governance committee satisfies the independence requirements of the New York Stock Exchange. Wen Hsieh will serve as the chair of our nominating and corporate governance committee. The functions of this committee include, among other things:

- identifying, reviewing, and making recommendations of candidates to serve on our board of directors;
- evaluating the performance of our board of directors, committees of our board of directors, and individual directors and determining whether continued service on our board is appropriate;
- evaluating nominations by stockholders of candidates for election to our board of directors;
- evaluating the current size, composition, and organization of our board of directors and its committees and making recommendations to our board of directors for approvals;

- developing a set of corporate governance policies and principles and recommending to our board of directors any changes to such policies and principles;
- reviewing issues and developments related to corporate governance and identifying and bringing to the attention of our board of directors current and emerging corporate governance trends; and
- reviewing periodically the nominating and corporate governance committee charter, structure, and membership requirements and recommending any proposed changes to our board of directors, including undertaking an annual review of its own performance.

We believe that the composition and functioning of our nominating and corporate governance committee complies with all applicable requirements of the Sarbanes-Oxley Act and all applicable SEC and New York Stock Exchange rules and regulations. We intend to comply with future requirements to the extent they become applicable to us.

Compensation Committee Interlocks and Insider Participation

None of our executive officers currently serve, or has served during the last completed fiscal year, on the compensation committee or board of directors of any other entity that has one or more executive officers serving as a member of our board of directors.

Limitation on Liability and Indemnification of Directors and Officers

Our amended and restated certificate of incorporation, which will be effective upon the closing of this offering, limits our directors' and officers' liability to the fullest extent permitted under Delaware General Corporation Law (the DGCL). The DGCL provides that directors of a corporation will not be personally liable for monetary damages for breach of their fiduciary duties as directors, except for liability:

- for any transaction from which the director derives an improper personal benefit;

- for any act or omission not in good faith or that involves intentional misconduct or a knowing violation of law;
- for any unlawful payment of dividends or redemption of shares; or
- for any breach of a director's duty of loyalty to the corporation or its stockholders.

If the DGCL is amended to authorize corporate action further eliminating or limiting the personal liability of directors or officers, then the liability of our directors and officers shall be eliminated or limited to the fullest extent permitted by the DGCL, as so amended.

Delaware law and our amended and restated bylaws provide that we will, in certain situations, indemnify our directors and officers and may indemnify other employees and other agents, to the fullest extent permitted by law. Any indemnified person is also entitled, subject to certain limitations, to advancement, direct payment, or reimbursement of reasonable expenses (including attorneys' fees and disbursements) in advance of the final disposition of the proceeding.

In addition, we have entered, and intend to continue to enter, into separate indemnification agreements with our directors and officers. These agreements, among other things, require us to indemnify our directors and officers for certain expenses, including attorneys' fees, judgments, fines, and settlement amounts incurred by a director or officer in any action or proceeding arising out of their services as one of our directors or officers or any other company or enterprise to which the person provides services at our request.

We maintain a directors' and officers' insurance policy pursuant to which our directors and officers are insured against liability for actions taken in their capacities as directors and officers. We believe that these provisions in our amended and restated certificate of incorporation and amended and restated bylaws and these indemnification agreements are necessary to attract and retain qualified persons as directors and officers.

Insofar as indemnification for liabilities arising under the Securities Act of 1933, as amended (Securities Act), may be permitted to directors, officers, or control persons, in the opinion of the SEC, such indemnification is against public policy as expressed in the Securities Act and is therefore unenforceable.

Code of Business Conduct and Ethics for Employees, Executive Officers and Directors

We intend to adopt a Code of Business Conduct and Ethics (the Code of Conduct), applicable to all of our employees, executive officers, and directors. The Code of Conduct will be available on our website at www.ambiq.com. Information contained on or accessible through our website is not a part of this prospectus, and the inclusion of our website address in this prospectus is an inactive textual reference only. The nominating and corporate governance committee of our board of directors is responsible for overseeing the Code of Conduct and must approve any waivers of the Code of Conduct for employees, executive officers, and directors. We expect that any amendments to the Code of Conduct, or any waivers of its requirements, will be disclosed on our website.

Non-Employee Director Compensation

During the year ended December 31, 2024, each of the following individuals served on our board of directors as non-employee directors: Kim Stevenson, Timothy Chen, Wen Hsieh, Ph.D., Edward Hill, Ker Zhang, Ph.D. and Huang Lee. Ms. Stevenson, Mr. Hill and Mr. Lee resigned from our board of directors, effective November 15, 2024. In connection with their resignations, each of Ms. Stevenson and Mr. Hill entered into advisor agreements with us.

The following table presents all of the compensation awards to or earned by or paid to our named non-employee directors during the year ended December 31, 2024.

Name	Fees Earned or Paid in Cash ($)	Stock Awards ($)[1][2]	Total ($)
Kim Stevenson	$45,833	300,000	345,833
Ker Zhang, Ph.D.	—	253,333	253,333
Joseph Tautges	50,000	300,000	350,000
Timothy Chen	—	300,000	300,000

(1) All of the stock awards were granted under the 2020 Equity Incentive Plan (the 2020 Plan), the terms of which are described below under "Executive Compensation—Equity Benefit Plans—2020 Equity Incentive Plan." The amounts shown represent the grant date fair values of RSUs granted in 2024 as computed in accordance with Financial Accounting Standards Board (FASB) Accounting Standard Codification (ASC) Topic 718. See Note 13, Stock Option Plan and Stock- Based Compensation, to our consolidated financial statements included elsewhere in this prospectus for a discussion of the assumptions used in the calculation. This amount does not reflect the actual economic value that may be realized by the non-employee director.

(2) As of December 31, 2024, the aggregate number of shares underlying outstanding RSUs under our 2020 Plan and options to purchase shares of common stock under our 2010 Equity Incentive Plan (the 2010 Plan), the terms of which are described below under "Executive Compensation—Equity Benefit Plans—2010 Equity Incentive Plan," and our 2020 Plan held by each of our non-employee directors was as follows:

Name	Number of Shares Underlying RSUs		Number of Shares Underlying Outstanding Options	
Edward Hill	—		102,000	(a)
Ker Zhang, Ph.D.	666,666	(b)	900,000	

Kim Stevenson	936,936	(c)	—
Joeseph Tautges	916,666	(d)	—
Timothy Chen	666,666	(e)	—

(a) In connection with Mr. Hill's provision of advisory services to us, his option awards continue to vest pursuant to their original vesting schedules until Mr. Hill's continuous service (as defined in the 2020 Plan) under the 2020 Plan is otherwise terminated or interrupted.

(b) Represents 666,666 RSUs, 1/4th of which vest on December 1, 2025, and 1/16 of which vest quarterly thereafter.

(c) Consists of (i) 270,270 RSUs, 1/4th of which vested on March 1, 2023, and 1/16 of which vest quarterly thereafter, and (ii) 666,666 RSUs, 1/4th of which will vest on June 1, 2025, and 1/16 of which vest quarterly thereafter. In connection with Ms. Stevenson's provision of advisory services to us, her RSUs continue to vest pursuant to their original vesting schedules until Ms. Stevenson's continuous service (as defined in the 2020 Plan) under the 2020 Plan is otherwise terminated or interrupted.

(d) Consists of (i) 250,000 RSUs, 1/4th of which vested on December 1, 2023, and 1/16 of which vest quarterly thereafter, and (ii) 666,666 RSUs, 1/4th of which vest on June 1, 2025, and 1/16 of which vest quarterly thereafter.

(e) Represents 666,666 RSUs, 1/4th of which vested on December 1, 2024, and 1/16 of which vest quarterly thereafter.

Wen Hsieh, Ph.D., Huang Lee and Edward Hill were not compensated for their service on our board of directors during the year ended December 31, 2024. Mr. Esaka, Mr. Sean Chen and Dr. Hanson each also served on our board of directors during the year ended December 31, 2024, but did not receive any additional compensation for their service as a director. See the section titled "Executive Compensation" for more information regarding the compensation earned by Mr. Esaka, Mr. Sean Chen and Dr. Hanson.

We have reimbursed and will continue to reimburse all of our non-employee directors for their reasonable expenses incurred in attending meetings of our board of directors and committees of our board of directors.

We expect to adopt a director compensation policy for non-employee director compensation that will become effective following the completion of this offering.

Executive Compensation

Our named executive officers, who consist of our principal executive officer and our two other most highly compensated executive officers, for the year ended December 31, 2024 were:

- Fumihide Esaka, our Chief Executive Officer;
- Sean Chen, our President and Chief Operating Officer; and
- Scott Hanson, Ph.D., our Chief Technology Officer.

Summary Compensation Table

The following table presents all of the compensation awarded to, or earned by, our named executive officers during the fiscal year ended December 31, 2024.

Name and Position	Year	Salary ($)	Bonus ($)	Option Awards ($)[1]	All Other Compensation ($)[2]	Total ($)
Fumihide Esaka	2024	614,000	518,000	1,434,119	178,557	2,744,676
Chief Executive Officer						
Sean Chen	2024	420,000	227,000	766,616	29,946	1,443,562
President and Chief Operating Officer						
Scott Hanson, Ph.D.	2024	410,000	184,000	618,281	4,100	1,216,381
Chief Technology Officer						

(1) The amounts disclosed represent the sum of (a) the aggregate grant date fair value of the awards granted to our named executive officers during 2024 under our 2020 Plan, as computed in accordance with FASB ASC Topic 718 and (b) the incremental fair values to the option awards resulting from the Repricing (defined below). The assumptions used in calculating are set forth in Note 13 to our consolidated financial statements included elsewhere in this prospectus. This amount does not reflect the actual economic value that may be realized by the named executive officer.

(2) The amounts disclosed represent: (i) for Mr. Esaka, $50,000 of US housing allowance, $69,480 for Singapore housing allowance, $42,000 for Singapore car allowance and $17,077 of company matching contributions to a 401(k) plan, and (ii) for Mr. Chen, $29,946 in Taiwan housing allowance and (iii) $4,100 of company matching contributions to a 401(k) plan.

Narrative to Summary Compensation Table

In setting executive base salaries and bonuses, we consider compensation for comparable positions in the market, the historical compensation levels of our executives, individual performance as compared to our expectations and objectives, our desire to motivate our employees to achieve short- and long-term results that are in the best interests of our stockholders, and a long-term commitment to us. We do not target a specific competitive position or a specific mix of compensation among base salary or bonus.

Annual Base Salary and Bonuses

Base salaries for our executive officers are initially established through arm's-length negotiations at the time of the executive officer's hiring, taking into account such executive officer's qualifications, experience, the scope of such executive officer's responsibilities and competitive market compensation paid by other companies for similar positions within the industry and geography. Base salaries are reviewed periodically and adjusted from time to time to realign salaries with market levels after taking into account individual responsibilities, performance, experience and adjustments to reflect cost of living increases. In making decisions regarding salary increases, we may also draw upon the experience of members of its board of directors with executives at other

companies. The base salaries for Messrs. Esaka and Chen and Dr. Hanson for 2024 were $614,000, $420,000 and $410,000, respectively.

In 2024, our named executive officers were eligible to receive discretionary annual performance bonuses based on individual performance, company performance or as otherwise deemed appropriate, as determined by our board of directors. As disclosed in the summary compensation table, our board of directors approved discretionary bonuses to each of Messrs. Esaka and Chen and Dr. Hanson of $518,000, $227,000 and $184,000, respectively for fiscal year 2024.

Equity-Based Incentive Awards

Our equity award program is the primary vehicle for offering long-term incentives to our executive officers. We believe that equity awards provide our executives with a strong link to our long-term performance, create an ownership culture and help align the interests of our executives and stockholders. To date, we have primarily used stock option grants for this purpose because we believe they are an effective means by which to align the long-term interests of our executive officers with those of stockholders. We believe that the equity awards are an important retention tool for executive officers, as well as for our other employees. Grants to executive officers and other employees have historically been made at the discretion of our board of directors and are not made at any specific time period during a year.

In July 2024, each of Messrs. Esaka and Chen and Dr. Hanson was granted an option to purchase 5,352,055, 2,631,427 and 2,000,000 shares of our common stock, respectively. The options have an exercise price of $0.45 per share and are subject to a four-year vesting schedule, with 25% of the shares vesting on the first anniversary of the vesting commencement date and the balance vesting monthly over 36 months thereafter, subject to the continued service of the respective executive officers.

In May 2024, our board of directors approved, effective May 13, 2024, a one-time repricing of certain stock options that had been granted to date under the 2020 Plan. The repricing impacted stock options with exercise prices equal to or greater than $0.61, and each such stock option was repriced to have a per share exercise price of $0.45, the fair market value of a share of our common stock as determined by our board of directors on the date of the repricing (the Repricing). No changes were made to the expiration dates of or number of shares underlying the repriced stock options. The incremental fair values from the Repricing to the impacted stock options for the named executive officers are included in the "Option Awards" column in the Summary Compensation Table above.

Outstanding Equity Awards as of December 31, 2024

The following table sets forth information concerning outstanding equity-based incentive awards held by our named executive officers as of December 31, 2024.

	Option Awards						
Name	Grant Date[1]	Vesting Commencement Date	Number of Securities Underlying Options (#) Exercisable	Number of Securities Underlying Options (#) Unexercisable		Option Exercise Price ($)	Option Expiration Date
Fumihide Esaka	01/12/2016	12/01/2015	1,502,502	–		0.21	01/11/2026
	02/21/2019	02/21/2019	5,725,163	–		0.29	02/20/2029
	05/26/2021	05/26/2021	3,058,751	355,569	(2)	0.45	05/25/2031
	07/21/2024	01/01/2023	2,564,526	2,787,529	(3)	0.45	07/20/2034
Sean Chen	03/11/2016	03/11/2016	1,559,229	–		0.21	03/10/2026
	02/21/2019	02/21/2019	4,603,855	–		0.29	02/20/2029
	05/26/2021	05/26/2021	2,863,699	332,989	(2)	0.45	05/25/2031
	07/21/2024	01/01/2023	1,260,892	1,370,535	(3)	0.45	07/20/2034

	Option Awards						
Name	Grant Date[1]	Vesting Commencement Date	Number of Securities Underlying Options (#) Exercisable	Number of Securities Underlying Options (#) Unexercisable	Option Exercise Price ($)		Option Expiration Date
Scott Hanson, Ph.D.	10/02/2018	08/16/2018	2,452,219	—	0.29		10/01/2028
	02/21/2019	02/21/2019	3,879,645	—	0.29		02/20/2029
	05/26/2021	05/26/2021	2,931,708	340,897	0.45	(2)	05/25/2031
	07/21/2024	01/01/2023	958,333	1,041,667	0.45	(3)	07/20/2034

(1) Options shown in the table with grant dates before 2020 were granted under our 2010 Plan and options shown in the table with grant dates from and after 2021 were granted under our 2020 Plan, the terms of which are described below under "—Equity Incentive Plans."

(2) The options vest in equal monthly installments over 48 months measured from the vesting commencement date, subject to the named executive officer's continuous service with us. If a change in control (as defined in the 2020 Plan) occurs and the named executive officer's continuous service to us has not terminated as of the effective time of the change in control, then, as of the effective time of the change in control, the vesting and exercisability of the option will be accelerated in full, subject to the named executive officer timely executing and allowing to become irrevocable a general release of claims in our favor. Pursuant to the Repricing, the exercise price of the options was modified to be $0.45 per share, the fair market value of our common stock on the repricing date (May 13, 2024), as determined by our board of directors. There was no change to the expiration dates of or number of shares underlying the options made in connection with the Repricing.

(3) 25% of the options vest on the one-year anniversary of the vesting commencement date and thereafter the remaining 75% of the options vest in equal monthly installments over thirty-six months, subject to the named executive officer's continuous service with us. If we are subject to a change in control (as defined in the 2010 Plan) and the named executive officer's continuous service to us has not terminated as of, or immediately prior to, the effective time of such change in control, then, as of the effective time of such change in control, the vesting and exercisability of the option will be accelerated in full, subject to the named executive officer timely executing and allowing to become irrevocable a general release of claims in our favor.

Agreements with Our Named Executive Officers and Potential Payments Upon Termination or Change of Control

We have entered into offer letters with Mr. Esaka and Mr. Chen, the material terms of which are described below. We have not entered into an offer letter with Dr. Hanson. Each of our named executive officers has also executed our standard confidentiality and inventions assignment agreement.

Agreement with Fumihide Esaka

In November 2015, we entered into an offer letter with Fumihide Esaka, our Chief Executive Officer. The offer letter has no specific term and provides for at-will employment. Mr. Esaka's current annual base salary is $614,000. In addition, under the terms of Mr. Esaka's offer letter, we are required to reimburse him up to $50,000 per year (prorated for any partial year of employment) for out-of-pocket expenses he incurs in connection with his primary housing in Austin, Texas, subject to his timely submission of substantiation for such expenses to the Company in accordance with its reimbursement policies.

Agreement with Sean Chen

In September 2015, we entered into an offer letter with Sean Chen, our President and Chief Operating Officer. The offer letter has no specific term and provides for at-will employment. Mr. Chen's current annual base salary is $420,000. In addition, under the terms of Mr. Chen's offer letter, if he elects to attend an advanced

management program with a leading business school, we are required to reimburse him for reasonable out-of-pocket expenses he incurs in connection with attending such program, subject to his continued employment with the company and timely submission of substantiation for such expenses to the Company in accordance with its reimbursement policies.

Potential Payments upon Termination or Change of Control

We do not currently maintain any agreements or arrangements under which our name executive officers are eligible to receive cash severance benefits. Certain options granted to our named executive officers are subject to acceleration of vesting and exercisability upon a change in control or a termination of the named executive officer's employment under certain circumstances, as described in more detail above in the section titled "Outstanding Equity Awards as of December 31, 2024."

Health, Welfare and Retirement Benefits

All our current named executive officers are eligible to participate in our employee benefit plans, including our life, medical, dental, and vision insurance plans and 401(k) plan (as described below), in each case on the same basis as all of our other employees.

Nonqualified Deferred Compensation

None of our named executive officers participates in or has account balances in nonqualified defined contribution plans or other nonqualified deferred compensation plans maintained by us. Our board of directors may elect to provide our officers and other employees with nonqualified defined contribution or other nonqualified deferred compensation benefits in the future if it determines that doing so is in our best interests.

401(k) Plan

We sponsor a qualified retirement plan that is intended to qualify for favorable tax treatment under Section 401(a) of the Code and contains a cash or deferred feature that is intended to meet the requirements of Section 401(k) of the Code. Participants may make pre-tax and certain after-tax (Roth) salary deferral contributions to the plan from their eligible compensation up to the statutorily prescribed annual limit under the Code. Participants who are 50 years of age or older may contribute additional amounts based on the statutory limits for catch-up contributions. The plan also permits us to make annual discretionary matching contributions to the plan accounts of eligible participants. Contributions are held in trust as required by law. No minimum benefit is provided under the plan. An employee's interest in his or her salary deferral contributions are 100% vested when contributed.

Equity Incentive Plans

2025 Equity Incentive Plan

In , our board of directors adopted, and our stockholders approved the 2025 Plan. We expect our 2025 Plan will become effective on the date of this prospectus. Our 2025 Plan came into existence upon its adoption by our board of directors, but no grants will be made under our 2025 Plan prior to its effectiveness. Once our 2025 Plan becomes effective, no further grants will be made under our 2020 Plan.

Awards. Our 2025 Plan provides for the grant of incentive stock options (ISOs), within the meaning of Section 422 of the Code, to our employees and our parent and subsidiary corporations' employees, and for the grant of nonstatutory stock options (NSOs), stock appreciation rights, restricted stock awards, restricted stock unit awards, performance awards and other forms of awards to our employees, directors and consultants and any of our affiliates' employees and consultants.

Authorized shares. Initially, the maximum number of shares of our common stock that may be issued under our 2025 Plan after it becomes effective will not exceed shares of our common stock, which is the sum of

(i) new shares, plus (ii) an additional number of shares not to exceed shares, consisting of (a) shares that remain available for the issuance of awards under our 2020 Plan as of immediately prior to the time our 2025 Plan becomes effective and (b) any shares of our common stock subject to outstanding stock options or other stock awards granted under our 2020 Plan that, on or after our 2025 Plan becomes effective, terminate or expire prior to exercise or settlement; are not issued because the award is settled in cash; are forfeited because of the failure to vest; or are reacquired or withheld (or not issued) to satisfy a tax withholding obligation or the purchase or exercise price. In addition, the number of shares of our common stock reserved for issuance under our 2025 Plan will automatically increase on January 1 of each year for a period of ten years, beginning on January 1, 2026 and continuing through January 1, 2035, in an amount equal to (1) % of the total number of shares of our common stock outstanding on December 31 of the immediately preceding year, or (2) a lesser number of shares determined by our board of directors no later than December 31 of the immediately preceding year. The maximum number of shares of our common stock that may be issued on the exercise of ISOs under our 2025 Plan is shares.

Shares subject to stock awards granted under our 2025 Plan that expire or terminate without being exercised in full or that are paid out in cash rather than in shares will not reduce the number of shares available for issuance under our 2025 Plan. Shares withheld under a stock award to satisfy the exercise, strike or purchase price of a stock award or to satisfy a tax withholding obligation will not reduce the number of shares available for issuance under our 2025 Plan. If any shares of our common stock issued pursuant to a stock award are forfeited back to or repurchased or reacquired by us (i) because of a failure to meet a contingency or condition required for the vesting of such shares; (ii) to satisfy the exercise, strike or purchase price of a stock award; or (iii) to satisfy a tax withholding obligation in connection with a stock award, the shares that are forfeited or repurchased or reacquired will revert to and again become available for issuance under our 2025 Plan.

Plan administration. Our board of directors, or a duly authorized committee of our board of directors, administers our 2025 Plan. Our board of directors may delegate to one or more of our officers the authority to (i) designate employees (other than officers) to receive specified stock awards; and (ii) determine the number of shares subject to such stock awards. Under our 2025 Plan, our board of directors has the authority to determine stock award recipients, the types of stock awards to be granted, grant dates, the number of shares subject to each stock award, the fair market value of our common stock, and the provisions of each stock award, including the period of exercisability and the vesting schedule applicable to a stock award.

Under our 2025 Plan, our board of directors also generally has the authority to effect, with the consent of any materially adversely affected participant, (i) the reduction of the exercise, purchase, or strike price of any outstanding option or stock appreciation right; (ii) the cancellation of any outstanding option or stock appreciation right and the grant in substitution therefor of other awards, cash, or other consideration; or (iii) any other action that is treated as a repricing under generally accepted accounting principles.

Stock options. ISOs and NSOs are granted under stock option agreements adopted by the administrator. The administrator determines the exercise price for stock options, within the terms and conditions of our 2025 Plan, except the exercise price of a stock option generally will not be less than 100% of the fair market value of our common stock on the date of grant. Options granted under our 2025 Plan will vest at the rate specified in the stock option agreement as determined by the administrator.

The administrator determines the term of stock options granted under our 2025 Plan, up to a maximum of 10 years. Unless the terms of an optionholder's stock option agreement, or other written agreement between us and the optionholder, provide otherwise, if an optionholder's service relationship with us or any of our affiliates ceases for any reason other than disability, death, or cause, the optionholder may generally exercise any vested options for a period of three months following the cessation of service. This period may be extended in the event that exercise of the option is prohibited by applicable securities laws. If an optionholder's service relationship with us or any of our affiliates ceases due to death, or an optionholder dies within a certain period following cessation of service, the optionholder or a beneficiary may generally exercise any vested options for a period of

18 months following the date of death. If an optionholder's service relationship with us or any of our affiliates ceases due to disability, the optionholder may generally exercise any vested options for a period of 12 months following the cessation of service. In the event of a termination for cause, options generally terminate upon the termination date. In no event may an option be exercised beyond the expiration of its term.

Acceptable consideration for the purchase of common stock issued upon the exercise of a stock option will be determined by the administrator and may include (i) cash, check, bank draft or money order; (ii) a broker- assisted cashless exercise; (iii) the tender of shares of our common stock previously owned by the optionholder; (iv) a net exercise of the option if it is an NSO; or (v) other legal consideration approved by the administrator.

Unless the administrator provides otherwise, options or stock appreciation rights generally are not transferable except by will or the laws of descent and distribution. Subject to approval of the administrator or a duly authorized officer, an option may be transferred pursuant to a domestic relations order, official marital settlement agreement, or other divorce or separation instrument.

Tax limitations on ISOs. The aggregate fair market value, determined at the time of grant, of our common stock with respect to ISOs that are exercisable for the first time by an award holder during any calendar year under all of our stock plans may not exceed $100,000. Options or portions thereof that exceed such limit will generally be treated as NSOs. No ISO may be granted to any person who, at the time of the grant, owns or is deemed to own stock possessing more than 10% of our total combined voting power or that of any of our parent or subsidiary corporations unless (i) the option exercise price is at least 110% of the fair market value of the stock subject to the option on the date of grant; and (ii) the term of the ISO does not exceed five years from the date of grant.

Restricted stock unit awards. Restricted stock unit awards are granted under restricted stock unit award agreements adopted by the administrator. Restricted stock unit awards may be granted in consideration for any form of legal consideration that may be acceptable to our board of directors and permissible under applicable law. A restricted stock unit award may be settled by cash, delivery of stock, a combination of cash and stock as deemed appropriate by the administrator, or in any other form of consideration set forth in the restricted stock unit award agreement. Additionally, dividend equivalents may be credited in respect of shares covered by a restricted stock unit award. Except as otherwise provided in the applicable award agreement, or other written agreement between us and the recipient, restricted stock unit awards that have not vested will be forfeited once the participant's continuous service ends for any reason.

Restricted stock awards. Restricted stock awards are granted under restricted stock award agreements adopted by the administrator. A restricted stock award may be awarded in consideration for cash, check, bank draft or money order, past or future services to us, or any other form of legal consideration that may be acceptable to our board of directors and permissible under applicable law. The administrator determines the terms and conditions of restricted stock awards, including vesting and forfeiture terms. If a participant's service relationship with us ends for any reason, we may receive any or all of the shares of common stock held by the participant that have not vested as of the date the participant terminates service with us through a forfeiture condition or a repurchase right.

Stock appreciation rights. Stock appreciation rights are granted under stock appreciation right agreements adopted by the administrator. The administrator determines the purchase price or strike price for a stock appreciation right, which generally will not be less than 100% of the fair market value of our common stock on the date of grant. A stock appreciation right granted under our 2025 Plan will vest at the rate specified in the stock appreciation right agreement as determined by the administrator. Stock appreciation rights may be settled in cash or shares of our common stock or in any other form of payment as determined by our board of directors and specified in the stock appreciation right agreement.

The administrator determines the term of stock appreciation rights granted under our 2025 Plan, up to a maximum of 10 years. If a participant's service relationship with us or any of our affiliates ceases for any reason other than cause, disability, or death, the participant may generally exercise any vested stock appreciation right for a period of three months following the cessation of service. This period may be further extended in the event that exercise of the stock appreciation right following such a termination of service is prohibited by applicable securities laws. If a participant's service relationship with us, or any of our affiliates, ceases due to disability or death, or a participant dies within a certain period following cessation of service, the participant or a beneficiary may generally exercise any vested stock appreciation right for a period of 12 months in the event of disability and 18 months in the event of death. In the event of a termination for cause, stock appreciation rights generally terminate upon the termination date. In no event may a stock appreciation right be exercised beyond the expiration of its term.

Performance awards. Our 2025 Plan permits the grant of performance awards that may be settled in stock, cash or other property. Performance awards may be structured so that the stock or cash will be issued or paid only following the achievement of certain pre-established performance goals during a designated performance period. Performance awards that are settled in cash or other property are not required to be valued in whole or in part by reference to, or otherwise based on, our common stock.

The performance goals may be based on any measure of performance selected by our board of directors. The performance goals may be based on company-wide performance or performance of one or more business units, divisions, affiliates, or business segments, and may be either absolute or relative to the performance of one or more comparable companies or the performance of one or more relevant indices. Unless specified otherwise by our board of directors at the time the performance award is granted, our board will appropriately make adjustments in the method of calculating the attainment of performance goals as follows: (i) to exclude restructuring or other nonrecurring charges; (ii) to exclude exchange rate effects; (iii) to exclude the effects of changes to generally accepted accounting principles; (iv) to exclude the effects of any statutory adjustments to corporate tax rates; (v) to exclude the effects of items that are "unusual" in nature or occur "infrequently" as determined under generally accepted accounting principles; (vi) to exclude the dilutive effects of acquisitions or joint ventures; (vii) to assume that any business divested by us achieved performance objectives at targeted levels during the balance of a performance period following such divestiture; (viii) to exclude the effect of any change in the outstanding shares of our common stock by reason of any stock dividend or split, stock repurchase, reorganization, recapitalization, merger, consolidation, spin-off, combination or exchange of shares or other similar corporate change, or any distributions to common stockholders other than regular cash dividends; (ix) to exclude the effects of stock based compensation

and the award of bonuses under our bonus plans; (x) to exclude costs incurred in connection with potential acquisitions or divestitures that are required to be expensed under generally accepted accounting principles; and (xi) to exclude the goodwill and intangible asset impairment charges that are required to be recorded under generally accepted accounting principles.

Other stock awards. The administrator may grant other awards based in whole or in part by reference to our common stock. The administrator will set the number of shares under the stock award (or cash equivalent) and all other terms and conditions of such awards.

Non-employee director compensation limit. The aggregate value of all compensation granted or paid to any non-employee director with respect to any calendar year, including awards granted and cash fees paid by us to such non-employee director, will not exceed $ in total value, except such amount will increase to $ for the first year for newly appointed or elected non-employee directors.

Changes to capital structure. In the event there is a specified type of change in our capital structure, such as a stock split, reverse stock split, or recapitalization, appropriate adjustments will be made to (i) the class and maximum number of shares reserved for issuance under our 2025 Plan, (ii) the class and maximum number of shares by which the share reserve may increase automatically each year, (iii) the class and maximum number of shares that may be issued on the exercise of ISOs, and (iv) the class and number of shares and exercise price, strike price, or purchase price, if applicable, of all outstanding stock awards.

Corporate transactions. In the event of a corporate transaction (as defined in the 2025 Plan), unless otherwise provided in a participant's stock award agreement or other written agreement with us or one of our affiliates or unless otherwise expressly provided by the administrator at the time of grant, any stock awards outstanding under our 2025 Plan may be assumed, continued or substituted for by any surviving or acquiring corporation (or its parent company), and any reacquisition or repurchase rights held by us with respect to the stock award may be assigned to the successor (or its parent company). If the surviving or acquiring corporation (or its parent company) does not assume, continue or substitute for such stock awards, then (i) with respect to any such stock awards that are held by participants whose continuous service has not terminated prior to the effective time of the corporate transaction, or current participants, the vesting (and exercisability, if applicable) of such stock awards will be accelerated in full (or, in the case of performance awards with multiple vesting levels depending on the level of performance, vesting will accelerate at 100% of the target level) to a date prior to the effective time of the corporate transaction (contingent upon the effectiveness of the corporate transaction), and such stock awards will terminate if not exercised (if applicable) at or prior to the effective time of the corporate transaction, and any reacquisition or repurchase rights held by us with respect to such stock awards will lapse (contingent upon the effectiveness of the corporate transaction); and (ii) any such stock awards that are held by persons other than current participants will terminate if not exercised (if applicable) prior to the effective time of the corporate transaction, except that any reacquisition or repurchase rights held by us with respect to such stock awards will not terminate and may continue to be exercised notwithstanding the corporate transaction.

In the event a stock award will terminate if not exercised prior to the effective time of a corporate transaction, the administrator may provide, in its sole discretion, that the holder of such stock award may not exercise such stock award but instead will receive a payment equal in value to the excess (if any) of (i) the value of the property the participant would have received upon the exercise of the stock award, over (ii) any per share exercise price payable by such holder, if applicable. In addition, any escrow, holdback, earn out or similar provisions in the definitive agreement for the corporate transaction may apply to such payment to the same extent and in the same manner as such provisions apply to the holders of our common stock.

Change in control. Stock awards granted under our 2025 Plan may be subject to acceleration of vesting and exercisability upon or after a change in control (as defined in the 2025 Plan) as may be provided in the applicable stock award agreement or in any other written agreement between us or any affiliate and the participant, but in the absence of such provision, no such acceleration will automatically occur.

Plan amendment or termination. Our board of directors has the authority to amend, suspend, or terminate our 2025 Plan at any time, provided that such action does not materially impair the existing rights of any participant without such participant's written consent. Certain material amendments also require the approval of our stockholders. No ISOs may be granted after the tenth anniversary of the date our board of directors adopted our 2025 Plan. No stock awards may be granted under our 2025 Plan while it is suspended or after it is terminated.

2025 Employee Stock Purchase Plan

In , our board of directors adopted, and our stockholders approved, our ESPP. We expect our ESPP will become effective on the date of this prospectus. The purpose of our ESPP is to secure the services of new employees, to retain the services of existing employees, and to provide incentives for such individuals to exert maximum efforts toward our success and that of our affiliates. Our ESPP includes two components. One component is designed to allow eligible U.S. employees to purchase our common stock in a manner that may qualify for favorable tax treatment under Section 423 of the Code. The other component permits the grant of purchase rights that do not qualify for such favorable tax treatment in order to allow deviations necessary to permit participation by eligible employees who are foreign nationals or employed outside of the United States while complying with applicable foreign laws.

Share reserve. Our ESPP authorizes the issuance of shares of our common stock under purchase rights granted to our employees or to employees of any of our designated affiliates. The number of shares of our

common stock reserved for issuance will automatically increase on January 1 of each year for a period of ten years, beginning on January 1, 2026 and continuing through January 1, 2035, by the lesser of (i) % of the total number of shares of our common stock outstanding on December 31 of the immediately preceding year; and (ii) shares, except before the date of any such increase, our board of directors may determine that such increase will be less than the amount set forth in clauses (i) and (ii).

Administration. Our board of directors administers our ESPP and may delegate its authority to administer our ESPP to our compensation committee. Our ESPP is implemented through a series of offerings under which eligible employees are granted purchase rights to purchase shares of our common stock on specified dates during such offerings. Under our ESPP, our board of directors may specify offerings with durations of not more than 27 months and to specify shorter purchase periods within each offering. Each offering will have one or more purchase dates on which shares of our common stock will be purchased for employees participating in the offering. Our ESPP provides that an offering may be terminated under certain circumstances.

Payroll deductions. Generally, all regular employees, including executive officers, employed by us or by any of our designated affiliates, may participate in our ESPP and to contribute, normally through payroll deductions, up to 15% of their earnings (as defined in our ESPP) for the purchase of our common stock under our ESPP. Unless otherwise determined by our board of directors, common stock will be purchased for the accounts of employees participating in our ESPP at a price per share that is not less than the lesser of (i) 85% of the fair market value of a share of our common stock on the first day of an offering; or (ii) 85% of the fair market value of a share of our common stock on the date of purchase.

Limitations. Employees may have to satisfy one or more of the following service requirements before participating in our ESPP, as determined by our board of directors: (i) being customarily employed for more than 20 hours per week; (ii) being customarily employed for more than five months per calendar year; or (iii) continuous employment with us or one of our affiliates for a period of time (not to exceed two years). No employee may purchase shares under our ESPP at a rate in excess of $25,000 worth of our common stock (based on the fair market value per share of our common stock at the beginning of an offering) for each calendar year such a purchase right is outstanding. Finally, no employee will be eligible for the grant of any purchase rights under our ESPP if immediately after such rights are granted, such employee has voting power over 5% or more of our outstanding capital stock measured by vote or value under Section 424(d) of the Code.

Changes to capital structure. Our ESPP provides that in the event there occurs a change in our capital structure through such actions as a stock split, merger, consolidation, reorganization, recapitalization, reincorporation, stock dividend, dividend in property other than cash, large nonrecurring cash dividend, liquidating dividend, combination of shares, exchange of shares, change in corporate structure, or similar transaction, our board of directors will make appropriate adjustments to: (i) the class(es) and maximum number of shares reserved under our ESPP; (ii) the class(es) and maximum number of shares by which the share reserve may increase automatically each year; (iii) the class(es) and number of shares subject to, and purchase price applicable to, outstanding offerings and purchase rights; and (iv) the class(es) and number of shares that are subject to purchase limits under ongoing offerings.

Corporate transactions. Our ESPP provides that in the event of a corporate transaction (as defined in the ESPP), any then-outstanding rights to purchase our common stock under our ESPP may be assumed, continued, or substituted for by any surviving or acquiring entity (or its parent company). If the surviving or acquiring entity (or its parent company) elects not to assume, continue, or substitute for such purchase rights, then the participants' accumulated payroll contributions will be used to purchase shares of our common stock within 10 business days before such corporate transaction, and such purchase rights will terminate immediately after such purchase.

Plan amendment or termination. Our board of directors has the authority to amend or terminate our ESPP, except in certain circumstances such amendment or termination may not materially impair any outstanding

purchase rights without the holder's consent. We will obtain stockholder approval of any amendment to our ESPP as required by applicable law or listing requirements.

Clawback. All awards granted under the 2025 Plan will be subject to recoupment in accordance with any clawback policy that we are required to adopt pursuant to the listing standards of any national securities exchange or association on which our securities are listed or as is otherwise required by the U.S. Dodd-Frank Wall Street Reform and Consumer Protection Act or other applicable law. In addition, our board of directors may impose such other clawback, recovery or recoupment provisions in a stock award agreement as our board of directors determines necessary or appropriate.

2020 Equity Incentive Plan

In October 2020, our board of directors adopted, and our stockholders approved, our 2020 Plan. Our 2020 Plan was last amended in May 2024. No further stock awards will be granted under our 2020 Plan on or after the effectiveness of our 2025 Plan; however, awards outstanding under our 2020 Plan will continue to be governed by their existing terms.

As of December 31, 2024, options to purchase 37,713,893 shares of our common stock and 3,186,934 RSUs were outstanding under our 2020 Plan, and 11,134,315 shares of our common stock remained available for future awards under our 2020 Plan.

Awards. Our 2020 Plan provides for the grant of ISOs to our employees and our parent and subsidiary corporations' employees, and for the grant of NSOs, stock appreciation rights, restricted stock awards, restricted stock unit awards and other forms of awards to our employees, directors and consultants and any of our affiliates' employees and consultants.

Plan administration. Our board of directors, or a duly authorized committee of our board of directors, administers our 2020 Plan. The administrator has the authority to determine stock award recipients, the types of stock awards to be granted, grant dates, the number of shares subject to each stock award, the fair market value of our common stock, and the provisions of each stock award, including the period of exercisability and the vesting schedule applicable to a stock award. Under our 2020 Plan, the administrator also generally has the authority to effect, with the consent of any adversely affected participant, (i) the reduction of the exercise, purchase, or strike price of any outstanding stock award; (ii) the cancellation of any outstanding stock award and the grant in substitution therefor of other awards, cash, or other consideration; or (iii) any other action that is treated as a repricing under generally accepted accounting principles.

Stock options. Stock options granted under the 2020 Plan are subject to substantially similar terms as options that may be granted under the 2025 Plan once effective, except with respect to the treatment of options in the event of a corporate transaction, as described in more detail below.

Restricted Stock Units. Restricted stock units granted under our 2020 Plan have terms substantially similar to restricted stock units that may be granted under our 2025 Plan once it becomes effective.

Changes to capital structure. In the event there is a specified type of change in our capital structure, such as a stock split, reverse stock split, or recapitalization, appropriate adjustments will be made to (i) the class and maximum number of shares reserved for issuance under our 2020 Plan, (ii) the class and maximum number of shares that may be issued on the exercise of ISOs, and (iii) the class and number of shares and exercise price, strike price, or purchase price, if applicable, of all outstanding stock awards.

Corporate transaction. Our 2020 Plan provides that in the event of a corporate transaction (as defined in our 2020 Plan), unless otherwise provided in an award agreement or other written agreement between us and the participant, the administrator may take one or more of the following actions with respect to outstanding stock awards:

- arrange for the assumption, continuation, or substitution of a stock award by the surviving or acquiring corporation or its parent company;

- arrange for the assignment of any reacquisition or repurchase rights held by us to the surviving or acquiring corporation or its parent company;
- accelerate the vesting, in whole or in part, of the stock award and, if applicable, the time at which the stock award may be exercised, to a date prior to the effective time of the corporate transaction and provide for its termination if not exercised (if applicable) at or prior to the effective time of the corporate transaction;
- arrange for the lapse, in whole or in part, of any reacquisition or repurchase rights held by us;
- cancel the stock award, to the extent not vested or not exercised prior to the effective time of the corporate transaction, in exchange for such cash consideration, if any, as the administrator deems appropriate; and
- make a payment, in such form as determined by the administrator, equal to the excess, if any, of the value of the property the participant would have received upon the exercise of the stock award immediately prior to the effective time of the corporate transaction over any exercise price payable by the holder in connection with such exercise.

The administrator is not obligated to treat all stock awards or portions of stock awards in the same manner and is not obligated to treat all participants in the same manner.

Change in control. Stock awards granted under our 2020 Plan may be subject to acceleration of vesting and exercisability upon or after a change in control (as defined in the 2020 Plan) as may be provided in the applicable stock award agreement or in any other written agreement between us or any affiliate and the participant, but in the absence of such provision, no such acceleration will automatically occur.

Plan amendment or termination. Our board of directors has the authority to amend, suspend, or terminate our 2020 Plan at any time, provided that such action does not materially impair the existing rights of any participant without such participant's written consent. Certain material amendments also require the approval of our stockholders. As noted above, no further stock awards will be granted under our 2020 Plan on or after the effectiveness of our 2025 Plan.

2010 Equity Incentive Plan

In October 2010, our board of directors adopted, and our stockholders approved, our 2010 Plan. Our 2010 Plan was amended on several occasions and was most recently amended in February 2019. On and after our 2020 Plan became effective, no further stock awards could be granted under our 2010 Plan; however, stock awards outstanding under our 2010 Plan continue to be governed by their existing terms.

As of December 31, 2024, options to purchase 24,693,977 shares of our common stock were outstanding under our 2010 Plan.

Awards. Our 2010 Plan provided for the grant of ISOs to our employees and our parent and subsidiary corporations' employees, and for the grant of NSOs, stock appreciation rights, restricted stock awards, and restricted stock unit awards to our employees, directors and consultants and any of our affiliates' employees and consultants.

Plan administration. Our board of directors, or a duly authorized committee of our board of directors, administers our 2010 Plan. The administrator has the authority to construe and interpret our 2010 Plan and stock awards granted under it and to make all other determinations necessary or expedient for the administration of our 2010 Plan. Under our 2010 Plan, the administrator also has the authority to effect, with the consent of any adversely affected optionholder, (i) the reduction of the exercise price of any outstanding option; (ii) the cancellation of any outstanding option and the grant in substitution therefor of other awards, cash, or other consideration; or (iii) any other action that is treated as a repricing under generally accepted accounting principles.

Stock options. Stock options granted under our 2010 Plan are subject to substantially similar terms as stock options that may be granted under our 2025 Plan once effective.

Changes to capital structure. If there is a specified type of change in our capital structure, such as a stock split, reverse stock split, or recapitalization, the administrator will appropriately and proportionately adjust the class(es) and number of shares and price per share, if applicable, of stock subject to outstanding stock awards.

Corporate transactions. In the event of a corporate transaction (as defined in the 2010 Plan), unless otherwise provided in a participant's stock award agreement or other written agreement with us or one of our affiliates or unless otherwise expressly provided by the administrator at the time of grant, any stock awards outstanding under our 2010 Plan may be assumed, continued or substituted for by any surviving or acquiring corporation (or its parent company), and any reacquisition or repurchase rights held by us with respect to the stock award may be assigned to the successor (or its parent company). If the surviving or acquiring corporation (or its parent company) does not assume, continue or substitute for such stock awards, then (i) with respect to any such stock awards that are held by participants whose continuous service has not terminated prior to the effective time of the corporate transaction, or current participants, the vesting (and exercisability, if applicable) of such stock awards will (contingent upon the effectiveness of the corporate transaction) be accelerated in full to a date prior to the effective time of the corporate transaction, and such stock awards will terminate if not exercised (if applicable) at or prior to the effective time of the corporate transaction, and any reacquisition or repurchase rights held by us with respect to such stock awards will lapse (contingent upon the effectiveness of the corporate transaction); and (ii) any such stock awards that are held by persons other than current participants will terminate if not exercised (if applicable) prior to the effective time of the corporate transaction, except that any reacquisition or repurchase rights held by us with respect to such stock awards will not terminate and may continue to be exercised notwithstanding the corporate transaction.

In the event a stock award will terminate if not exercised prior to the effective time of a corporate transaction, the administrator may provide, in its sole discretion, that the holder of such stock award may not exercise such stock award but instead will receive a payment equal in value to the excess (if any) of (i) the value of the property the participant would have received upon the exercise of the stock award, over (ii) any per share exercise price payable by such holder, if applicable.

Change in control. A stock award under our 2010 Plan may be subject to additional acceleration of vesting and exercisability upon or after a change in control (as defined in our 2010 Plan) as may be provided in the award agreement or any other written agreement between us and the participant, but in the absence of such provision, no such acceleration will occur.

Plan amendment and termination. The administrator may amend, suspend, or terminate our 2010 Plan at any time, provided that such action does not impair the existing rights of any participant without such participant's written consent. Certain material amendments of our 2010 Plan also require the approval of our stockholders. As noted above, no further stock awards could be granted under our 2010 Plan on or after the effectiveness of our 2020 Plan.

Rule 10b5-1 Sales Plans

Our directors and executive officers may adopt written plans, known as Rule 10b5-1 plans (Rule 10b5-1 Plans), in which they will contract with a broker to buy or sell shares of our common stock on a periodic basis. Under a Rule 10b5-1 Plan, a broker executes trades pursuant to parameters established by the director or officer when entering into the plan, without further direction from them. The director or officer may amend a Rule 10b5-1 Plan in some circumstances and may terminate a plan at any time. Our directors and executive officers also may buy or sell additional shares outside of a Rule 10b5-1 Plan when they are not in possession of material nonpublic information subject to compliance with the terms of our insider trading policy. Prior to 180 days after the date of this offering, subject to early termination, the sale of any shares under such plan would be prohibited by the lock-up agreement that the director or officer has entered into with the underwriters.

Certain Relationships and Related Party Transactions

The following includes a summary of transactions since January 1, 2022 to which we have been a party, in which the amount involved in the transaction exceeded the lesser of $120,000 or one percent of the average of our total assets at year-end for the last two completed years, and in which any of our directors, executive officers or, to our knowledge, beneficial owners of more than 5% of our capital stock or any member of the immediate family of any of the foregoing persons had or will have a direct or indirect material interest, other than equity and other compensation, termination, change of control, and other arrangements, which are described under "Executive Compensation."

Series F-1 Redeemable Convertible Preferred Stock Financing

In multiple closings held between February 2022 and March 2022, we issued and sold an aggregate of 23,664,027 shares of our Series F-1 redeemable convertible preferred stock at a purchase price of $2.091505 per share for an aggregate purchase price of approximately $49.49 million.

The following table summarizes the Series F-1 redeemable convertible preferred stock purchased by holders of more than 5% of our capital stock and entities affiliated with our executive officers and members of our board of directors.

Participants (1)	Shares of Series F-1 Redeemable Convertible Preferred Stock Purchased (#)	Aggregate Purchase Price ($)
Entities affiliated with KPCB[2]	14,343,738	$29,999,999.75
El Camino Fund, L.P.[3]	956,249	$ 1,999,999.56
Total	15,299,987	$31,999,999.31

(1) Additional details regarding these stockholders and their equity holdings are included in this prospectus under the section titled "Principal Stockholders."

(2) KPCB is a holder of 5% or more of our capital stock, and is affiliated with Wen Hsieh, Ph.D. and Ker Zhang, Ph.D., two of our non-employee directors.

(3) El Camino Fund, L.P. is affiliated with Ker Zhang, Ph.D., one of our non-employee directors.

In connection with the issuance of our Series F-1 redeemable convertible preferred stock, on October 19, 2022, we issued a warrant to purchase up to 100,000 shares of our common stock at an exercise price of $1.11 per share to EDB Investments Ptd Ltd.

Series G Redeemable Convertible Preferred Stock Financing

In multiple closings held between August 2023 and July 2024, we issued and sold an aggregate of 99,229,883 shares of our Series G redeemable convertible preferred stock at a purchase price of $0.904179 per share for an aggregate purchase price of approximately $89.7 million.

The following table summarizes the Series G redeemable convertible preferred stock purchased by holders of more than 5% of our capital stock and entities affiliated with our executive officers and members of our board of directors.

Participants (1)	Shares of Series F Redeemable Convertible Preferred Stock Purchased (#)	Aggregate Purchase Price ($)
EDB Investments Pte Ltd[2]	11,059,756	$9,999,999.12
Matter Venture Partners Fund I, L.P.[3]	11,059,756	$9,999,999.12
El Camino Fund, L.P.[4]	1,105,975	$ 999,999.37
Total	23,225,487	$20,999,997.61

(1) Additional details regarding these stockholders and their equity holdings are included in this prospectus under the section titled “Principal Stockholders.”

(2) EDB Investments PTE Ltd is a holder of 5% or more of our capital stock.

(3) Matter Venture Partners Fund I, L.P. is affiliated with Wen Hsieh, Ph.D., one of our non-employee directors.

(4) El Camino Fund, L.P. is affiliated with Ker Zhang, Ph.D., one of our non-employee directors.

As a result of our issuance of our Series G redeemable convertible preferred stock, subject to the terms of our amended and restated certificate of incorporation then in effect, the conversion prices of our Series E redeemable convertible preferred stock, Series F redeemable convertible preferred stock and Series F-1 redeemable convertible preferred stock were adjusted from $1.033800, $1.827990 and $2.091505, respectively, to $1.002756, $1.606742 and $1.807146, respectively.

In connection with the issuance of our Series G redeemable convertible preferred stock, on June 1, 2024, we issued a warrant to purchase up to 6,690,069 shares of our common stock at an exercise price of $0.45 per share to Matter Venture Partners Fund I, L.P., which is affiliated with Wen Hsieh, Ph.D., our Chairman.

Stock Option Repricing

In May 2024, our board of directors approved a stock option repricing whereby the exercise prices of previously granted and unexercised options held by certain employees and directors with exercise prices between $0.61 and $1.20 per share, were adjusted to equal the fair market value of our common stock as of December 31, 2023. Options to purchase a total of 22,267,883 shares of our common stock are subject to the repricing. No other modifications were made to these options.

The following table sets forth the stock options held by current executive officers and directors that are subject to the repricing:

Name	Grant Date	Number of Securities Underlying Each Option	Exercise Price Pre-Stock Option Repricing	Exercise Price Post-Stock Option Repricing[1]
Executive Officers				
Fumihide Esaka	5/26/2021	3,414,420	$0.61	$0.45
Sean Chen	5/26/2021	3,195,688	$0.61	$0.45
Scott Hanson, Ph.D.	5/26/2021	3,272,605	$0.61	$0.45

Directors				
Ker Zhang, Ph.D.	3/3/2021	500,000	$0.61	$0.45

Employment Agreements

We have entered into employment agreements and offer letter agreements with certain of our executive officers. See "Executive Compensation—Agreements with our Named Executive Officers and Potential Payments Upon Termination or Change of Control."

Investor Rights Agreement

In August 2023, we entered into an Amended and Restated Investor Rights Agreement (the Rights Agreement) with certain holders of more than 5% of our outstanding capital stock and certain affiliates of our directors. The Rights Agreement grants to the holders of our outstanding convertible preferred stock certain rights, including certain registration rights (subject to lock-up agreements entered into in connection with this offering, market standoff agreements and federal securities laws) with respect to the registrable securities held by them, which would allow such holders to trade these shares when the applicable registration statement is declared effective. Certain holders are entitled to demand registration rights, which allows them to request that we register all or a portion of their shares under certain circumstances; piggyback registration rights, which allows them to include their shares (subject to certain limitations) in the same registration if, after this offering, we propose to register any of our securities under the Securities Act; and form S-3 registration rights, which allows them to request that we register their shares on Form S-3 if we are qualified to file a registration statement on Form S-3 and net proceeds would equal or exceed $1.0 million. See the section titled "Description of Capital Stock—Registration Rights" for additional information.

In addition, the Rights Agreement imposes certain affirmative obligations on us, including our obligation to, among other things, (i) grant certain holders, including each holder who holds at least 3% of the fully diluted shares of common stock on an as-converted basis (the Major Investors) a right of first offer with respect to future sales of our equity, excluding the shares to be offered and sold in this offering, and (ii) grant certain information and inspection rights to such Major Investors. Each of these obligations will terminate in connection with the closing of this offering.

Voting Agreement

In August 2023, we entered into an Amended and Restated Voting Agreement (the Voting Agreement) with certain holders of more than 5% of our outstanding capital stock and certain affiliates of our directors.

Pursuant to the Voting Agreement, certain holders of preferred stock, collectively, have the right to designate four members to be elected to our board of directors. See the section titled "Management—Board Composition." The Voting Agreement will terminate by its terms in connection with the closing of this offering and none of our stockholders will have any continuing rights regarding the election or designation of members of our board of directors following this offering.

Right of First Refusal and Co-Sale Agreement

In August 2023, we entered into an Amended and Restated Right of First Refusal and Co-Sale Agreement (the Co-Sale Agreement) with certain holders of more than 5% of our outstanding capital stock and certain affiliates of our directors. Pursuant to the Co-Sale Agreement, we have a right of first refusal in respect of certain sales of securities by certain holders of our common stock and convertible preferred stock. To the extent we do not exercise such right in full, certain holders, including the Major Investors, are granted certain rights of first refusal and co-sale in respect of such sale. The Co-Sale Agreement will terminate in connection with the closing of this offering.

Indemnification Agreements

We have entered, and intend to continue to enter, into separate indemnification agreements with our directors and executive officers, in addition to the indemnification provided for in our amended and restated

certificate of incorporation and amended and restated bylaws. These agreements, among other things, require us to indemnify our directors and executive officers for certain expenses, including attorneys' fees, judgments, fines, and settlement amounts incurred by a director or executive officer in any action or proceeding arising out of their services as one of our directors or executive officers or as a director or executive officer of any other company or enterprise to which the person provides services at our request. For more information regarding these indemnification arrangements, see "Management—Limitation on Liability and Indemnification of Directors and Officers." We believe that these charter provisions and indemnification agreements are necessary to attract and retain qualified persons as directors and officers.

The limitation of liability and indemnification provisions in our amended and restated certificate of incorporation and amended and restated bylaws may discourage stockholders from bringing a lawsuit against directors for breach of their fiduciary duties. They may also reduce the likelihood of derivative litigation against directors and officers, even though an action, if successful, might benefit us and our stockholders. A stockholder's investment may decline in value to the extent we pay the costs of settlement and damage awards against directors and officers pursuant to these indemnification provisions.

Policies and Procedures for Transactions with Related Persons

We will adopt a written Related Person Transactions Policy that sets forth our policies and procedures regarding the identification, review, consideration, and oversight of "related person transactions." For purposes of our policy only, a "related person transaction" is a transaction, arrangement, or relationship (or any series of similar transactions, arrangements or relationships) in which we or any of our subsidiaries are participants involving an amount that

exceeds $120,000, in which any “related person” has a material interest.

Transactions involving compensation for services provided to us as an employee, consultant, or director are not considered related person transactions under this policy. A related person is any executive officer, director, nominee to become a director or a holder of more than 5% of any class of our voting securities (including our common stock), including any of their immediate family members and affiliates, including entities owned or controlled by such persons.

Under the policy, the related person in question or, in the case of transactions with a holder of more than 5% of any class of our voting securities, an officer with knowledge of the proposed transaction, must present information regarding the proposed related person transaction to our audit committee (or, where review by our audit committee would be inappropriate, to another independent body of our board of directors) for review. To identify related person transactions in advance, we rely on information supplied by our executive officers, directors, and certain significant stockholders. In considering related person transactions, our audit committee takes into account the relevant available facts and circumstances, which may include, but not limited to:

- the risks, costs, and benefits to us;
- the impact on a director’s independence in the event the related person is a director, immediate family member of a director or an entity with which a director is affiliated;
- the terms of the transaction;
- the availability of other sources for comparable services or products; and
- the terms available to or from, as the case may be, unrelated third parties.

Our audit committee will approve only those transactions that it determines are fair to us and in our best interests. All of the transactions described above were entered into prior to the adoption of such policy.

Principal Stockholders

The following table sets forth information regarding beneficial ownership of our capital stock as of , 2025 by:

- each person, or group of affiliated persons, known by us to beneficially own more than 5% of our common stock;
- each of our directors;
- each of our named executive officers; and
- all of our current executive officers and directors as a group.

The percentage ownership information under the column “Shares beneficially owned before offering” is based on shares of common stock outstanding as of , 2025 assuming (i) the automatic conversion of all 341,496,158 shares of our outstanding redeemable convertible preferred stock into an aggregate of 356,423,629 shares of our common stock, which will occur upon the closing of this offering, (ii) the issuance of shares of our common stock upon the exercise of warrants to purchase our common stock outstanding as of December 31, 2024 at a weighted average exercise price of $ per share that are being exercised in connection with this offering, and (iii) the filing of our amended and restated certificate of incorporation upon the closing of this offering. The percentage ownership information under the column “Shares beneficially owned after offering” is based on the sale of shares of common stock in this offering by us, based on an assumed initial public offering price of $ per share (the midpoint of the price range set forth on the cover page of this prospectus).

Information with respect to beneficial ownership has been furnished by each director, officer, or beneficial owner of more than 5% of our common stock. We have determined beneficial ownership in accordance with the rules of the SEC. These rules generally attribute beneficial ownership of securities to persons who possess sole or shared voting power or investment power with respect to those securities. In addition, the rules include shares of our common stock issuable pursuant to the exercise of stock options or warrants that are either immediately exercisable or exercisable within 60 days of , 2025. These shares are deemed to be outstanding and beneficially owned by the person holding those options or warrants for the purpose of computing the percentage ownership of that person, but they are not treated as outstanding for the purpose of computing the percentage ownership of any other person. Unless otherwise indicated, the persons or entities identified in this table have sole voting and investment power with respect to all shares shown as beneficially owned by them, subject to applicable community property laws.

Except as otherwise noted below, the address for each person or entity listed in the table is c/o Ambiq Micro Inc., 6500 River Place Blvd., Building 7, Suite 200, Austin, Texas 78730.

		Percentage of Shares Beneficially Owned	
	Beneficially Owned (#)		

Name of Beneficial Owner	Before Offering (%)	After Offering (%)
Greater than 5% Holders:		
Entities affiliated with KPCB		
Entities affiliated with EDB Investments Pte Ltd.		
Directors and Named Executive Officers:		
Fumihide Esaka		
Scott Hanson, Ph.D.		
Sean Chen		
Wen Hsieh, Ph.D.		
Joseph Tautges		
Ker Zhang, Ph.D.		
Timothy Chen		
All directors and executive officers as a group (8 persons)		

Description of Capital Stock

The following is a summary of the rights of our common and preferred stock and some of the provisions of our amended and restated certificate of incorporation and amended and restated bylaws, which will become effective upon the closing of this offering, and of the DGCL. This summary is not complete. For more detailed information, please see our amended and restated certificate of incorporation and amended and restated bylaws, which are filed as exhibits to the registration statement of which this prospectus is a part, as well as the relevant provisions of the DGCL.

General

Upon the closing of this offering and upon the filing of our amended and restated certificate of incorporation, our authorized capital stock will consist of 500,000,000 shares of common stock, $0.000001 par value per share and 10,000,000 shares of preferred stock, $0.000001 par value per share. All of our authorized preferred stock upon the closing of this offering will be undesignated.

Common Stock

Outstanding Shares

As of March 31, 2025, there were 12,560,868 shares of common stock outstanding and 341,496,158 shares of convertible preferred stock outstanding. Upon the closing of this offering and assuming no exercise by the underwriters of their over-allotment option to purchase additional shares, shares of common stock will be outstanding.

Voting

Our common stock is entitled to one vote for each share held of record on all matters submitted to a vote of the stockholders, including the election of directors, and does not have cumulative voting rights. Accordingly, the holders of a majority of the shares of our common stock entitled to vote in any election of directors can elect all of the directors standing for election.

Dividends

Subject to preferences that may be applicable to any then outstanding preferred stock, the holders of common stock are entitled to receive dividends, if any, as may be declared from time to time by our board of directors out of legally available funds.

Liquidation

In the event of our liquidation, dissolution, or winding up, holders of our common stock will be entitled to share ratably in the net assets legally available for distribution to stockholders after the payment of all of our debts and other liabilities, subject to the satisfaction of any liquidation preference granted to the holders of any outstanding shares of preferred stock.

Rights and Preferences

Holders of our common stock have no preemptive, conversion, or subscription rights, and there are no redemption or sinking fund provisions applicable to our common stock. The rights, preferences, and privileges of the holders of our common stock are subject to, and may be adversely affected by, the rights of the holders of shares of any series of our preferred stock that we may designate and issue in the future.

Fully Paid and Nonassessable

All of our outstanding shares of common stock are, and the shares of common stock to be issued in this offering will be, fully paid and nonassessable.

Preferred Stock

Under our amended and restated certificate of incorporation, our board of directors will have the authority, without further action by the stockholders, to issue up to 10,000,000 shares of preferred stock in one or more series, to establish from time to time the number of shares to be included in each such series, to fix the rights, preferences, and privileges of the shares of each wholly unissued series and any qualifications, limitations, or restrictions thereon, and to increase or decrease the number of shares of any such series, but not below the number of shares of such series then outstanding.

Our board of directors may authorize the issuance of preferred stock with voting or conversion rights that could adversely affect the voting power or other rights of the holders of the common stock. The issuance of preferred stock, while providing flexibility in connection with possible acquisitions and other corporate purposes, could, among other things, have the effect of delaying, deferring, or preventing a change in our control that may otherwise benefit holders of our common stock and may adversely affect the market price of our common stock and the voting and other rights of the holders of common stock. We have no current plans to issue any shares of preferred stock.

Stock Options

As of March 31, 2025, 61,795,893 shares of common stock were issuable upon the exercise of outstanding stock options, at a weighted-average exercise price of $0.38 per share. For additional information regarding terms of our equity incentive plans, see the section titled "Executive Compensation — Equity Benefit Plans."

Warrants

Series C Preferred Stock Warrants

As of March 31, 2025, we had 290,563 outstanding warrants to purchase 290,563 shares of our Series C preferred stock, with an exercise price of $0.59 per share (the Series C Warrants). 153,827 of the Series C Warrants expired on May 14, 2025 and 136,736 of the Series C Warrants are exercisable in whole or in part, at any time before their expiration on November 30, 2026, *provided*, however, that if we complete this offering within the one-year period immediately prior to November 30, 2026, the Series C Warrants will expire on the first anniversary of the effective date of this offering. If the Series C Warrants are not exercised prior to the completion of this offering, they will automatically convert into warrants to purchase shares of our common stock.

Common Stock Warrants

As of March 31, 2025, we had 6,690,069 outstanding warrants to purchase 6,690,069 shares of our common stock, with an exercise price of $0.45 per share. These warrants are exercisable in whole or in part, on the earlier of (i) June 1, 2028, (ii) immediately prior to the completion of this offering or (iii) immediately prior to the consummation of an acquisition or asset transfer, until their expiration on June 1, 2034. These warrants also include a cashless exercise feature allowing the holder to elect to receive upon exercise of the holder's warrant (either in whole or in part) the net number of shares determined according to a formula set forth in such warrants.

As of March 31, 2025, we also had 170,689 outstanding warrants to purchase 170,689 shares of our common stock, with an exercise price of $0.29 per share. These warrants are exercisable in whole or in part, at any time prior to their expiration on November 5, 2028.

Restricted Stock Units

As of March 31, 2025, restricted stock units to purchase an aggregate of 5,186,932 shares of common stock were outstanding.

Registration Rights

Upon the closing of this offering and subject to the lock-up agreements entered into in connection with this offering, market standoff agreements and federal securities laws, certain holders of shares of our common stock, including those shares of our common stock that will be issued upon the conversion of our convertible preferred stock in connection with this offering, will initially be entitled to certain rights with respect to registration of such shares under the Securities Act. These shares are referred to as registrable securities. The holders of these registrable securities possess registration rights pursuant to the terms of our amended and restated investor rights agreement and are described in additional detail below. The registration of shares of our common stock pursuant to the exercise of the registration rights described below would enable the holders to trade these shares without restriction under the Securities Act when the applicable registration statement is declared effective. We will pay the registration expenses, other than underwriting discounts, selling commissions and stock transfer taxes, of the shares registered pursuant to the demand, piggyback and Form S-3 registrations described below.

Generally, in an underwritten offering, the managing underwriter, if any, has the right, subject to specified conditions and limitations, to limit the number of shares the holders may include. The demand, piggyback and Form S-3 registration rights described below will expire no later than five years after the closing of this offering.

Demand Registration Rights

Upon the closing of this offering, holders of an aggregate of shares of our common stock will be entitled to certain demand registration rights. At any time beginning 180 days after the closing of this offering, the holders of a majority of these shares may request that we register all or a portion of their shares. We are not required to effect more than two registration statements which are declared or ordered effective. Such request for registration must cover

shares with an anticipated aggregate offering price of at least $10.0 million. With certain exceptions, we are not required to effect the filing of a registration statement during the period starting with the date of the filing of, and ending on a date 180 days following the effective date of the registration statement for this offering.

Piggyback Registration Rights

In connection with this offering, the holders of an aggregate of shares of our common stock were entitled to, and the necessary percentage of holders waived, their rights to notice of this offering and to include their shares of registrable securities in this offering. After this offering, in the event that we propose to register any of our securities under the Securities Act, either for our own account or for the account of other security holders, the holders of these shares will be entitled to certain piggyback registration rights allowing the holder to include their shares in such registration, subject to certain marketing and other limitations.

Form S-3 Registration Rights

Upon the closing of this offering, holders of an aggregate of shares of common stock will be entitled to certain Form S-3 registration rights. Holders of these shares can make a request that we register their shares on Form S-3 if we are qualified to file a registration statement on Form S-3 and if the reasonably anticipated aggregate net proceeds of the shares offered would equal or exceed $1.0 million. We will not be required to effect more than two registrations on Form S-3 within any 12-month period.

Anti-Takeover Effects of Provisions of Our Certificate of Incorporation, Our Bylaws and Delaware Law

Some provisions of Delaware law, our amended and restated certificate of incorporation, and our amended and restated bylaws contain or will contain provisions that could make the following transactions more difficult:

an acquisition of us by means of a tender offer; an acquisition of us by means of a proxy contest or otherwise; or the removal of our incumbent officers and directors. It is possible that these provisions could make it more difficult to accomplish or could deter transactions that stockholders may otherwise consider to be in their best interest or in our best interests, including transactions which provide for payment of a premium over the market price for our shares.

These provisions, summarized below, are intended to discourage coercive takeover practices and inadequate takeover bids. These provisions are also designed to encourage persons seeking to acquire control of us to first negotiate with our board of directors. We believe that the benefits of the increased protection of our potential ability to negotiate with the proponent of an unfriendly or unsolicited proposal to acquire or restructure us outweigh the disadvantages of discouraging these proposals because negotiation of these proposals could result in an improvement of their terms.

Preferred Stock

Our board of directors will have the authority, without further action by our stockholders, to issue up to 10,000,000 shares of undesignated preferred stock with rights and preferences, including voting rights, designated from time to time by our board of directors. The existence of authorized but unissued shares of preferred stock would enable our board of directors to render more difficult or to discourage an attempt to obtain control of us by means of a merger, tender offer, proxy contest, or other means.

Stockholder Meetings

Our amended and restated bylaws will provide that a special meeting of stockholders may be called only by our chairperson of the board, chief executive officer or president, or by a resolution adopted by a majority of our board of directors.

Requirements for Advance Notification of Stockholder Nominations and Proposals

Our amended and restated bylaws will establish advance notice procedures with respect to stockholder proposals to be brought before a stockholder meeting and the nomination of candidates for election as directors, other than nominations made by or at the direction of the board of directors, or a committee of the board of directors.

Elimination of Stockholder Action by Written Consent

Our amended and restated certificate of incorporation and amended and restated bylaws will eliminate the right of stockholders to act by written consent without a meeting.

Staggered Board

Our board of directors will be divided into three classes. The directors in each class will serve for a three- year term, one class being elected each year by our stockholders. For more information on the classified board, see “Management—Board Composition.” This system of electing and removing directors may tend to discourage a third-party from making a tender offer or otherwise attempting to obtain control of us, because it generally makes it more difficult for stockholders to replace a majority of the directors.

Removal of Directors

Our amended and restated certificate of incorporation will provide that no member of our board of directors may be removed from office by our stockholders except for cause and, in addition to any other vote required by law, upon the approval of not less than two-thirds of the total voting power of all of our outstanding voting stock then entitled to vote in the election of directors.

Stockholders Not Entitled to Cumulative Voting

Our amended and restated certificate of incorporation will not permit stockholders to cumulate their votes in the election of directors. Accordingly, the holders of a majority of the outstanding shares of our common stock entitled to vote in any election of directors can elect all of the directors standing for election, if they choose, other than any directors that holders of our preferred stock may be entitled to elect.

Choice of Forum

Our amended and restated certificate of incorporation will provide that the Court of Chancery of the State of Delaware is, unless we consent in writing to the selection of an alternate forum, the sole and the exclusive forum for the following types of actions or proceedings under Delaware statutory or common law: (1) any derivative action or proceeding brought on our behalf; (2) any action asserting a claim of breach of a fiduciary duty or other wrongdoing by any of our directors, officers, employees, or agents to us or our stockholders; (3) any action asserting a claim against us arising pursuant to any provision of the General Corporation Law of the State of Delaware or our amended and restated certificate of incorporation or amended and restated bylaws; (4) any action to interpret, apply, enforce, or determine the validity of our amended and restated certificate of incorporation or amended and restated bylaws; or (5) any action asserting a claim governed by the internal affairs doctrine. The provisions would not apply to suits brought to enforce a duty or liability created by the Securities Act, the Exchange Act, or any other claim for which the U.S. federal courts have exclusive jurisdiction. However, as Section 22 of the Securities Act creates concurrent jurisdiction for federal and state courts over all suits brought to enforce any duty or liability created by the Securities Act, and an investor cannot waive compliance with the federal securities laws and the rules and regulations thereunder, there is uncertainty as to whether a court would enforce such a provision. To prevent having to litigate claims in multiple jurisdictions and the threat of inconsistent or contrary rulings by different courts, among other considerations, our amended and restated certificate of incorporation provides that the U.S. federal district courts will be the exclusive forum for resolving any complaint asserting a cause or causes of action arising under the Securities Act, including all causes of action asserted against any defendant to such complaint. For the avoidance of doubt, this provision is intended to benefit and may be enforced by us, our officers and directors, the underwriters to any offering giving rise to such complaint, and any other professional entity whose profession gives authority to statement made by that person or entity and who has prepared or certified any part of the documents underlying the offering.

While the Delaware courts have determined that such choice of forum provisions are facially valid, a stockholder may nevertheless seek to bring a claim in a venue other than those designated in the exclusive forum provisions. In such instance, we would expect to vigorously assert the validity and enforceability of the exclusive forum provisions of our amended and restated certificate of incorporation. This may require significant additional costs associated with resolving such action in other jurisdictions and there can be no assurance that the provisions will be enforced by a court in those other jurisdictions.

These exclusive forum provisions may limit a stockholder's ability to bring a claim in a judicial forum that it finds favorable for disputes with us or our directors, officers, or other employees, which may discourage lawsuits against us and our directors, officers and other employees. If a court were to find either exclusive-forum provision in our amended and restated certificate of incorporation to be inapplicable or unenforceable in an action, we may incur further significant additional costs associated with resolving the dispute in other jurisdictions, all of which could seriously harm our business.

Amendment of Charter Provisions

The amendment of any of the above provisions, except for the provision making it possible for our board of directors to issue preferred stock, would require approval by holders of at least two-thirds of the total voting power of all of our outstanding voting stock.

The provisions of Delaware law, our amended and restated certificate of incorporation, and our amended and restated bylaws could have the effect of discouraging others from attempting hostile takeovers and, as a consequence, they may also inhibit temporary fluctuations in the market price of our common stock that often result from actual or rumored hostile takeover attempts. These provisions may also have the effect of preventing changes in the composition of our board and management. It is possible that these provisions could make it more difficult to accomplish transactions that stockholders may otherwise deem to be in their best interests.

Delaware Anti-Takeover Law

We are subject to Section 203 of the DGCL (Section 203). Section 203 generally prohibits a public Delaware corporation from engaging in a "business combination" with an "interested stockholder" for a period of three years after the date of the transaction in which the person became an interested stockholder, unless:

- prior to the date of the transaction, the board of directors of the corporation approved either the business combination or the transaction which resulted in the stockholder becoming an interested stockholder;
- the interested stockholder owned at least 85% of the voting stock of the corporation outstanding at the time the transaction commenced, excluding for purposes of determining the number of shares outstanding (but not the outstanding voting stock owned by the interested stockholder) shares owned (a) by persons who are directors and also officers, and (b) by employee stock plans in which employee participants do not have the right to determine confidentially whether shares held subject to the plan will be tendered in a tender or exchange offer; or
- upon or subsequent to the consummation of the transaction, the business combination is approved by the board and authorized at an annual or special meeting of stockholders, and not by written consent, by the affirmative vote of at least 66 2/3% of the outstanding voting stock which is not owned by the interested stockholder.

Section 203 defines a business combination to include:

- any merger or consolidation involving the corporation and the interested stockholder;

- any sale, transfer, pledge or other disposition of 10% or more of the assets of the corporation to or with the interested stockholder;
- subject to exceptions, any transaction involving the corporation that has the effect of increasing the proportionate share of the stock of any class or series of the corporation owned by the interested stockholder;
- subject to exceptions, any transaction that results in the issuance or transfer by the corporation of any stock of the corporation to the interested stockholder; and
- the receipt by the interested stockholder of the benefit of any loans, advances, guarantees, pledges or other financial benefits provided by or through the corporation.

In general, Section 203 defines an interested stockholder as any entity or person beneficially owning 15% or more of the outstanding voting stock of the corporation and any entity or person affiliated with or controlling or controlled by the entity or person.

Listing

We intend to apply to list our common stock on the New York Stock Exchange under the symbol "AMBQ".

Transfer Agent and Registrar

The transfer agent and registrar for our common stock is Computershare Trust Company, N.A.. The transfer agent and registrar's address is 150 Royall Street, Canton, MA 02021 and the telephone number is 1-800-736-3001.

Shares Eligible for Future Sale

Immediately prior to this offering, there has been no public market for our common stock. Future sales of substantial amounts of common stock in the public market could adversely affect prevailing market prices. Furthermore, since only a limited number of shares will be available for sale shortly after this offering because of contractual and legal restrictions on resale described below, sales of substantial amounts of common stock in the public market after the restrictions lapse could adversely affect the prevailing market price for our common stock as well as our ability to raise equity capital in the future.

Based on the number of shares of common stock outstanding as of March 31, 2025, upon the closing of this offering, shares of common stock will be outstanding, assuming no exercise of the underwriters' option to purchase additional shares to cover any over-allotments. All of the shares sold in this offering will be freely tradable unless purchased by our "affiliates" as that term is defined in Rule 144 under the Securities Act. The remaining shares of common stock outstanding after this offering will be restricted as a result of securities laws, market standoff agreements or lock-up agreements. These remaining shares will generally become available for sale in the public market as follows:

- No restricted shares will be eligible for immediate sale upon the closing of this offering; and
- Up to restricted shares will be eligible for sale under Rule 144, subject to the volume limitations, manner-of-sale, and notice provisions described below under "Rule 144," upon expiration of lock-up agreements or market standoff agreements at least six months after the date of this offering.

Registration Rights

Pursuant to the Rights Agreement, after the closing of this offering, the holders of up to shares of our common stock, or certain transferees, will be entitled to certain rights with respect to the registration of the offer and sale of those shares under the Securities Act. See the section titled "Description of Capital Stock— Registration Rights" for a description of these registration rights. If the offer and sale of these shares of our common stock are registered, the shares will be freely tradable without restriction under the Securities Act, subject to the Rule 144 limitations applicable to affiliates, and a large number of shares may be sold into the public market.

Lock-Up Agreements and Market Standoff Agreements

We, all of our directors and executive officers and holders that represent an aggregate of approximately % of our outstanding shares of common stock and other equity securities, including options to acquire shares of our common stock have agreed, for the Lock-Up Period, subject to certain limited exceptions, not to sell or transfer any common stock or securities convertible into or exercisable or exchangeable for our common stock, without first obtaining the written consent of BofA Securities, Inc. and UBS Securities LLC.

Furthermore, (i) an additional approximately % of our outstanding common stock and other equity securities are subject to the market standoff provisions in our Rights Agreement, pursuant to which such holders agreed to not sell, lend, offer, pledge, transfer or dispose (directly or indirectly), make any short sale of, grant any option, warrant or right for the purchase of, or enter into any hedging or similar transaction with the same economic effect as a sale of our common stock (or other securities) held by such holder during the Lock-Up Period, subject to certain limited exceptions and (ii) an additional approximately % of our outstanding common stock and other equity securities are subject to restrictions contained in market standoff agreements with us that include restrictions on the sale, transfer or other disposition of shares for the Lock-Up Period. The forms and specific restrictive provisions within these market standoff provisions may vary among security holders.

As a result of the foregoing, substantially all of our outstanding shares of common stock and securities directly or indirectly convertible into or exercisable or exchangeable for our common stock are subject to a lock-up agreement or market standoff provisions during the Lock-Up Period. We have agreed to enforce all such

market standoff restrictions on behalf of the underwriters and not to amend or waive any such market standoff provisions during the lock-up period without the prior written consent of BofA Securities, Inc. and UBS Securities LLC, on behalf of the underwriters.

For a further description of these lock-up agreements and market stand-off agreements, please see "Underwriting."

Rule 144

In general, under Rule 144 as currently in effect, beginning 90 days after the effective date of the registration statement of which this prospectus is a part, any person who is not an affiliate of ours and has held their shares for at least six months, including the holding period of any prior owner other than one of our affiliates, may sell shares without restriction, provided current public information about us is available. In addition, under Rule 144, any person who is not an affiliate of ours and has held their shares for at least one year, including the holding period of any prior owner other than one of our affiliates, would be entitled to sell an unlimited number of shares immediately upon the closing of this offering without regard to whether current public information about us is available.

Beginning 90 days after the effective date of the registration statement of which this prospectus is a part, a person who is an affiliate of ours and who has beneficially owned restricted securities for at least six months, including the holding period of any prior owner other than one of our affiliates, is entitled to sell a number of restricted shares within any three-month period that does not exceed the greater of:

- 1% of the number of shares of our common stock then outstanding, which will equal approximately shares immediately after this offering; or
- the average weekly trading volume of our common stock on during the four calendar weeks preceding the filing of a notice on Form 144 with respect to the sale.

Sales of restricted shares under Rule 144 held by our affiliates are also subject to requirements regarding the manner-of-sale, notice, and the availability of current public information about us. Rule 144 also provides that affiliates relying on Rule 144 to sell shares of our common stock that are not restricted shares must nonetheless comply with the same restrictions applicable to restricted shares, other than the holding period requirement.

Rule 701

Under Rule 701, shares of common stock acquired upon the exercise of outstanding options or pursuant to other rights granted under compensatory stock plans may be resold by:

- persons other than affiliates, beginning 90 days after the effective date of the registration statement of which this prospectus is a part, subject only to the manner-of-sale provisions of Rule 144; and
- our affiliates, beginning 90 days after the effective date of the registration statement of which this prospectus is a part, subject to the manner-of-sale and volume limitations, current public information, and filing requirements of Rule 144, in each case, without compliance with the six-month holding period requirement of Rule 144.

Form S-8 Registration Statements

As soon as practicable after the completion of this offering, we intend to file with the SEC one or more registration statements on Form S-8 under the Securities Act to register the offer and sale of shares of our common stock that are issuable pursuant to the 2025 Plan and the ESPP. These registrations statements will become effective immediately upon filing. Shares covered by these registration statements will then be eligible for sale in the public markets, subject to vesting restrictions, any applicable lock-up agreements and market standoff agreements described below and Rule 144 limitations applicable to affiliates.

Certain Material U.s. Federal Income Tax Consequences

To Non-u.s. Holders of Our Common Stock

The following is a summary of the material U.S. federal income tax considerations relating to the ownership and disposition of common stock acquired pursuant to this offering by non-U.S. holders (as defined below). This summary deals only with common stock held as a capital asset within the meaning of Section 1221 of the Code (generally, property held for investment) and does not discuss all of the U.S. federal income tax considerations applicable to a non-U.S. holder in light of such non-U.S. holder's particular circumstances. This discussion also does not consider the specific facts or circumstances that may be relevant to a non-U.S. holder that is subject to special treatment under U.S. federal income tax laws, including, but not limited to: a dealer in securities; a broker-dealer; a bank, investment fund, insurance company, or other financial institution; a qualified retirement plan, individual retirement plan, or other tax-deferred account; a regulated investment company; a real estate investment trust; a tax-exempt organization; a governmental organization; a person holding common stock as part of a hedging, integrated, conversion, or straddle transaction or a person deemed to sell common stock under the constructive sale provisions of the Code; a trader in securities that has elected the mark-to-market method of tax accounting; an entity that is treated as a partnership or other pass-through entity for U.S. federal income tax purposes; a person that received such common stock through the exercise of an option or otherwise as compensation for services; a corporation that accumulates earnings to avoid U.S. federal income tax; a corporation organized outside the United States, any

state thereof or the District of Columbia that is nonetheless treated as a U.S. taxpayer for U.S. federal income tax purposes; a person that is not a non-U.S. holder; a "controlled foreign corporation;" a "passive foreign investment company;" a qualified pension fund as defined in Section 897(l)(2) of the Code or an entity all of the interests of which are held by qualified foreign pension funds; or a U.S. expatriate.

This summary is based upon provisions of the Code, its legislative history, applicable U.S. Treasury Regulations promulgated thereunder, published rulings, and judicial decisions, all as in effect as of the date hereof. We have not sought, and will not seek, any ruling from the Internal Revenue Service (IRS), with respect to the tax consequences discussed herein, and there can be no assurance that the IRS will not take a position contrary to the tax consequences discussed below or that any position taken by the IRS would not be sustained. Those authorities may be repealed, revoked, or modified, perhaps retroactively, or may be subject to differing interpretations, which could result in U.S. federal income tax consequences different from those discussed below. In particular, the U.S. House of Representatives recently passed a tax bill that, if enacted into law in its current form, would add new Section 899 to the Code. Proposed Section 899 could significantly increase the aggregate tax liability of certain non-U.S. holders with respect to the ownership and disposition of common stock acquired pursuant to this offering. Non-U.S. holders are urged to consult their tax advisors regarding the potential application of proposed Section 899 with respect to the ownership and disposition of common stock acquired pursuant to this offering. This summary does not address all aspects of U.S. federal income tax, does not deal with all tax considerations that may be relevant to stockholders in light of their personal circumstances, and does not address the Medicare tax imposed on certain investment income or any state, local, foreign, gift, estate, or alternative minimum tax considerations, or the special tax accounting rules under Section 451(b) of the Code.

For purposes of this discussion, a "non-U.S. holder" is a beneficial owner of common stock that is neither a U.S. holder nor a partnership (or any other entity or arrangement that is treated as a partnership) for U.S. federal income tax purposes regardless of its place of organization or formation. A "U.S. holder" is a beneficial holder of common stock that is for U.S. federal income tax purposes: (a) an individual citizen or resident of the United States; (b) a corporation (or any other entity treated as a corporation for U.S. federal income tax purposes) created or organized in or under the laws of the United States, any state thereof or the District of Columbia; (c) an estate the income of which is subject to U.S. federal income taxation regardless of its source; or (d) a trust if it (1) is subject to the primary supervision of a court within the United States and one or more U.S. persons have the authority to control all substantial decisions of the trust, or (2) was in existence on August 20, 1996 and has a valid election in effect under applicable U.S. Treasury Regulations to be treated as a U.S. person.

If a partnership (or an entity or arrangement that is treated as a partnership for U.S. federal income tax purposes) holds common stock, the tax treatment of a partner will generally depend upon the status of the partner and the activities of the partnership. A partner of a partnership holding common stock is urged to consult its own tax advisors.

THIS DISCUSSION IS FOR INFORMATIONAL PURPOSES ONLY AND IS NOT TAX ADVICE. PROSPECTIVE INVESTORS ARE URGED TO CONSULT THEIR OWN TAX ADVISORS CONCERNING THE U.S. FEDERAL INCOME TAX CONSEQUENCES OF OWNING AND DISPOSING OF OUR COMMON STOCK IN LIGHT OF THEIR SPECIFIC SITUATIONS, AS WELL AS THE TAX CONSEQUENCES ARISING UNDER ANY STATE, LOCAL, OR NON-U.S. TAX LAWS AND ANY OTHER U.S. FEDERAL TAX LAWS (INCLUDING THE U.S. FEDERAL ESTATE AND GIFT TAX LAWS).

Distributions on Our Common Stock

As described under the section titled "Dividend Policy," we do not anticipate declaring or paying, in the foreseeable future, any cash dividends on our capital stock. Distributions with respect to common stock, if any, generally will constitute dividends for U.S. federal income tax purposes to the extent paid out of current or accumulated earnings and profits, as determined for U.S. federal income tax purposes. Any portion of a distribution in excess of current or accumulated earnings and profits will be treated as a return of capital and will first be applied to reduce the holder's tax basis in its common stock, but not below zero. Any remaining amount will then be treated as gain from the sale or exchange of the common stock and will be treated as described under "— Disposition of Our Common Stock" below.

Subject to the discussions below regarding effectively connected income, backup withholding and FATCA (as defined below), distributions treated as dividends that are paid to a non-U.S. holder, if any, with respect to shares of our common stock generally will be subject to U.S. federal withholding tax at a rate of 30% (or such lower rate as may be specified in an applicable income tax treaty) of the gross amount of the dividends. A non-U.S. holder who wishes to claim the benefit of a reduced rate of withholding tax under an applicable treaty must furnish to us or our paying agent (or to a financial institution through which such non-U.S. holder holds common stock) a valid IRS Form W-8BEN or IRS Form W-8BEN-E (or applicable successor form) certifying such holder's qualification for the exemption or reduced rate. If a non-U.S. holder is eligible for a reduced rate of U.S. withholding tax pursuant to an income tax treaty and does not timely furnish the required certification, it may obtain a refund of any excess amounts withheld by timely filing an appropriate claim for refund with the IRS. Non-U.S. holders are urged to consult their tax advisors regarding their entitlement to benefits under a relevant income tax treaty.

If a non-U.S. holder is engaged in a trade or business in the United States and dividends with respect to the common stock are effectively connected with the conduct of that trade or business and, if required by an applicable income tax treaty, are attributable to a U.S. permanent establishment or fixed base maintained by such holder in the United States, then the non-U.S. holder will generally be exempt from the 30% U.S. federal withholding tax, provided certain certification requirements are satisfied. However, the non-U.S. holder will be subject to U.S. federal income tax on those dividends on a net income basis at regular U.S. federal income tax rates in the same manner as if such holder were a resident of the United States. Any such effectively connected income received by a foreign corporation may, under certain circumstances, be subject to an additional branch profits tax equal to 30% (or lower applicable income tax treaty rate) of its effectively connected earnings and profits for the taxable year, as adjusted under the Code. To claim the exemption from withholding with respect to any such effectively connected income, the non-U.S. holder must generally furnish to us or our paying agent a properly executed IRS Form W-8ECI (or applicable successor form).

In the case of a non-U.S. holder that is an entity, Treasury Regulations and the relevant tax treaty provide rules to determine whether, for purposes of determining the applicability of a tax treaty, dividends will be treated as paid to the entity or to those holding an interest in that entity. If a non-U.S. holder holds stock through a financial institution or other agent acting on the holder's behalf, the holder will be required to provide appropriate documentation to such agent. Such holder's agent will then be required to provide certification to us or our paying agent.

Disposition of Our Common Stock

Subject to the discussions below regarding backup withholding and FATCA, a non-U.S. holder generally will not be subject to U.S. federal income tax on any gain from a sale, exchange or other disposition of our common stock unless: (a) that gain is effectively connected with the non-U.S. holder's conduct of a trade or business in the United States (and, if required by an applicable income tax treaty, is attributable to a U.S. permanent establishment or fixed base maintained by the non-U.S. holder); (b) the non-U.S. holder is a nonresident alien individual who is present in the United States for 183 days or more in the taxable year of that disposition, and certain other conditions are met; or (c) we are or have been a "United States real property holding corporation" within the meaning of Code Section 897(c)(2) for U.S. federal income tax purposes at any time during the shorter of the five-year period preceding the date of disposition or the non-U.S. holder's holding period for our common stock, and certain other requirements are met. Determining whether we are a United States real property holding corporation depends on the fair market value of our "United States real property interests relative to fair market value of our other trade or business assets and our foreign real property interests. Although there can be no assurance, we believe that we are not, and we do not anticipate becoming, a United States real property holding corporation for U.S. federal income tax purposes. Even if we are treated as a United States real property holding corporation, gain realized by a non-U.S. holder on a disposition of our common stock will not be subject to U.S. federal income tax so long as (1) the non-U.S. holder owned, directly, indirectly and constructively, no more than five percent of our common stock at all times within the shorter of (x) the five-year period preceding the disposition, or (y) the holder's holding period, and (2) our common stock is regularly traded on an established securities market. There can be no assurance that our common stock will qualify as regularly traded on an established securities market. If any gain on a non-U.S. holder's disposition is taxable because we are a United States real property holding corporation and either (x) such non-U.S. holder's ownership of our common stock exceeds five percent during the relevant period described above, or (y) our common stock is not regularly traded on an established securities market at any time during the calendar year in which the disposition occurs, such non-U.S. holder will be taxed on such gain generally in the same manner as gain that is effectively connected with the conduct of a U.S. trade or business, except that the branch profits tax generally will not apply. In addition, a purchaser of your common stock may be required to withhold tax with respect to that obligation.

If gain recognized by a non-U.S. holder is described in clause (a) of the preceding paragraph, the non-U.S. holder will generally be subject to tax on the net gain derived from the disposition at the regular U.S. federal income tax rates in the same manner as if such non-U.S. holder were a U.S. person. In addition, a non-U.S. holder that is a corporation may be subject to the branch profits tax at a rate equal to 30% (or lower applicable income tax treaty rate) of its effectively connected earnings and profits. If a non-U.S. holder is an individual described in clause (b) of the preceding paragraph, the non-U.S. holder will generally be subject to a flat 30% tax on the gain derived from the disposition, which may be offset by certain U.S.-source capital losses even though the non-U.S. holder is not considered a resident of the United States, provided that the non-U.S. holder has timely filed U.S. federal income tax returns with respect to such losses.

Information Reporting and Backup Withholding Tax

We report to our non-U.S. holders and the IRS certain information with respect to any dividends we pay on our common stock, including the amount of dividends paid during each year, the name and address of the recipient, and the amount, if any, of tax withheld. Information reporting requirements apply even if no withholding was required because the distributions were effectively connected with the non-U.S. holder's conduct of a U.S. trade or business or withholding was reduced by an applicable income tax treaty. This information also may be made available under a specific treaty or agreement with the tax authorities in the country in which the non-U.S. holder resides or is established. Under U.S. federal income tax law, interest, dividends, and other reportable payments may, under certain circumstances, be subject to "backup withholding" at the then applicable rate (currently, 24%). Backup withholding, however, generally will not apply to distributions on our common stock to a non-U.S. holder, provided the non-U.S. holder furnishes to us or our paying agent the required certification as to its non-U.S. status, such as by providing a valid IRS Form W-8BEN, IRS Form W-8BEN-E or IRS Form W-8ECI, or certain other requirements are met.

Notwithstanding the foregoing, backup withholding may apply if either we or our paying agent has actual knowledge, or reason to know, that the holder is a U.S. person that is not an exempt recipient. Backup withholding is not an additional tax. It may be credited against the tax liability of persons subject to backup withholding or refunded to the extent it results in an overpayment of tax and the appropriate information is timely supplied to the IRS.

Foreign Accounts

Under Sections 1471 through 1474 of the Code, which are commonly referred to as FATCA, certain withholding taxes may apply to certain types of payments made to "foreign financial institutions" (as specially defined under these rules) and certain other non-U.S. entities if certification, information reporting and other specified requirements are not met. A 30% withholding tax may apply to "withholdable payments" if they are paid to a foreign financial institution or to a non-financial foreign entity, unless (a) the foreign financial institution undertakes certain diligence and reporting obligations and other specified requirements are satisfied, or (b) the non-financial foreign entity either certifies it does not have any substantial U.S. owners or furnishes identifying information regarding each substantial U.S. owner and other specified requirements are satisfied. "Withholdable payment" generally means any payment of interest, dividends, rents, and certain other types of generally passive income if such payment is from sources within the United States. Treasury Regulations proposed in December 2018 (and upon which taxpayers and withholding agents are entitled to rely) eliminate possible withholding under these rules on the gross proceeds from any sale or other disposition of our common stock, previously scheduled to apply beginning January 1, 2019. If the payee is

a foreign financial institution, it must enter into an agreement with the U.S. Treasury requiring, among other things, that it undertake to identify accounts held by certain U.S. persons or U.S.-owned foreign entities, annually report certain information about such accounts and withhold 30% on payments to account holders whose actions prevent it from complying with these reporting and other requirements, or comply with comparable requirements under an applicable inter- governmental agreement between the United States and the foreign financial institution's home jurisdiction. If an investor does not provide us with the information necessary to comply with these rules, it is possible that distributions to such investor that are attributable to withholdable payments, such as dividends, will be subject to the 30% withholding tax. Holders should consult their own tax advisors regarding the implications of these rules for their investment in our common stock.

Underwriting

BofA Securities, Inc. and UBS Securities LLC are acting as representatives of each of the underwriters named below. Subject to the terms and conditions set forth in an underwriting agreement among us and the underwriters, we have agreed to sell to the underwriters, and each of the underwriters has agreed, severally and not jointly, to purchase from us, the number of shares of common stock set forth opposite its name below.

Underwriter	Number of Shares
BofA Securities, Inc.	
UBS Securities LLC	
Needham & Company LLC	
Stifel, Nicolaus & Company, Incorporated	
Total	

Subject to the terms and conditions set forth in the underwriting agreement, the underwriters have agreed, severally and not jointly, to purchase all of the shares sold under the underwriting agreement if any of these shares are purchased. If an underwriter defaults, the underwriting agreement provides that the purchase commitments of the nondefaulting underwriters may be increased or the underwriting agreement may be terminated.

We have agreed to indemnify the underwriters against certain liabilities, including liabilities under the Securities Act, or to contribute to payments the underwriters may be required to make in respect of those liabilities.

The underwriters are offering the shares, subject to prior sale, when, as and if issued to and accepted by them, subject to approval of legal matters by their counsel, including the validity of the shares, and other conditions contained in the underwriting agreement, such as the receipt by the underwriters of officer's certificates and legal opinions. The underwriters reserve the right to withdraw, cancel or modify offers to the public and to reject orders in whole or in part.

Commissions and Discounts

The representatives have advised us that the underwriters propose initially to offer the shares to the public at the public offering price set forth on the cover page of this prospectus and to dealers at that price less a concession not in excess of $ per share. After the initial offering, the public offering price, concession or any other term of the offering may be changed.

The following table shows the public offering price, underwriting discount and proceeds before expenses to us. The information assumes either no exercise or full exercise by the underwriters of their option to purchase additional shares.

	Per Share	Without Option	With Option
Public offering price		$	$
Underwriting discount		$	$
Proceeds, before expenses, to us		$	$

The expenses of the offering, not including the underwriting discount, are estimated at $ and are payable by us. We have agreed to reimburse the underwriters for expenses of up to $ relating to clearance of this offering with the Financial Industry Regulatory Authority (FINRA).

Option to Purchase Additional Shares

We have granted an option to the underwriters, exercisable for 30 days after the date of this prospectus, to purchase up to additional shares at the public offering price, less the underwriting discount. If the underwriters exercise this option, each will be obligated, subject to conditions contained in the underwriting agreement, to purchase a number of additional shares proportionate to that underwriter's initial amount reflected in the above table.

No Sales of Similar Securities

We, our executive officers and directors and holders that represent an aggregate of approximately % of our outstanding shares of common stock and other equity securities, including options to acquire shares of our common stock (such persons, the Lock-Up Parties) have agreed that, for a period of 180 days after the date of this prospectus (such 180-day period, the Lock-Up Period) subject to certain limited exceptions as described below, we and they will not directly or indirectly, without the prior written consent of BofA Securities, Inc. and UBS Securities LLC, (1) offer for sale, sell, pledge, lend or otherwise dispose of (or enter into any transaction or device that is designed to, or could be expected to, result in the disposition by any person at any time in the future of) any shares of common stock (including, without limitation, shares of common stock that may be deemed to be beneficially owned by us or them in accordance with the rules and regulations of the SEC and shares of common stock that may be issued upon exercise of any options or warrants) or securities convertible into or exercisable or exchangeable for common stock (other than the stock and shares issued pursuant to employee benefit plans, qualified stock option plans, or other employee compensation plans existing on the date of this prospectus or pursuant to currently outstanding options, warrants or rights not issued under one of those plans), or sell or grant options, rights or warrants with respect to any shares of common stock or securities convertible into or exchangeable for common stock (other than the grant of options pursuant to option plans existing on the date of this prospectus), (2) enter into any swap or other derivatives transaction that transfers to another, in whole or in part, any of the economic benefits or risks of ownership of shares of common stock, whether any such transaction described in clause (1) or (2) above is to be settled by delivery of common stock or other securities, in cash or otherwise, (3) make any demand for or exercise any right or confidentially submit or publicly file or cause a registration statement to be filed or confidentially submitted, including any amendments thereto, with respect to the registration of any shares of common stock or securities convertible, exercisable or exchangeable into common stock or any of our other securities (other than any registration statement on Form S-8), or (4) publicly disclose the intention to do any of the foregoing.

The restrictions above do not apply to: (a) transactions relating to shares of common stock or other securities acquired in the offering or in the open market, in each case after the completion of the offering, (b) transfer of shares of common stock or any security convertible into common stock as a bona fide gift or charitable contribution or for bona fide estate planning purposes, (c) sales or other dispositions of shares of any class of our capital stock or any security convertible into shares of our capital stock, in each case that are made exclusively between and among (i) the Lock-Up Parties or members of their family (including to a trust, limited partnership, limited liability company or other entity for the direct or indirect benefit of the Lock-Up Parties or members of the Lock-Up Parties' family, or (ii) affiliates of the Lock-Up Parties, including their partners (if a partnership) or members (if a limited liability company), stockholders (if a corporation) or any investment fund or other entity controlling, controlled by, managing, or managed by or under common control with the Lock-Up Parties or affiliates of them (including, for the avoidance of doubt, where the Lock-Up Parties are partnerships, to their general partner or successor partnership or fund, or any other funds managed by such partnership), or to a nominee or custodian of a person or entity to whom a disposition or transfer would be permissible under clause (b) and this clause (c)(i), (d) transfers of shares of common stock or any security convertible into common stock by will, testamentary document or intestate succession upon the death of the Lock-Up Parties, by operation of law or pursuant to a court order or settlement agreement related to the distribution of assets in connection with the dissolution of a marriage or civil union; (e) transfers to the Company (i) from an employee or other service provider of the Company upon death, disability or termination of employment or other service relationship, in each case, of such employee or service provider or (ii) in connection with the vesting, settlement, or exercise of

restricted stock units, options, warrants or other rights to purchase shares of common stock (including, in each case, by way of "net" or "cashless" exercise), including for the payment of exercise price and tax and remittance payments due as a result of the vesting, settlement, or exercise of such restricted stock units, options, warrants or rights, pursuant to any plan or agreement granting such an award to an employee or other service provider of the Company, which plan or agreement is described in the registration statement relating to the offering or the final prospectus relating to the offering; provided further that it shall be a condition to any transaction or transfer pursuant to clause (a), (b), (c), (d) and (e), that each party (the Lock-Up Parties, donor, donee, transferor or transferee, as applicable) shall not be required by law (including without limitation the disclosure requirements of the Securities Act and the Exchange Act to make, and shall agree to not voluntarily make, any filing or public announcement of the transfer or disposition prior to the expiration of the 180-day period referred to above (other than a filing on Form 4 in the case of a transfer pursuant to clause (b), (d), or (e) required by Section 16(a) of the Exchange Act clearly stating the nature and conditions of such transfer and that, if applicable, such shares remain subject to the terms of the lock-up agreement and that, in the case of clause (e)(i), no common stock was sold by the Lock-Up Parties other than such transfer to the Company as described above); *provided further* that it shall be a condition to any transfer pursuant to clause (b), (c), (d) and (e) that such transfer shall not involve a disposition for value and the transferee/donee agrees to be bound by the terms of the lock-up agreement (including, without limitation, the restrictions set forth in the preceding sentence) to the same extent as if the transferee/donee were a party thereto, (f) the exercise of warrants or the exercise of stock options granted pursuant to our stock option/incentive plans or otherwise outstanding on the date hereof; *provided*, that the restrictions shall apply to shares of common stock issued upon such exercise or conversion, (g) the establishment of any contract, instruction or plan that satisfies all of the requirements of Rule 10b5-1 under the Exchange Act; *provided*, *however*, that no sales of common stock or securities convertible into, or exchangeable or exercisable for, common stock, shall be made pursuant to a Rule 10b5-1 Plan prior to the expiration of the Lock-Up Period (as the same may be extended pursuant to the provisions hereof); *provided further*, that, to the extent a public report or filing under the Exchange Act, if any, is required to be made by or on behalf of the Lock-Up Parties or us to report the establishment of such Rule 10b5-1 Plan, such report or filing shall include a statement to the effect that no transfer of shares pursuant to such Rule 10b5-1 Plan during the Lock-Up Period and no public report or filing under the Exchange Act regarding the establishment of such Rule 10b5-1 Plan shall otherwise be voluntarily effected by or on behalf of the Lock-Up Parties or us during the Lock-Up Period, (h) any demands or requests for, exercises of any right with respect to, or taking of any action in preparation of, the registration by us under the Securities Act of the Lock-Up Parties' shares of common stock, provided that no transfer of the Lock-Up Parties' shares of common stock registered pursuant to the exercise of any such right and no registration statement shall be filed under the Securities Act with respect to any of the Lock-Up Parties' shares of common stock during the Lock-Up Period, and (i) sales or other dispositions to a bona fide third party pursuant to a merger, consolidation, tender offer or other similar transaction made to all holders of common stock and involving a Change of Control of the Company and approved by the Company's board of directors; provided that, in the event that such Change of Control is not completed, the Lock-Up Parties' common stock and any security convertible into common stock shall remain

subject to the restrictions contained herein. (“Change of Control” shall mean the transfer (whether by tender offer, merger, consolidation or other similar transaction), in one transaction or a series of related transactions, to a person or group of affiliated persons, of the Company’s voting securities if, after such transfer, such person or group of affiliated persons would hold more than 50% of the outstanding voting securities of the Company or the surviving entity).

BofA Securities, Inc. and UBS Securities LLC may, at their sole discretion and at any time, release the common stock and other securities subject to the lock-up agreements described above in whole or in part at any time. At least three business days before the effectiveness of any release or waiver of any of the restrictions described above with respect to an officer or director of the Company, BofA Securities, Inc. and UBS Securities LLC will notify us of the impending release or waiver and we have agreed to announce the impending release or waiver in accordance with any method permitted by applicable law or regulation (which may include a press release), except where the release or waiver is effected solely to permit a transfer of common stock that is not for consideration and where the transferee has agreed in writing to be bound by the same terms as the lock-up agreements described above to the extent and for the duration that such terms remain in effect at the time of transfer.

Furthermore, (i) an additional approximately % of our outstanding common stock and other equity securities are subject to the market standoff provisions in our Rights Agreement, pursuant to which such holders agreed to not sell, lend, offer, pledge, transfer or dispose (directly or indirectly), make any short sale of, grant any option, warrant or right for the purchase of, or enter into any hedging or similar transaction with the same economic effect as a sale of our common stock (or other securities) held by such holder during the Lock-Up Period, subject to certain limited exceptions and (ii) an additional approximately % of our outstanding common stock and other equity securities are subject to restrictions contained in market standoff agreements with us that include restrictions on the sale, transfer or other disposition of shares for the Lock-Up Period. The forms and specific restrictive provisions within these market standoff provisions may vary among security holders.

As a result of the foregoing, substantially all of our outstanding shares of common stock and securities directly or indirectly convertible into or exercisable or exchangeable for our common stock are subject to a lock-up agreement or market standoff provisions during the Lock-Up Period. We have agreed to enforce all such market standoff restrictions on behalf of the underwriters and not to amend or waive any such market standoff provisions during the lock-up period without the prior written consent of BofA Securities, Inc. and UBS Securities LLC, on behalf of the underwriters.

New York Stock Exchange Listing

We expect the shares to be approved for listing on the New York Stock Exchange under the symbol “AMBQ.” In order to meet the requirements for listing on that exchange, the underwriters have undertaken to sell a minimum number of shares to a minimum number of beneficial owners as required by that exchange.

Before this offering, there has been no public market for our common stock. The initial public offering price will be determined through negotiations between us and the representatives. In addition to prevailing market conditions, the factors to be considered in determining the initial public offering price are

- the valuation multiples of publicly traded companies that the representatives believe to be comparable to us,
- our financial information,
- the history of, and the prospects for, our company and the industry in which we compete,
- an assessment of our management, its past and present operations, and the prospects for, and timing of, our future revenues,
- the present state of our development, and
- the above factors in relation to market values and various valuation measures of other companies engaged in activities similar to ours.

An active trading market for the shares may not develop. It is also possible that after the offering the shares will not trade in the public market at or above the initial public offering price.

The underwriters do not expect to sell more than 5% of the shares in the aggregate to accounts over which they exercise discretionary authority.

Price Stabilization, Short Positions and Penalty Bids

Until the distribution of the shares is completed, SEC rules may limit underwriters and selling group members from bidding for and purchasing our common stock. However, the representatives may engage in transactions that stabilize the price of the common stock, such as bids or purchases to peg, fix or maintain that price.

In connection with the offering, the underwriters may purchase and sell our common stock in the open market. These transactions may include short sales, purchases on the open market to cover positions created by short sales and stabilizing transactions. Short sales involve the sale by the underwriters of a greater number of

shares than they are required to purchase in the offering. “Covered” short sales are sales made in an amount not greater than the underwriters’ option to purchase additional shares described above. The underwriters may close out any covered short position by either exercising their option to purchase additional shares or purchasing shares in the open market. In determining the source of shares to close out the covered short position, the underwriters will consider, among other things, the price of shares available for purchase in the open market as compared to the price at which they may purchase shares through the option granted to them. “Naked” short sales are sales in excess of such option. The underwriters must close out any naked short position by purchasing shares in the open market. A naked short position is more likely to be created if the underwriters are concerned that there may be downward pressure on the price of our common stock in the open market after pricing that could adversely affect investors who purchase in the offering. Stabilizing transactions consist of various bids for or purchases of shares of common stock made by the underwriters in the open market prior to the completion of the offering.

The underwriters may also impose a penalty bid. This occurs when a particular underwriter repays to the underwriters a portion of the underwriting discount received by it because the representatives have repurchased shares sold by or for the account of such underwriter in stabilizing or short covering transactions.

Similar to other purchase transactions, the underwriters’ purchases to cover the syndicate short sales may have the effect of raising or maintaining the market price of our common stock or preventing or retarding a decline in the market price of our common stock. As a result, the price of our common stock may be higher than the price that might otherwise exist in the open market. The underwriters may conduct these transactions on the New York Stock Exchange, in the over-the-counter market or otherwise.

Neither we nor any of the underwriters make any representation or prediction as to the direction or magnitude of any effect that the transactions described above may have on the price of our common stock. In addition, neither we nor any of the underwriters make any representation that the representatives will engage in these transactions or that these transactions, once commenced, will not be discontinued without notice.

Electronic Distribution

In connection with the offering, certain of the underwriters or securities dealers may distribute prospectuses by electronic means, such as e-mail.

Other Relationships

Some of the underwriters and their affiliates have engaged in, and may in the future engage in, investment banking and other commercial dealings in the ordinary course of business with us or our affiliates. They have received, or may in the future receive, customary fees and commissions for these transactions.

In addition, in the ordinary course of their business activities, the underwriters and their affiliates may make or hold a broad array of investments and actively trade debt and equity securities (or related derivative securities) and financial instruments (including bank loans) for their own account and for the accounts of their customers. Such investments and securities activities may involve securities and/or instruments of ours or our affiliates. The underwriters and their affiliates may also make investment recommendations and/or publish or express independent research views in respect of such securities or financial instruments and may hold, or recommend to clients that they acquire, long and/or short positions in such securities and instruments.

Selling Restrictions

Notice to Prospective Investors in the European Economic Area

In relation to each member state of the European Economic Area (each a “Relevant State”), no shares have been offered or will be offered pursuant to the offering to the public in that Relevant State prior to the publication of a prospectus in relation to the shares which has been approved by the competent authority in that Relevant

State or, where appropriate, approved in another Relevant State and notified to the competent authority in that Relevant State, all in accordance with the Prospectus Regulation, except that offers of shares may be made to the public in that Relevant State at any time under the following exemptions under the Prospectus Regulation:

a. to any legal entity which is a qualified investor as defined under the Prospectus Regulation;

b. to fewer than 150 natural or legal persons (other than qualified investors as defined under the Prospectus Regulation), subject to obtaining the prior consent of the representatives for any such offer; or

c. in any other circumstances falling within Article 1(4) of the Prospectus Regulation,

provided that no such offer of shares shall require the Company or any underwriter to publish a prospectus pursuant to Article 3 of the Prospectus Regulation or supplement a prospectus pursuant to Article 23 of the Prospectus Regulation.

Each person in a Relevant State who initially acquires any shares or to whom any offer is made will be deemed to have represented, acknowledged and agreed to and with the Company and the underwriters that it is a qualified investor within the meaning of the Prospectus Regulation.

In the case of any shares being offered to a financial intermediary as that term is used in Article 5(1) of the Prospectus Regulation, each such financial intermediary will be deemed to have represented, acknowledged and agreed that the shares acquired by it in the offer have not been acquired on a non-discretionary basis on behalf of, nor have they been acquired with a view to their offer or resale to, persons in circumstances which may give rise to an offer to the public other than their offer or resale in a Relevant State to qualified investors, in circumstances in which the prior consent of the representatives has been obtained to each such proposed offer or resale.

The Company, the underwriters and their affiliates will rely upon the truth and accuracy of the foregoing representations, acknowledgements and agreements.

For the purposes of this provision, the expression an "offer to the public" in relation to any shares in any Relevant State means the communication in any form and by any means of sufficient information on the terms of the offer and any shares to be offered so as to enable an investor to decide to purchase or subscribe for any shares, and the expression "Prospectus Regulation" means Regulation (EU) 2017/1129.

The above selling restriction is in addition to any other selling restrictions set out below.

In connection with the offering, the representatives are not acting for anyone other than the Company and will not be responsible to anyone other than the Company for providing the protections afforded to their clients nor for providing advice in relation to the offering.

Notice to Prospective Investors in the United Kingdom

In relation to the United Kingdom (UK), no shares have been offered or will be offered pursuant to the offering to the public in the UK prior to the publication of a prospectus in relation to the shares which has been approved by the Financial Conduct Authority in the UK in accordance with the UK Prospectus Regulation and the FSMA, except that offers of shares may be made to the public in the UK at any time under the following exemptions under the UK Prospectus Regulation and the FSMA:

a. to any legal entity which is a qualified investor as defined under the UK Prospectus Regulation;

b. to fewer than 150 natural or legal persons (other than qualified investors as defined under the UK Prospectus Regulation), subject to obtaining the prior consent of the representatives for any such offer; or

c. at any time in other circumstances falling within section 86 of the FSMA,

provided that no such offer of shares shall require the Company or any underwriter to publish a prospectus pursuant to Section 85 of the FSMA or Article 3 of the UK Prospectus Regulation or supplement a prospectus pursuant to Article 23 of the UK Prospectus Regulation.

Each person in the UK who initially acquires any shares or to whom any offer is made will be deemed to have represented, acknowledged and agreed to and with the Company and the underwriters that it is a qualified investor within the meaning of the UK Prospectus Regulation.

In the case of any shares being offered to a financial intermediary as that term is used in Article 5(1) of the UK Prospectus Regulation, each such financial intermediary will be deemed to have represented, acknowledged and agreed that the shares acquired by it in the offer have not been acquired on a non-discretionary basis on behalf of, nor have they been acquired with a view to their offer or resale to, persons in circumstances which may give rise to an offer to the public other than their offer or resale in the UK to qualified investors, in circumstances in which the prior consent of the representatives has been obtained to each such proposed offer or resale.

The Company, the underwriters and their affiliates will rely upon the truth and accuracy of the foregoing representations, acknowledgements and agreements.

For the purposes of this provision, the expression an "offer to the public" in relation to any shares in the UK means the communication in any form and by any means of sufficient information on the terms of the offer and any shares to be offered so as to enable an investor to decide to purchase or subscribe for any shares, the expression "UK Prospectus Regulation" means Regulation (EU) 2017/1129 as it forms part of domestic law by virtue of the European Union (Withdrawal) Act 2018, and the expression "FSMA" means the Financial Services and Markets Act 2000.

In connection with the offering, the representatives are not acting for anyone other than the Company and will not be responsible to anyone other than the Company for providing the protections afforded to their clients nor for providing advice in relation to the offering.

This document is for distribution only to persons who (i) have professional experience in matters relating to investments and who qualify as investment professionals within the meaning of Article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) Order 2005 (as amended, the "Financial Promotion Order"), (ii) are persons falling within Article 49(2)(a) to (d) ("high net worth companies, unincorporated associations, etc.") of the Financial Promotion Order, (iii) are outside the United Kingdom, or (iv) are persons to whom an invitation or inducement to engage in investment activity (within the meaning of Section 21 of the Financial Services and Markets Act 2000, as amended (FSMA)) in connection with the issue or sale of any securities may otherwise lawfully be communicated or caused to be communicated (all such persons together being referred to as "relevant persons"). This document is directed only at relevant persons and must not be acted on or relied on by persons who are not relevant persons. Any investment or investment activity to which this document relates is available only to relevant persons and will be engaged in only with relevant persons.

Notice to Prospective Investors in Switzerland

The shares may not be publicly offered in Switzerland and will not be listed on the SIX Swiss Exchange (SIX) or on any other stock exchange or regulated trading facility in Switzerland. This document has been prepared without regard to the disclosure standards for issuance prospectuses under art. 652a or art. 1156 of the Swiss Code of Obligations or the disclosure standards for listing prospectuses under art. 27 ff. of the SIX Listing Rules or the listing rules of any other stock exchange or regulated trading facility in Switzerland. Neither this document nor any other offering or marketing material relating to the shares or the offering may be publicly distributed or otherwise made publicly available in Switzerland.

Neither this document nor any other offering or marketing material relating to the offering, the Company, or the shares have been or will be filed with or approved by any Swiss regulatory authority. In particular, this

document will not be filed with, and the offer of shares will not be supervised by, the Swiss Financial Market Supervisory Authority (FINMA), and the offer of shares has not been and will not be authorized under the Swiss Federal Act on Collective Investment Schemes (CISA). The investor protection afforded to acquirers of interests in collective investment schemes under the CISA does not extend to acquirers of shares.

Notice to Prospective Investors in the Dubai International Financial Centre

This prospectus relates to an Exempt Offer in accordance with the Offered Securities Rules of the Dubai Financial Services Authority (DFSA). This prospectus is intended for distribution only to persons of a type specified in the Offered Securities Rules of the DFSA. It must not be delivered to, or relied on by, any other person. The DFSA has no responsibility for reviewing or verifying any documents in connection with Exempt Offers. The DFSA has not approved this prospectus nor taken steps to verify the information set forth herein and has no responsibility for the prospectus. The shares to which this prospectus relates may be illiquid and/or subject to restrictions on their resale. Prospective purchasers of the shares offered should conduct their own due diligence on the shares. If you do not understand the contents of this prospectus you should consult an authorized financial advisor.

Notice to Prospective Investors in Australia

No placement document, prospectus, product disclosure statement or other disclosure document has been lodged with the Australian Securities and Investments Commission (ASIC), in relation to the offering. This prospectus does not constitute a prospectus, product disclosure statement or other disclosure document under the Corporations Act 2001 (the Corporations Act), and does not purport to include the information required for a prospectus, product disclosure statement or other disclosure document under the Corporations Act.

Any offer in Australia of the shares may only be made to persons (the Exempt Investors) who are "sophisticated investors" (within the meaning of section 708(8) of the Corporations Act), "professional investors" (within the meaning of section 708(11) of the Corporations Act) or otherwise pursuant to one or more exemptions contained in section 708 of the Corporations Act so that it is lawful to offer the shares without disclosure to investors under Chapter 6D of the Corporations Act.

The shares applied for by Exempt Investors in Australia must not be offered for sale in Australia in the period of 12 months after the date of allotment under the offering, except in circumstances where disclosure to investors under Chapter 6D of the Corporations Act would not be required pursuant to an exemption under section 708 of the Corporations Act or otherwise or where the offer is pursuant to a disclosure document which complies with Chapter 6D of the Corporations Act. Any person acquiring shares must observe such Australian on-sale restrictions.

This prospectus contains general information only and does not take account of the investment objectives, financial situation or particular needs of any particular person. It does not contain any securities recommendations or financial product advice. Before making an investment decision, investors need to consider whether the information in this prospectus is appropriate to their needs, objectives and circumstances, and, if necessary, seek expert advice on those matters.

Notice to Prospective Investors in Hong Kong

The shares have not been offered or sold and will not be offered or sold in Hong Kong, by means of any document, other than (a) to "professional investors" as defined in the Securities and Futures Ordinance (Cap. 571) of Hong Kong and any rules made under that Ordinance; or (b) in other circumstances which do not result in the document being a "prospectus" as defined in the Companies Ordinance (Cap. 32) of Hong Kong or which do not constitute an offer to the public within the meaning of that Ordinance. No advertisement, invitation or document relating to the shares has been or may be issued or has been or may be in the possession of any person for the purposes of issue, whether in Hong Kong or elsewhere, which is directed at, or the contents of which are

likely to be accessed or read by, the public of Hong Kong (except if permitted to do so under the securities laws of Hong Kong) other than with respect to shares which are or are intended to be disposed of only to persons outside Hong Kong or only to "professional investors" as defined in the Securities and Futures Ordinance and any rules made under that Ordinance.

Notice to Prospective Investors in Japan

The shares have not been and will not be registered under the Financial Instruments and Exchange Law of Japan (Law No. 25 of 1948, as amended) and, accordingly, will not be offered or sold, directly or indirectly, in Japan, or for the benefit of any Japanese Person or to others for re-offering or resale, directly or indirectly, in Japan or to any Japanese Person, except in compliance with all applicable laws, regulations and ministerial guidelines promulgated by relevant Japanese governmental or regulatory authorities in effect at the relevant time. For the purposes of this paragraph, "Japanese Person" shall mean any person resident in Japan, including any corporation or other entity organized under the laws of Japan.

Notice to Prospective Investors in Singapore

This prospectus has not been registered as a prospectus with the Monetary Authority of Singapore. Accordingly, the shares have not been offered or sold or caused to be made the subject of an invitation for subscription or purchase and will not be offered or sold or caused to be made the subject of an invitation for subscription or purchase, and this prospectus or any other document or material in connection with the offer or sale, or invitation for subscription or purchase, of the shares, has not been circulated or distributed, nor will it be circulated or distributed, whether directly or indirectly, to any person in Singapore other than (i) to an institutional investor (as defined in Section 4A of the Securities and Futures Act (Chapter 289) of Singapore, as modified or amended from time to time (the SFA)) pursuant to Section 274 of the SFA, (ii) to a relevant person (as defined in Section 275(2) of the SFA) pursuant to Section 275(1) of the SFA, or any person pursuant to Section 275(1A) of the SFA, and in accordance with the conditions specified in Section 275 of the SFA, or (iii) otherwise pursuant to, and in accordance with the conditions of, any other applicable provision of the SFA.

Where the shares are subscribed or purchased under Section 275 of the SFA by a relevant person which is:

(a)	a corporation (which is not an accredited investor (as defined in Section 4A of the SFA)) the sole business of which is to hold investments and the entire share capital of which is owned by one or more individuals, each of whom is an accredited investor; or
(b)	a trust (where the trustee is not an accredited investor) whose sole purpose is to hold investments and each beneficiary of the trust is an individual who is an accredited investor,

securities or securities-based derivatives contracts (each term as defined in Section 2(1) of the SFA) of that corporation or the beneficiaries' rights and interest (howsoever described) in that trust shall not be transferred within six months after that corporation or that trust has acquired the shares pursuant to an offer made under Section 275 of the SFA except:

(a)	to an institutional investor or to a relevant person, or to any person arising from an offer referred to in Section 275(1A) or Section 276(4)(i)(B) of the SFA;
(b)	where no consideration is or will be given for the transfer;
(c)	where the transfer is by operation of law;
(d)	as specified in Section 276(7) of the SFA; or
(e)	as specified in Regulation 32 of the Securities and Futures (Offers of Investments) (Shares and Debentures) Regulations of 2005 of Singapore (Regulation 32).

Singapore SFA *product classification*. In connection with Section 309B of the SFA and the Capital Markets Products (the CMP) Regulations 2018, unless otherwise specified before an offer of shares, we have determined,

and hereby notify all relevant persons (as defined in Section 309A(1) of the SFA), that the shares are "prescribed capital markets products" (as defined in the CMP Regulations 2018) and Excluded Investment Products (as defined in Monetary Authority of Singapore Notice SFA 04-N12: Notice on the Sale of Investment Products and Monetary Authority of Singapore Notice FAA-N16: Notice on Recommendations on Investment Products).

Notice to Prospective Investors in Canada

The shares may be sold only to purchasers purchasing, or deemed to be purchasing, as principal that are accredited investors, as defined in National Instrument 45-106 *Prospectus Exemptions* or subsection 73.3(1) of the *Securities Act* (Ontario), and are permitted clients, as defined in National Instrument 31-103 Registration Requirements, Exemptions and Ongoing Registrant Obligations. Any resale of the shares must be made in accordance with an exemption from, or in a transaction not subject to, the prospectus requirements of applicable securities laws.

Securities legislation in certain provinces or territories of Canada may provide a purchaser with remedies for rescission or damages if this prospectus (including any amendment thereto) contains a misrepresentation, provided that the remedies for rescission or damages are exercised by the purchaser within the time limit prescribed by the securities legislation of the purchaser's province or territory. The purchaser should refer to any applicable provisions of the securities legislation of the purchaser's province or territory for particulars of these rights or consult with a legal advisor.

Pursuant to section 3A.3 (or, in the case of securities issued or guaranteed by the government of a non-Canadian jurisdiction, section 3A.4) of National Instrument 33-105 *Underwriting Conflicts* (NI 33-105), the underwriters are not required to comply with the disclosure requirements of NI 33-105 regarding underwriter conflicts of interest in connection with this offering.

Legal Matters

The validity of the shares of common stock being offered by this prospectus will be passed upon for us by Cooley LLP, Chicago, Illinois. Davis Polk & Wardwell LLP, Redwood City, California, is acting as counsel to the underwriters in connection with certain legal matters relating to the shares of common stock offered by this prospectus.

Experts

The consolidated financial statements of Ambiq Micro, Inc. as of December 31, 2024 and 2023, and for the years then ended, have been included herein and in the registration statement in reliance upon the report of KPMG LLP, independent registered public accounting firm, appearing elsewhere herein, and upon the authority of said firm as experts in accounting and auditing.

Where You Can Find Additional Information

We have filed with the SEC a registration statement on Form S-1, including exhibits and schedules, under the Securities Act, with respect to the shares of common stock being offered by this prospectus. This prospectus, which constitutes part of the registration statement, does not contain all of the information in the registration statement and its exhibits. For further information with respect to us and the common stock offered by this prospectus, we refer you to the registration statement and its exhibits. Statements contained in this prospectus as to the contents of any contract or any other document referred to are not necessarily complete, and in each instance, we refer you to the copy of the contract or other document filed as an exhibit to the registration statement. Each of these statements is qualified in all respects by this reference.

You can read our SEC filings, including the registration statement, over the Internet at the SEC's website at www.sec.gov. You may also request a copy of these filings, at no cost, by writing us at 6500 River Place Blvd., Building 7, Suite 200, Austin, Texas 78730.

Upon the closing of this offering, we will be subject to the information reporting requirements of the Exchange Act and we will file reports, proxy statements and other information with the SEC. These reports, proxy statements and other information will be available at the web site of the SEC referred to above. We also maintain a website at www.ambiq.com, at which, following the completion of this offering, you may access these materials free of charge as soon as reasonably practicable after they are electronically filed with, or furnished to, the SEC. Information contained on or accessible through our website is not a part of this prospectus, and the inclusion of our website address in this prospectus is an inactive textual reference only.

Table of Contents

Index to Financial Statements

Audited Financial Statements:	
Report of Independent Registered Public Accounting Firm	F-2
Consolidated Balance Sheets as of December 31, 2024 and 2023	F-3
Consolidated Statements of Operations and Comprehensive Loss for the year ended December 31, 2024 and 2023	F-4
Consolidated Statements of Changes in Redeemable Convertible Preferred Stock and Stockholders' Deficit for the year ended December 31, 2024 and 2023	F-5
Consolidated Statements of Cash Flows for the year ended December 31, 2024 and 2023	F-6
Notes to Consolidated Financial Statements	F-7
Unaudited Condensed Financial Statements:	
Condensed Consolidated Balance Sheets as of March 31, 2025 and December 31, 2024	F-35
Condensed Consolidated Statements of Operations and Comprehensive Loss for the three months ended March 31, 2025 and 2024	F-36
Condensed Consolidated Statements of Changes in Redeemable Convertible Preferred Stock and Stockholders' Deficit for the three months ended March 31, 2025 and 2024	F-37
Condensed Consolidated Statements of Cash Flows for the three months ended March 31, 2025 and 2024	F-38
Notes to Condensed Consolidated Financial Statements	F-39

Table of Contents

Report of Independent Registered Public Accounting Firm

To the Stockholders and the Board of Directors

Ambiq Micro, Inc.:

Opinion on the Consolidated Financial Statements

We have audited the consolidated financial statements and the related notes (collectively, the consolidated financial statements) of Ambiq Micro, Inc. and subsidiaries (the Company) as listed in the accompanying index. In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company as of December 31, 2024 and 2023, and the results of its operations and its cash flows for the years then ended, in conformity with U.S. generally accepted accounting principles.

Basis for Opinion

These consolidated financial statements are the responsibility of the Company's management. Our responsibility is to express an opinion on these consolidated financial statements based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audit to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud. Our audits included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing

procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. We believe that our audits provide a reasonable basis for our opinion.

/s/ KPMG LLP

We have served as the Company's auditor since 2021.

Austin, Texas

March 31, 2025

AMBIQ MICRO, INC.

Consolidated Balance Sheets

As of December 31, 2024 and 2023

(in thousands, except share amounts)

	December 31, 2024	December 31, 2023
Assets		
Current assets:		
Cash and cash equivalents	$60,981	$27,321
Accounts receivable, net	10,401	12,275
Inventories, net	15,008	21,477
Prepaid expenses and other current assets	2,566	2,789
Total current assets	$88,956	$63,862
Property, equipment and software, net of accumulated depreciation and amortization of $13,158 and $11,565, respectively	2,616	1,983
Right-of-use assets, net	928	990
Intangible assets, net of accumulated amortization of $5,082 and $5,537, respectively	11,729	5,557
Other assets	49	40
Total assets	$104,278	$72,432
Liabilities, Redeemable Convertible Preferred Stock and Stockholders' Deficit		
Current liabilities:		
Accounts payable	$2,933	$2,410
Accrued and other current liabilities	8,202	9,703
Short-term lease liabilities	633	804
Total current liabilities	$11,768	$12,917
Long-term lease liabilities	333	219
Warrant liabilities	112	163
Other long-term liabilities	6,317	—
Total liabilities	$18,530	$13,299
Commitments and contingencies (Note 9)		
Redeemable convertible preferred stock, $0.000001 par value; 347,983,712 shares authorized; 341,496,158 shares and 277,238,979 shares issued and outstanding at December 31, 2024 and 2023, respectively	$378,150	$320,161
Stockholders' (deficit) equity:		

Common stock, $0.000001 par value; 500,000,000 shares authorized; 12,173,766 shares and 9,979,217 shares issued and outstanding at December 31, 2024 and 2023, respectively	—	—
Additional paid-in-capital	28,368	17,676
Accumulated deficit	(320,250)	(277,865)
Accumulated other comprehensive loss	(520)	(839)
Total stockholders' deficit	(292,402)	(261,028)
Total liabilities, redeemable convertible preferred stock and stockholders' deficit	$104,278	$72,432

See accompanying notes to consolidated financial statements.

AMBIQ MICRO, INC.

Consolidated Statements of Operations and Comprehensive Loss

For the years ended December 31, 2024 and 2023

(in thousands)

	2024	2023
Net sales	$76,067	$65,537
Cost of sales	51,776	46,096
Gross profit	24,291	19,441
Operating expenses:		
Research and development	37,168	44,020
Selling, general, and administrative	27,736	26,743
Total operating expenses	64,904	70,763
Loss from operations	(40,613)	(51,322)
Other income, net	980	1,020
Loss before income taxes	(39,633)	(50,302)
Provision for income taxes	28	30
Net loss	$(39,661)	$(50,332)
Deemed dividends	(2,724)	(1,941)
Net loss attributable to common stockholders	$(42,385)	$(52,273)
Net loss per share attributable to common stockholders, basic and diluted	$(4.06)	$(5.38)
Weighted-average shares used in computing net loss per share attributable to common stockholders, basic and diluted	10,427,915	9,722,970
Unaudited pro forma net loss per share attributable to common stock holders, basic and diluted	$(0.12)	$(0.18)
Weighted-average shares used in computing unaudited pro forma net loss per share attributable to common stockholders, basic and diluted	366,851,544	292,732,177
Comprehensive loss:		
Currency translation adjustment	319	(508)
Comprehensive loss	$(39,342)	$(50,840)
Deemed dividends	(2,724)	(1,941)

Comprehensive loss attributable to common stockholders	$(42,066)	$(52,781)

See accompanying notes to consolidated financial statements.

Consolidated Statements of Changes in Redeemable Convertible Preferred Stock and Stockholders' Deficit

(in thousands, except share amounts)

	Redeemable Convertible Preferred Stock		Common Stock		Additional Paid-In-Capital	Accumulated Deficit	Accumulated Other Comprehensive Income (Loss)	Total Stockholders' Deficit
	Shares	Amount	Shares	Amount				
Balance, January 1, 2023	241,828,956	$288,304	9,332,402	$—	$12,836	$(225,592)	$(331)	$(213,087)
Issuance of Series G preferred stock, net of offering costs of $280	34,972,704	31,258	—	—	—	—	—	—
Exercise of warrants	437,319	599	—	—	—	—	—	—
Exercise of stock options	—	—	646,815	—	276	—	—	276
Stock-based compensation	—	—	—	—	2,623	—	—	2,623
Deemed dividend	—	—	—	—	1,941	(1,941)	—	—
Currency translation adjustment	—	—	—	—	—	—	(508)	(508)
Net loss	—	—	—	—	—	(50,332)	—	(50,332)
Balance, December 31, 2023	277,238,979	$320,161	9,979,217	$—	$17,676	$(277,865)	$(839)	$(261,028)
Issuance of Series G preferred stock, net of offering costs of $121	64,257,179	57,989	—	—	—	—	—	—
Issuance of warrants	—	—	—	—	1,940	—	—	1,940
Exercise of stock options	—	—	2,194,549	—	854	—	—	854
Stock-based compensation	—	—	—	—	5,174	—	—	5,174
Deemed dividend	—	—			2,724	(2,724)	—	—
Currency translation adjustment	—	—	—	—	—	—	319	319
Net loss	—	—	—	—	—	(39,661)	—	(39,661)
Balance, December 31, 2024	341,496,158	$378,150	12,173,766	$—	$28,368	$(320,250)	$(520)	$(292,402)

See accompanying notes to consolidated financial statements.

Consolidated Statements of Cash Flows

For the years ended December 31, 2024 and 2023

(in thousands)

	2024	2023
Cash flows from operating activities		
Net loss	$(39,661)	$(50,332)
Adjustments to reconcile net loss to net cash used in operating activities:		

Depreciation and amortization	6,246	7,716
Stock-based compensation	5,174	2,623
Change in right-of-use assets	987	1,176
Non-cash issuance of warrants	1,940	—
Gain on receipt of nonmonetary tangible asset	(1,600)	—
Change in warrant valuations and cancellations	(51)	(581)
Inventory allowance	428	839
Other non-cash	—	(1,309)
Changes in operating assets and liabilities		
Accounts receivable	1,754	(2,983)
Inventories	6,041	15,162
Prepaid expenses and other assets	521	2,735
Accounts payable	821	(3,938)
Accrued and other current liabilities	(3,565)	2,721
Other long-term liabilities	(463)	(787)
Net cash used in operating activities	(21,428)	(26,958)
Cash flows from investing activities		
Purchase of intangible assets	(3,073)	(2,562)
Purchases of property, equipment and software	(658)	(459)
Net cash used in investing activities	(3,731)	(3,021)
Cash flows from financing activities		
Proceeds from issuance of preferred stock, net of issuance costs	57,989	31,258
Proceeds from exercise of stock options	854	276
Proceeds from exercise of warrants	—	240
Net cash provided by financing activities	58,843	31,774
Effect of exchange rate changes on cash and cash equivalents	(24)	(254)
Net increase in cash and cash equivalents	33,660	1,541
Cash and cash equivalents at beginning of period	27,321	25,780
Cash and cash equivalents at end of period	$60,981	$27,321
Supplemental disclosure of non-cash investing and financing activities		
Intangible assets in accrued and other long-term liabilities	10,328	2,169
Deemed dividends	2,724	1,941
Gain on receipt of nonmonetary tangible asset	1,600	—
Right-of-use assets obtained in exchange for new operating lease liabilities	1,207	51

See accompanying notes to consolidated financial statements.

Notes to the Consolidated Financial Statements

As of and for the years ended December 31, 2024 and 2023

1. Description of Business

Ambiq Micro, Inc. (the Company) is a fabless semiconductor company that has developed technology based on a patented Sub-threshold Power Optimized Technology (SPOT™) platform that significantly reduces the amount of power consumed by integrated circuits. The following subsidiaries were formed by the Company and are wholly owned:

Ambiq Micro Singapore Private Ltd.
Shenzhen DeKean Electronics Co.
Ambiq Micro Asia Ltd.
Ambiq Micro Asia Ltd. Taiwan
Ambiq (Shenzhen) Electronics Co., Ltd.

Global Economic Considerations

The global macroeconomic environment is uncertain, and could be negatively affected by, among other things, increased U.S. disputes with countries who are existing trade partners, instability in the global capital and credit markets, supply chain weaknesses, and instability in the geopolitical environment in Asia, Europe and the Middle East, and other tensions. Deterioration in economic factors arising from trade disputes between the United States and China, in particular, could have an adverse impact on our financial results given our customer concentrations in both countries. Such challenges have caused, and may continue to cause, recession fears, rising interest rates, foreign exchange volatility and inflationary pressures through the imposition of tariffs on sensitive technologies, including semiconductors.

Despite the uncertainty of these challenges, the Company believes it has considered all appropriate factors when making financial estimates and will continue to evaluate the impact of any significant changes resulting from these and other factors affecting the Company's business performance.

Liquidity and Capital Resources

During the year ended December 31, 2024, the Company reported a net loss of $39.7 million and had an operating cash flow deficit of $21.4 million. As of December 31, 2024, the Company had cash and cash equivalents totaling $61.0 million. Since inception, the Company has had negative cash flows and losses from operations which it has funded primarily through the issuances of common and preferred stock. The Company anticipates incurring additional losses and negative cash flows from operations until such time, if ever, that it can generate significant sales of its products currently in development and is highly dependent on its ability to find additional sources of funding in the form of debt and equity financing. The Company's ability to fund its planned operations, research and development, and commercialization of its products significantly depends on the amount and timing of cash receipts from these funding sources. Adequate additional funding may not be available to the Company on acceptable terms or at all. The failure to raise funds as and when needed and generate adequate sales could have a negative impact on the Company's financial condition and ability to satisfy its obligations over the next twelve months.

Notes to the Consolidated Financial Statements

As of and for the years ended December 31, 2024 and 2023

2. Summary of Significant Accounting Policies

Basis of Accounting and Consolidation

The accompanying consolidated financial statements have been prepared on the accrual basis of accounting in accordance with accounting principles generally accepted in the United States of America (U.S. GAAP) and the applicable rules and regulations of the Securities and Exchange Commission (SEC) and include the accounts of the Company and its wholly owned subsidiaries. All significant intercompany balances and transactions have been eliminated in consolidation.

Use of Estimates

The preparation of financial statements in conformity with U.S. GAAP requires management to make estimates and assumptions that affect the amounts reported in the financial statements and accompanying notes. Estimates are used for, but not limited to, write-down for excess and obsolete inventories, useful lives for property, equipment and software, allowance for sales returns and discounts, allowance for credit losses, warranties, accrued liabilities, determination of the fair value of stock-based awards, warrants, and deemed dividends, and the realization of tax assets and estimates of tax reserves. These estimates are subjective in nature and involve judgments that affect the reported amounts of assets and liabilities, the disclosure of contingent assets and liabilities at fiscal year-end and the reported amounts of revenue and expenses during the reporting period. Actual results could differ from those estimates. In the current macroeconomic environment affected by geopolitical uncertainty, these estimates require increased judgment and carry a higher degree of variability and volatility. As events continue to evolve and additional information becomes available, these estimates may change materially in future periods.

Nonmonetary Transactions

The Company accounts for nonmonetary transactions in accordance with ASC 845, *Nonmonetary Transactions*. These transactions, if determined to have commercial substance, are generally recorded at the fair value of the assets or services received. During the fiscal year ended December 31, 2024, the Company received nonreciprocal transfer of assets with a vendor. The total fair value of nonmonetary transactions recorded during the period was approximately $1.6 million, which was recognized as a gain in cost of sales. Management evaluates nonmonetary transactions on a case-by-case basis to ensure compliance with applicable accounting guidance and to reflect the economic substance of the exchange.

Unaudited Pro Forma Information

Immediately prior to the closing of a qualified initial public offering (IPO), all of the Company's outstanding redeemable convertible preferred stock will automatically convert into 356,423,629 shares of common stock based on the number of shares outstanding and the applicable conversion price as of December 31, 2024. In the accompanying statements of operations and comprehensive loss, unaudited pro forma basic and diluted net loss per share of common stock has been prepared to give effect to the automatic conversion of all outstanding shares of redeemable convertible preferred stock as if they had been converted at the beginning of the reporting period. The shares of the Company's common stock expected to be issued, and the related net proceeds expected to be received, in connection with the planned IPO are excluded from such pro forma information.

Cash and Cash Equivalents

The Company considers all highly liquid investments acquired with an original maturity of three months or less at the date of purchase to be cash equivalents. Cash equivalents are stated at cost, which approximates fair value, due to the short maturity of these instruments.

Accounts Receivable

Accounts receivable are recorded at the invoice amount, net of an allowance for expected credit losses. The Company assesses the collectability of outstanding customer invoices, and if deemed necessary maintains an allowance for estimated losses resulting from uncollectible customer receivables. In estimating this allowance, the Company considers factors such as historical collection experience, a customer's current creditworthiness, customer concentrations, age of the receivable balance, both individually and in the aggregate, and general macroeconomic conditions that may affect a customer's ability to pay. Actual customer collections could differ from the Company's estimates. For the years ended December 31, 2024 and 2023, the Company did not record an allowance for expected credit losses as all amounts outstanding were deemed collectible.

Concentration of Risks

Financial Instruments

Financial instruments that potentially subject the Company to concentrations of credit risk consist primarily of cash and cash equivalents and trade accounts receivable. The Company maintains its cash and cash equivalent balances in highly rated financial institutions, which at times may exceed federally insured limits or be held in foreign jurisdictions. The Company has not experienced any loss relating to cash and cash equivalents in these accounts and believes no significant concentration risk exists with respect to cash. The Company performs periodic credit evaluations of its customers' financial conditions and generally does not require collateral.

Demand

The Company had three customers representing 49%, 19% and 13%, respectively, of total accounts receivable as of December 31, 2024. There were two customers representing 60% and 31%, respectively, of total accounts receivable as of December 31, 2023.

The Company had three customers representing 41%, 24% and 21%, respectively, of total net sales for the year ended December 31, 2024. There were three customers representing 30%, 32% and 17%, respectively, of total net sales for the year ended December 31, 2023.

The loss of one or more of these customers could have a material adverse impact on the Company's results of operations and financial position.

Supply

The Company's products depend on a sole supplier of wafers and a limited number of third-party manufacturers. The continued and timely supply of input materials and the availability of manufacturing capacity and packaging and testing services impact the Company's ability to meet customer demand. Supply chain disruptions, shortages of raw materials, and manufacturing limitations could limit the Company's ability to meet customer demand and result in delayed, reduced, or canceled orders. The Company has established relationships with leading suppliers

and partners, and it believes these relationships increase the resiliency of our supply chain for its customers. From time to time, subject to inventory disruptions, the Company's customers may buy and hold excess inventories. Consequently, the Company may be subject to resulting fluctuations in the demand for its products.

Fair Value of Financial Instruments

The Company defines fair value as the price that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants at the measurement date. When determining the fair value measurements for assets and liabilities required or permitted to be recorded at fair value, the Company considers the principal market or the most advantageous market in which it would transact.

The level of the fair value hierarchy in which the fair value measurement falls is determined by the lowest level input that is significant to the fair value measurement. The fair value hierarchy is as follows:

- Level 1 applies to assets or liabilities for which there are quoted prices in active markets for identical assets or liabilities that the Company can access at the measurement date.
- Level 2 applies to assets or liabilities for which there are inputs other than quoted prices included in Level 1 that are observable for the asset or liability, either directly or indirectly, such as quoted prices for similar assets or liabilities in active markets; quoted prices for identical assets or liabilities in markets with insufficient volume or infrequent transactions (less active markets).
- Level 3 applies to assets or liabilities for which fair value is derived from valuation techniques in which one or more significant inputs are unobservable, including the Company's own assumptions.

The Company's financial instruments consist principally of cash and cash equivalents, accounts receivable, accounts payable, accrued expenses and warrants. The carrying amounts of cash and cash equivalents, accounts receivable, accounts payable, and accrued expenses are considered to approximate their respective fair values due to the short-term nature of such financial instruments. Cash equivalents, measured at fair value on a recurring basis, are categorized as Level 1 based on quoted prices in active markets. On a recurring basis, the Company measures its liability classified warrant obligations at fair value based on Level 3 inputs, which requires a higher degree of judgment (refer to Note 8).

There have been no changes in Level 1, Level 2, and Level 3 and no changes in valuation techniques for these assets or liabilities for the years ended December 31, 2024 or 2023.

Cloud Computing Implementation Costs

Under the guidance in ASU 2018-15, *Intangibles—Goodwill and Other—Internal-Use Software (Subtopic 350-40),* we capitalize the implementation costs for cloud computing arrangements, mainly for our inventory management system and accounting system. These cloud-based computing implementation costs are recorded in prepaid expenses and other current assets on our consolidated balance sheets. The capitalized costs are amortized based on the estimated useful life of the software arrangement. Capitalized cloud-based computing implementation costs as of December 31, 2024 and 2023 are:

	2024	2023
Capitalized cloud-based computing implementation costs	$1,137	$1,125
Less: Accumulated amortization	380	13
Capitalized cloud-based computing implementation costs, net	$757	$1,112

Inventories

Inventories are stated at the lower of cost or net realizable value. Cost is calculated on a weighted moving average basis. In estimating net realizable value, the Company considers (among other things) the age of its inventories and its product market acceptance based on known business factors and conditions by comparing forecasted customer unit demand for products over a specific future period, or demand horizon, to quantities on hand at the end of each accounting period.

Property, Equipment and Software, Net

Property, equipment and software are carried at cost less accumulated depreciation and amortization. Depreciation and amortization are computed using the straight-line method over the estimated useful lives of the related assets, generally three to five years. The Company capitalizes costs for the fabrication of mask sets used by foundry partners to manufacture the Company's product. Major additions and betterments that extend the estimated useful lives of assets are capitalized. Repairs, maintenance, and minor replacements that do not materially improve or extend the lives of the respective assets are charged to operating expense as incurred.

Intangible Assets

Intangible assets represent costs to acquire licensed intellectual property and software tools. The Company accounts for a noncancelable internal-use software license as the acquisition of an intangible asset and the incurrence of a liability to the extent that all or a portion of the software licensing fees are not paid on or before the license acquisition. The acquired intangibles are subject to amortization on a straight-line basis over the shorter of their estimated useful lives or term of the license arrangement. Costs incurred to renew or extend the term of a recognized intangible asset are capitalized as part of the intangible and amortized over its revised estimated useful life.

Long-Lived Assets

Long-lived assets, which consist primarily of property, equipment, software and intangible assets, are reviewed for impairment whenever events or circumstances indicate their carrying value may not be recoverable.

Conditions that would necessitate an impairment assessment include a significant decline in the observable market value of an asset, a significant change in the extent or manner in which an asset is used, or any other significant adverse change that would indicate that the carrying amount of an asset or group of assets may not be recoverable. When such events or circumstances arise, an estimate of future undiscounted cash flow produced by the asset, or the appropriate grouping of assets, is compared to the asset's carrying value to determine if impairment exists. If the asset is determined to be impaired, the impairment loss is measured based on the excess of its carrying value over its fair value. The Company determines fair value based on discounted cash flows using a discount rate commensurate with the risk inherent in the Company's current business model for the specific asset or asset group being valued.

No long-lived assets were impaired for the years ended December 31, 2024 and 2023.

Leases

At the commencement date of a lease, the Company recognizes a liability to make lease payments and an asset representing the right to use the underlying asset during the lease term. The lease liability is measured at the present value of lease payments over the lease term. The Company does not combine lease components with non-lease components and as such, the non-lease components are accounted for separately. As its leases typically do not provide an implicit rate, the Company uses its incremental borrowing rate based on the information available at the commencement date taking into consideration necessary adjustments for collateral, depending on the facts and circumstances of the lessee and the leased asset, and term to match the lease term. The right-of-use (ROU) asset is measured at cost, which includes the initial measurement of the lease liability and initial direct costs incurred by the Company and excludes lease incentives. Lease liabilities are recorded in other current liabilities and other non-current liabilities. ROU assets are recorded in other assets, net. The Company has elected not to recognize ROU assets and lease liabilities that arise from short-term (12 months or less).

Lease terms may include options to extend or terminate the lease when it is reasonably certain that the Company will exercise that option. Operating lease costs are recognized on a straight-line basis over the lease term. Lease agreements that contain both lease and non-lease components are generally accounted for separately.

Redeemable Convertible Preferred Stock

The Company accounts for its redeemable convertible preferred stock as temporary equity in its consolidated balance sheets due to the nature of the terms of ownership.

Preferred securities that become redeemable upon a contingent event that is not solely within the control of the Company should be classified outside of permanent equity, as in this case a change in control of the Company. Since it is not probable that the instruments will become redeemable, subsequent adjustment to fair value is not required. As such, the redeemable convertible preferred stock was recorded at cost as of December 31, 2024 and 2023, and will remain in temporary equity without further adjustment unless it becomes probable that the redemption feature will become applicable.

Revenue Recognition

The Company's revenues consist of the sale of its products to the Company's customers recognized on a point-in-time basis. The Company's customers are distributors, large global original equipment manufacturers and other direct customers who design and manufacture devices for the consumer and industrial markets.

Revenues are recognized when control of the product is transferred to a customer in an amount that reflects the consideration the Company is expected to be entitled to in exchange for those products, in accordance with Topic 606 (ASC 606), *Revenue from Contracts with Customers*. For direct customers and large global customer contracts, revenue is recognized either when the product is shipped to the customer or upon delivery depending on the prevailing

contract terms. For distributor contracts, revenues are recognized when the product is shipped to the distributor as in these cases the distributor is the contracted customer. Revenues are recognized net of sales credits and allowances for returns and discounts, when applicable. Revenue is recognized net of any taxes collected from customers, which are subsequently remitted to governmental authorities.

Transaction Price

The transaction price is the amount of consideration to which the Company expects to be entitled in exchange for transferring the promised goods to the customer. The transaction price is specified in the standard price list or contract and is confirmed at the time the order for products is placed by the customer and acknowledged by the Company. Certain distributor contracts include variable consideration, such as limited price protection, return and stock rotation provisions, while direct customer contracts include general right of return provisions. The Company estimates variable consideration using all available information, including historical experience and current expectations of return and pricing credits and records an allowance to reduce revenue recognized. These estimates take into consideration a range of possible outcomes that are probability-weighted in accordance with the expected value method in ASC 606 for relevant factors such as current contractual and statutory requirements, specific known market events and trends, industry data, and forecasted customer buying and payment patterns. Overall, these reserves reflect the Company's best estimates of the amount of consideration to which it is entitled based on the terms of the respective underlying contracts. The Company recorded a product returns allowance of approximately $1.0 million and $1.4 million as of December 31, 2024 and 2023, respectively, as reductions of accounts receivable. These allowances are classified as reductions of accounts receivable when the right of offset exists in accordance with ASU 2013-1, *Balance Sheet (Topic 210): Clarifying the Scope of Disclosures about Offsetting Assets and Liabilities*, or as a current liability.

Performance Obligations

Under most of the Company's contracts, the Company's single performance obligation is the delivery of promised goods to the customer. The promised goods are explicitly stated in the customer contract and comprise of a single type of good or goods that are substantially the same and are neither capable of being distinct nor separable from the other promised goods in the contract. This performance obligation is satisfied upon transfer of control of the promised goods to the customer, as defined per the shipping terms within the customer's contract. While the Company's customers purchase products on a purchase order basis, the vast majority of the Company's master contracts with customers have an original expected term of one year or less with automatic renewal features. As allowed by ASC 606, the Company has not disclosed the value of any unsatisfied performance obligations related to these contracts.

The Company's standard terms and conditions provide for a one-year warranty coverage on all its products by guaranteeing the product will be free from defects and conform to the product's specifications. Customers may remedy the warranty claim for a replacement of the product or credit. This assurance-type warranty is not considered a separate performance obligation under ASC 606 and thus no transaction price is allocated to it. The warranty reserve is calculated using historical claim information to project future warranty claims activity.

Contract Balances

Payments are typically due within 60 to 90 days of invoicing and terms do not include a significant financing component or non-cash consideration. There are no contract assets, other than accounts receivable, or contract liabilities recorded on the consolidated balance sheets. The Company expenses sales commissions when incurred and these costs are recorded within selling, general and administrative expenses on the consolidated statement of operations and comprehensive loss.

Accrued Contract Costs

The Company's accrued sales commissions are immaterial as of December 31, 2024 and 2023, and are recorded within accrued liabilities on the accompanying consolidated balance sheets.

End Customer Concentration

Although the Company recognizes revenue and directly invoice distributors for sales of its products, the timing and uncertainty of its revenue and cash flows are most impacted by the ultimate end customer. The following is a summary of net sales for the years ended December 31, 2024 and 2023, based on the country of the corporate headquarters of the ultimate end customer (in thousands):

	2024	2023
China	$38,016	$43,113
United States	36,554	17,024
Netherlands	34	3,458
Rest of the World*	1,463	1,942
Total	$76,067	$65,537

* Other countries individually less than 10%

Cost of Sales

The Company's cost of sales includes the cost of purchasing finished wafers manufactured by independent foundries and costs associated with the assembly, testing, shipping and handling of products along with allocated costs for salary, stock compensation and related benefits for personnel involved in the manufacturing of our products. Cost of sales also includes depreciation for equipment and photomasks supporting the manufacturing process, write downs of inventory, sell-through of products previously reserved for, IP royalties, licensing fees, logistics, quality assurance, warranty, and other costs incurred by the Company.

Segment Information

Segment reporting is based on the "management approach," following the method that management uses to organize the Company's reportable segments for which separate financial information is made available to, and evaluated regularly by, the chief operating decision maker (CODM) in allocating resources and in assessing performance. The CODM is the Company's Chief Executive Officer (CEO), who makes resource allocation decisions and assesses performance based on financial information presented on a consolidated basis. The CEO reviews the Company's consolidated results as one operating segment. Refer to Note 15 for more information.

Stock-Based Compensation

The Company records stock-based compensation expense based upon the grant date fair value for all stock options issued to all persons to the extent such options vest. That expense is determined under the fair value method using the Black-Scholes option pricing model. The Company recognizes the compensation cost for awards on a straight-line basis over the vesting period.

The Black-Scholes option pricing model used to compute stock-based compensation expense requires extensive use of accounting judgment and financial estimates. Items requiring estimation include the fair value of the Company's common stock, the expected term over which option holders will retain their vested stock options before exercising them, and the estimated volatility of the Company's common stock price over the expected term of a stock option. Application of alternative assumptions could result in significantly different stock-based compensation amounts being recorded in the consolidated financial statements.

The absence of an active market for the Company's common stock requires its board of directors, the members of which the Company believes have extensive business, finance and venture capital experience, to determine the fair value of its common stock for purposes of granting options and for calculating stock-based compensation expense for the periods presented. The Company obtained contemporaneous third-party valuations to assist the board of directors in determining fair value. These contemporaneous third-party valuations used the methodologies, approaches and assumptions consistent with the American Institute of Certified Public Accountants Practice Guide, Valuation of Privately Held Company Equity Securities Issued as Compensation. All options granted were intended to be exercisable at a price per share not less than the fair value of the shares underlying those options on their respective dates of grant.

As there is no public market for its common stock, the Company estimated the volatility of its common stock periodically based on the volatility of publicly traded stock of comparable companies and consistent with authoritative guidance, as such is deemed more representative of future stock price trends than the Company's historical volatility. The risk-free interest rate assumption is based on observed market interest rates appropriate for the term of the Company's employee options. As allowed by authoritative guidance, the Company has elected to apply the practical expedient in developing its estimate of the expected term for stock options by using the mid-point between the vesting date and contractual termination date of the award. The Company has not paid and does not anticipate paying cash dividends on its common stock; therefore, the expected dividend yield is assumed to be zero. The Company has elected to recognize forfeitures as they occur, and therefore has not estimated potential forfeitures of stock grants and adjusted compensation cost for this estimate. All options granted were intended to be exercisable at a price per share not less than the fair value of the shares underlying those options on their respective dates of grant.

Transactions with non-employees in which stock-based payment awards are granted in exchange for the receipt of goods or services may involve a contemporaneous exchange of the stock-based payment awards for goods or services or may involve an exchange that spans several financial reporting periods. Judgment is required in determining the period over which to recognize cost, otherwise known as the non-employee's vesting period.

Research and Development Costs

Research and development costs are expensed as incurred. Research and development costs consist primarily of personnel-related expenses, including stock-based compensation, external consulting and services costs, photomasks not directly attributable to a finished product, equipment tooling and allocations of other costs

incurred by the Company. Assets purchased to support the Company's ongoing research and development activities are capitalized when related to products that have achieved technological feasibility and have an alternative future use and are amortized over their estimated useful lives. The Company also has hosted software subscriptions, which are stated at cost and amortized as part of research and development expense based on usage of the asset or on a straight-line basis over the related period of benefit. Research and development costs were $37.2 million and $44.0 million for the years ended December 31, 2024 and 2023, respectively.

Foreign Currency Translation

The functional currency of the Company's foreign subsidiaries was determined to be the local currency; therefore, assets and liabilities are translated at the current exchange rate at the balance sheet date and statement of operations items are translated at the average exchange rate prevailing during the reporting period. Translation adjustments are included in accumulated other comprehensive loss and as a separate component of stockholders' deficit. Transaction gains and losses arising from transactions denominated in a currency other than the functional currency of the entity involved are included in other income and expense in the accompanying consolidated statement of operations and comprehensive loss.

Income Taxes

The Company uses the asset and liability method of accounting for income taxes as set forth in the applicable guidance. Under this method, deferred tax assets and liabilities are recognized for the expected future tax consequences of temporary differences between the carrying amounts and the tax bases of assets and liabilities. Deferred tax assets and liabilities are measured using enacted tax rates expected to apply to taxable income in the years in which those temporary differences are expected to be recovered or settled. The effect of a change in tax rates on deferred tax assets and liabilities will be recognized in the period that includes the enactment date. A valuation allowance is established against the deferred tax assets to reduce their carrying value to an amount that is more likely than not to be realized.

The Company accounts for uncertainty in income taxes based on a "more-likely-than-not" threshold for the recognition and de-recognition of tax positions, which includes the accounting for interest and penalties relating to tax positions. At December 31, 2024, the Company has no unrecognized tax benefits. At December 31, 2023, the Company has unrecognized tax benefits of $1.0 million. The Company recognizes interest and penalties related to unrecognized tax benefits as a component of income tax expense. No such interest or penalties were recognized during the periods presented. The Company had no accruals for interest and penalties at December 31, 2024 and 2023.

Net Loss Per Share

Basic net loss per share is based on the weighted effect of common stock issued and outstanding and is calculated by dividing net loss adjusted for dividends declared on preferred stock (whether or not paid), cumulative undeclared dividends, accretion or decretion of mezzanine equity, redemption or induced conversion of preferred stock, amortization of beneficial conversion features and value of the effect of a downround feature, where applicable, by the weighted average number of shares of common stock outstanding during the period without consideration of the effect of securities that could be converted into shares of common stock.

The Company recorded deemed dividends of $2.7 million and $1.9 million for the years ended December 31, 2024 and 2023, respectively, as a result of a down round feature in the Company's Series E, F, and F-1 preferred stock triggered during the issuance by the Company of Series G preferred stock (see Note 11).

Diluted net loss per share is calculated by adjusting the weighted average number of shares of common stock outstanding for the dilutive effect of additional shares of common stock that would have been outstanding if potentially dilutive securities had been issued, as determined using the treasury-stock method for stock options and warrants and the if-converted method for redeemable convertible preferred stock. Potential dilutive securities consist of outstanding preferred stock, stock options, and warrants to purchase the Company's common and preferred stock. Diluted net loss per share excludes potentially dilutive securities for all periods presented as their effect would be anti-dilutive.

Accordingly, basic and diluted net loss per share is the same for the years ended December 31, 2024 and 2023, and is presented in the accompanying consolidated statements of operations and comprehensive loss.

Recently Issued Accounting Pronouncements

Not Yet Adopted

In November 2024, the FASB issued ASU No. 2024-03, *Income Statement—Reporting Comprehensive Income—Expense Disaggregation Disclosures (Subtopic 220-40)*, to enhance transparency and comparability in the reporting of expense information for public entities. The update requires public business entities to disclose disaggregated expense details, such as purchases of inventory, employee compensation, depreciation, and amortization, within each relevant income statement caption. These disclosures must be presented in a tabular format for both interim and annual reporting periods. Additionally, entities are

required to include qualitative descriptions of expense components which are not separately disclosed. Early adoption is permitted. The Company will adopt the updates in this amendment on January 1, 2027 for annual periods and January 1, 2028 for interim periods. The Company does not expect a material impact of the disclosure as a result of adoption.

Adopted

In June 2016, the FASB issued ASU No. 2016-13, *Financial Instruments–Credit Losses (Topic 326)*, which requires that all financial assets measured at amortized cost to be evaluated for current expected credit losses. Under the new guidance, immediate recognition of credit losses expected over the life of a financial instrument is required. The pronouncement eliminates the incurred credit loss impairment methodology and replaces it with an expected credit loss concept based on historical experience, current conditions, and reasonable and supportable forecasts. Early adoption is permitted. Topic 326 requires a modified retrospective approach by recording a cumulative-effect adjustment to retained earnings as of the beginning of the period of adoption. The Company adopted ASC 326 on January 1, 2023 and determined that the impact was immaterial.

In November 2023, the FASB issued ASU No. 2023-7, *Segment Reporting: Improvements to Reportable Segment Disclosures (Topic 280)*, which requires disclosure of incremental segment information on an annual and interim basis for all public entities to enable investors to develop more decision-useful financial analyses. Under the new guidance, public entities must disclose significant segment expenses, provided regularly to the CODM, that are included in each reported measure of segment profit or loss. Entities must also disclose any other segment items category for each reportable segment, representing differences in various profit and loss measures, along with descriptions of their composition. Additionally, annual disclosures about a segment's profit or loss and assets are now required in interim financial reports as well. Entities must disclose the CODM's title and position, along with an explanation of how they use reported profit and loss measures. Early adoption is permitted. Topic 280 requires amendments to be applied retrospectively to all prior periods presented in the financial statements. The Company adopted ASC 280 on January 1, 2024 for annual periods and January 1, 2025 for interim periods and has included the required updated disclosures within these financial statements.

3. Net Loss Per Share

The table below sets forth the computation of basic and diluted loss per share for the periods presented (in thousands, except share and per share amounts):

	Years Ended December 31,	
	2024	**2023**
Numerator:		
Net loss	$(39,661)	$(50,332)
Deemed dividends	(2,724)	(1,941)
Net loss attributable to common stockholder	$(42,385)	$(52,273)
Denominator:		
Weighted average shares outstanding	10,427,915	9,722,970
Effect of dilutive shares	–	–
Basic and diluted loss per share	$(4.06)	$(5.38)

Since the Company incurred a net loss for the years ended December 31, 2024 and 2023, the diluted net loss per share calculation excludes potentially dilutive securities. The following table summarizes the securities that were excluded from the calculation of diluted earnings per share because their inclusion would have been anti-dilutive:

	Years Ended December 31,	
	2024	**2023**
Redeemable convertible preferred stock	341,496,158	277,238,979
Common warrants	18,833,721	12,143,652
Preferred warrants	290,563	290,563
Restricted stock units	3,186,934	520,270
Stock options	62,407,870	49,307,550
Total shares	426,215,246	339,501,014

The following table summarizes the calculation of unaudited pro forma basic and diluted earnings per share for the period presented (in thousands, except share and per share amounts):

	Year Ended December 31, 2024
	(unaudited)
Numerator:	
Net loss used to compute pro forma net loss per share	$(39,661)
Deemed dividend	(2,724)
Net loss attributable to common stockholder used to compute pro forma net loss per share	$(42,385)
Denominator:	
Weighted average shares outstanding	10,427,915
Pro forma adjustment to reflect the automatic conversion of all convertible preferred stock to common stock upon an IPO	356,423,629
Pro forma weighted average number of shares outstanding-basic and diluted	366,851,544
Unaudited pro forma net loss per share-basic and diluted	$(0.12)

4. Inventories

The following table represents the components of inventories as of December 31, 2024 and 2023 (in thousands):

	2024	2023
Raw materials	$45	$1,928
Work in progress	9,547	8,641
Finished goods	5,416	10,908
Total inventories	$15,008	$21,477

The Company recorded inventory valuation adjustments of $0.4 million and $0.8 million during the twelve months ended December 31, 2024 and 2023, respectively. The charges and reversals were included in cost of sales in the consolidated statements of operations and comprehensive loss. The inventory valuation allowance was primarily due to the sell-through or scrap of inventory that was reserved for in a prior period.

Changes in the Company's inventory allowance were as follows (in thousands):

Balance at December 31, 2022	$23,365
Valuation allowances	839
Balance at December 31, 2023	$24,204
Valuation allowances	428
Sell-through or scrap of inventory	(7,213)
Balance at December 31, 2024	$17,419

5. Property, Equipment and Software

Property, equipment and software consisted of the following as of December 31, 2024 and 2023 (in thousands):

	2024	2023
Probe cards and photomasks	$10,083	$8,046
Equipment	4,386	4,327
Software	298	303
Leasehold improvements	728	589
Furniture and fixtures	279	283

Total	15,774	13,548
Less: Accumulated depreciation and amortization	(13,158)	(11,565)
Property, equipment and software, net	$2,616	$1,983

Depreciation and amortization expense relating to the Company's property, equipment and software was approximately $1.6 million and $3.3 million for the years ended December 31, 2024 and 2023, respectively, and are allocated to cost of sales, selling, general and administrative, and research and development costs in the accompanying consolidated statements of operations and comprehensive loss.

During the year ended December 31, 2024, the Company received a mask set in exchange for no consideration from a vendor in lieu of reimbursement. The mask set was capitalized and a gain was recorded within cost of sales in accordance with ASC 845: *Nonmonetary Transactions*.

6. Intangible Assets

The following is a summary of the Company's intangible assets, net (in thousands):

	Term	2024	2023
Intellectual property	1-5 years	$8,491	$5,970
Software licenses	3 years	8,320	5,124
Accumulated amortization		(5,082)	(5,537)
Intangible assets, net		$11,729	$5,557

During the years ended December 31, 2024 and 2023, the Company purchased and capitalized certain license software tooling costs and intellectual property licensing agreements of $10.8 million and $0.4 million, respectively to support research and development efforts.

Amortization expense relating to the Company's intangible assets was approximately $4.7 million and $4.4 million for the years ended December 31, 2024 and 2023, respectively, and was included in research and development expense in the accompanying consolidated statement of operations and comprehensive loss.

The following table represents the total estimated amortization of intangible assets for the succeeding years (in thousands):

2025	$5,336
2026	4,145
2027	2,248
Total amortization expense	$11,729

7. Accrued and Other Current Liabilities

Accrued liabilities and other current liabilities consisted of the following as of December 31, 2024 and 2023 (in thousands):

	2024	2023
Payroll and related benefits	$2,063	$4,035
Accrued expenses	4,443	4,114
Warranty liability	843	843
Other	853	711
Total accrued liabilities and other current liabilities	$8,202	$9,703

8. Fair Value Measurements

Certain non-financial assets, such as intangible assets and property, equipment and software, are remeasured at fair value only if an impairment is recognized in the current period.

Financial Instruments Recorded at Fair Value on a Recurring Basis

The Company's warrant liabilities are remeasured at each reporting period.

Financial Instruments Not Recorded at Fair Value on a Recurring Basis

Some of the Company's financial instruments are not measured at fair value on a recurring basis but are recorded at amounts that are approximate fair value due to their liquid or short-term nature. Such financial assets and financial liabilities include cash and cash equivalents, including money market deposits, accounts receivables, certain other assets, accounts payable, accrued expenses, and other current liabilities.

The following summarizes the fair value of our financial instruments as of December 31, 2024 (in thousands):

	Quoted Prices In Active Markets for Identical Assets Level 1	Significant Other Observable Inputs Level 2	Significant Unobservable Inputs Level 3	Total
Common stock warrants	—	—	45	$45
Preferred stock warrants	—	—	67	$67
Total	—	—	$112	$112

The following summarizes the fair value of our financial instruments as of December 31, 2023 (in thousands):

	Quoted Prices In Active Markets for Identical Assets Level 1	Significant Other Observable Inputs Level 2	Significant Unobservable Inputs Level 3	Total
Common stock warrants	—	—	64	$64
Preferred stock warrants	—	—	99	$99
Total	—	—	$163	$163

The following table shows the Company warrant activity for the years ended December 31, 2024 and 2023 (in thousands):

	Common Stock warrants	Preferred Stock warrants	Total
Balance at December 31, 2022	218	885	1,103
Change in fair value	(154)	(346)	(500)
Exercises	—	(359)	(359)
Cancellations	—	(81)	(81)
Balance at December 31, 2023	64	99	163
Change in fair value	(19)	(32)	(51)
Balance at December 31, 2024	45	67	112

9. Commitments and Contingencies

Contract Manufacturer Commitments

The Company relies on a third-party foundry and contract manufacturer for the manufacturing of its products. Generally, its foundry agreements do not have volume purchase commitments and primarily provide for purchase commitments based on purchase orders. Purchase orders are placed in advance with consideration of estimates of future demand. These purchase orders can be canceled and rescheduled upon agreement of the Company and the contract manufacturer. As of December 31, 2024, the Company had total manufacturing purchase commitments of $4.3 million.

Litigation

From time to time, the Company may become involved in various legal actions arising in the ordinary course of business. As of December 31, 2024, management was not aware of any existing, pending, or threatened legal actions that would have a material impact on the financial position, results of operations, or cash flows of the Company.

10. Leases

The Company has entered various operating leases for its worldwide office buildings and research and development facilities and equipment, which are accounted for in accordance with ASC 842.

For the years ended December 31, 2024 and 2023, the Company recorded approximately $1.0 million and $1.2 million of operating lease expense, respectively, under ASC 842. All of the Company's leases have been classified as operating leases.

Supplemental cash flow information for the years ending December 31, 2024 and 2023, were as follows (in thousands):

	2024	2023
Cash paid for operating leases included in operating cash flow	$1,068	$1,251
Supplemental non-cash information related to lease liabilities arising from obtaining right-of-use assets	1,207	51
Lease assets obtained in exchange for operating lease liabilities arising from lease modifications	—	(4)

The aggregate future lease payments of operating leases as of December 31, 2024, are as follows (in thousands):

2025	$622
2026	255
2027	121
Total future annual minimum lease payments	998
Less: interest	(32)
Total lease liabilities	$966

As of December 31, 2024, the weighted average remaining lease term for the Company's operating leases is 1.71 years and the weighted average discount rate used to determine the present value of the Company's operating leases is 4.4%.

11. Warrants, Redeemable Convertible Preferred Stock and Stockholders' Deficit

Common Stock

The Company has one class of authorized common stock and has authorized for issuance up to 500,000,000 shares of common stock with a par value of $0.000001 per share.

The holders of common stock are entitled to receive dividends at the discretion of the board of directors, subject to preferences that may apply to shares of preferred stock outstanding at the time.

All holders of common stock are entitled to one vote per share on all matters to be voted on by the Company's stockholders. Upon liquidation, dissolution or winding up, the holders of common stock are entitled to share equally in all the Company's assets remaining after payment of all liabilities and the liquidation preferences of any outstanding preferred stock.

At December 31, 2024, the Company has reserved 452,277,032 shares of common stock for the conversion or exercise, as applicable, of the following securities:

Redeemable convertible preferred stock	341,496,158
Preferred warrants	290,563
Common warrants	18,833,721
Restricted stock units	3,186,934
Stock options	62,407,870
Total shares	426,215,246

Common Stock Warrants

The Company issued warrants to certain consultants and service providers. The warrants, which were issued on various dates between July 2017 and June 2024, expire between July 2027 and June 2034 and grant the holders the right to purchase shares of common stock at exercise prices ranging from $0.23 to $1.11. At the time of issuance, the Company valued these warrants using the Black-Scholes option pricing model using the following assumptions:

Risk-free interest rate	1.47% – 4.80%
Expected life	10 years
Dividend rate	0%
Volatility	35.14% – 65.00%
Fair value per share of common stock	$0.29 – $2.59

The Company determined that the common stock warrants issued in November 2018 had characteristics of a liability. As a result, the warrants have been recorded within long-term liabilities and were valued at $0.0 million and $0.1 million as of December 31, 2024 and 2023, respectively. The warrant liabilities are remeasured at fair value at each reporting period. Refer to Note 8 for more information.

Redeemable Convertible Preferred Stock

During the year ending December 31, 2024, the Company issued 64,257,179 shares of Series G preferred stock with an original issuance price of $0.90 per share for cash proceeds of $58.1 million. The Company had issuance costs of approximately $0.1 million associated with this transaction.

During the year ending December 31, 2023, the Company issued 34,972,704 shares of Series G preferred stock with an original issuance price of $0.90 per share for cash proceeds of $31.6 million. The Company had issuance costs of approximately $0.3 million associated with this transaction.

All classes of preferred stock have a par value of $0.000001 per share. The Company's preferred stock is recorded in the temporary equity section of the Company's consolidated balance sheets.

See the table below for a roll forward of the total preferred stock from January 1, 2023 to December 31, 2024, which details the total redeemable convertible preferred stock presented in the consolidated statements of changes in redeemable convertible preferred stock and stockholders' deficit (in thousands, except share amounts):

	Series Seed Redeemable convertible preferred stock		Series A Redeemable convertible preferred stock		Series B Redeemable convertible preferred stock		Series C Redeemable convertible preferred stock		Series D Redeemable convertible preferred stock		Series E Redeemable convertible preferred stock
	Shares	Amount	Shares	Amount	Shares	Amount	Shares	Amount	Shares	Amount	Shares
Balance at January 1, 2023	5,045,368	$2,473	8,255,614	$5,179	18,456,996	$10,228	42,890,840	$24,908	39,656,811	$33,411	29,019,153
Issuance of Series G Preferred stock, net	–	–	–	–	–	–	–	–	–	–	–
Exercise of warrants	–	–	–	–	437,319	599	–	–	–	–	–
Balance at December 31, 2023	5,045,368	$2,473	8,255,614	$5,179	18,894,315	$10,827	42,890,840	$24,908	39,656,811	$33,411	29,019,153
Issuance of Series G Preferred stock, net	–	–	–	–	–	–	–	–	–	–	–
Balance at December 31, 2024	**5,045,368**	**$2,473**	**8,255,614**	**$5,179**	**18,894,315**	**$10,827**	**42,890,840**	**$24,908**	**39,656,811**	**$33,411**	**29,019,153**

A general summary of the rights with respect to the preferred stock is provided below.

Dividends

The holders of preferred stock, in preference to the holders of common stock, shall be entitled to receive on a pari passu basis, out of funds that are legally available, cash dividends at the rate of 8% of the applicable original issue price per annum on each outstanding share of preferred stock. All original issue prices are subject to adjustment for any stock dividends, combinations, splits, recapitalizations and similar events.

Dividends are non-cumulative, and the Company has no obligation to pay dividends until declared by the Company's board of directors. Through December 31, 2024, no dividends have been declared by the Company's board of directors.

Liquidation Preference

Upon any liquidation, dissolution, or winding up of the Company, whether voluntary or involuntary, before any distribution or payment shall be made to the holders of any common stock, the holders of preferred stock shall be entitled to be paid on a pari passu basis out of the assets of the Company legally available for distribution (or the consideration received by the Company or its stockholders in an acquisition) for each share of preferred stock held by them, an amount per share of preferred stock equal to the original issue price for such share of preferred stock plus all declared and unpaid dividends on such share of preferred stock for an aggregate liquidation value of approximately (in millions):

Series Seed	$2.5
Series A	$4.5
Series B	$10.4
Series C	$25.3
Series D	$34.9
Series E	$29.9
Series F-1	$137.0
Series F	$49.5
Series G	$178.6
Total	$472.6

If upon liquidation, the assets of the Company are insufficient to make payment in full to all holders of preferred stock, then such assets shall be distributed among the holders of preferred stock at the time outstanding, ratably in proportion to the full amounts to which they would otherwise be respectively entitled.

Conversion

Each share of preferred stock shall be convertible at the option of the holder, at any time, into fully paid and nonassessable shares of common stock. The number of shares of common stock to which a holder of preferred stock shall be entitled upon conversion shall be the product obtained by multiplying the applicable "Series Preferred Conversion Rate", by the number of shares of such preferred stock being converted, as defined in the Company's latest amended and restated certificate of incorporation. The Series Preferred Conversion Rate is the

quotient obtained by dividing the Original Issue Price of such series of Series Preferred by the applicable "Series Preferred Conversion Price". The Series Preferred Conversion Price is generally the Original Issuance Price for all series of Series Seed, A, B, C, D and G preferred stock, subject to certain adjustments.

At December 31, 2024 and 2023, the preferred stock Original Issuance Prices were $0.50 per share for Series Seed preferred stock, $0.55 per share for Series A and Series B preferred stock, $0.59 per share for Series C preferred stock, $0.88 per share for Series D preferred stock, $1.03 per share for Series E preferred stock, $1.83 per share for Series F preferred stock, $2.09 per share for Series F-1 preferred stock and $0.90 for Series G preferred stock. These conversion prices are subject to adjustment in accordance with the Company's latest amended and restated certificate of incorporation.

The issuance of the Series G Preferred Stock triggered adjustments to the Series Preferred Conversion Price for Series E through Series F-1, such that at December 31, 2024 and December 31, 2023, the Series Preferred Conversion Prices were $1.00 per share and $1.02 per share, respectively, for Series E preferred stock; $1.61 per share and $1.74 per share, respectively, for Series F preferred stock; and, $1.81 per share and $1.97 per share, respectively, for

Series F-1 preferred stock. These conversion prices are subject to adjustment in accordance with the Company's latest amended and restated certificate of incorporation.

Voting

The holders of preferred stock shall vote together with the holders of common stock and not as a separate class and may act by written consent in the same manner as the holders of common stock. Each holder of preferred stock shall be entitled to the number of votes equal to the number of shares of common stock into which such shares of preferred stock could be converted immediately after the close of business on the record date fixed for such meeting or the effective date of such written consent and shall have voting rights and powers equal to the voting rights and powers of the holders of common stock and shall be entitled to notice of any stockholders' meeting in accordance with the bylaws of the Company.

Redemption

The Company's preferred stock is not generally redeemable except in situations set forth in the certificate of incorporation, including when more than fifty percent of the Company's voting power is transferred, a triggering event is created that is outside the control of the Company, and a change in control of Company ownership.

Preferred Stock Warrants

In connection with the issuance of promissory notes later converted by the Company to various series of preferred stock, the Company issued warrants to the convertible promissory note holders. The warrants, which were issued on various dates between February 2012 and February 2017, expire between February 2022 and November 2026 and grant the holders the right to purchase in the aggregate the following shares of preferred stock:

	Shares	Exercise Price
Series A Preferred	797,351	$0.55
Series B Preferred	733,047	$0.55
Series C Preferred	290,563	$0.59

The Company determined that these preferred stock warrants had characteristics of a liability. As a result, the warrants have been recorded within long-term liabilities and were valued at $0.1 million and $0.2 million at December 31, 2024 and 2023, respectively. The warrant liabilities are remeasured at fair value at each reporting period. Refer to Note 8 for more information.

At the time of issuance, the Company has valued these warrants using the Black-Scholes option pricing model using the following assumptions:

Risk-free interest rate	1.74% – 2.37%
Expected life	10 years
Dividend rate	0%
Volatility	33.00% – 59.00%
Fair value per share of preferred stock	$0.55 – $0.59

During the year ending December 31, 2023, certain holders of our preferred stock warrants exercised 437,319 shares at an exercise price of $0.55. Cash proceeds of approximately $0.2 million were received in connection with these exercises. The Company had minimal issuance costs associated with these transactions. There were no preferred stock warrants exercised during the year ended December 31, 2024.

12. Stock Option Plan and Stock-Based Compensation

In October 2010, the Company's board of directors and its stockholders adopted the 2010 Equity Incentive Plan (the 2010 Plan) under which the Company may grant incentive stock options (ISOs), nonqualified stock options, restricted stock awards, restricted stock unit awards (RSUs), and stock appreciation rights of common stock to employees, non-employee members of the Company's board of directors and consultants who provide services to the Company. In 2019, the Company increased the number of shares issuable under the 2010 Plan to 40,837,342. In October 2020, the Company's board of directors and its stockholders adopted the 2020 Equity Incentive Plan (the 2020 Plan) as the successor to the 2010 Plan (collectively, the Plans) and increased the number of shares issuable under the 2020 Plan to 49,056,404. In 2023, the Company increased the number of shares issuable under the 2020 Plan to 64,800,980. The maximum term of options granted under the Plans is ten years from the date of grant. Options outstanding under the Plans generally vest over four years, with 25% vesting on the one-year anniversary and equal monthly vesting installments thereafter. The fair values of the associated common stock underlying the option, rate of exercisability and expiration dates of the options granted are determined by the board of directors at the time of grant. RSUs granted under the Plans are generally subject to service-based and performance-based vesting conditions. The Company currently uses authorized and unissued shares to satisfy stock award exercises.

The following table summarizes the activity in total shares of common stock available for issuance under the 2020 Plan:

	Shares Available for Grant
Balance, December 31, 2022	5,313,590
Shares added	6,170,507
Granted	(478,000)
Forfeited	789,751
Balance, December 31, 2023	11,795,848
Shares added	17,516,000
Granted	(19,059,324)
Forfeited	881,791
Balance, December 31, 2024	11,134,315

The following table summarizes the effects of stock-based compensation on cost of sales, research and development, and sales, general and administrative expenses granted under the Plans (in thousands) for the years ended December 31, 2024 and 2023, respectively:

	2024	2023
Cost of sales	$356	$147
Research and development	2,416	1,350
Sales, general and administrative	2,402	1,126
Total	$5,174	$2,623

The fair value for the Company's options was estimated at the date of grant using a Black-Scholes option pricing model with the following weighted-average assumptions:

	Years Ended December 31,	
	2024	2023
Risk-free interest rate	4.16%	3.74%
Expected life of the options	5.47 years	6.01 years
Dividend rate	0%	0%
Volatility	54.51%	52.74%
Fair value per share of common stock	$0.45	$1.09

Additional information regarding outstanding options that are vesting, expected to vest, or exercisable as of December 31, 2024, is as follows:

	Number of Options	Weighted-Average Price Per Share	Weighted-Average Exercise Price	Weighted-Average Remaining Contractual Life (Years)
Options outstanding at December 31, 2022	50,675,200	$0.27	$0.54	7.03
Options granted	228,000	$0.59	$1.09	—
Options exercised	(646,815)	$0.21	$0.43	—
Options forfeited	(948,835)	$0.49	$0.97	—
Options outstanding at December 31, 2023	49,307,550	$0.26	$0.53	6.03
Options granted	16,392,660	$0.24	$0.45	—

Options exercised	(2,194,549)	$0.31	$0.39	–
Options forfeited	(1,097,791)	$0.37	$0.62	–
Options outstanding at December 31, 2024	62,407,870	$0.28	$0.38	6.23
Vested and expected to vest	62,264,932	$0.28	$0.38	6.23
Exercisable	50,174,761	$0.27	$0.36	5.53

The weighted-average fair value of options granted during the years ended December 31, 2024 and 2023, was $0.38 and $0.59 per share, respectively. The aggregate intrinsic value of options exercised during the years ended December 31, 2024 and 2023 was $0.3 million and $0.5 million, respectively. Intrinsic value represents the difference between the market value of the Company's common stock at the time of exercise and the strike price of the stock option. The total unrecognized stock-based compensation expense related to unvested stock options and subject to recognition in future periods was approximately $4.2 million and $4.1 million as of December 31, 2024 and 2023, respectively. As of December 31, 2024, the Company anticipates this expense to be recognized over a weighted-average period of approximately 1.7 years.

During the years ended December 31, 2024 and 2023, the Company received $0.9 million and $0.3 million, respectively, from the exercise of stock options granted under the Plans and granted 2,666,664 and 250,000 RSUs, respectively, to several non-employee directors which vest upon satisfaction of a service-based condition and performance condition based on a liquidity event.

In May 2024, our board of directors approved a one-time repricing of certain stock options that had been granted to date under the 2020 Plan, effective May 13, 2024. The repricing impacted stock options with exercise prices greater than or equal to $0.61, and each such stock option was repriced to have a per-share exercise price of $0.45, the fair market value of a share of our common stock as determined by our board of directors on the date of the repricing. No changes were made to the expiration dates of or number of shares underlying the repriced stock options.

The incremental compensation cost resulting from the repricing was measured as the excess of the fair value of the modified options over the fair value of the original options immediately before modification. As a result, we recognized additional stock-based compensation expense of approximately $984,000 during the fiscal year ended December 31, 2024 for vested awards. The remaining incremental expense is being amortized over the remaining vesting period of the repriced options.

13. Income Taxes

The income tax amount for each of the years ended December 31, 2024 and 2023, differs from the amount that would be expected after applying the statutory U.S. federal income tax rate primarily due to an increase in the valuation allowance.

In assessing the realizability of deferred tax assets, management considers whether it is more likely than not that some portion or all of the deferred tax assets will not be realized. The ultimate realization of deferred tax assets is dependent upon the generation of future taxable income during the periods in which those temporary differences become deductible. Due to the uncertainty of the Company's ability to realize the benefit of the deferred tax assets, the net deferred tax assets are fully offset by a valuation balance at December 31, 2024 and 2023.

Provision for Income Taxes

Loss before provision for income taxes, includes the components for the years ended December 31, 2024 and 2023, as follows (in thousands):

	2024	2023
Domestic	(35,548)	$(45,548)
Foreign	(4,086)	(4,754)
Loss before provision for income taxes	(39,634)	$(50,302)

Deferred Tax Assets

The components of the deferred tax assets and liabilities at December 31, 2024 and 2023, are as follows (in thousands):

	2024	2023
Deferred tax assets:		
Net operating loss carryforwards	$37,760	$31,513
Depreciable assets	223	276
Intangible assets	843	559
Research and development capitalization	17,883	14,841
Inventory allowance	3,675	4,926

Accrued liabilities and other	1,787	1,459
Uniform capitalization	304	959
Sales return allowance	204	284
Warrants	1,019	606
Total deferred tax assets	63,698	55,423
Less: valuation allowance	(63,698)	(55,423)
Net deferred tax assets	–	–

As of December 31, 2024 and 2023, the Company had net operating loss carryforwards (NOL) of approximately $160.5 million and $139.4 million, respectively. For federal tax reporting purposes, NOL carryforwards of $51.6 million will expire in fiscal years 2032 through 2037 if not utilized prior to that time. The remaining

$108.9 million carries forward indefinitely. As of December 31, 2024, the Company had $17.7 million foreign NOL carryforwards. The foreign NOL will begin to expire after 2026 if not utilized.

Utilization of NOL carryforwards may be subject to substantial annual limitation due to the “change in ownership” provisions of the Internal Revenue Code. The annual limitation may result in the expiration of NOL and other tax attributes before utilization. The NOL carryforwards are subject to Internal Revenue Service adjustments until the statute closes on the year the NOL is utilized.

Uncertain Tax Position

The Company recognizes the tax benefit from an uncertain tax position only if it is more likely than not that the tax position will be sustained on examination by the taxing authorities. At December 31, 2024, the Company has no unrecognized tax benefits. At December 31, 2023, the Company has unrecognized tax benefits of $1.0 million. Additionally, the Company does not expect any unrecognized tax benefits to change significantly over the next twelve months.

Effective Tax Rate

For the years ended December 31, 2024 and 2023, the income tax provision related to continuing operations differs from the amounts computed by applying the statutory income tax rate of 21% to pre-tax loss as follows (in thousands):

	2024	2023
U.S. Federal benefit	$(8,323)	$(10,563)
Permanent items	90	156
Foreign tax differential	126	(108)
Other	(139)	(4)
Change in valuation allowance	8,274	10,549
Total	$28	$30

The Company has established a valuation allowance equal to the net deferred tax assets due to uncertainties regarding the realization of the deferred tax assets based on the Company’s lack of earnings history. The valuation allowance increased by approximately $8.3 million and $10.5 million during the years ended December 31, 2024 and 2023, respectively, primarily due to net losses from operations.

The Company provides for U.S. income taxes on the earnings of its foreign subsidiaries to the extent required by the Tax Cuts and Jobs Act (TCJA). However, the Company does not provide for withholding taxes on any portion of the undistributed earnings of its foreign subsidiaries because it intends to permanently reinvest those earnings indefinitely outside the United States. At the present time it is not practicable to estimate the amount of income or withholding taxes that might be payable if these earnings were repatriated.

The Company files foreign and U.S. federal and various state income tax returns. For U.S. federal and state income tax purposes, the statute of limitations currently remains open for the years ending December 31, 2019 to present. In addition, prior year NOL carryforwards, including those originated from 2010, that may be utilized in future years, may be subject to examination. The Company is not currently under examination by income tax authorities in any jurisdiction.

14. Related Party Transactions

The Company defines related parties as any party that controls or can significantly influence the management or operating policies of the Company to the extent that the Company may be prevented from fully pursuing its own interests, such as directors, executive officers, and stockholders, including a greater than 10% beneficial owner of the Company's capital and their affiliates or immediate family members. For the year ended December 31, 2024, the Board of Directors approved the issuance of 6,690,069 warrants to purchase common stock to a related party for consulting services at an exercise price of $0.45 per share. The warrants are exercisable on the earlier of (i) June 1, 2028, (ii) immediately prior to the consummation of a qualified initial public offering, (iii) immediately prior to the consummation of an asset transfer or acquisition as defined in the Company's charter. The warrants expire after June 1, 2034. The Company did not have any related party transactions in the year ended December 31, 2023.

15. Segment and Geographic Information

The Company's CODM is the Company's CEO and reviews the financial information presented on a consolidated basis for purposes of making operating decisions, assessing financial performance, and allocating resources. The CODM uses revenue, gross margin, operating expenses and net loss by its single operating and reportable segment to make strategic business decisions.

The following table sets forth the Company's disaggregation of operating expenses which were reviewed by the CODM for the years ended December 31, 2024 and 2023 (in thousands):

	2024	2023
Research and development	$37,168	$44,020
Sales and marketing	10,731	11,512
General and administrative	17,005	15,231
Total operating expenses	$64,904	$70,763

The following is a summary of net sales for the years ended December 31, 2024 and 2023, based on the country to which the products were shipped, which may be different from the geographic locations of the ultimate end customers (in thousands):

	2024	2023
China	$54,728	$57,692
Taiwan	18,190	57
Rest of the World*	3,149	7,788
Total	$76,067	$65,537

* Other countries individually less than 10%

The following illustrates property, equipment and software, net, and right-of-use assets, net by geographic locations based on physical location (in thousands):

	2024	2023
North America	$786	$1,400
Taiwan	2,139	762
China	593	774
Rest of the World*	26	37
Total	$3,544	$2,973

* Other countries individually less than 10%

16. Subsequent Events

The Company performed a review of events subsequent to the balance sheet date through the date the consolidated financial statements are available to be issued on March 31, 2025, and has determined there are no such events that require disclosure.

AMBIQ MICRO, INC.

Condensed Consolidated Balance Sheets

As of March 31, 2025 and December 31, 2024

(Unaudited and in thousands, except share amounts)

	March 31, 2025	December 31, 2024
Assets		
Current assets:		
Cash and cash equivalents	$61,418	$60,981
Accounts receivable, net	4,056	10,401
Inventories, net	15,294	15,008
Prepaid expenses and other current assets	2,502	2,566
Total current assets	$83,270	$88,956
Property, equipment and software, net of accumulated depreciation and amortization of $13,588 and $13,158, respectively	4,026	2,616
Right-of-use assets, net	1,067	928
Intangible assets, net of accumulated amortization of $6,615 and $5,082, respectively	10,883	11,729
Other assets	49	49
Total assets	$99,295	$104,278
Liabilities, Redeemable Convertible Preferred Stock and Stockholders' Deficit		
Current liabilities:		
Accounts payable	$5,911	$2,933
Accrued and other current liabilities	8,359	8,202
Short-term lease liabilities	579	633
Total current liabilities	$14,849	$11,768
Long-term lease liabilities	526	333
Warrant liabilities	170	112
Other long-term liabilities	5,317	6,317
Total liabilities	$20,862	$18,530
Commitments and contingencies (Note 6)		
Redeemable convertible preferred stock, $0.000001 par value; 347,983,712 shares authorized; 341,496,158 shares and 341,496,158 shares issued and outstanding at March 31, 2025 and December 31, 2024, respectively	$378,150	$378,150
Stockholders' deficit:		
Common stock, $0.000001 par value; 500,000,000 shares authorized; 12,560,868 shares and 12,173,766 shares issued and outstanding at March 31, 2025 and December 31, 2024, respectively	—	—
Additional paid-in-capital	29,367	28,368
Accumulated deficit	(328,534)	(320,250)
Accumulated other comprehensive loss	(550)	(520)

Total stockholders' deficit	(299,717)	(292,402)
Total liabilities, redeemable convertible preferred stock and stockholders' deficit	$99,295	$104,278

AMBIQ MICRO, INC.

Condensed Consolidated Statements of Operations and Comprehensive Loss

For the three months ended March 31, 2025 and 2024

(Unaudited and in thousands)

	March 31,	
	2025	2024
Net sales	$15,732	$15,210
Cost of sales	7,343	8,976
Gross profit	8,389	6,234
Operating expenses:		
Research and development	8,687	9,436
Selling, general, and administrative	8,443	5,988
Total operating expenses	17,130	15,424
Loss from operations	(8,741)	(9,190)
Other income (expense)	461	(204)
Loss before income taxes	(8,280)	(9,394)
Provision for income taxes	4	5
Net loss	$(8,284)	$(9,399)
Deemed dividends	—	(375)
Net loss attributable to common stockholders	$(8,284)	$(9,774)
Net loss per share attributable to common stockholders, basic and diluted	$(0.68)	$(0.97)
Weighted-average shares used in computing net loss per share attributable to common stockholders, basic and diluted	12,232,916	10,102,901
Unaudited pro forma net loss per share attributable to common stock holders, basic and diluted	$(0.02)	
Weighted-average shares used in computing unaudited pro forma net loss per share attributable to common stockholders, basic and diluted	368,656,545	
Comprehensive loss:		
Currency translation adjustment	(30)	150
Comprehensive loss	$(8,314)	$(9,249)
Deemed dividends	—	(375)
Comprehensive loss attributable to common stockholders	$(8,314)	$(9,624)

AMBIQ MICRO, INC.

CONDENSED CONSOLIDATED STATEMENTS OF CHANGES IN REDEEMABLE CONVERTIBLE PREFERRED STOCK AND STOCKHOLDERS' DEFICIT

(Unaudited and in thousands, except share amounts)

	Redeemable Convertible Preferred Stock		Common Stock		Additional Paid-In-Capital	Accumulated Deficit	Accumulated Other Comprehensive Income (Loss)	Total Stockholders' Deficit
	Shares	Amount	Shares	Amount				
Balance January 1, 2024	277,238,979	$320,161	9,979,217	$—	$17,676	$(277,865)	$(839)	$(261,028)
Issuance of Series G preferred shares, net of offering costs of $48	8,847,805	7,949	—	—	—	—	—	—
Exercise of stock options	—	—	262,064	—	202	—	—	202
Stock-based compensation	—	—	—	—	551	—	—	551
Deemed dividend	—	—	—	—	375	(375)	—	—
Currency translation adjustment	—	—	—	—	—	—	150	150
Net loss	—	—	—	—	—	(9,399)	—	(9,399)
Balance March 31, 2024	286,086,784	$328,110	10,241,281	$—	$18,804	$(287,639)	$(689)	$(269,524)
Balance January 1, 2025	341,496,158	$378,150	12,173,766	$—	$28,368	$(320,250)	$(520)	$(292,402)
Exercise of stock options	—	—	387,102	—	148	—	—	148
Stock-based compensation	—	—	—	—	851	—	—	851
Currency translation adjustment	—	—	—	—	—	—	(30)	(30)
Net loss	—	—	—	—	—	(8,284)	—	(8,284)
Balance March 31, 2025	341,496,158	$378,150	12,560,868	$—	$29,367	$(328,534)	$(550)	$(299,717)

AMBIQ MICRO, INC.

Condensed Consolidated Statements of Cash Flows

For the three months ended March 31, 2025 and 2024

(Unaudited and in thousands)

	For the three months ended March 31,	
	2025	**2024**
Cash flows from operating activities		
Net loss	$(8,284)	$(9,399)
Adjustments to reconcile net loss to net cash provided by (used in) operating activities:		
Depreciation and amortization	1,961	1,557
Stock-based compensation	851	551
Gain on receipt of nonmonetary tangible assets	(1,600)	(1,600)
Change in right-of-use assets	242	240
Other	(58)	—
Changes in operating assets and liabilities		
Accounts receivable	6,349	4,860
Inventories	(286)	1,196
Prepaid expenses and other assets	66	50
Other long-term assets	—	41
Accounts payable	2,255	1,707
Accrued and other current liabilities	(90)	(1,877)
Other long-term liabilities	—	(183)

Net cash provided by (used in) operating activities	1,406	(2,857)
Cash flows from investing activities		
Purchase of intangible assets	(1,000)	(640)
Purchases of property, equipment and software	(116)	(45)
Net cash used in investing activities	(1,116)	(685)
Cash flows from financing activities		
Proceeds from issuance of preferred stock, net of issuance costs	—	7,949
Proceeds from exercise of stock options	148	202
Net cash provided by financing activities	148	8,151
Effect of exchange rate changes on cash and cash equivalents	(1)	(18)
Net increase in cash and cash equivalents	437	4,591
Cash and cash equivalents at beginning of period	60,981	27,321
Cash and cash equivalents at end of period	$61,418	$31,912
Supplemental disclosure of non-cash investing and financing activities		
Intangible assets in accounts payable, accrued and other long-term liabilities	6,004	—
Gain on receipt of nonmonetary tangible asset	1,600	1,600
Right-of-use assets obtained in exchange for new operating lease liabilities	383	321
Deemed dividends	—	375

Ambiq Micro, Inc.

Notes to the Condensed Consolidated Financial Statements

(Unaudited)

1. Description of Business

Ambiq Micro, Inc. (the “Company”) is a fabless semiconductor company that has developed technology based on a patented Sub-threshold Power Optimized Technology (“SPOT”) platform that significantly reduces the amount of power consumed by integrated circuits. The following subsidiaries were formed by the Company and are wholly owned:

Ambiq Micro Singapore Private Ltd.
Shenzhen DeKean Electronics Co.
Ambiq Micro Asia Ltd.
Ambiq Micro Asia Ltd. Taiwan
Ambiq (Shenzhen) Electronics Co., Ltd.

Global Economic Considerations

The global macroeconomic environment is uncertain, and could be negatively affected by, among other things, increased U.S. disputes with countries who are existing trade partners, instability in the global capital and credit markets, supply chain weaknesses, and instability in the geopolitical environment in Asia, Europe and the Middle East, and other tensions. Deterioration in economic factors arising from trade disputes between the United States and China, in particular, could have an adverse impact on our financial results given our customer concentrations in both countries. Such challenges have caused, and may continue to cause, recession fears, rising interest rates, foreign exchange volatility and inflationary pressures through the imposition of tariffs on sensitive technologies, including semiconductors.

Despite the uncertainty of these challenges, the Company believes it has considered all appropriate factors when making financial estimates and will continue to evaluate the impact of any significant changes resulting from these and other factors affecting the Company’s business performance.

Liquidity and Capital Resources

During the three months ended March 31, 2025, the Company reported a net loss of $8.3 million and had an operating cash flow surplus of $1.4 million. As of March 31, 2025, the Company had cash and cash equivalents totaling $61.4 million. Since inception, the Company has had negative cash flows and losses from operations which it has funded primarily through the issuances of common and preferred stock. The Company anticipates incurring additional losses and negative cash flows from operations until such time, if ever, that it can generate significant sales of its products currently in development and is highly dependent on its ability to find additional sources of funding in the form of debt and equity financing. The Company's ability to fund its planned operations, research and development, and commercialization of its products significantly depends on the amount and timing of cash receipts from these funding sources. Adequate additional funding may not be available to the Company on acceptable terms or at all. The failure to raise funds as and when needed and generate adequate sales could have a negative impact on the Company's financial condition and ability to satisfy its obligations over the next twelve months.

2. Summary of Significant Accounting Policies

Basis of Accounting and Consolidation

The accompanying unaudited condensed consolidated financial statements have been prepared by the Company, and reflect all adjustments, consisting only of normal recurring adjustments, that are, in the opinion of

management, necessary for the fair presentation of our financial position, results of operations, cash flows and stockholders' equity for the interim periods presented. The statements have been prepared in accordance with accounting principles generally accepted in the United States of America ("U.S. GAAP") and the applicable rules and regulations of the Securities and Exchange Commission ("SEC") for interim financial information and include the accounts of the Company and its wholly owned subsidiaries. Accordingly, these statements do not include all information and footnotes required by U.S. GAAP for annual consolidated financial statements, and should be read in conjunction with our audited consolidated financial statements as of and for the fiscal year ended December 31, 2024. The results of operations for the three months ended March 31, 2025 are not necessarily indicative of the operating results expected for the full fiscal year or future operating periods.

The condensed consolidated balance sheet as of December 31, 2024 has been derived from the Company's audited consolidated financial statements. All significant intercompany balances and transactions have been eliminated in consolidation.

There have been no material changes to the Company's significant accounting policies described in Note 2—Summary of Significant Accounting Policies, of the notes to the Company's audited consolidated financial statements.

Unaudited Pro Forma Information

Immediately prior to the closing of a qualified initial public offering ("IPO"), all of the Company's outstanding redeemable convertible preferred stock will automatically convert into 356,423,629 shares of common stock based on the number of shares outstanding and the applicable conversion price as of March 31, 2025. In the accompanying statements of operations and comprehensive loss, unaudited pro forma basic and diluted net loss per share of common stock has been prepared to give effect to the automatic conversion of all outstanding shares of redeemable convertible preferred stock as if they had been converted at the beginning of the reporting period. The shares of the Company's common stock expected to be issued, and the related net proceeds expected to be received, in connection with the planned IPO are excluded from such pro forma information.

Concentration of Risks

Financial Instruments

Financial instruments that potentially subject the Company to concentrations of credit risk consist primarily of cash and cash equivalents and trade accounts receivable. The Company maintains its cash and cash equivalent balances in highly rated financial institutions, which at times may exceed federally insured limits or be held in foreign jurisdictions. The Company has not experienced any loss relating to cash and cash equivalents in these accounts and believes no significant concentration risk exists with respect to cash. The Company performs periodic credit evaluations of its customers' financial conditions and generally does not require collateral.

Demand

The Company had three customers representing 39%, 34% and 13%, respectively, of total accounts receivable as of March 31, 2025. There were three customers representing 38%, 25% and 23%, respectively, of total net sales for the three months ended March 31, 2025.

The Company had three customers representing 36%, 24% and 24%, respectively, of total accounts receivable as of March 31, 2024. There were four customers representing 28%, 26%, 19%, and 16%, respectively, of total net sales for the three months ended March 31, 2024.

The loss of one or more of these customers could have a material adverse impact on the Company's results of operations and financial position.

Supply

The Company's products depend on a sole supplier of wafers and a limited number of third-party manufacturers. The continued and timely supply of input materials and the availability of manufacturing capacity and packaging and testing services impact the Company's ability to meet customer demand. Supply chain disruptions, shortages of raw materials, and manufacturing limitations could limit the Company's ability to meet customer demand and result in delayed, reduced, or canceled orders. The Company has established relationships with leading suppliers and partners, and it believes these relationships increase the resiliency of our supply chain for its customers. From time to time, subject to inventory disruptions, the Company's customers may buy and hold excess inventories. Consequently, the Company may be subject to resulting fluctuations in the demand for its products.

End Customer Concentration

Although the Company recognizes revenue and directly invoice distributors for sales of its products, the timing and uncertainty of its revenue and cash flows are most impacted by the ultimate end customer. The following is a summary of net sales for the three months ended March 31, 2025 and 2024, based on the country of the corporate headquarters of the ultimate end customer (in thousands):

	Three months ended March 31,	
	2025	2024
United States	$13,752	$7,214
China	968	7,636
Rest of the World*	1,012	360
Total	$15,732	$15,210

* Other countries individually less than 10%

Recently Issued Accounting Pronouncements

In December 2023, the FASB issued ASU No. 2023-09, *Income Taxes (Topic 740): Improvements to Income Tax Disclosures*, which will require the Company to disclose specified additional information in its income tax rate reconciliation and provide additional information for reconciling items that meet a quantitative threshold. The update will also require the Company to disaggregate its income taxes paid disclosure by federal, state and foreign taxes, with further disaggregation required for significant individual jurisdictions. The Company will adopt this amendment effective with its annual financial statements for the year ending December 31, 2025. The Company is currently evaluating the potential impact of adopting this guidance on the notes to the consolidated financial statements.

3. Net Loss Per Share

The table below sets forth the computation of basic and diluted loss per share for the periods presented (in thousands, except share and per share amounts):

	Three months ended March 31,	
	2025	2024
Numerator:		
Net loss	$(8,284)	$(9,399)
Deemed dividends	—	(375)
Net loss attributable to common stockholder	$(8,284)	$(9,774)
Denominator:		
Weighted average shares outstanding	12,232,916	10,102,901
Effect of dilutive shares	—	—
Basic and diluted loss per share	$(0.68)	$(0.97)

Since the Company incurred a net loss for the three months ended March 31, 2025 and 2024, the diluted net loss per share calculation excludes potentially dilutive securities. The following table summarizes the securities that were excluded from the calculation of diluted earnings per share because their inclusion would have been anti-dilutive:

	Three months ended March 31,	
	2025	2024
Redeemable convertible preferred stock	341,496,158	286,086,785
Common warrants	18,833,721	12,143,652
Preferred warrants	290,563	290,563
Restricted stock units	5,186,932	520,270
Stock options	61,795,893	48,613,851
Total shares	427,603,267	347,655,121

The following table summarizes the calculation of unaudited pro forma basic and diluted earnings per share for the periods presented (in thousands, except share and per share amounts):

	Three months ended March 31,	
	2025	2024
Numerator:		
Net loss used to compute pro forma net loss per share	$(8,284)	$(9,399)
Deemed dividend	—	(375)
Net loss attributable to common stockholder used to compute pro forma net loss per share	$(8,284)	$(9,774)
Denominator:		
Weighted average shares outstanding	12,232,916	10,102,901
Pro forma adjustment to reflect the automatic conversion of all convertible preferred stock to common stock upon an IPO	356,423,629	356,423,629
Pro forma weighted average number of shares outstanding-basic and diluted	368,656,545	366,526,530
Unaudited pro forma net loss per share-basic and diluted	$(0.02)	$(0.03)

The Company recorded deemed dividends of $0.4 million for the three months ended March 31, 2024, as a result of a down round feature in the Company's Series E, F, and F-1 preferred stock triggered during the issuance by the Company of Series G preferred stock (see Note 7). No deemed dividends were recorded for the three months ended March 31, 2025.

4. Inventories

The following table represents the components of inventories as of March 31, 2025 and December 31, 2024 (in thousands):

	March 31, 2025	December 31, 2024
Raw materials	$297	$45
Work in progress	9,555	9,547
Finished goods	5,442	5,416
Total inventories	$15,294	$15,008

5. Property, Equipment and Software

Property, equipment and software consisted of the following as of March 31, 2025, and December 31, 2024 (in thousands):

	March 31, 2025	December 31, 2024
Probe cards and photomasks	$11,782	$10,083
Equipment	4,446	4,386
Software	302	298
Leasehold improvements	773	728
Furniture and fixtures	311	279
Total	17,614	15,774
Less: Accumulated depreciation and amortization	(13,588)	(13,158)
Property, equipment and software, net	$4,026	$2,616

Depreciation and amortization expense relating to the Company's property, equipment and software was approximately $0.4 million and $0.4 million for the three months ended March 31, 2025 and 2024, respectively, and are allocated to cost of sales, selling, general and administrative, and research and development costs in the accompanying unaudited condensed consolidated statements of operations and comprehensive loss.

During the three months ended March 31, 2025 and 2024, the Company received a mask set in exchange for no consideration from a vendor in lieu of reimbursement. The mask set was capitalized and a $1.6 million gain was recorded within cost of sales in accordance with ASC 845: *Nonmonetary Transactions*.

6. Commitments and Contingencies

Contract Manufacturer Commitments

The Company relies on a third-party foundry and contract manufacturer for the manufacturing of its products. Generally, its foundry agreements do not have volume purchase commitments and primarily provide for purchase commitments based on purchase orders. Purchase orders are placed in advance with consideration of estimates of future demand. These purchase orders can be canceled and rescheduled upon agreement of the Company and the contract manufacturer. As of March 31, 2025, the Company had total manufacturing purchase commitments of $7.6 million.

Litigation

From time to time, the Company may become involved in various legal actions arising in the ordinary course of business. As of March 31, 2025, management was not aware of any existing, pending, or threatened legal actions that would have a material impact on the financial position, results of operations, or cash flows of the Company.

7. Warrants, Redeemable Convertible Preferred Stock and Stockholders' Deficit

Common Stock

The Company has one class of authorized common stock and has authorized for issuance up to 500,000,000 shares of common stock with a par value of $0.000001 per share.

The holders of common stock are entitled to receive dividends at the discretion of the board of directors, subject to preferences that may apply to shares of preferred stock outstanding at the time.

All holders of common stock are entitled to one vote per share on all matters to be voted on by the Company's stockholders. Upon liquidation, dissolution or winding up, the holders of common stock are entitled to share equally in all the Company's assets remaining after payment of all liabilities and the liquidation preferences of any outstanding preferred stock.

At March 31, 2025, the Company has reserved 451,889,930 shares of common stock for the conversion or exercise, as applicable, of the following securities:

Redeemable convertible preferred stock	341,496,158
Preferred warrants	290,563
Common warrants	18,833,721
Restricted stock units	5,186,932
Stock options	61,795,893
Total shares	427,603,267

Redeemable Convertible Preferred Stock

During the three months ended March 31, 2025, the Company did not issue any additional shares of Series G preferred stock.

During the three months ended March 31, 2024, the Company issued 8,847,805 shares of Series G preferred stock with an original issuance price of $0.90 per share for cash proceeds of $7.9 million. The Company had issuance costs of approximately $48.4 thousand associated with this transaction.

All classes of preferred stock have a par value of $0.000001 per share. The Company's preferred stock is recorded in the temporary equity section of the Company's consolidated balance sheets.

A general summary of the rights with respect to the preferred stock is provided below.

Dividends

The holders of preferred stock, in preference to the holders of common stock, shall be entitled to receive on a pari passu basis, out of funds that are legally available, cash dividends at the rate of 8% of the applicable original issue price per annum on each outstanding share of preferred stock. All original issue prices are subject to adjustment for any stock dividends, combinations, splits, recapitalizations and similar events.

Dividends are non-cumulative, and the Company has no obligation to pay dividends until declared by the Company's board of directors. Through March 31, 2025, no dividends have been declared by the Company's board of directors.

Liquidation Preference

Upon any liquidation, dissolution, or winding up of the Company, whether voluntary or involuntary, before any distribution or payment shall be made to the holders of any common stock, the holders of preferred stock shall be

entitled to be paid on a pari passu basis out of the assets of the Company legally available for distribution (or the consideration received by the Company or its stockholders in an acquisition) for each share of preferred stock held by them, an amount per share of preferred stock equal to the original issue price for such share of preferred stock plus all declared and unpaid dividends on such share of preferred stock.

If upon liquidation, the assets of the Company are insufficient to make payment in full to all holders of preferred stock, then such assets shall be distributed among the holders of preferred stock at the time outstanding, ratably in proportion to the full amounts to which they would otherwise be respectively entitled.

Conversion

Each share of preferred stock shall be convertible at the option of the holder, at any time, into fully paid and nonassessable shares of common stock. The number of shares of common stock to which a holder of preferred stock shall be entitled upon conversion shall be the product obtained by multiplying the applicable "Series Preferred Conversion Rate", by the number of shares of such preferred stock being converted, as defined in the Company's latest amended and restated certificate of incorporation. The Series Preferred Conversion Rate is the quotient obtained by dividing the Original Issue Price of such series of Series Preferred by the applicable "Series Preferred Conversion Price". The Series Preferred Conversion Price is generally the Original Issuance Price for all series of Series Seed, A, B, C, D and G preferred stock, subject to certain adjustments.

Voting

The holders of preferred stock shall vote together with the holders of common stock and not as a separate class and may act by written consent in the same manner as the holders of common stock. Each holder of preferred stock shall be entitled to the number of votes equal to the number of shares of common stock into which such shares of preferred stock could be converted immediately after the close of business on the record date fixed for such meeting or the effective date of such written consent and shall have voting rights and powers equal to the voting rights and powers of the holders of common stock and shall be entitled to notice of any stockholders' meeting in accordance with the bylaws of the Company.

Redemption

The Company's preferred stock is not generally redeemable except in situations set forth in the certificate of incorporation, including when more than fifty percent of the Company's voting power is transferred, a triggering event is created that is outside the control of the Company, and a change in control of Company ownership.

Notes to the Condensed Consolidated Financial Statements

(Unaudited)

See the table below for a roll forward of the total preferred stock from January 1, 2024 to March 31, 2024 and January 1, 2025 to March 31, 2025, which details the total redeemable convertible preferred stock presented in the unaudited condensed consolidated statements of changes in redeemable convertible preferred stock and stockholders' deficit (in thousands, except share amounts):

	Series Seed Redeemable convertible preferred stock		Series A Redeemable convertible preferred stock		Series B Redeemable convertible preferred stock		Series C Redeemable convertible preferred stock		Series D Redeemable convertible preferred stock		Series E Redeemable convertible preferred stock
	Shares	Amount	Shares	Amount	Shares	Amount	Shares	Amount	Shares	Amount	Shares
Balance at January 1, 2024	5,045,368	$2,473	8,255,614	$5,179	18,894,315	$10,827	42,890,840	$24,908	39,656,811	$33,411	29,019,153
Issuance of Series G preferred stock, net	–	–	–	–	–	–	–	–	–	–	–
Balance at March 31, 2024	**5,045,368**	**$2,473**	**8,255,614**	**$5,179**	**18,894,315**	**$10,827**	**42,890,840**	**$24,908**	**39,656,811**	**$33,411**	**29,019,153**
Balance at January 1, 2025	5,045,368	$2,473	8,255,614	$5,179	18,894,315	$10,827	42,890,840	$24,908	39,656,811	$33,411	29,019,153
Balance at March 31, 2025	**5,045,368**	**$2,473**	**8,255,614**	**$5,179**	**18,894,315**	**$10,827**	**42,890,840**	**$24,908**	**39,656,811**	**$33,411**	**29,019,153**

8. Stock Option Plan and Stock-Based Compensation

The following table summarizes the activity in total shares of common stock available for issuance under the 2020 Plan:

	Shares Available for Grant
Balance, January 1, 2025	11,134,315
Shares added	–
Granted	(1,999,998)
Forfeited	224,875
Balance, March 31, 2025	9,359,192

The following table summarizes the effects of stock-based compensation on cost of sales, research and development, and sales, general and administrative expenses granted under the Plans (in thousands) for the three months ended March 31, 2025 and 2024:

	Three months ended March 31,	
	2025	2024
Cost of sales	$60	$30

Research and development	389	275
Sales, general and administrative	402	246
Total	$851	$551

The total unrecognized stock-based compensation expense related to unvested stock options and subject to recognition in future periods was approximately $3.3 million for the three months ended March 31, 2025. As of March 31, 2025, the Company anticipates this expense to be recognized over a weighted-average period of approximately 1.5 years.

During the three months ended March 31, 2025, the Company received $0.1 million from the exercise of stock options granted under the Plans and granted 1,999,998 RSUs, to several non-employee directors which vest upon satisfaction of a service-based condition and performance condition based on a liquidity event. No RSUs were granted during the three months ended March 31, 2024.

9. Income Taxes

Effective Tax Rate

The following table presents the provision for income taxes and the effective tax rates for the three months ended March 31, 2025 and 2024 (in thousands):

	Three months ended March 31,	
	2025	2024
Loss before income taxes	$(8,280)	$(9,394)
Income tax expense	(4)	(5)
Effective tax rate	0.0%	0.1%

The income tax amount for each of the quarters ended March 31, 2025 and 2024, differs from the amount that would be expected after applying the statutory U.S. federal income tax rate primarily due to an increase in the valuation allowance. The effective tax rate was less than 1% for both the three months ended March 31, 2025 and 2024. The provision for income taxes is primarily related to the foreign subsidiaries' local country obligations. There is no federal provision for income taxes as the Company has sufficient carryforward of net operating losses to offset any operating income earned since inception and has projected an operating loss in the current year.

In assessing the realizability of deferred tax assets, management considers whether it is more likely than not that some portion or all of the deferred tax assets will not be realized. The ultimate realization of deferred tax assets is dependent upon the generation of future taxable income during the periods in which those temporary differences become deductible. Due to the uncertainty of the Company's ability to realize the benefit of the deferred tax assets, the net deferred tax assets are fully offset by a valuation balance at March 31, 2025 and 2024.

10. Related Party Transactions

The Company defines related parties as any party that controls or can significantly influence the management or operating policies of the Company to the extent that the Company may be prevented from fully pursuing its own interests, such as directors, executive officers, and stockholders, including a greater than 10% beneficial owner of the Company's capital and their affiliates or immediate family members. For the three months ended March 31, 2025 and 2024, the Company did not have any related party transactions.

11. Segment and Geographic Information

The Company's chief operating decision maker ("CODM") is the Company's Chief Executive Officer ("CEO") and reviews the financial information presented on a consolidated basis for purposes of making operating decisions, assessing financial performance, and allocating resources. The CODM uses revenue, gross margin, operating expenses and net loss by its single operating and reportable segment to make strategic business decisions.

The following table sets forth the Company's disaggregation of operating expenses which were reviewed by the CODM for the three months ended March 31, 2025 and 2024 (in thousands):

	Three months ended March 31,	
	2025	2024
Research and development	$8,687	$9,436
Sales and marketing	2,549	2,725
General and administrative	5,894	3,263
Total operating expenses	$17,130	$15,424

The following is a summary of net sales for the three months ended March 31, 2025 and 2024, based on the country to which the products were shipped, which may be different from the geographic locations of the ultimate end customers (in thousands):

	Three months ended March 31,	
	2025	2024
Taiwan	$6,032	$2,649
China	5,590	11,823
Singapore	3,599	—
Rest of the World*	511	738
Total	$15,732	$15,210

* Other countries individually less than 10%

Ambiq Micro, Inc.

Notes to the Condensed Consolidated Financial Statements

(Unaudited)

The following illustrates property, equipment and software, net, and right-of-use assets, net by geographic locations based on physical location (in thousands):

	March 31, 2025	December 31, 2024
Taiwan	$3,617	$2,139
China	826	593
North America	626	786
Rest of the World*	24	26
Total	$5,093	$3,544

* Other countries individually less than 10%

12. Subsequent Events

The Company performed a review of events subsequent to the unaudited balance sheet date through the date the unaudited consolidated financial statements are available to be issued on June 13, 2025, and has determined there are no such events that require disclosure

Shares

AMBIQ MICRO, INC.

Common Stock

BofA Securities	UBS Investment Bank
Needham & Company	Stifel

, 2025

Part Ii

Information Not Required in Prospectus

Item 13. Other Expenses of Issuance and Distribution.

The following table sets forth all costs and expenses, other than underwriting discounts and commissions, payable by Ambiq Micro, Inc. or the Registrant, in connection with the sale of the common stock being registered. All amounts shown are estimates except for the Securities and Exchange Commission (the SEC), registration fee, the FINRA filing fee and the New York Stock Exchange listing fee.

	Amount
SEC registration fee	$ *
FINRA filing fee	*
New York Stock Exchange listing fee	*
Printing and engraving expenses	*
Legal fees and expenses	*
Accounting fees and expenses	*
Transfer agent and registrar fees and expenses	*
Miscellaneous fees and expenses	*
Total	$ *

* To be provided by amendment.

Item 14. Indemnification of Directors and Officers.

The Registrant is incorporated under the laws of the State of Delaware. Section 145 of the Delaware General Corporation Law (the DGCL), provides that a Delaware corporation may indemnify any persons who were, are, or are threatened to be made, parties to any threatened, pending, or completed action, suit, or proceeding, whether civil, criminal, administrative or investigative (other than an action by or in the right of such corporation), by reason of the fact that such person is or was an officer, director, employee, or agent of such corporation, or is or was serving at the request of such corporation as an officer, director, employee, or agent of another corporation or enterprise. The indemnity may include expenses (including attorneys' fees), judgments, fines, and amounts paid in settlement actually and reasonably incurred by such person in connection with such action, suit, or proceeding, provided that such person acted in good faith and in a manner he or she reasonably believed to be in or not opposed to the corporation's best interests and, with respect to any criminal action or proceeding, had no reasonable cause to believe that his or her conduct was illegal. A Delaware corporation may indemnify any persons who were, are, or are threatened to be made, a party to any threatened, pending, or completed action or suit by or in the right of the corporation by reason of the fact that such person is or was a director, officer, employee, or agent of such corporation, or is or was serving at the request of such corporation as a director, officer, employee, or agent of another corporation or enterprise. The indemnity may include expenses (including attorneys' fees) actually and reasonably incurred by such person in connection with the defense or settlement of such action or suit provided such person acted in good faith and in a manner he or she reasonably believed to be in or not opposed to the corporation's best interests, except that no indemnification is permitted without judicial approval if the officer or director is adjudged to be liable to the corporation. Where an officer or director is successful on the merits or otherwise in the defense of any action referred to above, the corporation must indemnify him or her against the expenses (including attorneys' fees) actually and reasonably incurred.

The Registrant's amended and restated certificate of incorporation provides for the indemnification of its directors to the fullest extent permitted under the DGCL. The Registrant's amended and restated bylaws provide for the indemnification of its directors and officers to the fullest extent permitted under the DGCL. Each of the Registrant's amended and restated certificate of incorporation and amended and restated bylaws will become effective upon the closing of this offering.

Section 102(b)(7) of the DGCL permits a corporation to provide in its certificate of incorporation that a director of the corporation shall not be personally liable to the corporation or its stockholders for monetary damages for breach of fiduciary duties as a director, except for liability for any:

- transaction from which the director derives an improper personal benefit;
- act or omission not in good faith or that involves intentional misconduct or a knowing violation of law;
- unlawful payment of dividends or redemption of shares; or
- breach of a director's duty of loyalty to the corporation or its stockholders.

The Registrant's amended and restated certificate of incorporation includes such a provision. Under the Registrant's amended and restated bylaws, expenses incurred by any director or officers in defending any such action, suit, or proceeding in advance of its final disposition shall be paid by the Registrant upon delivery to it of an undertaking, by or on behalf of such director or officer, to repay all amounts so advanced if it shall ultimately be determined that such director or officer is not entitled to be indemnified by the Registrant, as long as such undertaking remains required by the DGCL.

Section 174 of the DGCL provides, among other things, that a director who willfully or negligently approves of an unlawful payment of dividends or an unlawful stock purchase or redemption, may be held liable for such actions. A director who was either absent when the unlawful actions were approved or dissented at the time may avoid liability by causing his or her dissent to such actions to be entered in the books containing minutes of the meetings of the board of directors at the time such action occurred or immediately after such absent director receives notice of the unlawful acts.

As permitted by the DGCL, the Registrant has entered into indemnity agreements with each of its directors and officers that require the Registrant, among other things, to indemnify its directors and officers against certain liabilities which may arise by reason of their status or service as directors or officers to the fullest extent not prohibited by law. These indemnification agreements may be sufficiently broad to permit indemnification of the Registrant's officers and directors for liabilities, including reimbursement of expenses incurred, arising under the Securities Act. Under these agreements, the Registrant is not required to provide indemnification for certain matters. The indemnification agreements also set forth certain procedures that will apply in the event of a claim for indemnification thereunder.

There is at present no pending litigation or proceeding involving any of the Registrant's directors or executive officers as to which indemnification is required or permitted, and the Registrant is not aware of any threatened litigation or proceeding that may result in a claim for indemnification.

The Registrant intends to enter into an insurance policy that covers its officers and directors with respect to certain liabilities, including liabilities arising under the Securities Act or otherwise.

The Registrant plans to enter into an underwriting agreement which provides that the underwriters are obligated, under some circumstances, to indemnify the Registrant's directors, officers, and controlling persons against specified liabilities, including liabilities under the Securities Act.

Item 15. Recent Sales of Unregistered Securities.

Set forth below is information regarding unregistered securities issued by us since January 1, 2022.

`Equity Plan-Related Issuances`

1. We have granted to certain of our directors, employees and consultants options to purchase 51,378,579 shares of our common stock with per share exercise prices ranging from $0.10 to $0.33 under the 2010 Plan.

2. We have granted to certain of our directors, employees and consultants options to purchase 46,179,913 shares of our common stock with per share exercise prices ranging from $0.38 to $1.20 under the 2020 Plan.

3. We have issued to certain of our directors 5,186,932 shares of common stock having fair market values ranging from $0.38 to $1.20 per share.

4. We have issued to certain of our directors, employees and consultants an aggregate of 8,457,280 shares of our common stock at per share purchase prices ranging from $0.10 to $0.33 pursuant to exercises of options under the 2010 Plan for an aggregate purchase price of $1,694,358.

5. We have issued to certain of our directors, employees and consultants an aggregate of 1,663,562 shares of our common stock at per share purchase prices ranging from $0.45 to $1.20 pursuant to exercises of options under the 2020 Plan for an aggregate purchase price of $1,018,370.

`Other Issuances of Capital Stock`

6. In multiple closings held between February 2022 and March 2022, we issued and sold an aggregate of 23,664,027 shares of our Series F-1 redeemable convertible preferred stock at a purchase price of $2.091505 per share for an aggregate purchase price of approximately $49.5 million.

7. On multiple dates between February 2022 and July 2022, we issued an aggregate of 776,307 shares of our Series A redeemable convertible preferred stock upon the exercise of warrants at a purchase price of $0.55 per share for an aggregate purchase price of approximately $427,000.

8. On multiple dates between February 2022 and June 2023, we issued an aggregate of 637,113 shares of our Series B redeemable convertible preferred stock upon the exercise of warrants at a purchase price of $0.55 per share for an aggregate purchase price of approximately $350,412.

9. In multiple closings held between August 2023 and July 2024, we issued and sold an aggregate of 99,229,883 shares of our Series G redeemable convertible preferred stock at a purchase price of $0.904179 per share for an aggregate purchase price of approximately $89.7 million.

10. On June 1, 2024, we issued a warrant to purchase up to 6,690,069 shares of common stock at an exercise price of $0.45 per share. Such warrant is exercisable until June 1, 2034.

11. On October 19, 2022, we issued a warrant to purchase up to 100,000 shares of common stock at an exercise price of $1.11 per share. Such warrant is exercisable until October 18, 2032.

The offers, sales and issuances of the securities described in paragraphs (1) through (3) were deemed to be exempt from registration under Rule 701 promulgated under the Securities Act as transactions under compensatory benefit plans and contracts relating to compensation, or under Section 4(a)(2) of the Securities Act as a transaction by an issuer not involving a public offering. The recipients of such securities were our directors, employees or bona fide consultants and received the securities under our equity incentive plans. Appropriate legends were affixed to the securities issued in these transactions. Each of the recipients of securities in these transactions had adequate access, through employment, business or other relationships, to information about us.

The offers, sales and issuances of the securities described in paragraphs (4) through (9) were deemed to be exempt under Section 4(a)(2) of the Securities Act or Rule 506 of Regulation D under the Securities Act as a transaction by an issuer not involving a public offering. The recipients of securities in each of these transactions acquired the securities for investment only and not with a view to or for sale in connection with any distribution thereof and appropriate legends were affixed to the securities issued in these transactions. Each of the recipients of securities in these transactions was an accredited investor within the meaning of Rule 501 of Regulation D under the Securities Act and had adequate access, through employment, business or other relationships, to information about us. No underwriters were involved in these transactions.

Item 16. Exhibits and Financial Statement Schedules.

(a)	Exhibits.

The list of exhibits is set forth under "Exhibit Index" at the end of this registration statement and is incorporated herein by reference.

(b)	Financial Statement Schedules.

No financial statement schedules are provided because the information called for is not required or is shown either in the financial statements or the notes thereto.

Item 17. Undertakings.

Insofar as indemnification for liabilities arising under the Securities Act may be permitted to directors, officers, and controlling persons of the Registrant pursuant to the foregoing provisions, or otherwise, the Registrant has been advised that in the opinion of the SEC such indemnification is against public policy as expressed in the Securities Act and is, therefore, unenforceable. In the event that a claim for indemnification against such liabilities (other than the payment by the Registrant of expenses incurred or paid by a director, officer, or controlling person of the Registrant in the successful defense of any action, suit, or proceeding) is asserted by such director, officer, or controlling person in connection with the securities being registered, the Registrant will, unless in the opinion of its counsel the matter has been settled by controlling precedent, submit to a court of appropriate jurisdiction the question of whether such indemnification by it is against public policy as expressed in the Securities Act and will be governed by the final adjudication of such issue.

The undersigned Registrant hereby undertakes that:

(a) For purposes of determining any liability under the Securities Act, the information omitted from the form of prospectus filed as part of this registration statement in reliance upon Rule 430A and contained in a form of prospectus filed by the Registrant pursuant to Rule 424(b)(1) or (4) or 497(h) under the Securities Act shall be deemed to be part of this registration statement as of the time it was declared effective.

(b) For the purpose of determining any liability under the Securities Act, each post-effective amendment that contains a form of prospectus shall be deemed to be a new registration statement relating to the securities offered therein, and the offering of such securities at that time shall be deemed to be the initial bona fide offering thereof.

Exhibit Index

Exhibit No.	Description
1.1†	Form of Underwriting Agreement.
3.1	Amended and Restated Certificate of Incorporation, as amended and as currently in effect.
3.2	Certificate of Amendment to Amended and Restated Certificate of Incorporation, as amended and as currently in effect.
3.3†	Form of Amended and Restated Certificate of Incorporation, to be effective upon the closing of this offering.
3.4	Bylaws, as amended and as currently in effect.
3.5†	Form of Amended and Restated Bylaws, to be effective upon the closing of this offering.
4.1	Form of Common Stock Certificate of the Registrant.
4.2†	Eighth Amended and Restated Investor Rights Agreement, dated August 31, 2023, by and among the Registrant and certain of its stockholders.
5.1†	Opinion of Cooley LLP.
10.1 +	Form of Indemnification Agreement.
10.2 †+	Non-Employee Director Compensation Policy.
10.3 +	Ambiq Micro, Inc. 2010 Equity Incentive Plan, as amended.
10.4 +	Forms of Incentive Stock Option Award Notice, Incentive Stock Option Award Agreement, Exercise Notice and Investment Representation Statement under the 2010 Equity Incentive Plan.
10.5 +	Ambiq Micro, Inc. 2020 Equity Incentive Plan, as amended.
10.6 +	Forms of Incentive Stock Option Award Notice, Incentive Stock Option Award Agreement, Exercise Notice and Investment Representation Statement under the 2020 Equity Incentive Plan.
10.7†+	Ambiq Micro, Inc. 2025 Equity Incentive Plan.
10.8†+	Forms of Incentive Stock Option Award Notice, Incentive Stock Option Award Agreement, Exercise Notice and Investment Representation Statement under the 2025 Equity Incentive Plan.
10.9†+	Ambiq Micro, Inc. 2025 Employee Stock Purchase Plan.
10.10+	Offer of Employment, dated September 25, 2015, between Sean Chen and the Company.
10.11+	Offer of Employment, dated November 2, 2015, between Fumihide Esaka and the Company.
10.12+	Offer of Employment, dated April 16, 2025, between Jeff Winzeler and the Company.
10.13	TSMC Master Technology License Agreement, dated as of January 27, 2011, by and between the Company and Taiwan Semiconductor Manufacturing Co., Ltd.
10.14	Lease, dated November 11, 2016, by and between the Company and G&I VII River Place LP. and Confirmation of Lease Term, dated March 31, 2017, by and between the Company and G&I VII River Place LP.
10.15	First Amendment to Lease, dated September 6, 2019, by and between the Company and G&I VII River Place LP.
10.16	Second Amendment to Lease, dated December 11, 2020, by and between the Company and G&I River Place LP.

Exhibit No.	Description
10.17	Third Amendment to Lease, dated November 29, 2022, by and between the Company and G&I River Place LP.
10.18	Form of 2025 Indemnification Agreement.
21.1	Subsidiaries of the Registrant.
23.1	Consent of KPMG LLP, an Independent Registered Public Accounting Firm.
23.2†	Consent of Cooley LLP (included in Exhibit 5.1).
24.1	Power of Attorney (see signature page hereto)
107	Filing Fee Exhibit

† To be filed by amendment.

+ Indicates management contract or compensatory plan.

Signatures

Pursuant to the requirements of the Securities Act of 1933, as amended, the registrant has duly caused this registration statement to be signed on its behalf by the undersigned, thereunto duly authorized, in the City of Austin, State of Texas, on July 3, 2025.

AMBIQ MICRO, INC.
/s/ Fumihide Esaka
Fumihide Esaka
Chief Executive Officer

Power of Attorney

KNOW ALL BY THESE PRESENTS, that each person whose signature appears below constitutes and appoints Fumihide Esaka and Scott Hanson, Ph.D., and each of them, as his or her true and lawful attorneys-in-fact and agents, each with the full power of substitution, for him or her and in his or her name, place, or stead, in any and all capacities, to sign any and all amendments to this registration statement (including post-effective amendments), and to sign any registration statement for the same offering covered by this registration statement that is to be effective upon filing pursuant to Rule 462(b) promulgated under the Securities Act, and all post- effective amendments thereto, and to file the same, with exhibits thereto and other documents in connection therewith, with the Securities and Exchange Commission, granting unto said attorneys-in-fact and agents, and each of them, full power and authority to do and perform each and every act and thing requisite and necessary to be done in and about the premises, as fully to all intents and purposes as he or she might or could do in person, hereby ratifying and confirming all that said attorneys-in-fact and agents, or his or her substitute or substitutes, may lawfully do or cause to be done by virtue hereof.

Pursuant to the requirements of the Securities Act of 1933, as amended, this registration statement has been signed by the following persons in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Fumihide Esaka Fumihide Esaka	Chief Executive Officer and Director *(Principal Executive Officer)*	July 3, 2025
/s/ Jeff Winzeler Jeff Winzeler	Chief Financial Officer (*Principal Financial and Accounting Officer*)	July 3, 2025
/s/ Scott Hanson Scott Hanson, Ph.D.	Director	July 3, 2025
/s/ Wen Hsieh Wen Hsieh, Ph.D.	Director	July 3, 2025
/s/ Ker Zhang Ker Zhang, Ph.D.	Director	July 3, 2025
/s/ Joseph Tautges Joseph Tautges	Director	July 3, 2025
/s/ Timothy Chen Timothy Chen	Director	July 3, 2025